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HEAVY QUARKONIUM SUPPRESSION IN THE QUARK-GLUON PLASMA
USING REAL-TIME QUANTUM FIELD THEORY FOR AN OPEN QUANTUM SYSTEM

Dissertation Advisor: Prof. Michael Strickland

Numerous previous investigations of in-medium quarkonium suppression have tacitly relied on an adiabatic approximation, presuming that the potential governing heavy quark interactions evolves slowly with time. In this adiabatic scenario, one can decouple the calculation of the in-medium breakup rate and the temporal evolution of the medium, combining them only at the conclusion of the analysis. We relax this assumption by solving the 3d Schrödinger equation in real-time in order to compute quarkonium suppression dynamically.

We introduce a method for reducing anisotropic heavy-quark potentials to isotropic potentials by using an effective screening mass that depends on the quantum numbers l and m of a given state. We demonstrate that using the resulting 1D effective potential model, one can solve a 1D Schrödinger equation and reproduce the full 3D results for the energies and binding energies of low-lying heavy-quarkonium bound states to relatively high accuracy.

We introduce a framework called Heavy Quarkonium Quantum Dynamics (HQQD) which can be used to compute the dynamical suppression of heavy quarkonia propagating in the quark-gluon plasma using real-time in-medium quantum evolution. Using HQQD we compute large sets of real-time solutions to the Schrödinger equation using a realistic in-medium complex-valued potential. We sample 2 million quarkonia wave packet trajectories and evolve them through the QGP using HQQD to obtain their survival probabilities.

Using the potential non-relativistic quantum chromodynamics (pNRQCD) effective field theory, we derive a Lindblad equation for the evolution of the heavy-quarkonium reduced density matrix that is accurate to next-to-leading order (NLO) in the ratio of the binding energy

of the state to the temperature of the medium. The resulting NLO Lindblad equation can be used to more reliably describe heavy-quarkonium evolution in the quark-gluon plasma at low temperatures compared to the leading-order truncation. For phenomenological application, we numerically solve the resulting NLO Lindblad equation using the quantum trajectories algorithm. To achieve this, we map the solution of the three-dimensional Lindblad equation to the solution of an ensemble of one-dimensional Schrödinger evolutions with Monte-Carlo sampled quantum jumps. Averaging over the Monte-Carlo sampled quantum jumps, we obtain the solution to the NLO Lindblad equation without truncation in the angular momentum quantum number of the states considered. We demonstrate the importance of quantum jumps in the non-equilibrium evolution of bottomonium states in the quark-gluon plasma. We evolve the density matrix of color singlet and octet pairs. We show that quantum regeneration of singlet states from octet configurations is necessary to understand experimental results for the suppression of both bottomonium ground and excited states.

HEAVY QUARKONIUM SUPPRESSION IN THE QUARK-GLUON PLASMA USING REAL-TIME QUANTUM FIELD THEORY FOR AN OPEN QUANTUM SYSTEM

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by

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Table of Contents

List of Figures x

List of Publications xviii

Acknowledgments xx

1 **Introduction** 1

1.1 Big Bang 1

1.2 Little Bang 5

1.3 Standard Model of Particle Physics 6

1.4 QCD Phase Diagram 7

1.5 Background Knowledge for Heavy Ion Collisions 7

1.5.1 Participants 8

1.5.2 Spectators 8

1.5.3 Impact parameter 8

1.5.4 Transverse mass 9

1.5.5 Rapidity 10

1.5.6 Pseudo-rapidity 10

1.5.7 Centrality 12

1.6 Heavy quarkonium suppression as a probe of QGP 12

1.6.1 Heavy quarkonium states and their suppression in QGP 13

1.6.2 Suppression or Nuclear modification factor (R_{AA}) 15

1.6.3 Elliptic flow 16

1.7 Set up of the dissertation 18

2 **The real-time dynamics of heavy quarkonium in QGP** 19
2.1 Background 19
2.1.1 Unitary time evolution 19
2.1.2 Open and Closed Quantum Systems 20
2.1.3 Master equation and Lindblad Form 21
2.2 Real-time dynamics 23
2.2.1 The Heavy quark Lagrangian density 23
2.2.2 Real-time formalism 26
2.2.3 Influence functional 33
2.2.4 Renormalized effective Hamiltonian 42
2.2.5 Master equation for the reduced density matrix 50
2.2.6 Time evolution of the forward correlator 61
2.3 Conclusion 69

3 **Heavy Quarkonium Dynamics Beyond The Adiabatic Limit** 70
3.1 Complex KMS potential 70
3.2 Numerical algorithm 72
3.2.1 Time evolution algorithm 73
3.2.2 Time-dependent basis functions 74
3.3 Numerical results 75
3.3.1 Bottomonia 76
3.3.2 Charmonia 80
3.4 Discussion and conclusion 83

4 **Stochastic Potential** 87
4.1 Heavy quarkonium as an open quantum system 87
4.2 Stochastic evolution of heavy quarkonium 88

4.3 Stochastic potential model . . . 89
4.4 Stochastic potential with two noise sources . . . 98
4.5 Noise Correlation D-function for KMS potential . . . 103

5 **Isotropic Heavy Quarkonium Quantum Dynamics** . . . 105
5.1 Phenomenological potential . . . 106
5.2 Computation of R_{AA} including feed down . . . 107
5.3 Results . . . 113
5.4 Conclusions and outlook . . . 125

6 **Ansotropic Heavy Quarkonium Quantum Dynamics** . . . 129
6.1 Real part of the anisotropic KMS potential . . . 129
6.2 Imaginary part to leading-order in ξ . . . 130
6.2.1 Angle-averaged imaginary part to leading-order in small-ξ limit . . . 131
6.2.2 Imaginary part to leading-order in large-ξ limit . . . 135
6.3 Model for the imaginary part of the potential for general ξ . . . 137
6.4 The complete anisotropic complex potential . . . 137
6.5 Phenomenological Results . . . 138

7 **Effective Heavy Quark Potential** . . . 139
7.1 Background . . . 139
7.2 Isotropic Potential Model . . . 141
7.2.1 Perturbative Contribution . . . 141
7.2.2 Non-perturbative Contribution . . . 141
7.2.3 Total Isotropic potential . . . 142
7.3 3D Anisotropic Potential Model . . . 142
7.4 1D Effective Potential Model . . . 143

7.5 Static Results . . . 144
7.6 Dynamical Results . . . 146
7.6.1 Bottomonium . . . 146
7.6.2 Charmonium . . . 146
7.7 Conclusions . . . 147

8 **PNRQCD and Open Quantum System Approach** . . . 149
8.1 Non-relativistic Effective Field Theories . . . 150
8.1.1 Heavy Quark Effective Field Theory . . . 150
8.1.2 Non Relativistic QCD . . . 152
8.1.3 potential Non Relativistic QCD . . . 154
8.2 Derivation of the Lindblad Equation . . . 157
8.2.1 The pNRQCD master equation . . . 157
8.2.2 Lindblad equation to leading-order in E/T . . . 159
8.2.3 Lindblad equation to next-to-leading order in E/T . . . 161
8.3 The quantum trajectories algorithm at NLO . . . 163
8.3.1 Jump operators in the reduced spherical representation . . . 164
8.3.2 Width operators in the reduced spherical representation . . . 165
8.3.3 Effective Hamiltonian in the reduced spherical representation . . . 165
8.3.4 Description of the algorithm . . . 166
8.3.5 Evolution between jumps . . . 167
8.4 Phenomenological Results . . . 168
8.5 Conclusion . . . 173

A **SU(N) Generators** . . . 176

B **Derivation of the $f(\xi)$-function** . . . 177

Bibliography 180

List of Figures

1.1 Evolution of our universe after Big Bang [92] 1

1.2 Emergent of four fundamental forces after Big Bang [1] 2

1.3 An illustration of production and evolution of QGP [105, 96]. 6

1.4 Standard Model of Particle Physics [4]. 6

1.5 QCD phase diagram [5]. 7

1.6 Spectators and participants in heavy ion collision [2]. 8

1.7 Geometry of heavy ion collision [54]. 9

1.8 As polar angle approaches zero, pseudo-rapidity tends towards infinity [3]. . 11

1.9 A diagram depicts collision scenarios based on impact parameter and centrality in the transverse plane. It illustrates peripheral, semi-central, and central collisions from left to right.[68]. 12

1.10 Various states of charmonia and bottomonia [116]. 13

1.11 QGP thermometer [85]. 14

1.12 Experimental observation of bottomonium suppression inside QGP. 14

1.13 Pattern of sequential suppression of bottomonia [73]. 15

1.14 The geometry of colliding nuclei. Left: configuration of reaction plane and azimuthal angle in the lab frame. Right: The pressure gradients in momentum space [10]. 17

2.1 Closed-time contour in forward -backward integration paths (adapted from [58]). 28

2.2 Closed-time path $\mathcal{C}$ in complex time plane, parameterized by s $(-\infty < s < \infty)$ (adapted from [11]). 34

3.1 Panel (a) shows the survival probability of the $\Upsilon(1S)$ as a function of proper time. In this panel, the solid black line is the result obtained using real-time evolution and the dashed red line is the result obtained using the adiabatic approximation. Panel (b) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black, shorted-dashed red, and long-dashed blue lines are the overlaps computed for the 1s, 2s, and 3s states. In both panels the initial temperature was taken to be $T_0 = 0.6$ GeV at $\tau_0 = 1$ GeV^{-1} and the initial wave function consisted of a 1s state with the wave function determined self-consistently using the CKMS potential. 77

3.2 Panels (a) and (b) show the survival probability of the $\Upsilon(1S)$ and $\Upsilon(2S)$ as a function of proper time. In these panels, the solid black line is the result obtained using real-time evolution and the dashed red line is the result obtained using the adiabatic approximation. Panel (c) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black, shorted-dashed red, and long-dashed blue lines are the overlaps computed for the 1s, 2s, and 3s states. In both panels the initial temperature was taken to be $T_0 = 0.225$ GeV at $\tau_0 = 1$ GeV^{-1}. The initial wave function consisted of a linear combination of 1s and 2s components with the wave functions determined self-consistently using the CKMS potential. 78

3.3 Panel (a) shows the survival probability of the $\chi_b(1P)$ as a function of proper time. Panel (b) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black and shorted-dashed red lines are the overlaps computed for the 1p and 2p states, respectively. In both panels the initial temperature was taken to be $T_0 = 0.6$ GeV at $\tau_0 = 1$ GeV^{-1} and the initial wave function consisted of a 1s state with the wave function determined self-consistently using the CKMS potential. 79

3.4 Panel (a) shows the survival probability of the J/ψ as a function of proper time. In this panel, the solid black line is the result obtained using real-time evolution and the dashed red line is the result obtained using the adiabatic approximation. Panel (b) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black, shorted-dashed red, and long-dashed blue lines are the overlaps computed for the 1s, 2s, and 3s states. In both panels the initial temperature was taken to be $T_0 = 0.4$ GeV at $\tau_0 = 1$ GeV^{-1} and the initial wave function consisted of a 1s state with the wave function determined self-consistently using the CKMS potential. 81

3.5 Same as Fig. 3.4, but with $T_0 = 0.2$ GeV. 82

3.6 Panel (a) shows the survival probability of the $\chi_c(1P)$ as a function of proper time. In this panel, the solid black line is the result obtained using real-time evolution and the dashed red line is the result obtained using the adiabatic approximation. Panel (b) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black and shorted-dashed red lines are the overlaps computed for the 1p and 2p states. In both panels the initial temperature was taken to be $T_0 = 0.2$ GeV at $\tau_0 = 1$ GeV^{-1} and the initial wave function consisted of a 1p state with the wave function determined self-consistently using the CKMS potential. 83

4.1 Schematic view of the difference between a description on the level of the full system with a hermitian Hamiltonian $H = H_{sys} + H_{med} + H_{int}$ and a stochastic potential description of the heavy quarkonium system after tracing out the medium degrees of freedom (adapted from [93]). 89

4.2 Schematic view of the setup used to describe Bottomonium evolution inside QGP (adapted from [94]). 90

5.1 Nuclear suppression factor, R_{AA}, of bottomonium s-wave states as a function of $N_{\rm part}$ (left) and p_T (right). The solid, short-dashed, and dashed lines show the predictions of HQQD. The shaded bands indicate the variation of the HQQD result when changing the model parameters in the range specified in Eq. (5.22). The data points are from the ALICE [8], ATLAS [104], and CMS [102] collaborations. Experimental error bars shown were obtained by adding statistical and systematic uncertainties in quadrature. 114

5.2 Double ratio $[\Upsilon(2S)/\Upsilon(1S)]_{\mathrm{PbPb}}/[\Upsilon(2S)/\Upsilon(1S)]_{\mathrm{pp}}$ as a function of N_{part} (left panel) and p_T (right panel). The solid line shows the prediction of HQQD. The shaded bands indicate the variation of the HQQD result when changing the model parameters in the range specified in Eq. (5.22). The data points are from the ATLAS [104] and CMS [101] collaborations. Systematic and statistical experimental uncertainties are indicated by the green and black error bars, respectively. 115

5.3 Visualization of the starting positions of $\Upsilon(1S)$ (left) and $\Upsilon(2S)$ states (right). In both cases we selected all states with $0 < p_T < 40$ GeV. Color coding shows the R_{AA} for each particle. 117

5.4 Visualization of the initial probability distribution function for the starting positions of $\Upsilon(1S)$. Panels show distribution for states experiencing a given R_{AA} in 10% intervals, e.g. $0.4 < R_{AA} < 0.5$. For this visualization, we selected all states with $0 < p_T < 40$ GeV. Coloring encodes the normalized probability in each R_{AA} range. 118

5.5 Elliptic flow for s-wave bottomonium states as a function of centrality. Solid lines and bands show spline-interpolated results for the mean and statistical uncertainty of the mean obtained from HQQD.The upper (squares), central (circles), and lower (triangles) indicate the variation of the HQQD result when changing the model parameters in the range specified in Eq. (5.22). Points show results obtained in equally spaced bins of 10% centrality from 0-100%. 119

5.6 The elliptic flow $v_2[\Upsilon(1S)]$ as a function of p_T in three p_T-bins. Open red squares are HQQD predictions and the data are from the ALICE [7] and CMS [103] collaborations. The shaded red boxes associated with each HQQD prediction indicate the systematic variation observed when changing the model parameters in the range specified in Eq. (5.22). Note that, for the highest p_T bin, the systematic error estimate is so small that it is indistinguishable from the horizontal error bar. 120

5.7 Centrality dependence of $v_2[\Upsilon(1S)]$ shown in 10-30%, 30-50%, 50-90%, and 10-90% centrality bins. Open squares are predictions of HQQD. The shaded red boxes associated with each HQQD prediction indicate the systematic variation observed when changing the model parameters in the range specified in Eq. (5.22). 121

5.8 Centrality dependence of $v_2[\Upsilon(2S)]$ and $v_2[\Upsilon(3S)]$ in the same centrality bins as Fig. 5.7. Open symbols are predictions of HQQD. In the 10-90% class we include recent data reported by the CMS collaboration for integrated $v_2[\Upsilon(2S)]$ [103]. The shaded red and blue boxes associated with each HQQD prediction indicate the systematic variation observed when changing the model parameters in the range specified in Eq. (5.22). 122

5.9 HQQD predictions for the elliptic flow of $\Upsilon(2S)$ and $\Upsilon(3S)$ states in the 5-60% centrality bin. The transverse momentum bins are the same as used in Fig. 5.6. The shaded red and blue boxes associated with each HQQD prediction indicate the systematic variation observed when changing the model parameters in the range specified in Eq. (5.22). 123

6.1 The nuclear suppression factor R_{AA} of bottomonia states as a function of $N_{\rm part}$. The solid and dashed lines indicate results obtained with aniso-HQQD and with iso-HQQD potential model, respectively. The experimental measurements shown are from the ALICE [9], ATLAS [104], and CMS [100, 46] Collaborations. 138

7.1 The top row shows the overlaps of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ resulting from real-time solution of the Schrödinger equation. Here we initialized the wave function as pure $\Upsilon(1S)$ eigenstate. The bottom row shows the time evolution of the bottomonium p-wave overlaps resulting from initialization with different p-wave polarizations [48]. 147

7.2 The top row shows the overlaps of J/ψ, $\psi(2S)$, and $\psi(3S)$ resulting from real-time solution of the Schrödinger equation. Here we initialized the wave function as pure J/ψ eigenstate. The bottom row shows the time evolution of the charmonium p-wave overlaps resulting from initialization with different p-wave polarizations [48]. 148

8.1 Hierarchy of energy scales and EFTs for systems made of a heavy quark and antiquark pair near threshold. 151

8.2 The nuclear suppression factor R_{AA} of bottomonia states as a function of $N_{\rm part}$. The solid and dashed lines indicate results obtained with and without quantum jumps, respectively. The experimental measurements shown are from the ALICE [9], ATLAS [104], and CMS [100, 46] Collaborations. 169

8.3 R_{AA} of bottomonia states as a function of p_T. Line styles and experimental data sources are the same as Fig. 8.2. 170

8.4 The 2S/1S double ratio as a function of N_{part}. The solid blue lines and dashed orange lines show the QTraj results with and without quantum jumps, respectively. The experimental measurements shown are from the ATLAS [104] and CMS [100, 46] Collaborations. 172

8.5 The 3S/1S double ratio as a function of N_{part}. Line styles and experimental data sources are the same as Fig. 8.4. 173

8.6 The 3S/2S double ratio as a function of N_{part}. Line styles are the same as Fig. 8.4. The centrality classes used were 0-30, 30-50, 50-70, and 70-90%. Experimental data are from Ref. [46]. 174

8.7 The 2S/1S double ratio as a function of p_T. Line styles are the same as Fig. 8.4. The experimental data were inferred from the results of Refs. [100, 46]. . . . 175

8.8 The 3S/2S double ratio as a function of p_T. Line styles are the same as Fig. 8.4. The experimental data are from Ref. [46]. 175

List of Publications

Journal papers

1. Nora Brambilla, Miguel Angel Escobedo, **Ajaharul Islam**, Michael Strickland, Anurag Tiwari, Antonio Vairo, and Peter Vander Griend. *"Regeneration of bottomonia in an open quantum systems approach."*,Phys.Rev.D 108 (2023) 1, L011502.

2. Nora Brambilla, Miguel Angel Escobedo, **Ajaharul Islam**, Michael Strickland, Anurag Tiwari, Antonio Vairo, and Peter Vander Griend. *"Heavy quarkonium dynamics at next-to-leading order in the binding energy over temperature."*,JHEP, 08:303, 2022.

3. Lihua Donga, Yun Guo, **Ajaharul Islam**, Alexander Rothkopf, and Michael Strickland. *"The complex heavy-quark potential in an anisotropic quark-gluon plasma — Statics and dynamics"*, JHEP, 09:200, 2022.

4. Lihua Donga, Yun Guo, **Ajaharul Islam** and Michael Strickland. *Effective Debye screening mass in an anisotropic quark gluon plasma"*, Phys. Rev. D, 104(9):096017, 2021.

5. Hisham Ba Omar, Miguel Angel Escobedo, **Ajaharul Islam**, Michael Strickland, Sabin Thapa, Peter Vander Griend, and Johannes Heinrich Weber. *"QTRAJ 1.0: A Lindblad equation solver for heavy- quarkonium dynamics"*, Comput. Phys. Commun., 273:108266, 2022.

6. **Ajaharul Islam**, and Michael Strickland. *"Bottomonium suppression and elliptic flow using Heavy Quarkonium Quantum Dynamics"*, JHEP, 21:235, 2020.

7. **Ajaharul Islam**, and Michael Strickland. *"Bottomonium suppression and elliptic flow from real-time quantum evolution"*, Phys. Lett. B, 811:135949, 2020.

8. Jacob Boyd, Thomas Cook, **Ajaharul Islam**, and Michael Strickland. *"Heavy quarkonium suppression beyond the adiabatic limit"*, Phys. Rev. D, 100(7):076019, 2019.

Conference Proceedings

1. **Ajaharul Islam**. *"The complex heavy quark potential in an anisotropic quark-gluon plasma"*

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Dedication

To

Professor Michael Strickland

Thank you for being an incredible mentor and friend. Your belief in my abilities and your continuous support have inspired me to strive for excellence.

Chapter 1

Introduction

1.1 Big Bang

The Big Bang theory was originally formalized by Belgian cosmologist Georges Lemaître in 1927. It was developed from observations and theoretical considerations. According to the Big Bang model, the universe expanded from an extremely dense and hot point called singularity and continues to expand today. The evolution of the universe after the Big Bang is characterized by distinct epochs, each marked by unique physical conditions and processes. Here is an overview of some key epochs:

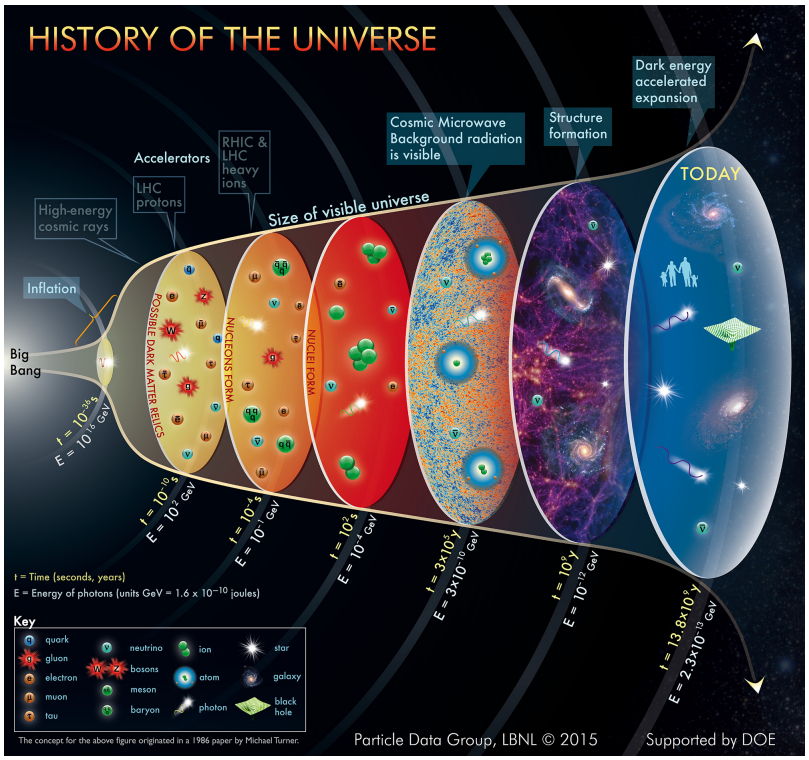


Figure 1.1: Evolution of our universe after Big Bang [92]

- **Planck epoch** ($t \sim 0$ to 10^{-43} s, $T > 10^{19}$ GeV)

  At this extremely brief moment, the universe was at its highest energy density. The Planck scale represents a boundary in the physical scale where existing physical theories may lose applicability and reliability in predicting events. In the Planck epoch, it is presumed that quantum effects of gravity predominantly governed both cosmology and physics.

- **Grand Unification Epoch** ($t \sim 10^{-43}$ to 10^{-36} s, $T > 10^{16}$ GeV)

  In this era, three out of the four fundamental interactions—electromagnetism, the strong interaction, and the weak interaction—coalesced into the unified force known as the electronuclear force. Gravity had already separated from the electronuclear force by the conclusion of the Planck era. At this moment, nature was described by a Grand Unified Theory. Throughout the grand unification epoch, physical attributes such as mass, charge, flavor, and color charge held no discernible meaning.

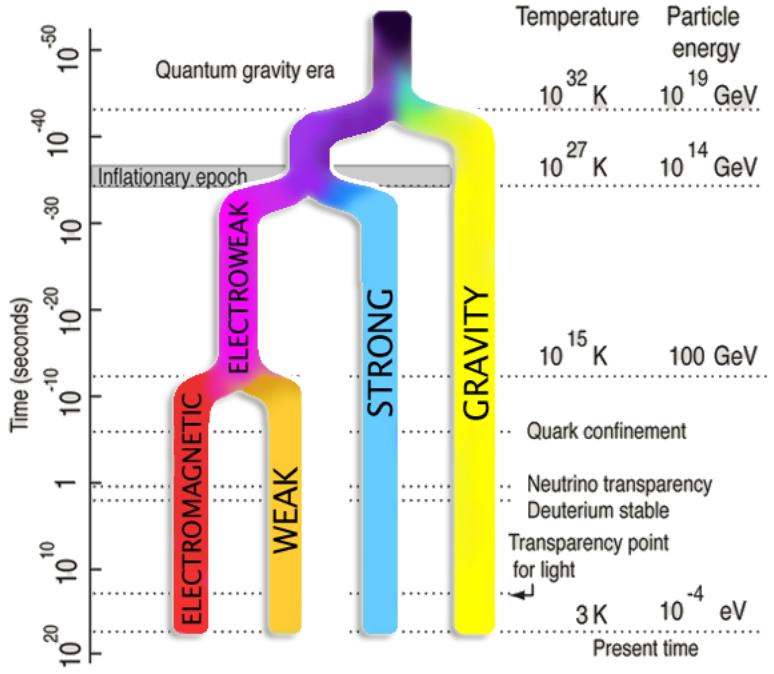


Figure 1.2: Emergent of four fundamental forces after Big Bang [1]

- **Inflationary Epoch** ($t \sim 10^{-36}$ to 10^{-32} s, $T \sim 10^{15}$ to 10^{9} GeV)

  The inflationary epoch denotes a phase in the early universe's evolution, as proposed by inflation theory, during which the universe experienced an extraordinarily rapid and exponential expansion.

- **Electroweak Epoch** ($t \sim 10^{-32}$ to 10^{-12} s, $T \sim 10^{9}$ GeV to 150 GeV)

  The electroweak epoch signifies a phase in the early universe's development during which the universe's temperature had decreased sufficiently for the strong force to separate from the electroweak interaction. However, the temperature remained high enough for electromagnetism and the weak interaction to persist as a unified electroweak interaction above the critical temperature for electroweak symmetry breaking.

- **Quark Epoch** ($t \sim 10^{-12}$ to 10^{-5} s, $T \sim 150$ GeV to 150 MeV)

  The quark epoch denotes a phase in the early universe's progression when the fundamental interactions of gravity, electromagnetism, the strong interaction, and the weak interaction had assumed their current configurations. However, the temperature of the universe remained too high for quarks to combine and form hadrons which are made of two or more quarks. Throughout the quark epoch, the entire universe was filled with a dense and hot Quark–Gluon Plasma (QGP), comprised of quarks, gluons, leptons, and their respective antiparticles.

- **Hadron Epoch** ($t \sim 10^{-5}$ to 1 s, $T \sim 150$ MeV to 1 MeV)

  The universe's temperature had sufficiently decreased, enabling the quarks from the preceding quark epoch to combine and form hadrons. This epoch marks the transition from a QGP to a state dominated by hadrons.

- **Neutrino decoupling** ($t \sim 1$ s, $T \sim 1$ MeV)

  During this epoch, neutrinos ceased interacting significantly with baryonic matter,

contributing to the emergence of the cosmic neutrino background.

- **Lepton Epoch** ($t \sim 1$ to 10 s, $T \sim 1$ MeV to 100 KeV)

  During this epoch, the universe was dominated by leptons, such as electrons, neutrinos, and their antiparticles.

- **Nucleosynthesis** ($t \sim 10$ to 10^3 s, $T \sim 100$ KeV to 1 KeV)

  In this period, protons and neutrons combine to form primordial atomic nuclei, predominantly hydrogen and helium-4. Additionally, trace amounts of deuterium, helium-3, and lithium-7 also come into existence.

- **Photon Epoch** ($t \sim 10$ s to 370 Kyr, $T \sim 100$ KeV to 0.4 eV)

  In this era, the universe was composed of a plasma containing nuclei, electrons, and photons. At this stage, the temperature was too low to generate electron-positron pairs but remained too high for electrons to bind with nuclei.

- **Recombination** ($t \sim 18$ Kyr to 370 Kyr, $T \sim 0.4$ eV)

  In this era, electrons and atomic nuclei initially combine to create neutral atoms, leading to a state where photons are no longer in thermal equilibrium with matter. This marks the point at which the universe first becomes transparent to light, allowing photons to travel freely, leading to the cosmic microwave background radiation.

- **Dark Ages** ($t \sim 370$ Kyr to 150 Kyr, $T \sim 4000$ to 60 K)

  The interval between recombination and the emergence of the first stars is characterized by a scarcity of visible light. During this period, the primary source of photons was hydrogen emitting radio waves. Freely propagating cosmic microwave background (CMB) photons underwent rapid redshift, transitioning to the infrared spectrum, rendering the universe devoid of visible light.

- **Star Formation** ($t \sim 300$ to 400 Myr, $T \sim 60$ K)

  In this era, primordial clouds of hydrogen and helium, enriched by trace amounts of lithium and deuterium, began to collapse under the influence of gravity. As these dense regions contracted, temperatures and pressures at their cores rose significantly. The conditions became ripe for nuclear fusion, marking the birth of the first stars.

- **Galaxy Formation and Evolution** ($t \sim 1$ to 10 Gyr, $T \sim 60$ K)

  Following the era of the first stars, their intense radiation began influencing the surrounding cosmic environment. The ultraviolet light emitted by these early stars initiated the process of cosmic reionization, ionizing the neutral hydrogen that permeated the cosmos. Within these ionized regions, the primordial gas underwent further gravitational collapse, giving rise to the formation of the first galaxies.

- **Present time** ($t \sim 13.8$ Gyr, $T \sim 2.7$ K)

  Billions of years after the Big Bang, the universe has transformed into a vast and intricate cosmic tapestry. Galaxies, comprising billions of stars, populate the cosmos, forming intricate structures that span unimaginable distances.

1.2 Little Bang

A few milliseconds just after the Big Bang, our entire universe was a super-hot plasma of quarks and gluons, known as Quark-Gluon Plasma (QGP). Scientists produce QGP experimentally using relativistic heavy-ion collision experiments at Brookhaven National Laboratory (BNL)'s Relativistic Heavy Ion Collider (RHIC) and European Organization for Nuclear Research (CERN)'s Large Hadron Collider (LHC). On February 10, 2000, some CERN scientists announced that "a compelling evidence now exists for the formation of a new state of matter at energy densities about 20 times larger than in the center of atomic nuclei and temperatures about 100,000 times higher than in the center of the sun". The process of the

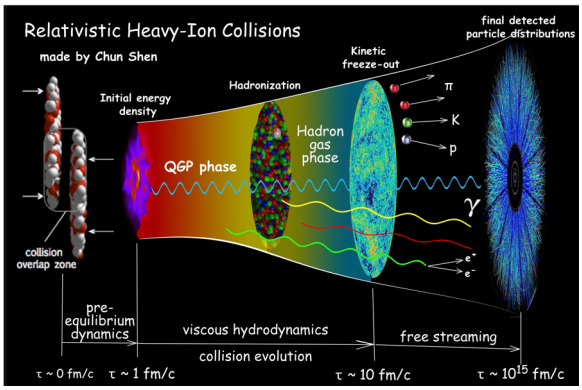


Figure 1.3: An illustration of production and evolution of QGP [105, 96].

production of hot and dense matter (QGP) in the URHICs is sometimes known as a "Little Bang".

1.3 Standard Model of Particle Physics

The quantum electro-weak theory together with QCD form the Standard Model(SM) of particle physics. The SM particles are shown in Fig.(1.4). They consist broadly of:

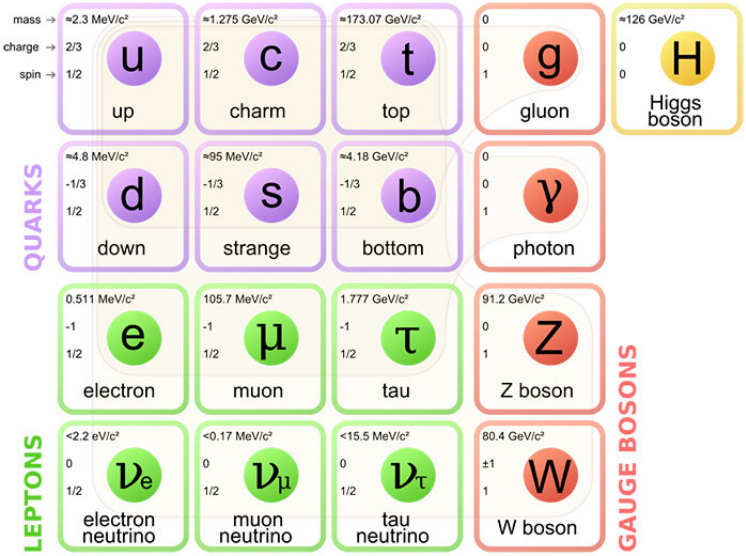


Figure 1.4: Standard Model of Particle Physics [4].

- Force carriers (bosons): intermediate vector bosons, photon & gluons.

- Matter particles (fermions): Leptons & their antiparticles, Quarks & their antiquarks.

The SM is reasonably elegant and very powerful to describe the large number of particles and their interactions.

1.4 QCD Phase Diagram

We know that the coupling constant of strong interaction decreases with increasing in energy. So, at a particular energy, the interaction among quarks becomes negligible and quarks behave as free particles. This phenomenon is known as asymptotic freedom. At

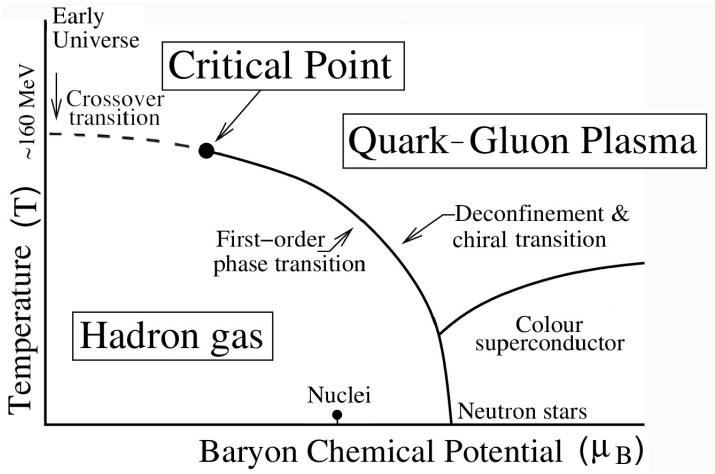


Figure 1.5: QCD phase diagram [5].

around 170 MeV, the deconfinement phase transition occurs, the quarks come out from hadronic state and make mixture of free quarks and gluons, we call this state as QGP.

1.5 Background Knowledge for Heavy Ion Collisions

Heavy-ion collisions refer to high-energy collisions between large atomic nuclei, typically involving ions like gold or lead. These collisions are conducted at facilities such as the LHC at CERN or the RHIC at Brookhaven National Laboratory. Unlike proton-proton collisions, heavy-ion collisions generate extremely high temperatures and energy densities, creating conditions similar to those believed to have existed in the early universe microseconds after

the Big Bang. During heavy-ion collisions, the extreme temperatures and energy densities lead to the formation of a state of matter called QGP.

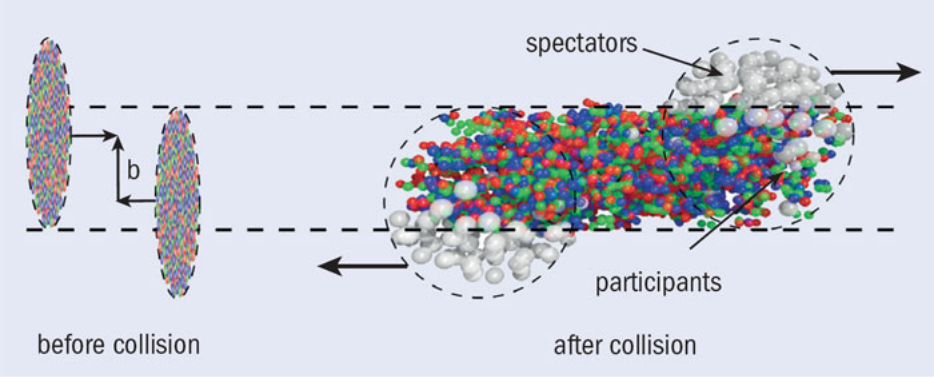


Figure 1.6: Spectators and participants in heavy ion collision [2].

1.5.1 Participants

Participants are the nucleons that actively take part in the collision process. They undergo interactions and are involved in the formation of the hot and dense region where extreme conditions, such as high temperatures and energy densities, can lead to the creation of QGP. The participants contribute to the observable signals that experimental detectors record.

1.5.2 Spectators

Spectators are the nucleons that do not directly participate in the collision process. Instead, they continue along their original trajectories without undergoing significant interactions.

1.5.3 Impact parameter

The impact parameter (b) is defined as the perpendicular distance between the trajectories of the centers of the two colliding nuclei (see Fig.1.6). It is measured in a plane perpendicular to the beam axis. It provides information about the relative orientation and

position of the colliding nuclei at the moment of collision. The impact parameter is a crucial parameter in understanding the dynamics of the collision and plays a significant role in determining the outcome of the interaction.

1.5.4 Transverse mass

Transverse mass is a useful quantity to define for use in particle physics as it is invariant under Lorentz boost along the z direction [1]. In natural units it is:

$$m_T^2 = m^2 + p_x^2 + p_y^2 = E^2 - p_z^2$$

where the z-direction is along the beam pipe and so p_x and p_y are the momentum perpendicular to the beam pipe and m is the (invariant) mass.

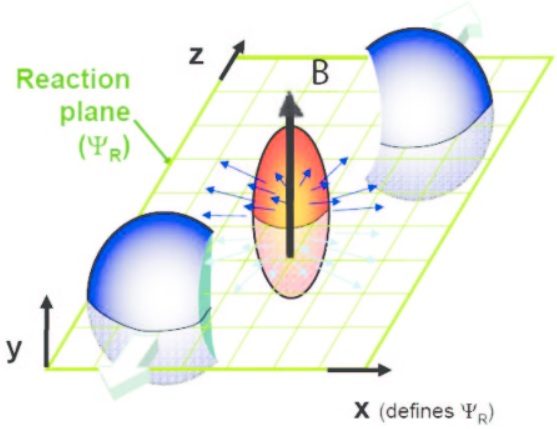


Figure 1.7: Geometry of heavy ion collision [54].

This definition of the transverse mass is used in conjunction with the definition of the (directed) transverse energy:

$$\mathbf{E}_T = E\frac{\mathbf{p}_T}{|\mathbf{p}|} = \frac{E}{\sqrt{E^2 - m^2}}\mathbf{p}_T$$

with the transverse momentum vector $\mathbf{p}_T = (p_x, p_y)$. It is easy to see that for vanishing mass ($m = 0$) the three quantities are the same: $E_T = p_T = m_T$.

[1] Beam line, $\parallel$, longitudinal direction, and z direction are equivalent where transverse direction, $\perp$, and $\{x, y\}$ are equivalent.

1.5.5 Rapidity

Rapidity (y) can be defined as the hyperbolic angle that differentiates two frames of reference in relative motion. For low speeds, rapidity and velocity are proportional, but for higher velocities, rapidity takes a larger value, the rapidity of light being infinite. Mathematically, rapidity defined as

$$y = \frac{1}{2} \ln \left(\frac{E + p_z}{E - p_z} \right) = \tanh^{-1} \left(\frac{p_z}{E} \right) = \tanh^{-1} (v_z) ,$$

Here E is the energy of a particle, $E = \sqrt{m^2 + p^2}$ and $v_z = \frac{p_z}{E}$ is the longitudinal component of the velocity. Rapidity is additive under Lorentz boosts along the z axis. For a non-relativistic particle, rapidity is the same as velocity along the z axis.

1.5.6 Pseudo-rapidity

One challenge with rapidity arises in its measurement for highly relativistic particles. It requires experimentally determining both energy and total momentum, which can be problematic to obtain accurately. Particularly at high rapidity values where the z component of momentum is significant, measuring the total momentum vector of a particle becomes challenging, often obstructed by the beam pipe. Nonetheless, an alternative quantity akin to rapidity exists, which proves much simpler to measure for highly energetic particles. This leads to the concept of pseudo–rapidity η. Pseudo-rapidity, η, is a commonly used spatial coordinate describing the angle of a particle relative to the beam axis (z axis). Mathematically, pseudo-rapidity defined as

$$\eta = \frac{1}{2} \ln \left(\frac{p + p_z}{p - p_z} \right) = \ln \left(\cot \frac{\theta}{2} \right) = - \ln \left(\tan \frac{\theta}{2} \right)$$

where θ is the angle between the particle three-momentum $\mathbf{p}$ and the positive direction of the beam axis (i.e., z axis). In the limit of small hadron mass, $m \to 0$, the rapidity and the pseudo-rapidity become equal. For highly relativistic particles, $y \simeq \eta$.

$$m \ll |\mathbf{p}| \Rightarrow E \approx |\mathbf{p}| \Rightarrow \eta \approx y$$

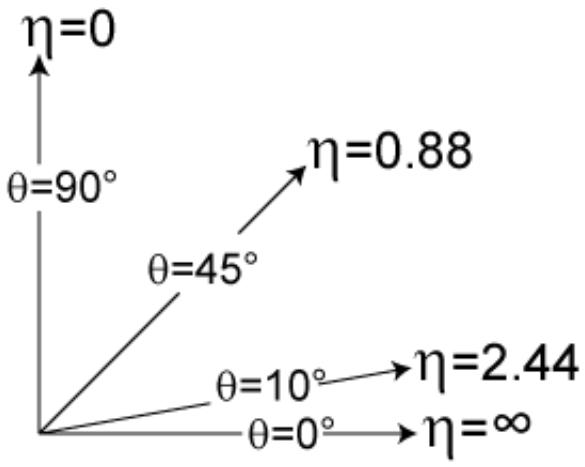


Figure 1.8: As polar angle approaches zero, pseudo-rapidity tends towards infinity [3].

. In this limit, the particle's only energy is its momentum-energy, (similar to the case of the photon), and hence the pseudo-rapidity converges to the definition of rapidity used in experimental particle physics. However, pseudo-rapidity depends only on the polar angle of the particle's trajectory, and not on the energy of the particle. For finite masses the relations between the rapidity and the pseudo-rapidity are more complicated:

$$y = \frac{1}{2}\ln\left[\frac{\sqrt{p_T^2\cosh^2\eta + m^2} + p_T\sinh\eta}{\sqrt{p_T^2\cosh^2\eta + m^2} - p_T\sinh\eta}\right],$$

$$\eta = \frac{1}{2}\ln\left[\frac{\sqrt{m_T^2\cosh^2 y - m^2} + m_T\sinh y}{\sqrt{m_T^2\cosh^2 y - m^2} - m_T\sinh y}\right].$$

- **Central-rapidity/mid-rapidity region:** In the center of mass frame, the region of phase-space where $y \approx \eta \approx 0$ is called the central-rapidity/mid-rapidity region.

- **Projectile fragmentation region:** The region corresponding to the initial rapidities of the projectile ($y \approx y_P$) is called projectile fragmentation region.

- **Target fragmentation region:** The region corresponding to the initial rapidities of the target ($y \approx y_T$) is called target fragmentation region.

1.5.7 Centrality

We have

Centrality $\propto$ Impact parameter $\propto$ Probability of peripheral collisions

Under general assumptions which hold very well in RHICs, we have the relation:

$$c(N) \simeq \frac{\pi b^2(N)}{\sigma_{in}^{AB}} \quad \text{for } b < \bar{R},$$

where σ_{in}^{AB} is the total inelastic nucleus-nucleus cross section and $\bar{R}$ is of the order of the sum of the radii of the colliding nuclei (for identical nuclei $\bar{R} = 2R$ and $c \approx \frac{b^2}{4R^2}$). The centrality $c(N)$ is the centrality of events with the multiplicity higher than N (number of participants), while $b(N)$ is the value of the impact parameter for which the average multiplicity $\bar{n}(b)$ is equal to N.

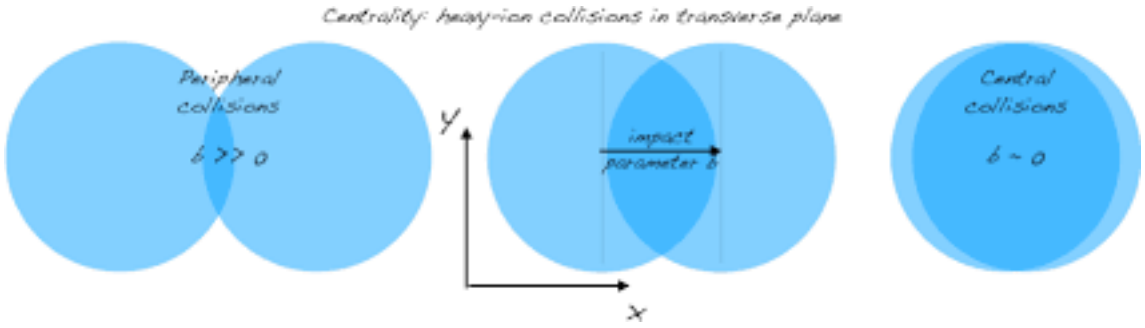


Figure 1.9: A diagram depicts collision scenarios based on impact parameter and centrality in the transverse plane. It illustrates peripheral, semi-central, and central collisions from left to right.[68].

1.6 Heavy quarkonium suppression as a probe of QGP

Heavy quarkonium suppression serves as an important probe for exploring the properties of QGP. In heavy-ion collisions, the extreme conditions generated momentarily recreate the early universe's environment, where quarks and gluons exist freely before hadronization

Bottomonium: $b\bar{b}$

$n^{2s+1}L_J$	J^{PC}	Particle	Mass (MeV)
1^3S_1	1^{--}	$\Upsilon(1S)$	9460.30
1^3P_0	0^{++}	$\chi_{b0}(1P)$	9859.44
1^3P_1	1^{++}	$\chi_{b1}(1P)$	9892.78
1^3P_2	2^{++}	$\chi_{b2}(1P)$	9912.21
2^3S_1	1^{--}	$\Upsilon(2S)$	10023.26
2^3P_0	0^{++}	$\chi_{b0}(2P)$	10232.50
2^3P_1	1^{++}	$\chi_{b1}(2P)$	10255.46
2^3P_2	2^{++}	$\chi_{b2}(2P)$	10268.65
3^3S_1	1^{--}	$\Upsilon(3S)$	10355.20

Charmonium: $c\bar{c}$

$n^{2s+1}L_J$	J^{PC}	Particle	Mass (MeV)
1^3S_1	1^{--}	J/ψ	3096.90
1^3P_0	0^{++}	$\chi_{c0}(1P)$	3414.71
1^3P_1	1^{++}	$\chi_{c1}(1P)$	3510.67
1^3P_2	2^{++}	$\chi_{c2}(1P)$	3556.17

Figure 1.10: Various states of charmonia and bottomonia [116].

occurs. Heavy quarkonium states, like charmonium and bottomonium, are particularly sensitive to the QGP medium due to their binding energies. The presence of QGP can disrupt the formation of quarkonium states through color screening and dissociation mechanisms, leading to a suppression in their production yields. By studying the suppression patterns of heavy quarkonium states across different collision systems and energies, researchers can gain insights into the QGP's temperature, density, and transport properties, thereby advancing our understanding of the strong force dynamics at extreme conditions and the nature of the early universe.

1.6.1 Heavy quarkonium states and their suppression in QGP

Here we will learn what we mean by quarkonium states. Quarkonium is a bound state of heavy quark and its antiquark. The bound state of bottom (b) and anti-bottom quark ($\bar{b}$) is known as a bottomonium ($b\bar{b}$). And the bound state of charm (c) and anti-charm quark ($\bar{c}$) is known as a charmonium ($c\bar{c}$). The left table and right table of Fig. 1.10 are showing spectral notations and masses of various bottomonia and charmonia respectively. Initially, inside QGP, there still remain some quarkonium states which are bound states of

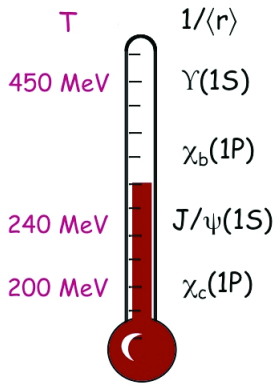


Figure 1.11: QGP thermometer [85].

heavy quark and its antiquark. Inside QGP, color charges are surrounded by opposite color charges, for example the ground bound state of a c and $\bar{c}$ is known as J/ψ. In the QGP, $\bar{c}$ is surrounded by opposite color charges that reduce the effective color charge of $\bar{c}$, this is known as Debye screening. And due to this Debye screening the interaction strength between c and $\bar{c}$ decreases. As the temperature increases c and $\bar{c}$ become unbound and J/ψ state melts in the QGP. In Fig. 1.11, the image of the thermometer is giving an idea about the melting temperatures of various quarkonium states. For example, the $\chi_c(1P)$ will melt first and the $\Upsilon(1S)$ will melt last in the QGP. So, now if we define a term "suppression" indicating

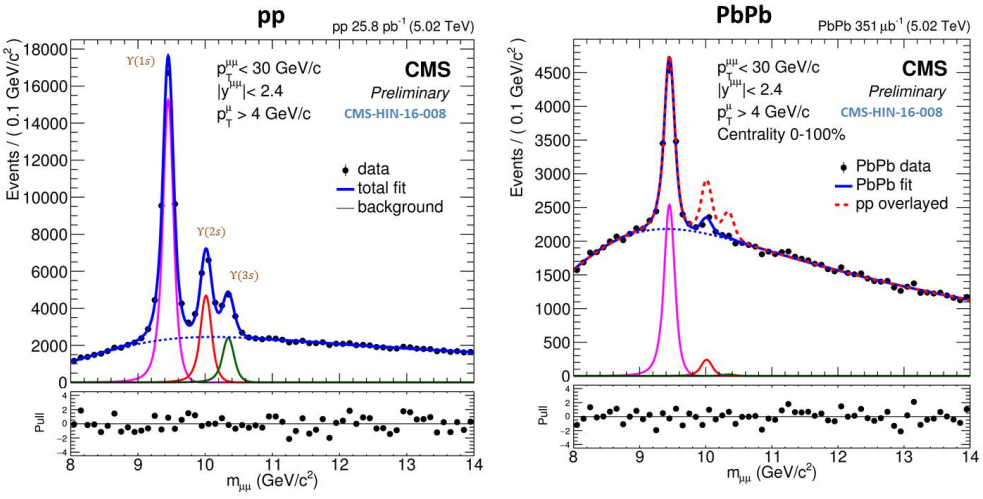


Figure 1.12: Experimental observation of bottomonium suppression inside QGP.

probability of melting of a particular state, then we can say $\chi_c(1P)$ is most suppressed and $\Upsilon(1S)$ is least suppressed in the QGP. In Fig. 1.12, a comparison of CMS data collected in pp collision (absence of QGP) and PbPb collision (presence of QGP) shows suppression of bottomonia inside QGP.

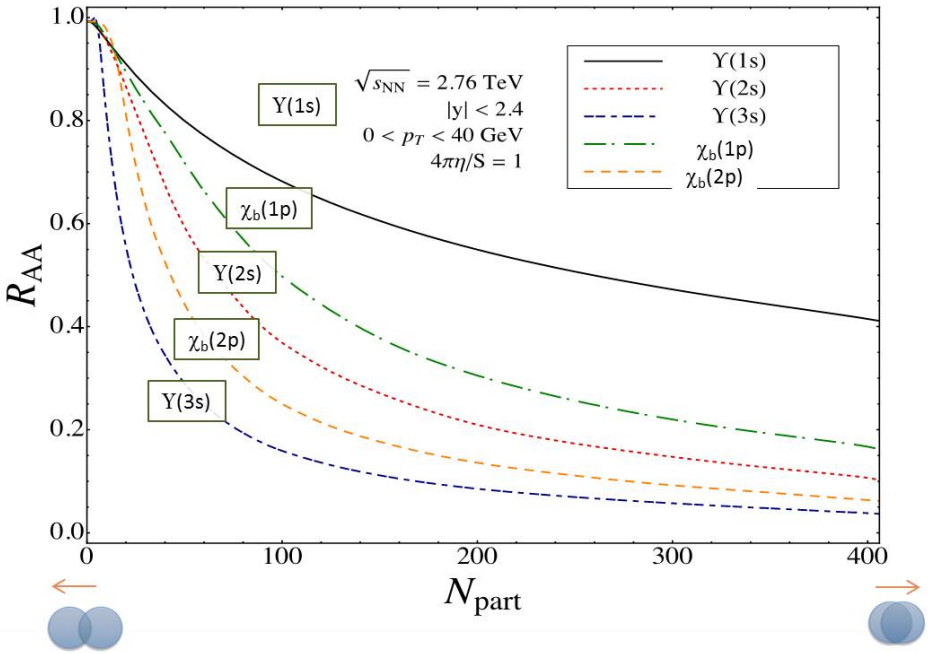


Figure 1.13: Pattern of sequential suppression of bottomonia [73].

1.6.2 Suppression or Nuclear modification factor (R_{AA})

In heavy-ion physics, the nuclear modification factor or suppression factor is a key parameter used to quantify the modification of particle production in nucleus-nucleus collisions compared to simpler reference systems, like proton-proton collisions. It is a measure of how the observed particle yield in heavy-ion collisions deviates from what would be expected based on a simple scaling of nucleon-nucleon interactions. The nuclear modification factor

(R_{AA}) is defined as

$$R_{AA} = \frac{\text{number produced in } AA}{\langle N_{\text{bin}} \rangle_b \text{ (number produced in } pp)} \tag{1.1}$$

$$= 1, \quad \text{no effect} \tag{1.2}$$

$$> 1, \quad \text{enhancement} \tag{1.3}$$

$$< 1, \quad \text{suppression} \tag{1.4}$$

where $\langle N_{\text{bin}} \rangle_b$ is the average number of binary nucleon-nucleon collisions. Fig. 1.13 shows a clear pattern of sequential suppression of bottomonia.

1.6.3 Elliptic flow

Elliptic flow in heavy-ion physics refers to one component of the anisotropic distribution of emitted particles resulting from the collective motion of the produced particles. The development of elliptic flow is influenced by the pressure gradients within the QGP medium, resulting in an anisotropic expansion. Experimental measurements of elliptic flow provide valuable information about the properties of the QGP, such as its viscosity, thermalization dynamics, and the nature of the strong force interactions in extreme conditions. To investigate the flow characteristics in heavy-ion collisions, a common approach involves performing Fourier analysis of the azimuthal distribution of particles in momentum space and calculating the corresponding Fourier coefficients. The first two coefficients are particularly significant. The first, termed directed flow, or v_1 describes the preferential particle emission along the direction defined by the impact parameter. The second coefficient, known as elliptic flow, or v_2 quantifies the anisotropy in the particle distribution relative to the reaction plane. Interestingly, regardless of the collision centrality, these flow fluctuations give rise to higher flow harmonics ($n > 2$). It is possible to express the triple differential invariant distribution of particles emitted in the final state as a Fourier series over the azimuthal angle as

$$E\frac{d^3N}{d^3\mathbf{p}} = \frac{d^3N}{p_T dp_T d\phi dy}\left(1 + 2\sum_{n=1}^{\infty} v_n \cos[n(\phi - \Psi_n)]\right), \tag{1.5}$$

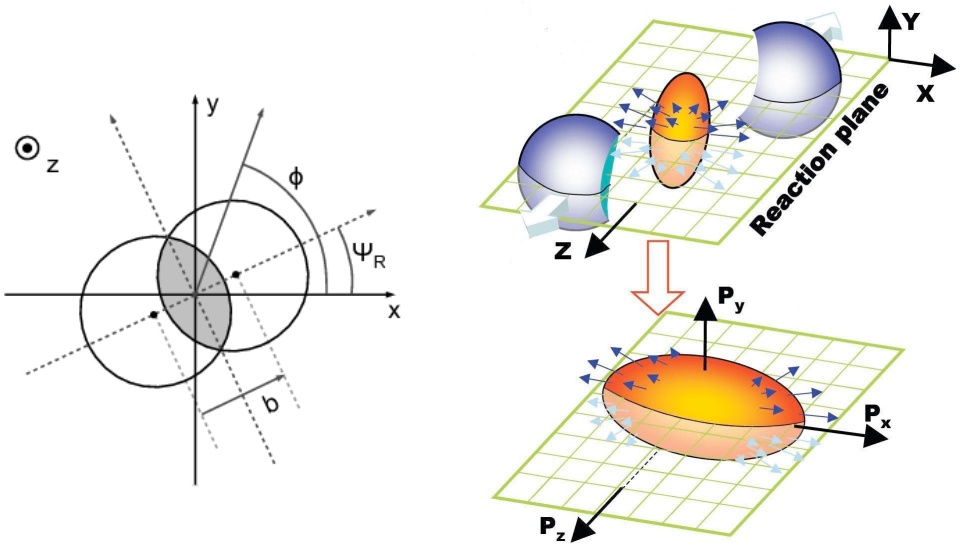


Figure 1.14: The geometry of colliding nuclei. Left: configuration of reaction plane and azimuthal angle in the lab frame. Right: The pressure gradients in momentum space [10].

where, $\mathbf{p}$ is the three vector momentum, p_T is the transverse momentum, ϕ is the azimuthal angle, y is the rapidity, and Ψ_n the reaction plane angle. The Fourier coefficients are given by

$$v_n(p_T, y) \quad = \quad \langle \cos[n(\phi - \Psi_R)] \rangle \, .$$

We assume $\Psi_n = \Psi_R$. The reaction plane Ψ_R is defined as the plane made by the beam direction and impact parameter vector Fig. 1.14.

1.7 Set up of the dissertation

In Chapter 2, I provide an overview of the real-time dynamics of heavy quarkonium in the QGP. Chapter 3 delves into the collaborative efforts and findings of Heavy Quarkonium Quantum Dynamics beyond the adiabatic limit. The details of stochastic potential which leads to a complex heavy quark potential, are explored in Chapter 4. The developed isotropic and anisotropic Heavy Quarkonium Quantum Dynamics (HQQD) models are presented in Chapters 5 and 6, respectively. A 1D Effective Heavy Quark Potential model, collaboratively developed, is detailed in Chapter 7. Finally, in Chapter 8, I discuss our research endeavors focused on determining bottomonium observables using the perspective of pNRQCD and an open quantum system approach.

Chapter 2

The real-time dynamics of heavy quarkonium in QGP

In this chapter, we will review the existing literature for real-time field theory at finite temperature and density and its application to the non-equilibrium quantum statistics [44, 58, 77, 82, 86]. In general, our plan is to apply real time formalism for the understanding of the dynamics of heavy quarks in the QGP. Theoretical framework of open quantum systems is appropriate for the investigation of the real-time dynamics of quarkonium quantum states and its suppression in the QGP [11, 12, 27, 28, 41]. The time evolution of the reduced density matrices governing a single quark (Q) and a quarkonium (QQ_c) within the Quark-Gluon Plasma (QGP) can be expressed in the Lindblad form[11, 12, 41]. Formulating the master equation in the Lindblad form ensures the complete positivity of the reduced density matrix throughout its temporal evolution [12]. Consequently, the derivation of the master equation within the Lindblad framework represents a significant theoretical advancement in elucidating the quantum dynamics of heavy quarks.

2.1 Background

Firstly, we will give an introduction to some topics that are necessary for understanding this chapter.

2.1.1 Unitary time evolution

Time evolution of a system means the system changes its state with the passage of time. Unitary time evolution is specific type of quantum time evolution where probability is conserved. Suppose we have a quantum state at time $t = 0$ given by $|\alpha(0)\rangle$. To find the

state at a later time $t = T$ given by $|\alpha(T)\rangle$, we apply the unitary time evolution operator $\hat{U}$:

$$\hat{U}|\alpha(0)\rangle = |\alpha(T)\rangle \ ,$$

where

$$\hat{U} = e^{-i\hat{H}T} \ ,$$

and $\hat{H}$ is the Hamiltonian of the system, which is Hermitian, $H = H^{\dagger}$.

Conservation of probability requires that

$$\langle\alpha(0)|\alpha(0)\rangle = \langle\alpha(T)|\alpha(T)\rangle.$$

Physically, this means that the total probability for the existence of the quantum system does not change with time. The total quantum probability is 1 . The state evolves in time from $t = 0$ to $t = T$ but no information about the quantum system "leaks out" during this time interval. For instance, in unstable particles or radioactive decay, there, as time goes on, the parent state decays into daughter states. If we observe only the parent state subsystem, it does not undergo unitary time evolution because it loses information about state subsystems. Probability of the parent state is not conserved; it decreases. But if we look at the full system, as a whole, the evolution is unitary, as expected.

2.1.2 Open and Closed Quantum Systems

In general, the time evolution of closed quantum systems is described by unitary operators acting on the system. And time evolution of closed quantum system is characterized by unitary time evolution. In physics, an open quantum system is a quantum-mechanical system which interacts with an external quantum system, the environment or bath. In general, these interactions significantly change the dynamics of the system and result in quantum dissipation, where the information contained in the system is lost to its environment. For open quantum systems, the interactions between the system and its environment make it

so that the dynamics of the system cannot be accurately described using unitary operators alone. The time evolution of open quantum systems can be determined by solving the effective equations of motion, also known as master equations, that govern how the density matrix describing the system changes over time and the dynamics of the observables that are associated with the system. The main assumption is that the entire system-environment combination is a large closed system. Therefore, its time evolution is governed by a unitary transformation generated by a global Hamiltonian. The global Hamiltonian can be separated into three components as follows:

$$H = H_S + H_B + H_{SB} \ ,$$

where H_S is the system's Hamiltonian, H_B is the bath Hamiltonian and H_{SB} is the system-bath interaction. The state of the system can then be obtained from a partial trace over the combined system and bath:

$$\rho_{\mathrm{S}}(t) = \mathrm{Tr}_{\mathrm{B}}\rho_{\mathrm{SB}}(t).$$

2.1.3 Master equation and Lindblad Form

Various mathematical techniques have been developed to address the interaction between an open quantum system and its environment. Among these techniques is the utilization of the density matrix, along with its associated master equation. A master equation comprises a set of first-order differential equations, which provide the temporal evolution—typically in terms of probabilities—of a system occupying various discrete states with regard to a continuous time variable, denoted as t. The Lindblad equation in quantum mechanics is a particular type of master equation describing the time evolution of a density matrix in the case of Markovian dynamics. The Lindblad equation governs not only the time evolution of probabilities (diagonal elements of the density matrix), but also of variables containing information about quantum coherence between the states of the system (non-diagonal elements of

the density matrix). A master equation in Lindblad form is the most general type of Markovian and time-homogeneous master equation describing (in general non-unitary) evolution of the density matrix ρ that preserves the laws of quantum mechanics (i.e., is trace-preserving and completely positive for any initial condition). More generally, the Lindblad master equation for an N-dimensional system's density matrix ρ can be written as:

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_{n,m=1}^{N^2-1} h_{nm} \left(A_n \rho A_m^\dagger - \frac{1}{2} \left\{ A_m^\dagger A_n, \rho \right\} \right) \ ,$$

where$H$ is a (Hermitian) Hamiltonian part, and A_m are an arbitrary orthonormal basis of the operators on the system's Hilbert space with the restriction that A_{N^2} is proportional to the identity operator. Our convention implies that the other A_m are traceless. Note that the summation only runs to $N^2 - 1$. The coefficient matrix h, together with the Hamiltonian, determines the system dynamics. It must be positive semidefinite to ensure that the equation is trace-preserving and completely positive. The anticommutator is defined as $\{a, b\} = ab + ba$. If the h_{mn} are all zero, then this is quantum Liouville equation (for a closed system), which is the quantum analog of the classical Liouville equation. Since the matrix h is positive semidefinite, it can be diagonalized with a unitary transformation u:

$$u^\dagger h u = \begin{bmatrix} \gamma_1 & 0 & \cdots & 0 \\ 0 & \gamma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \gamma_{N^2-1} \end{bmatrix} \ ,$$

where the eigenvalues γ_i are non-negative. If we define another orthonormal operator basis

$$L_i = \sum_{j=1}^{N^2-1} u_{ji} A_j \ ,$$

we can rewrite the Lindblad equation in diagonal form

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_{i=1}^{N^2-1} \gamma_i \left(L_i \rho L_i^\dagger - \frac{1}{2} \left\{ L_i^\dagger L_i, \rho \right\} \right) .$$

The new operators L_i are commonly called the Lindblad or jump operators of the system.

2.2 Real-time dynamics

2.2.1 The Heavy quark Lagrangian density

We will start with the Lagrangian density for a Dirac field

$$\mathcal{L} = \bar{\Psi}\left(i\gamma \cdot D - M\right)\Psi \ , \tag{2.1}$$

with

$$D_\mu = \partial_\mu - igA_\mu \ . \tag{2.2}$$

Also, in the Dirac-Pauli representation, we have

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \ . \tag{2.3}$$

So, we can write

$$\begin{aligned} \gamma \cdot D &\equiv \gamma^\mu D_\mu \\ &= \gamma^0 D_0 - \gamma \cdot \mathbf{D} \ , \end{aligned} \tag{2.4}$$

and

$$\begin{aligned} \gamma \cdot D - M &\equiv \gamma^0 D_0 - \gamma \cdot \mathbf{D} - MI \\ &= \begin{pmatrix} D_0 - M & -\sigma^i D_i \\ \sigma^i D_i & -D_0 - M \end{pmatrix} \ . \end{aligned} \tag{2.5}$$

Now considering

$$\Psi = \begin{pmatrix} Q \\ Q_C^\dagger \end{pmatrix} \ , \tag{2.6}$$

here Q and Q_C are the two-component Pauli spinors for a heavy quark and heavy antiquark.

Then we have

$$\begin{aligned} \mathcal{L} &= \begin{pmatrix} Q \\ Q_C^\dagger \end{pmatrix}^\dagger \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} iD_0 - M & -i\sigma^i D_i \\ i\sigma^i D_i & -iD_0 - M \end{pmatrix} \begin{pmatrix} Q \\ Q_C^\dagger \end{pmatrix} \\ &= Q^\dagger\left(iD_0 - M\right)Q - Q^\dagger\left(i\sigma^i D_i\right)Q_C^\dagger - Q_C\left(i\sigma^i D_i\right)Q + Q_C\left(iD_0 + M\right)Q_C^\dagger \ . \end{aligned} \tag{2.7}$$

From the Dirac equation, $(i\gamma \cdot D - M)\,\Psi = 0$, we have

$$\begin{pmatrix} iD_0 - M & -i\sigma^i D_i \\ i\sigma^i D_i & -iD_0 - M \end{pmatrix} \begin{pmatrix} Q \\ Q_C^\dagger \end{pmatrix} = 0 \ . \tag{2.8}$$

This gives us

$$(iD_0 - M)\,Q - i\sigma^i D_i Q_C = 0 \ , \tag{2.9}$$

and

$$i\sigma^i D_i Q - (iD_0 + M)\,Q_C^\dagger = 0 \ . \tag{2.10}$$

Rearranging these equations give

$$Q = (iD_0 - M)^{-1} \left(i\sigma^i D_i\right) Q_C^\dagger \ , \tag{2.11}$$

$$Q_C^\dagger = (iD_0 + M)^{-1} \left(i\sigma^i D_i\right) Q \ . \tag{2.12}$$

Substituting these expression into the Dirac Lagrangian Eq. (2.7) gives

$$\mathcal{L} = Q^\dagger\,(iD_0 - M)\,Q - Q^\dagger \left(i\sigma^i D_i\right) Q_C^\dagger - Q_C \left(i\sigma^i D_i\right) Q + Q_C\,(iD_0 + M)\,Q_C^\dagger \ .$$

Therefore,

$$\begin{aligned} \mathcal{L} = {} & Q^\dagger\,(iD_0 - M)\,Q - Q^\dagger \left(i\sigma^i D_i\right) (iD_0 + M)^{-1} \left(i\sigma^i D_i\right) Q \\ & - Q_C \left(i\sigma^i D_i\right) (iD_0 - M)^{-1} \left(i\sigma^i D_i\right) Q_C^\dagger + Q_C\,(iD_0 + M)\,Q_C^\dagger \ . \end{aligned} \tag{2.13}$$

Now, the operator $(iD_0 \pm M)^{-1}$ can be expanded in a Taylor series,

$$\begin{aligned} (iD_0 \pm M)^{-1} &= \frac{1}{2M} \left(\frac{iD_0 \mp M}{2M} \pm 1\right)^{-1} \\ &= \pm \frac{1}{2M} \left(1 \mp \frac{iD_0 \mp M}{2M} + ...\right) \ . \end{aligned} \tag{2.14}$$

Using Eq. (2.14) in Eq. (2.13), the heavy quark-antiquark Lagrangian density may be written as

$$\begin{aligned}\mathcal{L} = & Q^\dagger \left(iD_0 - M\right) Q + Q_C \left(iD_0 + M\right) Q_C^\dagger \\ & - Q^\dagger \left(\frac{1}{2M}\left(i\sigma^i D_i\right)^2 - \left(i\sigma^i D_i\right)\frac{iD_0 - M}{4M^2}\left(i\sigma^i D_i\right) + ...\right) Q \\ & + Q_C \left(\frac{1}{2M}\left(i\sigma^i D_i\right)^2 + \left(i\sigma^i D_i\right)\frac{iD_0 + M}{4M^2}\left(i\sigma^i D_i\right) + ...\right) Q_C^\dagger \ . \quad (2.15)\end{aligned}$$

To proceed, we will need the identity $\sigma^i\sigma^j = \delta^{ij} - \sigma^{ij}$ with $\sigma^{ij} = -i\epsilon^{ijk}\sigma_k$

$$\begin{aligned}\left(i\sigma^i D_i\right)^2 &= -\left(\delta^{ij} - \sigma^{ij}\right) D_i D_j \\ &= -\mathbf{D}^2 + \frac{1}{2}\sigma^{ij}[D_i, D_j] \\ &= -\mathbf{D}^2 + i\frac{g}{2}\sigma^{ij}G_{ij} \\ &= -\mathbf{D}^2 + \frac{g}{2}\epsilon^{ijk}\sigma_k\sigma^{ij}G_{ij} \\ &= -\mathbf{D}^2 - g\sigma.\mathbf{B} \ , \quad (2.16)\end{aligned}$$

where we have used $igG_{\mu\nu} = [D_\mu, D_\nu]$ and the chromo-magnetic field $B^k = -\frac{1}{2}\epsilon^{ijk}G_{ij}$

As a result, the Lagrangian may be written as

$$\mathcal{L} = Q^\dagger \left(iD_0 + \frac{1}{2M}\mathbf{D}^2 + \frac{g}{2M}\sigma.\mathbf{B} - M\right) Q + Q_C \left(iD_0 - \frac{1}{2M}\mathbf{D}^2 - \frac{g}{2M}\sigma.\mathbf{B} + M\right) Q_C^\dagger + ... \quad (2.17)$$

Replacing $D_0 \to \partial_0$ and $\mathbf{D}^2 \to \nabla^2$, Eq. (2.17) can be written as

$$\mathcal{L} = Q^\dagger \left(i\partial_0 - M + \frac{\nabla^2}{2M}\right) Q + Q_C \left(i\partial_0 + M - \frac{\nabla^2}{2M}\right) Q_C^\dagger + ... \quad (2.18)$$

The large separation of energy scales between the environment and the heavy quark system $M \gg T$, and small velocity of heavy quarks allows us to treat heavy quarks as non-relativistic particles. So, after the mass and kinetic terms expanded in terms of $\frac{1}{M}$, Eq. (2.18) can be viewed as:

$$\mathcal{L}_\Psi = \mathcal{L}_\Psi^{NR} + \mathcal{O}\left(\frac{T^2}{M}\right) \ , \quad (2.19)$$

where,

$$\mathcal{L}_{\Psi}^{NR} = Q^{\dagger}\left(i\partial_0 - M + \frac{\nabla^2}{2M}\right)Q + Q_C\left(i\partial_0 + M - \frac{\nabla^2}{2M}\right)Q_C^{\dagger} \ . \tag{2.20}$$

The non-relativistic limit of the currents is in turn obtained as:

$$\begin{aligned} j^{a0} &= \rho^a \\ &= \bar{\Psi}t^a\gamma^0\Psi \\ &= \Psi^{\dagger}t^a\Psi \\ &= \begin{pmatrix} Q^{\dagger} & Q_C \end{pmatrix}_{1\times 2} t^a \begin{pmatrix} Q \\ Q_C^{\dagger} \end{pmatrix}_{2\times 1} \\ &= Q^{\dagger}t^aQ + Q_Ct^aQ_C^{\dagger} \ . \end{aligned} \tag{2.21}$$

And three components of current are given by [11]

$$\mathbf{j}^a = Q^{\dagger}\left(\frac{\overrightarrow{\nabla} - \overleftarrow{\nabla}}{2iM}\right)t^aQ - Q_C\left(\frac{\overrightarrow{\nabla} - \overleftarrow{\nabla}}{2iM}\right)t^aQ_C^{\dagger} \ , \tag{2.22}$$

where t^a's are the generators of $SU(N)$ group and $a = 1, 2, 3, ..., (N^2 - 1)$.

2.2.2 Real-time formalism

Here, we will summarize, in general, the basic understanding of real time formalism. In this section will follow the theoretical literature of [58]. We will develop a perturbation theory for the time evolution of operators in a non-equilibrium situation. The perturbation theory for non-equilibrium quantum-statistical mechanics is known as closed-time path Green function formalism (CTPGF). This formalism was developed by Schwinger and Keldysh and has been applied successfully to many non-equilibrium problems.

We define thermal averages of one and two fields with respect to an arbitrary initial density operator $\hat{\rho}$:

$$\langle\hat{\phi}_H(x)\rangle_{\rho} = Tr\left[\hat{\rho}\ \hat{\phi}_H(x)\right] \ , \tag{2.23}$$

$$\langle\hat{\phi}_H(y)\hat{\phi}_H(x)\rangle_{\rho} = Tr\left[\hat{\rho}\ \hat{\phi}_H(x)\hat{\phi}_H(y)\right] \ . \tag{2.24}$$

The interaction gives rise to an interaction Hamiltonian $\hat{H}^{\rm int}(t)$. Then we introduce the field operators in Dirac's interaction picture $\hat{\phi}$. These are related to the Heisenberg operators via the free Hamiltonian $\hat{H}_0$, by

$$\hat{\phi}(x) \equiv e^{i\hat{H}_0(t-t_0)}\hat{\phi}_H(x,t_0)e^{-i\hat{H}_0(t-t_0)} \ . \tag{2.25}$$

The operators in Heisenberg and interaction pictures are equal to each other at a time t_0 at which the density operator $\hat{\rho}$ is known. We also introduce the interaction picture for the interaction Hamiltonian

$$\hat{H}_I^{\rm int} \equiv e^{i\hat{H}t}\hat{H}^{\rm int}e^{-i\hat{H}t} \ . \tag{2.26}$$

And the time evolution operator in the interaction picture is

$$\hat{U}(t,t_0) \equiv \hat{T}\exp\left[i\int_{t_0}^{t} dt' \hat{H}_I^{\rm int}(t')\right] \ , \tag{2.27}$$

where $\hat{T}$ is the time-ordering operator. The time dependence of the field operators $\hat{\phi}(x)$ can be expressed as follows:

$$\hat{\phi}(x)_H = \hat{U}(t_0,t)\hat{\phi}(x)\hat{U}(t,t_0) \ . \tag{2.28}$$

The averages of the Heisenberg fields in the interaction representation are (or Eq. 2.23 can be written as):

$$\langle\hat{\phi}_H(x)\rangle_\rho = Tr\left[\hat{\rho}\ \hat{U}(t_0,t)\hat{\phi}(x)\hat{U}(t,t_0)\right] \ , \tag{2.29}$$

and

$$\langle\hat{\phi}_H(x)\hat{\phi}_H(x')\rangle_\rho = \begin{cases} Tr[\hat{\rho}\ \hat{U}(t_0,t)\hat{\phi}(x)\hat{U}(t,t')\hat{\phi}(x')\hat{U}(t',t_0)], & t > t', \\ Tr[\hat{\rho}\ \hat{U}(t_0,t')\hat{\phi}(x')\hat{U}(t',t)\hat{\phi}(x)\hat{U}(t,t_0)], & t' > t \end{cases} \tag{2.30}$$

We let $t_0 \to -\infty$. In this limit, Eq. (2.29) can be written in terms of the scattering operator $\hat{S} \equiv \hat{U}(\infty,-\infty)$ of the system and the time ordering operator $\hat{T}$:

$$\langle\hat{\phi}_H(x)\rangle_\rho = Tr\left[\hat{\rho}\hat{S}^\dagger\hat{T}\hat{S}\hat{\phi}(x)\right] \ . \tag{2.31}$$

And Eq. (2.30) can be written as

$$\langle \hat{\phi}_H(x)\hat{\phi}_H(y)\rangle_\rho = Tr\left[\hat{\rho}\hat{S}^\dagger\hat{T}\hat{S}\hat{\phi}(x)\hat{\phi}_(y)\right] \ . \tag{2.32}$$

It is interesting to notice that

$$\begin{aligned}\hat{S}^\dagger\hat{T}\left(\hat{S}\hat{\phi}(x)\right) &= \hat{U}(-\infty,t)\hat{U}(t,\infty)\hat{T}\left(\hat{U}(\infty,t)\hat{\phi}(x)\hat{U}(t,-\infty)\right) \\ &= \hat{U}(-\infty,t)\hat{\phi}(x)\hat{U}(t,-\infty) \ . \end{aligned} \tag{2.33}$$

Now, we will consider a closed time contour which extends from $t = -\infty$ to $t = \infty$ and back. The contour is imagined to encircle the time axis in the complex $t-$ plane as shown in Fig. 2.1.

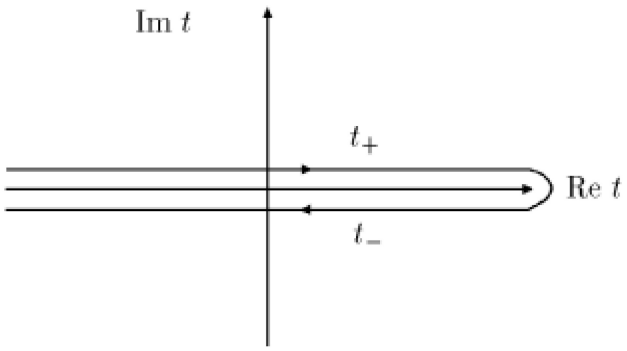


Figure 2.1: Closed-time contour in forward -backward integration paths (adapted from [58]).

The contour runs from $t = -\infty$ to $t = \infty$ above the real time axis and returns below it. We will denote t values on the upper branch as t_+ and on the lower branch as t_-. Similarly, we define $x(t_+) \equiv x_+$ and $x(t_-) \equiv x_-$. Now the operator

$$\hat{S}^\dagger\hat{T}\left(\hat{S}\hat{\phi}(x)\right) \ ,$$

can be written as

$$\hat{S}^\dagger\hat{T}_P\left(\hat{S}\hat{\phi}(x_+)\right) \ , \tag{2.34}$$

where $\hat{T}_P$ performs a path ordering along the closed-time contour.

Now, we can write down a generating functional for an arbitrary product of field operators ordered along the closed-time path

$$\hat{T}_P\left\{\hat{S}^\dagger\hat{S}\exp\left[i\int d^4x j(x_+)\hat{\phi}(x_+)\right]\right\}. \tag{2.35}$$

Functional differentiation with respect to source $j(x_+)$ produces $\hat{\phi}(x_+)$. Then a symmetric generating functional can be written as follows:

$$Z[j_P] = Tr\left\{\hat{\rho}\ \hat{T}_P\hat{S}^\dagger\hat{S}\exp\left[i\int_P d^4x j_P(x)\hat{\phi}_P(x)\right]\right\}, \tag{2.36}$$

with the subscript P distinguishing the time branches. Considering a useful formal expression for the interaction representation of the operator $\hat{S}^\dagger\hat{S}$:

$$\hat{S}^\dagger\hat{S} = \hat{T}_P\exp\left[-i\int_P dt\ \hat{H}_I^{\rm int}(t)\right], \tag{2.37}$$

and rewriting the closed-contour integral in the source term,

$$\begin{aligned}\int_P d^4x\ j_P(x)\hat{\phi}_P(x) &= \int_{-\infty}^{\infty} j(x_+)\hat{\phi}(x_+) + \int_{\infty}^{-\infty} j(x_-)\hat{\phi}(x_-) \\ &= \int_{-\infty}^{\infty} d^4x\ j(x_+)\hat{\phi}(x_+) - \int_{-\infty}^{\infty} d^4x\ j(x_-)\hat{\phi}(x_-)\ .\end{aligned} \tag{2.38}$$

Using Eq. (2.37) and Eq. (2.38) in Eq. (2.36), we obtain

$$Z[j_P] = Tr\left\{\hat{\rho}\ \hat{T}_P\exp\left[i\int_{-\infty}^{\infty} d^4x\ j(x_+)\hat{\phi}(x_+) - i\int_{-\infty}^{\infty} d^4x\ j(x_-)\hat{\phi}(x_-) - i\int_P dt\ \hat{H}_I^{\rm int}(t)\right]\right\}. \tag{2.39}$$

Taking the two fields $\hat{\phi}(x_+)$, $\hat{\phi}(x_+)$ as two components of a vector

$$\hat{\phi}(x) = \begin{pmatrix}\hat{\phi}(x_+)\\ \hat{\phi}(x_-)\end{pmatrix}, \tag{2.40}$$

with the associated current

$$\mathbf{j}(x) = \begin{pmatrix} j(x_+)\\ -j(x_-)\end{pmatrix}. \tag{2.41}$$

Eq. (2.38) can be rewritten as

$$\begin{aligned}\int_P d^4x\ j_P(x)\hat{\phi}_P(x) &= \int_{-\infty}^{\infty} d^4x\ j(x_+)\hat{\phi}(x_+) - \int_{-\infty}^{\infty} d^4x\ j(x_-)\hat{\phi}(x_-) \\ &= \int d^4x\ \mathbf{j}(x)\hat{\phi}(x)\ , \end{aligned} \tag{2.42}$$

and all closed-time path formulas go directly over into vector or matrix formulas whose integrals run only once along the positive time axis, for example

$$\int_P d^4x\ j_P(x)G_P(x,x')j_P(x') = \int d^4x\ \mathbf{j}(x)G(x,x')\mathbf{j}(x')\ , \tag{2.43}$$

where,

$$G_P(x,y) = G(x,y) = \begin{pmatrix} G_{++}(x,y) & G_{+-}(x,y) \\ G_{-+}(x,y) & G_{--}(x,y) \end{pmatrix} = \begin{pmatrix} G(x_+,y_+) & G(x_+,y_-) \\ G(x_-,y_+) & G(x_-,y_-) \end{pmatrix}\ . \tag{2.44}$$

If we functionally differentiate the generating functional with respect to j_P, we will obtain all Green functions. Two derivatives gives the two-point Green's function:

$$\begin{aligned} G_P(x,y) &= \frac{\delta}{i\delta j_P(x)}.\frac{\delta}{i\delta j_P(y)} Z[j_P]\Big|_{j_P=0} \\ &= Tr\left[\hat{\rho}\hat{T}_P\hat{S}^\dagger\hat{S}\hat{\phi}_P(x)\hat{\phi}_P(y)\right]\ , \end{aligned} \tag{2.45}$$

which we decompose according to the branches of the closed-time contour in the same way as the matrix (2.44). Here the system may be out of thermal equilibrium, so we will deal with ρ average rather than the thermal average, viz., ρ is an equilibrium density operator. Now:

$$G_P(x,y) = \langle \hat{T}_P\hat{\phi}_H(x_P)\hat{\phi}_H(y_P)\rangle_\rho\ , \tag{2.46}$$

and the four physically relevant time-dependent Green functions are:

- **Case-1:** If y_+ lies always before x_-, the path-ordering operator can be omitted so that

$$G_{-+}(x,y) = \langle \hat{\phi}_H(x)\hat{\phi}_H(y)\rangle_\rho\ . \tag{2.47}$$

- **Case-2:** In the opposite configuration, the path order is opposite. When reestablishing the original order, a negative sign arises for Fermion fields. Hence,

$$\begin{aligned} G_{+-}(x,y) &= \langle \hat{\phi}_H(y)\hat{\phi}_H(x)\rangle_\rho \\ &= \pm\langle \hat{\phi}_H(x)\hat{\phi}_H(y)\rangle_\rho \ . \end{aligned} \tag{2.48}$$

- **Case-3:** If both $x,$ and y lie on the upper branch, the path order coincides with the usual time order so that $G_{++}(x,y)$ is equal to the expectation.

$$\begin{aligned} G_{++}(x,y) &= \langle \hat{T}\hat{\phi}_H(x)\hat{\phi}_H(y)\rangle_\rho \\ &\equiv G(x,y) \ . \end{aligned} \tag{2.49}$$

- **Case-4:** Similarly, if $x,$ and y both lie on the lower branch, the path order coincides with the usual anti-time order:

$$\begin{aligned} G_{--}(x,y) &= \langle \hat{\bar{T}}\hat{\phi}_H(x)\hat{\phi}_H(y)\rangle_\rho \\ &\equiv G(y,x) \ . \end{aligned} \tag{2.50}$$

There exists a relationship among the four matrix elements of $G_P(x,y)$:

$$G_{++} + G_{--} = G_{+-} + G_{-+} \ . \tag{2.51}$$

As a consequence, we see that only three of the four matrix elements of $G_P(x,y)$ are linearly independent. Now, we will define the most convenient independent Green functions: the retarded and the advanced ones together with the expectation of the commutator (the anti-commutator for bosons) in the non-equilibrium case.

$$G^{\mathrm{R}}(x,y) = \Theta(x-y)\left\langle \left[\hat{\phi}_H(x),\hat{\phi}_H(y)\right]_\mp \right\rangle_\rho \ , \tag{2.52}$$

$$G^{\mathrm{A}}(x,y) = -\Theta(y-x)\left\langle \left[\hat{\phi}_H(x),\hat{\phi}_H(y)\right]_\mp \right\rangle_\rho \ , \tag{2.53}$$

$$A(x,y) = \left\langle \left[\hat{\phi}_H(x), \hat{\phi}_H(y)\right]_{\pm} \right\rangle_{\rho} . \tag{2.54}$$

The last expression coincides with the absorptive or dissipative part of the Green's function. The expectation of the commutator (the anti-commutator for fermions),

$$C(x,y) = \left\langle \left[\hat{\phi}_H(x), \hat{\phi}_H(y)\right]_{\mp} \right\rangle_{\rho} . \tag{2.55}$$

We have now the following relations [58]:

$$\begin{aligned} G^{\mathrm{R}} &= G_{-+} - G_{--} \\ &= G_{++} - G_{+-} \quad , \end{aligned} \tag{2.56}$$

$$\begin{aligned} G^{\mathrm{A}} &= G_{+-} - G_{--} \\ &= G_{++} - G_{-+} \quad , \end{aligned} \tag{2.57}$$

$$\begin{aligned} A &= G_{-+} + G_{+-} \\ &= G_{++} + G_{--} \quad , \end{aligned} \tag{2.58}$$

and

$$\begin{aligned} C &= G^{\mathrm{R}} - G^{\mathrm{A}} \\ &= G_{-+} - G_{+-} \quad . \end{aligned} \tag{2.59}$$

Expressing the matrix elements of the 2×2 Green functions in terms of the three independent quantities, we obtain [58],

$$\begin{aligned} G_{\mathrm{P}} &= \begin{pmatrix} G_{++} & G_{+-} \\ G_{-+} & G_{--} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2}\left(A + G^{\mathrm{R}} + G^{\mathrm{A}}\right) & \frac{1}{2}\left(A - G^{\mathrm{R}} + G^{A}\right) \\ \frac{1}{2}\left(A + G^{\mathrm{R}} - G^{\mathrm{A}}\right) & \frac{1}{2}\left(A - G^{\mathrm{R}} - G^{\mathrm{A}}\right) \end{pmatrix} . \end{aligned} \tag{2.60}$$

Now, we will use a similarity transformation (Keldysh's approach):

$$\tilde{G} = QG_PQ^{-1}, \qquad \text{with } \; Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = (Q^T)^{-1} \; . \tag{2.61}$$

This similarity transformation gives us [58]:

$$\tilde{G}(x,y) = \begin{pmatrix} 0 & G^{\rm A} \\ G^{\rm R} & A \end{pmatrix} \; ; \tag{2.62}$$

the source vectors:

$$\tilde{\hat{\phi}}(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} \hat{\phi}(x_+) - \hat{\phi}(x_-) \\ \hat{\phi}(x_+) + \hat{\phi}(x_-) \end{pmatrix}_{2\times 1} \; ; \tag{2.63}$$

and the field vectors:

$$\tilde{j}(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} j(x_+) - j(x_-) \\ j(x_+) + j(x_-) \end{pmatrix}_{2\times 1} \; . \tag{2.64}$$

Then the quadratic source term becomes

$$\begin{aligned} &\int dx^4 dx'^4 J_P(x) G_P(x,x') j_P(x') \\ &= \int dx^4 dx'^4 \begin{pmatrix} j(x_+) & -j(x_-) \end{pmatrix}_{1\times 2} \begin{pmatrix} G_{++} & G_{+-} \\ G_{-+} & G_{--} \end{pmatrix}_{2\times 2} (x,x') \begin{pmatrix} j(x'_+) \\ -j(x'_-) \end{pmatrix}_{2\times 1} \\ &= \int d^4x d^4x' \tilde{j}^T(x) \tilde{G}(x,x') \tilde{x}' \; . \end{aligned} \tag{2.65}$$

2.2.3 Influence functional

In this section, we will apply real-time formalism (with change in convention slightly) of previous section in the heavy quark systems in order to obtain the Feynman-Vernon influence functional. In the case of heavy quark system in the QGP, the open quantum system consists of heavy quarks and the environment is composed of light quarks and gluons. We assume that the time-evolution of the system starts at $t = -\infty$ and continues forever. Therefore, the contour $\mathcal{C}$ returns at $t = \infty$ and has both ends at $t = -\infty$.

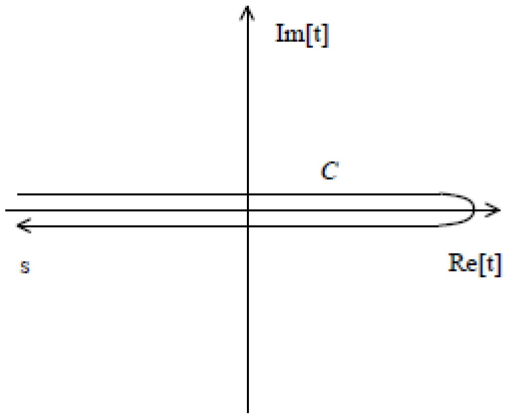


Figure 2.2: Closed-time path $\mathcal{C}$ in complex time plane, parameterized by s $(-\infty < s < \infty)$ (adapted from [11]).

In QCD at finite temperature, the generating functional is given by

$$Z\left[\eta,\bar{\eta}\right] = \int \mathcal{D}\left[\bar{\psi},\psi,\bar{q},q,A\right] \langle \bar{\psi},\bar{q},A(s=-\infty)|\rho_{\rm tot}|\psi,q,A(s=\infty)\rangle \times \exp\left[i\int_{\mathcal{C}} ds \int d^3x \Big\{ \mathcal{L}_{\rm QCD}(\bar{\psi},\psi,\bar{q},q,A) + \bar{\eta}\psi + \bar{\psi}\eta \Big\}\right] \quad , \quad (2.66)$$

where $(\eta,\bar{\eta})$ are sources for heavy quark fields, $\left(\bar{\psi},\psi\right)$ are the heavy quark fields, $(\bar{q},q)$ are light quark fields, $A = A^{a\mu}$ are gluon fields, and $\mathcal{L}_{\rm QCD}$ is the QCD Lagrangian density. Accounting for individual heavy quarks, Eq. (2.66) can be written as:

$$\begin{aligned} Z\left[\eta_1,\bar{\eta}_1,\eta_2,\bar{\eta}_2\right] = \int \mathcal{D}\left[\bar{\psi}_1,\psi_1,\bar{q}_1,q_1,A_1\right] \int \mathcal{D}\left[\bar{\psi}_2,\psi_2,\bar{q}_2,q_2,A_2\right] \\ \times \left\langle \bar{\psi}_1,\bar{q}_1,A_1(t=-\infty)|\rho_{tot}|\psi_2,q_2,A_2(t=-\infty)\right\rangle \\ \times \exp\left[i\int d^4x \Big\{ \mathcal{L}_{\rm QCD}(\bar{\psi}_1,\psi_1,\bar{q}_1,q_1,A_1) + \bar{\eta}_1\psi_1 + \bar{\psi}_1\eta_1 \Big\}\right] \\ \times \exp\left[-i\int d^4x \Big\{ \mathcal{L}_{\rm QCD}(\bar{\psi}_2,\psi_2,\bar{q}_2,q_2,A_2) + \bar{\eta}_2\psi_2 + \bar{\psi}\eta_2 \Big\}\right] \quad . \quad (2.67) \end{aligned}$$

Now, the matrix element of the density matrix is given by

$$\begin{aligned}
&\left\langle \bar{\psi}_1, \bar{q}_1, A_1(t=-\infty)|\rho_{\rm tot}|\psi_2, q_2, A_2(t=-\infty)\right\rangle \\
&\qquad = \left\langle \bar{q}_1, A_1(t=-\infty)|\rho_E^{eq}|q_2, A_2(t=-\infty)\right\rangle . \left\langle \bar{\psi}_1(t=-\infty)|\rho_S|\psi_2(t=-\infty)\right\rangle \ , \quad (2.68)
\end{aligned}$$

where, we have used the fact that

$$\rho_{tot} = \rho_{\rm E}^{\rm eq} \otimes \rho_S \ . \tag{2.69}$$

Here, $\rho_{\rm E}^{\rm eq}$ is the equilibrium density matrix for the environment degrees of freedom (i.e. light quarks and gluons), and ρ_S is an arbitrary density matrix for the system degrees of freedom (i.e. heavy quarks). Using Eq. (2.68) in Eq. (2.67) and integrating out the gluon and light quark fields:

$$\begin{aligned}
Z\left[\eta_1, \bar{\eta}_1, \eta_2, \bar{\eta}_2\right] = &\int \mathcal{D}\left[\bar{\psi}_1, \psi_1, \bar{\psi}_2, \psi_2\right] \left\langle \bar{\psi}_1(-\infty)\middle|\rho_S\middle|\psi_2(-\infty)\right\rangle \\
&\times \exp\left[i\int d^4x \left\{\mathcal{L}_\psi\left(\bar{\psi}_1, \psi_1\right) + \bar{\eta}_1\psi_1 + \bar{\psi}_1\eta_1\right\}\right] \\
&\times \exp\left[-i\int d^4x \left\{\mathcal{L}_\psi\left(\bar{\psi}_2, \psi_2\right) + \bar{\eta}_2\psi_2 + \bar{\psi}_2\eta_2\right\}\right] \\
\times \int \mathcal{D}\left[\bar{q}_1, q_1, A_1, \bar{q}_2, q_2, A_2, \right] &\left\langle \bar{q}_1, A_1(-\infty)\middle|\rho_{\rm E}^{\rm eq}\middle|\bar{q}_2, A_2(-\infty)\right\rangle \\
&\times \exp\left[i\int d^4x \left\{\mathcal{L}_{g+q}\left(\bar{q}_1, q_1, A_1\right) - g j_1^{a\mu} A_{1\mu}^a\right\}\right] \\
&\times \exp\left[-i\int d^4x \left\{\mathcal{L}_{g+q}\left(\bar{q}_2, q_2, A_2\right) - g j_2^{a\mu} A_{2\mu}^a\right\}\right] \\
&= \int \mathcal{D}\left[\bar{\psi}_1, \psi_1, \bar{\psi}_2, \psi_2\right] \left\langle \bar{\psi}_1(-\infty)\middle|\rho_S\middle|\psi_2(-\infty)\right\rangle \\
&\times \exp\left[i\int d^4x \left\{\mathcal{L}_\psi\left(\bar{\psi}_1, \psi_1\right) + \bar{\eta}_1\psi_1 + \bar{\psi}_1\eta_1\right\}\right] \\
&\times \exp\left[-i\int d^4x \left\{\mathcal{L}_\psi\left(\bar{\psi}_2, \psi_2\right) + \bar{\eta}_2\psi_2 + \bar{\psi}_2\eta_2\right\}\right] . \ Z_{\rm FV} \quad , \quad (2.70)
\end{aligned}$$

where, $\mathcal{L}_\psi$ and $\mathcal{L}_{g+q}$ are the Lagrangian densities of free heavy quarks and interacting gluons and light quarks.

The couplings between heavy quarks and gluons appear in the Feynman-Vernon influence functional i.e. $Z_{\rm FV}$ as source terms for gluons.

$$\begin{aligned} Z_{\rm FV} &= Z_{\rm FV}\left[j_1^{a\mu}, j_2^{a\mu}\right] \\ &= Z_{\rm FV}\left[\bar{\psi}_1 t^a \gamma^\mu \psi_1, \bar{\psi}_2 t^a \gamma^\mu \psi_2\right] \ , \end{aligned} \tag{2.71}$$

where,

$$\begin{aligned} Z_{\rm FV} = \int \mathcal{D}\left[\bar{q}_1, q_1, A_1, \bar{q}_2, q_2, A_2,\right] \left\langle \bar{q}_1, A_1(-\infty)\right|\rho_E^{eq}\left|\bar{q}_2, A_2(-\infty)\right\rangle \\ \times \exp\left[i\int d^4x \left\{\mathcal{L}_{g+q}\left(\bar{q}_1, q_1, A_1\right) - g j_1^{a\mu} A_{1\mu}^a\right\}\right] \\ \times \exp\left[-i\int d^4x \left\{\mathcal{L}_{g+q}\left(\bar{q}_2, q_2, A_2\right) - g j_2^{a\mu} A_{2\mu}^a\right\}\right] \ . \end{aligned} \tag{2.72}$$

Now, we will define two-point functions as follows:

$$G^{\rm F}_{ab,\mu\nu}(x-y) \equiv \langle T \mathbf{A}^a_\mu(x) \mathbf{A}^b_\nu(y)\rangle \ , \tag{2.73}$$

$$G^{\tilde{\rm F}}_{ab,\mu\nu}(x-y) \equiv \langle \tilde{T} \mathbf{A}^a_\mu(x) \mathbf{A}^b_\nu(y)\rangle \ , \tag{2.74}$$

$$G^{>}_{ab,\mu\nu}(x-y) \equiv \langle \mathbf{A}^a_\mu(x) \mathbf{A}^b_\nu(y)\rangle \ , \tag{2.75}$$

$$G^{<}_{ab,\mu\nu}(x-y) \equiv \langle \mathbf{A}^b_\nu(y) \mathbf{A}^a_\mu(x)\rangle \ . \tag{2.76}$$

The low frequency behavior of the gluonic two-point functions allows us to write

$$G(\omega, \mathbf{x}-\mathbf{y}) \approx \bar{G}(\mathbf{x}-\mathbf{y}) + \omega \bar{G}'(\mathbf{x}-\mathbf{y}) \ , \tag{2.77}$$

where

$$\begin{aligned} \bar{G}(\mathbf{x}-\mathbf{y}) &\equiv G(\omega=0, \mathbf{x}-\mathbf{y}) \\ &= \int_{-\infty}^{\infty} d(x^0-y^0) G(x-y) \ , \end{aligned} \tag{2.78}$$

and

$$\begin{aligned} \bar{G}'(\mathbf{x}-\mathbf{y}) &\equiv \frac{\partial}{\partial\omega} G(\omega, \mathbf{x}-\mathbf{y})\Big|_{\omega=0} \\ &= i\int_{-\infty}^{\infty} d(x^0-y^0)(x^0-y^0) G(x-y) \ . \end{aligned} \tag{2.79}$$

This is equivalent to the instantaneous approximation:

$$G(x-y) \approx \bar{G}(\mathbf{x}-\mathbf{y})\delta(x^0-y^0) + i\bar{G}'(\mathbf{x}-\mathbf{y})\frac{\partial}{\partial(x^0-y^0)}\delta(x^0-y^0) \ , \tag{2.80}$$

with which the interaction term can be approximated as:

$$\begin{aligned}
&\int d^4x d^4y j(x) G(x-y) j(y) \\
&\equiv \int dt \int d^3x d^3y \Big[\bar{G}(\mathbf{x}-\mathbf{y}) j(t,\mathbf{x}) j(t,\mathbf{y}) - \frac{i}{2}\bar{G}'(\mathbf{x}-\mathbf{y}) \Big\{ \partial_0 j(t,\mathbf{x}) j(t,\mathbf{y}) - j(t,\mathbf{x}) \partial_0 j(t,\mathbf{y}) \Big\} \Big] \ .
\end{aligned} \tag{2.81}$$

The expansion of $\ln\left\{ Z_{\rm FV}\left[j_1^{a\mu}, j_2^{a\mu} \right] \right\}$ in terms of $j_1^{a\mu}$ and $j_2^{a\mu}$ (considering leading order only) is [11]

$$\begin{aligned}
&\ln\left\{ Z_{\rm FV}\left[j_1^{a\mu}, j_2^{a\mu} \right] \right\} \\
&= -\frac{g^2}{2} \int d^4x d^4y \begin{pmatrix} j_1^{a\mu}(x) & j_2^{a\mu}(x) \end{pmatrix}_{1\times 2} \begin{pmatrix} G^F_{ab,\mu\nu}(x-y) & -G^<_{ab,\mu\nu}(x-y) \\ -G^>_{ab,\mu\nu}(x-y) & G^{\tilde{F}}_{ab,\mu\nu}(x-y) \end{pmatrix}_{2\times 2} \begin{pmatrix} j_1^{b\nu}(y) \\ j_2^{b\nu}(y) \end{pmatrix}_{2\times 1} \\
&\equiv i S_{\rm FV}\left[j_1^{a\mu}, j_2^{a\mu} \right] \ .
\end{aligned} \tag{2.82}$$

As said before, here we are using slightly different sign convention. If we compare Eq. (2.65) and Eq. (2.82), we will find:

$$\begin{aligned}
&j(x_+) \equiv j_1^{a\mu}(x), \quad -j(x_-) \equiv j_2^{a\mu}(x) \\
&G_{++} \equiv G^F_{ab,\mu\nu}(x-y), \quad G_{--} \equiv G^{\tilde{F}}_{ab,\mu\nu}(x-y), \quad G_{+-} \equiv -G^<_{ab,\mu\nu}(x-y), \quad G_{-+} \equiv -G^>_{ab,\mu\nu}(x-y) \\
&j(x'_+) \equiv j_1^{b\mu}(y), \quad -j(x'_-) \equiv j_2^{b\mu}(y) \ .
\end{aligned} \tag{2.83}$$

On simplifying, Eq. (2.82) yields

$$iS_{\rm FV}\left[j_1^{a\mu},j_2^{a\mu}\right]=\ln\left\{Z_{\rm FV}\left[j_1^{a\mu},j_2^{a\mu}\right]\right\}$$

$$=-\frac{g^2}{2}\int d^4xd^4y\begin{pmatrix} j_1^{a\mu}(x) & j_2^{a\mu}(x)\end{pmatrix}_{1\times2}\begin{pmatrix} G^{\rm F}_{ab,\mu\nu}(x-y) & -G^{<}_{ab,\mu\nu}(x-y)\\ -G^{>}_{ab,\mu\nu}(x-y) & G^{\tilde{\rm F}}_{ab,\mu\nu}(x-y)\end{pmatrix}_{2\times2}\begin{pmatrix} j_1^{b\nu}(y)\\ j_2^{b\nu}(y)\end{pmatrix}_{2\times1}$$

$$=-\frac{g^2}{2}\int d^4xd^4y\begin{pmatrix} j_1^{a\mu}(x) & j_2^{a\mu}(x)\end{pmatrix}_{1\times2}\begin{pmatrix} G^{F}_{ab,\mu\nu}(x-y).j_1^{b\nu}(y)-G^{<}_{ab,\mu\nu}(x-y).j_2^{b\nu}(y)\\ -G^{>}_{ab,\mu\nu}(x-y).j_1^{b\nu}(y)+G^{\tilde{F}}_{ab,\mu\nu}(x-y).j_2^{b\nu}(y)\end{pmatrix}_{2\times1}$$

$$=-\frac{g^2}{2}\int d^4xd^4y\begin{pmatrix} j_1^{a0}(x) & j_2^{a0}(x)\end{pmatrix}_{1\times2}\begin{pmatrix} G^{F}_{ab,00}(x-y)j_1^{b0}(y)-G^{<}_{ab,00}(x-y)j_2^{b0}(y)\\ -G^{>}_{ab,00}(x-y)j_1^{b0}(y)+G^{\tilde{F}}_{ab,00}(x-y)j_2^{b0}(y)\end{pmatrix}_{2\times1}$$

$$=-\frac{g^2}{2}\int d^4xd^4y\Big[j_1^{a0}(x)G^{F}_{ab,00}(x-y)j_1^{b0}(y)-j_1^{a0}(x)G^{<}_{ab,00}(x-y)j_2^{b0}(y)$$
$$-j_2^{a0}(x)G^{>}_{ab,00}(x-y)j_1^{b0}(y)+j_2^{a0}(x)G^{\tilde{F}}_{ab,00}(x-y)j_2^{b0}(y)\Big]\ . \quad (2.84)$$

Hence,

$$S_{\rm FV}\left[j_1^{a\mu},j_2^{a\mu}\right]=-\frac{g^2}{2}\int d^4xd^4y\ \ (-i)\Big[j_1^{a0}(x)G^{F}_{ab,00}(x-y)j_1^{b0}(y)-j_1^{a0}(x)G^{<}_{ab,00}(x-y)j_2^{b0}(y)$$
$$-j_2^{a0}(x)G^{>}_{ab,00}(x-y)j_1^{b0}(y)+j_2^{a0}(x)G^{\tilde{F}}_{ab,00}(x-y)j_2^{b0}(y)\Big]\ . \quad (2.85)$$

Now, applying Eq. (2.81) on $G^{\rm F}_{ab,00},\ G^{<}_{ab,00},\ G^{>}_{ab,00}$, and $G^{\tilde{\rm F}}_{ab,00}$ separately, Eq. (2.85) can be written as

$S_{\rm FV}\left[j_1^{a\mu},j_2^{a\mu}\right]$

$$
\begin{aligned}
&= -\frac{g^2}{2}\int dt \int d^3x d^3y \;\; (-i)\Bigg[\\
&+ \bar{G}^F_{ab,00}(\mathbf{x}-\mathbf{y}) j_1^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) - \frac{i}{2}\bar{G}'^F_{ab,00}(\mathbf{x}-\mathbf{y})\Big\{\partial_0 j_1^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) - j_1^{a0}(t,\mathbf{x})\partial_0 j_1^{b0}(t,\mathbf{y})\Big\} \\
&- \bar{G}^<_{ab,00}(\mathbf{x}-\mathbf{y}) j_1^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) + \frac{i}{2}\bar{G}'^<_{ab,00}(\mathbf{x}-\mathbf{y})\Big\{\partial_0 j_1^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) - j_1^{a0}(t,\mathbf{x})\partial_0 j_2^{b0}(t,\mathbf{y})\Big\} \\
&- \bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) j_2^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) + \frac{i}{2}\bar{G}'^>_{ab,00}(\mathbf{x}-\mathbf{y})\Big\{\partial_0 j_2^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) - j_2^{a0}(t,\mathbf{x})\partial_0 j_1^{b0}(t,\mathbf{y})\Big\} \\
&+ \bar{G}^{\tilde{F}}_{ab,00}(\mathbf{x}-\mathbf{y}) j_2^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) - \frac{i}{2}\bar{G}'^{\tilde{F}}_{ab,00}(\mathbf{x}-\mathbf{y})\Big\{\partial_0 j_2^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) - j_2^{a0}(t,\mathbf{x})\partial_0 j_2^{b0}(t,\mathbf{y})\Big\}\Bigg]
\end{aligned}
\tag{2.86}
$$

Also, all of the gluonic two-point functions in instantaneous approximation can be expressed in terms of two real function $\bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y})$ and $\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y})$ [11]:

$$
\bar{G}^F_{ab00}(\mathbf{x}-\mathbf{y}) = -i\bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y}) + \bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \;, \tag{2.87}
$$

$$
\begin{aligned}
\bar{G}^{\tilde{F}}_{ab,00}(\mathbf{x}-\mathbf{y}) &= \Big(\bar{G}^F_{ab,00}(\mathbf{x}-\mathbf{y})\Big)^* \\
&= i\bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y}) + \bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \;,
\end{aligned}
\tag{2.88}
$$

$$
\bar{G}'^F_{ab,00}(\mathbf{x}-\mathbf{y}) = 0 \;, \tag{2.89}
$$

$$
\bar{G}'^{\tilde{F}}_{ab,00}(\mathbf{x}-\mathbf{y}) = 0 \;, \tag{2.90}
$$

$$
\bar{G}^<_{ab,00}(\mathbf{x}-\mathbf{y}) = \bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \;, \tag{2.91}
$$

$$
\bar{G}'^>_{ab,00}(\mathbf{x}-\mathbf{y}) = \frac{1}{2T}\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \;, \tag{2.92}
$$

$$
\bar{G}'^<_{ab,00}(\mathbf{x}-\mathbf{y}) = -\frac{1}{2T}\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \;, \tag{2.93}
$$

where we define the retarded propagator

$$
G^R_{ab,00}(\mathbf{x}-\mathbf{y}) \equiv i\theta(x^0-y^0)\Big\langle \Big[\mathbf{A}^a_0(x), \mathbf{A}^b_0(y)\Big] \Big\rangle \;. \tag{2.94}
$$

In the instantaneous approximation, $\bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y})$ and $\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y})$ are expressed in terms of an Euclidean correlator and a spectral function:

$$
\bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y}) = \int_0^\beta d\tau \Big\langle \mathrm{A}_{a0}(t=-i\tau,\mathbf{x})\mathrm{A}_{b0}(t=0,\mathbf{y})\Big\rangle \;, \tag{2.95}
$$

$$\bar{G}^{>}_{ab,00}(\mathbf{x}-\mathbf{y}) = T\frac{\partial}{\partial\omega}\sigma_{ab,00}(\omega,\mathbf{x}-\mathbf{y})\Big|_{\omega=0} \quad , \tag{2.96}$$

$$\sigma_{ab,00}(\omega,\mathbf{x}-\mathbf{y}) \equiv \int_{-\infty}^{\infty} dx^0 e^{i\omega(x^0-y^0)}\Big\langle \mathrm{A}_{a0}(x)\mathrm{A}_{b0}(y)\Big\rangle \ . \tag{2.97}$$

One can find the detailed derivation of these relations in Appendix A of [11]. Now, inserting Eq. (2.87) to Eq. (2.93) in Eq. (2.86), we obtain

$$\begin{aligned}
&S_{\mathrm{FV}}\left[j_1^{a\mu}, j_2^{a\mu}\right] \\
&= -\frac{g^2}{2}\int dt \int d^3x d^3y \ \ (-i)\Bigg[\\
&\qquad + \Big\{ -i\bar{G}^{R}_{ab,00}(\mathbf{x}-\mathbf{y}) + \bar{G}^{>}_{ab,00}(\mathbf{x}-\mathbf{y})\Big\} j_1^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) \\
&- \bar{G}^{>}_{ab,00}(\mathbf{x}-\mathbf{y}) j_1^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) - \frac{i}{4T}\bar{G}^{>}_{ab,00}(\mathbf{x}-\mathbf{y})\Big\{\partial_0 j_1^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) - j_1^{a0}(t,\mathbf{x})\partial_0 j_2^{b0}(t,\mathbf{y})\Big\} \\
&- \bar{G}^{>}_{ab,00}(\mathbf{x}-\mathbf{y}) j_2^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) + \frac{i}{4T}\bar{G}^{>}_{ab,00}(\mathbf{x}-\mathbf{y})\Big\{\partial_0 j_2^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) - j_2^{a0}(t,\mathbf{x})\partial_0 j_1^{b0}(t,\mathbf{y})\Big\} \\
&\qquad + \Big\{ i\bar{G}^{R}_{ab,00}(\mathbf{x}-\mathbf{y}) + \bar{G}^{>}_{ab,00}(\mathbf{x}-\mathbf{y})\Big\} j_2^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y})\Bigg] \ .
\end{aligned} \tag{2.98}$$

Rearranging Eq. (2.98), we get

$S_{\mathrm{FV}}\left[j_1^{a\mu}, j_2^{a\mu}\right]$

$$
\begin{aligned}
&= -\frac{1}{2}\int dt \int d^3x d^3y \left[(-g^2)\Big\{ \bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y}) + i\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \Big\} j_1^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) \right. \\
&\qquad \left. + (-g^2)\Big\{ \bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y}) - i\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \Big\} j_2^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) \right] \\
&\quad + \frac{i}{2}\int dt \int d^3x d^3y \; (-g^2)\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \left[j_1^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) + j_2^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) \right] \\
&\quad - \frac{1}{8T}\int dt \int d^3x d^3y \; (-g^2)\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \left[\Big\{ \partial_0 j_1^{a0}(t,\mathbf{x}) j_2^{b0}(t,\mathbf{y}) - j_1^{a0}(t,\mathbf{x}) \partial_0 j_2^{b0}(t,\mathbf{y}) \Big\} \right. \\
&\qquad \left. - \Big\{ \partial_0 j_2^{a0}(t,\mathbf{x}) j_1^{b0}(t,\mathbf{y}) - j_2^{a0}(t,\mathbf{x}) \partial_0 j_1^{b0}(t,\mathbf{y}) \Big\} \right] .
\end{aligned}
\tag{2.99}
$$

Now, we will define a complex potential $V(\mathbf{x}-\mathbf{y})$ and a dissipative coupling $D(\mathbf{x}-\mathbf{y})$ through:

$$
-g^2 \left\{ \bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y}) + i\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \right\} \equiv V(\mathbf{x}-\mathbf{y})\delta_{ab} \; , \tag{2.100}
$$

$$
-g^2 \left\{ \bar{G}^R_{ab,00}(\mathbf{x}-\mathbf{y}) - i\bar{G}^>_{ab,00}(\mathbf{x}-\mathbf{y}) \right\} \equiv V^*(\mathbf{x}-\mathbf{y})\delta_{ab} \; , \tag{2.101}
$$

$$
-g^2 \; \bar{G}^>_{ab,00} \equiv D(\mathbf{x}-\mathbf{y})\delta_{ab} \; . \tag{2.102}
$$

Using Eq. (2.100), Eq. (2.101), and Eq. (2.102); Eq. (2.99) becomes

$$
\begin{aligned}
S_{\rm FV}\left[j_1^{a\mu}, j_2^{a\mu}\right] &= -\frac{1}{2}\int dt \; d^3x d^3y \left[V(\mathbf{x}-\mathbf{y}) j_1^{a0}(t,\mathbf{x}) j_1^{a0}(t,\mathbf{y}) + V^*(\mathbf{x}-\mathbf{y}) j_2^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) \right] \\
&\quad + i\int dt \; d^3x d^3y \; D(\mathbf{x}-\mathbf{y}) j_1^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) \\
&\quad - \frac{1}{4T}\int dt \; d^3x d^3y \; \nabla_x D(\mathbf{x}-\mathbf{y}). \left[\mathbf{j}^a_{1,\rm NR}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) + j_1^{a0}(t,\mathbf{x}) \mathbf{j}^a_{2,\rm NR}(t,\mathbf{y}) \right] ,
\end{aligned}
\tag{2.103}
$$

where

$$
\mathbf{j}^a_{\rm NR} = Q^\dagger \left(\frac{\overrightarrow{\nabla} - \overleftarrow{\nabla}}{2iM} \right) t^a Q - Q_C \left(\frac{\overrightarrow{\nabla} - \overleftarrow{\nabla}}{2iM} \right) t^a Q_C^\dagger \quad . \tag{2.104}
$$

In summary, we have derived an effective action S_{1+2} defined as follows:

$$\begin{aligned} S_{1+2} &\equiv S_{\rm kin} + S_{\rm FV} \\ &= \int d^4x \left[\mathcal{L}_\psi(\bar{\psi}_1, \psi_1) - \mathcal{L}_\psi(\bar{\psi}_2, \psi_2)\right] + S_{\rm FV} \ , \end{aligned} \tag{2.105}$$

at leading order of the perturbative expansion and in the non-relativistic limit:

$$S_{1+2} = S_{\rm kin}^{\rm NR} + S_{\rm FV}^{\rm LONR} + ... \tag{2.106}$$

$$S_{\rm kin}^{\rm NR} = \int d^4x \left[\mathcal{L}_\psi^{\rm NR}(\psi_1'^\dagger, \psi_1') - \mathcal{L}_\psi^{\rm NR}(\psi_2'^\dagger, \psi_2')\right] \ , \tag{2.107}$$

$$\begin{aligned} S_{\rm FV}^{\rm LONR} = -\frac{1}{2}\int dt\ d^3x d^3y\ &\left[V(\mathbf{x}-\mathbf{y}) j_1^{a0}(t,\mathbf{x}) j_1^{a0}(t,\mathbf{y}) - V^*(\mathbf{x}-\mathbf{y}) j_2^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y})\right] \\ &+ i\int dt\ d^3x d^3y\ D(\mathbf{x}-\mathbf{y}) j_1^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) \\ -\frac{1}{4T}\int dt\ d^3x d^3y\ \nabla_x D(\mathbf{x}-\mathbf{y}).&\left[\mathbf{j}_{1,\rm NR}^a(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) + j_1^{a0}(t,\mathbf{x})\mathbf{j}_{2,\rm NR}^a(t,\mathbf{y})\right] \ . \end{aligned} \tag{2.108}$$

Here, NR stands for non-relativistic and LONR stands for leading order non-relativistic terms.

2.2.4 Renormalized effective Hamiltonian

In this section, we will construct a renormalized effective Hamiltonian from the effective action (2.106) which we have derived in the previous section. The effective action is:

$$S_{1+2} = S_{\rm kin}^{\rm NR} + S_{\rm FV}^{\rm LONR} + ... \tag{2.109}$$

And the corresponding Hamiltonian is:

$$\mathbf{H}_{1+2} = \mathbf{H}_{\rm kin}^{\rm NR} + \mathbf{H}_{\rm FV}^{\rm LONR} + ... \tag{2.110}$$

Now, we will derive $\mathbf{H}_{\rm kin}^{\rm NR}$ from $S_{\rm kin}^{\rm NR}$ i.e.,

$$S_{\rm kin}^{\rm NR} = \int d^4x \left[\mathcal{L}_\psi^{\rm NR}(\psi_1^\dagger, \psi_1) - \mathcal{L}_\psi^{\rm NR}(\psi_2^\dagger, \psi_2)\right] \ . \tag{2.111}$$

We have also derived Lagrangian density for heavy quarks in non-relativistic limit (2.20):

$$\mathcal{L}_{\Psi}^{NR} = Q^{\dagger}\left(i\partial_0 - M + \frac{\nabla^2}{2M}\right)Q + Q_C\left(i\partial_0 + M - \frac{\nabla^2}{2M}\right)Q_C^{\dagger}\ . \tag{2.112}$$

So,

$$\mathcal{L}_{\psi}^{\rm NR}(\psi_1^{\dagger},\psi_1) = Q_1^{\dagger}\left(i\partial_0 - M + \frac{\nabla^2}{2M}\right)Q_1 + Q_{1C}\left(i\partial_0 + M - \frac{\nabla^2}{2M}\right)Q_{1C}^{\dagger}\ , \tag{2.113}$$

$$\mathcal{L}_{\psi}^{\rm NR}(\psi_2^{\dagger},\psi_2) = Q_2^{\dagger}\left(i\partial_0 - M + \frac{\nabla^2}{2M}\right)Q_2 + Q_{2C}\left(i\partial_0 + M - \frac{\nabla^2}{2M}\right)Q_{2C}^{\dagger}\ . \tag{2.114}$$

Inserting Eq. (2.113) and Eq. (2.114) , in Eq. (2.111) we get:[1]

$$\begin{aligned}
S_{\rm kin}^{\rm NR} &= \int d^4x\left[\mathcal{L}_{\psi}^{\rm NR}(\psi_1^{\dagger},\psi_1) - \mathcal{L}_{\psi}^{\rm NR}(\psi_2^{\dagger},\psi_2)\right] \\
&= \int d^4x\Bigg[\left\{Q_1^{\dagger}\left(-M + \frac{\nabla^2}{2M}\right)Q_1 + Q_{1C}\left(M - \frac{\nabla^2}{2M}\right)Q_{1C}^{\dagger}\right\} \\
&\qquad - \left\{Q_2^{\dagger}\left(-M + \frac{\nabla^2}{2M}\right)Q_2 + Q_{2C}\left(M - \frac{\nabla^2}{2M}\right)Q_{2C}^{\dagger}\right\}\Bigg] \\
&= \int d^4x\Bigg[\left\{ - Q_1^{\dagger}MQ_1 + Q_1^{\dagger}\frac{\nabla^2}{2M}Q_1 + Q_{1C}MQ_{1C}^{\dagger} - Q_{1C}\frac{\nabla^2}{2M}Q_{1C}^{\dagger}\right\} \\
&\qquad - \left\{ - Q_2^{\dagger}MQ_2 + Q_2^{\dagger}\frac{\nabla^2}{2M}Q_2 + Q_{2C}MQ_{2C}^{\dagger} - Q_{2C}\frac{\nabla^2}{2M}Q_{2C}^{\dagger}\right\}\Bigg] \\
&= \int d^4x\Bigg[\left\{\left(-Q_1^{\dagger}MQ_1 + Q_{1C}MQ_{1C}^{\dagger}\right) + \left(Q_2^{\dagger}MQ_2 - Q_{2C}MQ_{2C}^{\dagger}\right)\right\} \\
&\quad + \left\{\left(Q_1^{\dagger}\frac{\nabla^2}{2M}Q_1 - Q_{1C}\frac{\nabla^2}{2M}Q_{1C}^{\dagger}\right) + \left(-Q_2^{\dagger}\frac{\nabla^2}{2M}Q_2 + Q_{2C}\frac{\nabla^2}{2M}Q_{2C}^{\dagger}\right)\right\}\Bigg]\ .
\end{aligned} \tag{2.115}$$

So, we get

$$\begin{aligned}
S_{\rm kin}^{\rm NR} &= \int d^4x\Bigg[\left\{\left(-Q_1^{\dagger}MQ_1 + Q_{1C}MQ_{1C}^{\dagger}\right) + \left(Q_2^{\dagger}MQ_2 - Q_{2C}MQ_{2C}^{\dagger}\right)\right\} \\
&\quad + \left\{\left(Q_1^{\dagger}\frac{\nabla^2}{2M}Q_1 - Q_{1C}\frac{\nabla^2}{2M}Q_{1C}^{\dagger}\right) + \left(-Q_2^{\dagger}\frac{\nabla^2}{2M}Q_2 + Q_{2C}\frac{\nabla^2}{2M}Q_{2C}^{\dagger}\right)\right\}\Bigg]\ .
\end{aligned} \tag{2.116}$$

Now, we obtain the corresponding term in the Hamiltonian $\mathbf{H}_{kin}^{NR}$. The sign is determined as follows:

[1] We are not including first terms of Eq. (2.113) , and Eq. (2.114) as these will not appear in the Hamiltonian

- Take a term from the action $S^{\rm NR}_{\rm kin}$
- Multiply the term with (-1)
- And finally enter that term in the Hamiltonian $\mathbf{H}^{NR}_{kin}$

By applying the above steps, we obtain the Hamiltonian $\mathbf{H}^{NR}_{kin}$:

$$\mathbf{H}^{\rm NR}_{\rm kin} = \int d^3x Z_M \Bigg\{ \left(Q_1^\dagger M Q_1 - Q_{1C} M Q_{1C}^\dagger\right) + \left(-Q_2^\dagger M Q_2 + Q_{2C} M Q_{2C}^\dagger\right) \Bigg\}$$
$$+ \int d^3x Z_\psi \Bigg\{ \left(-Q_1^\dagger \frac{\nabla^2}{2M} Q_1 + Q_{1C} \frac{\nabla^2}{2M} Q_{1C}^\dagger\right) + \left(Q_2^\dagger \frac{\nabla^2}{2M} Q_2 - Q_{2C} \frac{\nabla^2}{2M} Q_{2C}^\dagger\right) \Bigg\} . \quad (2.117)$$

We consider

$$\psi_1 = \begin{pmatrix} Q_1 \\ Q_{1C}^\dagger \end{pmatrix}_{2\times 1} , \quad \psi_1^\dagger \equiv \begin{pmatrix} Q_1^\dagger & Q_{1C} \end{pmatrix}_{1\times 2} \quad (2.118)$$

$$\psi_2 = \begin{pmatrix} Q_2 \\ Q_{2C}^\dagger \end{pmatrix}_{2\times 1} , \quad \psi_2^\dagger \equiv \begin{pmatrix} Q_2^\dagger & Q_{2C} \end{pmatrix}_{1\times 2} \quad (2.119)$$

$$\tilde{\psi}_2 = \psi_2^\dagger = \begin{pmatrix} Q_2^\dagger \\ Q_{2C} \end{pmatrix}_{2\times 1} , \quad \tilde{\psi}_2^\dagger \equiv \begin{pmatrix} Q_2 & Q_{2C}^\dagger \end{pmatrix}_{1\times 2} \quad (2.120)$$

Now,

$$\psi_1^\dagger \begin{pmatrix} M & 0 \\ 0 & -M \end{pmatrix}_{2\times 2} \psi_1 = \begin{pmatrix} Q_1^\dagger & Q_{1C} \end{pmatrix}_{1\times 2} \begin{pmatrix} M & 0 \\ 0 & -M \end{pmatrix}_{2\times 2} \begin{pmatrix} Q_1 \\ Q_{1C}^\dagger \end{pmatrix}_{2\times 1}$$
$$= \begin{pmatrix} Q_1^\dagger & Q_{1C} \end{pmatrix}_{1\times 2} \begin{pmatrix} MQ_1 \\ -MQ_{1C}^\dagger \end{pmatrix}_{2\times 1}$$
$$= Q_1^\dagger M Q_1 - Q_{1C} M Q\dagger_{1C} \quad (2.121)$$

$$\tilde{\psi}_2^\dagger \begin{pmatrix} M & 0 \\ 0 & -M \end{pmatrix}_{2\times 2} \tilde{\psi}_2 = \begin{pmatrix} Q_2 & Q_{2C}^\dagger \end{pmatrix}_{1\times 2} \begin{pmatrix} M & 0 \\ 0 & -M \end{pmatrix}_{2\times 2} \begin{pmatrix} Q_2^\dagger \\ Q_{2C} \end{pmatrix}_{2\times 1}$$
$$= \begin{pmatrix} Q_2 & Q_{2C}^\dagger \end{pmatrix}_{1\times 2} \begin{pmatrix} MQ_2^\dagger \\ -MQ_{2C} \end{pmatrix}_{2\times 1}$$
$$= Q_2 M Q_2^\dagger - Q_{2C}^\dagger M Q_{2C}$$
$$= -Q_2^\dagger M Q_2 + Q_{2C} M Q_{2C}^\dagger \quad , \tag{2.122}$$

where in the last step we use $Q_2 Q_2^\dagger = -Q_2^\dagger Q_2 \ \ , Q_{2C}^\dagger Q_{2C} = -Q_{2C} Q_{2C}^\dagger$.

$$\psi_1^\dagger \begin{pmatrix} -\frac{\nabla^2}{2M} & 0 \\ 0 & \frac{\nabla^2}{2M} \end{pmatrix}_{2\times 2} \psi_1 = \begin{pmatrix} Q_1^\dagger & Q_{1C} \end{pmatrix}_{1\times 2} \begin{pmatrix} -\frac{\nabla^2}{2M} & 0 \\ 0 & \frac{\nabla^2}{2M} \end{pmatrix}_{2\times 2} \begin{pmatrix} Q_1 \\ Q_{1C}^\dagger \end{pmatrix}_{2\times 1}$$
$$= \begin{pmatrix} Q_1^\dagger & Q_{1C} \end{pmatrix}_{1\times 2} \begin{pmatrix} -\frac{\nabla^2}{2M} Q_1 \\ \frac{\nabla^2}{2M} Q_{1C}^\dagger \end{pmatrix}_{2\times 1}$$
$$= -Q_1^\dagger \frac{\nabla^2}{2M} Q_1 + Q_{1C} \frac{\nabla^2}{2M} Q\dagger_{1C} \tag{2.123}$$

$$\tilde{\psi}_2^\dagger \begin{pmatrix} -\frac{\nabla^2}{2M} & 0 \\ 0 & \frac{\nabla^2}{2M} \end{pmatrix}_{2\times 2} \tilde{\psi}_2 = \begin{pmatrix} Q_2 & Q_{2C}^\dagger \end{pmatrix}_{1\times 2} \begin{pmatrix} -\frac{\nabla^2}{2M} & 0 \\ 0 & \frac{\nabla^2}{2M} \end{pmatrix}_{2\times 2} \begin{pmatrix} Q_2^\dagger \\ Q_{2C} \end{pmatrix}_{2\times 1}$$
$$= \begin{pmatrix} Q_2 & Q_{2C}^\dagger \end{pmatrix}_{1\times 2} \begin{pmatrix} -\frac{\nabla^2}{2M} Q_2^\dagger \\ \frac{\nabla^2}{2M} Q_{2C} \end{pmatrix}_{2\times 1}$$
$$= -Q_2 \frac{\nabla^2}{2M} Q_2^\dagger + Q_{2C}^\dagger \frac{\nabla^2}{2M} Q_{2C}$$
$$= Q_2^\dagger \frac{\nabla^2}{2M} Q_2 - Q_{2C} \frac{\nabla^2}{2M} Q_{2C}^\dagger \quad , \tag{2.124}$$

where in the last step we use $Q_2 Q_2^\dagger = -Q_2^\dagger Q_2 \ \ , Q_{2C}^\dagger Q_{2C} = -Q_{2C} Q_{2C}^\dagger$.

Inserting Eq. (2.121), Eq. (2.122), Eq. (2.123), and Eq. (2.124) in Eq. (2.117), we obtain the Hamiltonian $\mathbf{H}_{kin}^{NR}$:

$$\mathbf{H}^{\rm NR}_{\rm kin} = \int d^3x Z_M \Bigg\{ \psi_1^\dagger \begin{pmatrix} M & 0 \\ 0 & -M \end{pmatrix}_{2\times 2} \psi_1 + \tilde{\psi}_2^\dagger \begin{pmatrix} M & 0 \\ 0 & -M \end{pmatrix}_{2\times 2} \tilde{\psi}_2 \Bigg\}$$
$$+ \int d^3x Z_\psi \Bigg\{ \psi_1^\dagger \begin{pmatrix} -\frac{\nabla^2}{2M} & 0 \\ 0 & \frac{\nabla^2}{2M} \end{pmatrix}_{2\times 2} \psi_1 + \tilde{\psi}_2^\dagger \begin{pmatrix} -\frac{\nabla^2}{2M} & 0 \\ 0 & \frac{\nabla^2}{2M} \end{pmatrix}_{2\times 2} \tilde{\psi}_2 \Bigg\} \ . \quad (2.125)$$

Now, we will derive the interaction Hamiltonian $\mathbf{H}^{\rm LONR}_{\rm FV}$ from $S^{\rm LONR}_{\rm FV}$ (108) i.e.,

$$S^{\rm LONR}_{\rm FV} = -\frac{1}{2}\int dt\ d^3x d^3y\ \Big[V(\mathbf{x}-\mathbf{y}) j_1^{a0}(t,\mathbf{x}) j_1^{a0}(t,\mathbf{y}) + V^*(\mathbf{x}-\mathbf{y}) j_2^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) \Big]$$
$$+ i\int dt\ d^3x d^3y\ D(\mathbf{x}-\mathbf{y})\ j_1^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y})$$
$$- \frac{1}{4T}\int dt\ d^3x d^3y\ \nabla_x D(\mathbf{x}-\mathbf{y}) \cdot \Big[\mathbf{j}^a_{1,\rm NR}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) + j_1^{a0}(t,\mathbf{x}) \mathbf{j}^a_{2,\rm NR}(t,\mathbf{y}) \Big] \ . \quad (2.126)$$

And we obtain the corresponding terms in the interaction Hamiltonian $\mathbf{H}^{\rm LONR}_{\rm FV}$ as follows:

$$\mathbf{H}^{\rm LONR}_{\rm FV} = \mathbf{H}_{11} + \mathbf{H}_{22} + \mathbf{H}_{21} \ , \quad (2.127)$$
$$\mathbf{H}_{11} = \frac{Z_g^2}{2}\int d^3x d^3y\ V(\mathbf{x}-\mathbf{y}) \mathrm{T}_{sym}\Big\{ j_1^{a0}(t,\mathbf{x}) j_1^{a0}(t,\mathbf{y}) \Big\} \ , \quad (2.128)$$
$$\mathbf{H}_{22} = -\frac{Z_g^2}{2}\int d^3x d^3y\ V^*(\mathbf{x}-\mathbf{y}) \mathrm{T}_{sym}\Big\{ j_2^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) \Big\} \ , \quad (2.129)$$

$$\mathbf{H}_{21} = -iZ_g^2\int d^3x d^3y\ D(\mathbf{x}-\mathbf{y}) \mathrm{T}_{sym}\Big\{ j_2^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) \Big\}$$
$$+ \frac{Z_g^2}{4T}\int d^3x d^3y\ \nabla_x D(\mathbf{x}-\mathbf{y}) . \mathrm{T}_{sym}\Big\{ \mathbf{j}^a_{1,\rm NR}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) + j_1^{a0}(t,\mathbf{x}) \mathbf{j}^a_{2,\rm NR}(t,\mathbf{y}) \Big\} \ . \quad (2.130)$$

The definition of of the notation T_{sym} is:

$$\mathrm{T}_{sym}\Big\{ \prod_{i=1}^n j(t,\mathbf{x}_i) \Big\} \equiv \frac{1}{n!}\sum_{\sigma\in} \mathrm{T}\Big\{ \prod_{i=1}^n j(t_{\sigma(i)},\mathbf{x}_i) \Big\} \ , \quad (2.131)$$

where, S_n is the permutation group of n elements and σ is one of these permutations. The time ordered currents are:

$$\mathrm{T}j_1^{a0} = \psi_1^\dagger t^a \psi_1 \ , \ \mathrm{T}\mathbf{j}^a_{1,\mathrm{NR}} = \psi_1^\dagger \left[t^a \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \left(\frac{\overrightarrow{\nabla} - \overleftarrow{\nabla}}{2iM} \right) \psi_1 \ , \tag{2.132}$$

$$\mathrm{T}j_2^{a0} = -\tilde{\psi}_2^\dagger t^a \tilde{\psi}_2 \ , \ \mathrm{T}\mathbf{j}^a_{2,\mathrm{NR}} = \tilde{\psi}_2^\dagger \left[t^a \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \left(\frac{\overrightarrow{\nabla} - \overleftarrow{\nabla}}{2iM} \right) \tilde{\psi}_1 \ . \tag{2.133}$$

Here $\psi^\dagger$, ψ , M , and g are renormalized fields and parameters and Z_M, Z_ψ, and Z_g are determined by the normalized conditions. Using Eq. (2.117), we can obtain the normal-ordered product of the Hamiltonian $\mathbf{H}^{\mathrm{NR}}_{\mathrm{kin}}$:

$$\begin{aligned} \mathbf{H}^{\mathrm{NR}}_{\mathrm{kin}} = & \int d^3x Z_M \Big\{ \left(Q_1^\dagger M Q_1 - Q_{1C} M Q_{1C}^\dagger \right) + \left(-Q_2^\dagger M Q_2 + Q_{2C} M Q_{2C}^\dagger \right) \Big\} \\ & + \int d^3x Z_\psi \Big\{ \left(-Q_1^\dagger \frac{\nabla^2}{2M} Q_1 + Q_{1C} \frac{\nabla^2}{2M} Q_{1C}^\dagger \right) + \left(Q_2^\dagger \frac{\nabla^2}{2M} Q_2 - Q_{2C} \frac{\nabla^2}{2M} Q_{2C}^\dagger \right) \Big\} \ . \end{aligned} \tag{2.134}$$

From Eq. (2.120), we have

$$\tilde{\psi}_2 = \begin{pmatrix} Q_2^\dagger \\ Q_{2C} \end{pmatrix}_{2\times 1} \ , \ \tilde{\psi}_2^\dagger \equiv \begin{pmatrix} Q_2 & Q_{2C}^\dagger \end{pmatrix}_{1\times 2} \tag{2.135}$$

using new variables as:

$$\tilde{\psi}_2 = \begin{pmatrix} \tilde{Q}_2 \\ \tilde{Q}_{2C}^\dagger \end{pmatrix}_{2\times 1} \ , \ \tilde{\psi}_2^\dagger \equiv \begin{pmatrix} \tilde{Q}_2^\dagger & \tilde{Q}_{2C} \end{pmatrix}_{1\times 2} \tag{2.136}$$

and doing a replacement like $Q_2 \to \tilde{Q}_2^\dagger$, $Q_2^\dagger \to \tilde{Q}_2$, $Q_{2C} \to \tilde{Q}_{2C}^\dagger$, and $Q_{2C}^\dagger \to \tilde{Q}_{2C}$ in (134), we get

$$\begin{aligned} \mathbf{H}^{\mathrm{NR}}_{\mathrm{kin}} = & \int d^3x Z_M \ M \Big\{ Q_1^\dagger Q_1 + Q_{1C}^\dagger Q_{1C} - \tilde{Q}_2 \tilde{Q}_2^\dagger + \tilde{Q}_{2C}^\dagger \tilde{Q}_{2C} \Big\} \\ & + \int d^3x Z_\psi \Big\{ \left(-Q_1^\dagger \frac{\nabla^2}{2M} Q_1 + Q_{1C} \frac{\nabla^2}{2M} Q_{1C}^\dagger \right) + \left(\tilde{Q}_2 \frac{\nabla^2}{2M} \tilde{Q}_2^\dagger - \tilde{Q}_{2C}^\dagger \frac{\nabla^2}{2M} \tilde{Q}_{2C} \right) \Big\} \ . \end{aligned} \tag{2.137}$$

If N is normal ordering operator than

$$N\left[\tilde{Q}_2\tilde{Q}_2^\dagger\right] = -\tilde{Q}_2^\dagger\tilde{Q}_2 \tag{2.138}$$

$$N\left[Q_{1C}Q_{1C}^\dagger\right] = -Q_{1C}^\dagger Q_{1C} \tag{2.139}$$

Using Eq. (2.138) and Eq. (2.139) in Eq. (2.137), we get the normal-ordered product of the Hamiltonian $\mathbf{H}_{\rm kin}^{\rm NR}$:

$$\mathbf{H}_{\rm kin}^{\rm NR} = \int d^3x Z_M\ M\Big\{Q_1^\dagger Q_1 + Q_{1C}^\dagger Q_{1C} + \tilde{Q}_2^\dagger\tilde{Q}_2 + \tilde{Q}_{2C}^\dagger\tilde{Q}_{2C}\Big\}$$
$$+\int d^3x Z_\psi\Big\{Q_1^\dagger\left(-\frac{\nabla^2}{2M}\right)Q_1 + Q_{1C}^\dagger\left(-\frac{\nabla^2}{2M}\right)Q_{1C} + \tilde{Q}_2^\dagger\left(-\frac{\nabla^2}{2M}\right)\tilde{Q}_2 + \tilde{Q}_{2C}^\dagger\left(-\frac{\nabla^2}{2M}\right)\tilde{Q}_{2C}\Big\}. \tag{2.140}$$

As an alternative if we use Eq. (2.118), and Eq. (2.136) in Eq. (2.125), we will also arrive at Eq. (2.140). The normal-ordered product of the interaction terms are [11]:

$$\mathbf{H}_{11} = \frac{Z_g^2}{2}\int d^3x d^3y\ V(\mathbf{x}-\mathbf{y})\mathrm{N}\Big\{j_1^{a0}(t,\mathbf{x})j_1^{a0}(t,\mathbf{y})\Big\}$$
$$+\frac{Z_g^2 C_F}{2}\int d^3x\ V^*(0)\left(Q_1^\dagger Q_1 + Q_{1C}^\dagger Q_{1C}\right) + \text{constant}\ , \tag{2.141}$$

$$\mathbf{H}_{22} = -\frac{Z_g^2}{2}\int d^3x d^3y\ V^*(\mathbf{x}-\mathbf{y})\mathrm{N}\Big\{j_2^{a0}(t,\mathbf{x})j_2^{a0}(t,\mathbf{y})\Big\}$$
$$+\frac{Z_g^2 C_F}{2}\int d^3x\ V^*(0)\left(\tilde{Q}_2^\dagger\tilde{Q}_1 + \tilde{Q}_{2C}^\dagger\tilde{Q}_{2C}\right) + \text{constant}\ , \tag{2.142}$$

$$\mathbf{H}_{21} = -iZ_g^2\int d^3x d^3y\ D(\mathbf{x}-\mathbf{y})\mathrm{N}\Big\{j_2^{a0}(t,\mathbf{x})j_2^{a0}(t,\mathbf{y})\Big\}$$
$$+\frac{Z_g^2}{4T}\int d^3x d^3y\ \nabla_x D(\mathbf{x}-\mathbf{y}).\mathrm{N}\Big\{\mathbf{j}_{1,\rm NR}^a(t,\mathbf{x})j_2^{a0}(t,\mathbf{y}) + j_1^{a0}(t,\mathbf{x})\mathbf{j}_{2,\rm NR}^a(t,\mathbf{y})\Big\}\ , \tag{2.143}$$

where, $C_F = \frac{N_C^2-1}{2N_C} = \frac{4}{3}$. And the complex potential at the origin $V(0)$ consists of a vacuum part and a finite-temperature part i.e., $V(0) = V_{\rm vac} + V_{\rm med}$. After renormalization, the renormalized effective Hamiltonian is finally obtained as [11] :

$$\begin{aligned}
\mathbf{H}_{1+2} = & \int d^3x \ \left[aM \left(Q_1^\dagger Q_1 + Q_{1C}^\dagger Q_{1C} \right) + \left\{ Q_1^\dagger \left(-\frac{\nabla^2}{2M} \right) Q_1 + Q_{1C}^\dagger \left(-\frac{\nabla^2}{2M} \right) Q_{1C} \right\} \right] \\
& + \int d^3x \ \left[a^* M \left(\tilde{Q}_2^\dagger \tilde{Q} + \tilde{Q}_{2C}^\dagger \tilde{Q}_{2C} \right) + \left\{ \tilde{Q}_2^\dagger \left(-\frac{\nabla^2}{2M} \right) \tilde{Q}_2 + \tilde{Q}_{2C}^\dagger \left(-\frac{\nabla^2}{2M} \right) \tilde{Q}_{2C} \right\} \right] \\
& + \frac{1}{2} \int d^3x d^3y \ \Bigg[v(\mathbf{x}-\mathbf{y}) \mathrm{N}\Big\{ j_1^{a0}(t,\mathbf{x}) j_1^{a0}(t,\mathbf{y}) \Big\} - v^*(\mathbf{x}-\mathbf{y}) \mathrm{N}\Big\{ j_2^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) \Big\} \\
& \qquad - 2id(\mathbf{x}-\mathbf{y}) \mathrm{N}\Big\{ j_1^{a0}(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) \Big\} \\
& \qquad + \frac{1}{2T} \overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y}) . \mathrm{N}\Big\{ \mathbf{j}_{1,\mathrm{NR}}^a(t,\mathbf{x}) j_2^{a0}(t,\mathbf{y}) + j_1^{a0}(t,\mathbf{x}) \mathbf{j}_{2,\mathrm{NR}}^a(t,\mathbf{y}) \Big\} \Bigg] \\
& + ..., \quad (2.144)
\end{aligned}$$

where

$$\begin{aligned}
a &= 1 + \frac{Z_g^2 C_F V_{\rm med}}{2M} \\
&= 1 + \frac{C_F}{2M} \frac{g(T)^2}{4\pi} (-\omega_D - iT) \ , \qquad (2.145)
\end{aligned}$$

$$\begin{aligned}
v(\mathbf{r}) &= Z_g^2 V(\mathbf{r}) \\
&= \frac{g(T)^2}{4\pi} \left\{ \frac{e^{-\omega_D r}}{r} - 2iT\omega_D^2 \int_0^\infty dk \ \frac{\sin(kr)}{r(k^2+\omega_D^2)^2} \right\} \ , \qquad (2.146)
\end{aligned}$$

$$\begin{aligned}
d(\mathbf{r}) &= Z_g^2 D(\mathbf{r}) \\
&= \mathrm{Im}[v(\mathbf{r})] \\
&= -\frac{g(T)^2 \omega_D^2 \ T}{2\pi} \int_0^\infty dk \ \frac{\sin(kr)}{r(k^2+\omega_D^2)^2} \ , \qquad (2.147)
\end{aligned}$$

$$\omega_D^2 = g(T)^2 \ T^2 \left(\frac{N_c + \frac{N_f}{2}}{3} \right) \ , \qquad (2.148)$$

and,

$$\mathbf{r} = \mathbf{x} - \mathbf{y} \ . \qquad (2.149)$$

2.2.5 Master equation for the reduced density matrix

In this section, we will derive the master equation for the reduced density matrix for a single heavy quark system. The matrix element of $\rho_S(t)$ between the heavy quark coherent states $\langle Q_1^*, Q_{1C}^*|$ and $\langle \tilde{Q}_2^*, \tilde{Q}_{2C}^*|$ is given as a functional of the heavy quark fields by

$$\rho_S\left[t, Q_1^*, Q_{1C}^*, \tilde{Q}_2^*, \tilde{Q}_{2C}^*\right] = \langle Q_1^*, Q_{1C}^*|\rho_S(t)|\tilde{Q}_2^*, \tilde{Q}_{2C}^*\rangle \tag{2.150}$$

$$\langle Q_1^*, Q_{1C}^*| \equiv \langle\Omega| \exp\left[-\int d^3x \Big\{Q(\mathbf{x})Q_1^*(\mathbf{x}) + Q_C(\mathbf{x})Q_{1C}^*(\mathbf{x})\Big\}\right] , \tag{2.151}$$

$$|\tilde{Q}_2^*, \tilde{Q}_{2C}^*\rangle \equiv \exp\left[-\int d^3x \Big\{\tilde{Q}_2^*(\mathbf{x})Q^\dagger(\mathbf{x}) + \tilde{Q}_{2C}^*(\mathbf{x})Q_C^\dagger(\mathbf{x})\Big\}\right] |\Omega\rangle \ . \tag{2.152}$$

The time-evolution of $\rho_S\left[t, Q_1^*, Q_{1C}^*, \tilde{Q}_2^*, \tilde{Q}_{2C}^*\right]$ is obtained by the functional Schrödinger equation:

$$\begin{aligned} & i\frac{\partial}{\partial t}\rho_S\left[t, Q_1^*, Q_{1C}^*, \tilde{Q}_2^*, \tilde{Q}_{2C}^*\right] \\ & \quad = \mathrm{H}_{1+2}\left[Q^\dagger{}_{1(C)} \to Q_{1(C)}^*,\ Q_{1(C)} \to \frac{\delta}{\delta Q_{1(C)}^*},\ \tilde{Q}_{2(C)}^\dagger \to \tilde{Q}_{2(C)}^*,\ \tilde{Q}_{2(C)} \to -\frac{\delta}{\delta \tilde{Q}_{2(C)}^*}\right] \\ & \qquad \times \rho_S\left[t, Q_1^*, Q_{1C}^*, \tilde{Q}_2^*, \tilde{Q}_{2C}^*\right] . \end{aligned} \tag{2.153}$$

By functionally differentiating $\rho_S\left[t, Q_1^*, Q_{1C}^*, \tilde{Q}_2^*, \tilde{Q}_{2C}^*\right]$, we obtain, for example, the reduced density matrix for a single heavy quark system:

$$\begin{aligned} \rho_Q^{ij}(t, \mathbf{x}, \mathbf{y}) &= \langle \mathbf{x}, i|\rho_Q|\mathbf{y}, j\rangle \\ &\propto \langle\Omega|Q^i(\mathbf{x})\rho_S(t)Q^{j\dagger}(\mathbf{y})|\Omega\rangle \\ &= -\frac{\delta}{\delta Q_1^{i*}(\mathbf{x})}\frac{\delta}{\delta \tilde{Q}_2^{j*}(\mathbf{y})}\rho_S\left[t, Q_1^*, Q_{1C}^*, \tilde{Q}_2^*, \tilde{Q}_{2C}^*\right]\Big|_{Q_{1C}^*=\tilde{Q}_{2C}^*=0} . \end{aligned} \tag{2.154}$$

We will split the renormalized effective Hamiltonian (2.114) as

$$\mathbf{H}_{1+2} = H_1 + H_2 + H_3 + H_4 \ , \tag{2.155}$$

and we will find the time evolution of the reduced density matrix (ρ_S) with H_1, H_2, H_3, and H_4 separately and at the end we will add these to get the time evolution of the reduced density matrix with the renormalized effective Hamiltonian ($\mathbf{H}_{1+2}$).

Mathematically,

$$\begin{aligned} i\frac{\partial}{\partial t}\rho_S &= H_{1+2}\ \rho_S \\ \Rightarrow -\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(i\frac{\partial}{\partial t}\rho_S\right) &= -\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_{1+2}\ \rho_S\right) \\ \Rightarrow i\frac{\partial}{\partial t}\left(-\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\rho_S\right) &= -\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_{1+2}\ \rho_S\right) \\ \Rightarrow i\frac{\partial}{\partial t}\rho_Q^{ij} &= -\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_{1+2}\ \rho_S\right)\ , \end{aligned} \tag{2.156}$$

we used Eq. (2.154) i.e.,

$$-\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\rho_S = \rho_Q^{ij}\ , \tag{2.157}$$

in LHS of Eq. (2.156).

$$\begin{aligned} \Rightarrow i\frac{\partial}{\partial t}\rho_Q^{ij} = &-\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_1\ \rho_S\right) - \frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_2\ \rho_S\right) \\ &- \frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_3\ \rho_S\right) - \frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_4\ \rho_S\right)\ , \end{aligned} \tag{2.158}$$

where, we used Eq. (2.155) in RHS of Eq. (2.156). At first we start the evolution with H_1:

$$\begin{aligned} H_1 &= \int d^3x\ \left[aM\left(Q_1^\dagger Q_1 + Q_{1C}^\dagger Q_{1C}\right) + a^*M\left(\tilde{Q}_2^\dagger \tilde{Q}_2 + \tilde{Q}_{2C}^\dagger \tilde{Q}_{2C}\right)\right] \\ &= \int d^3x\ \left[aM\left(Q_1^* Q_1 + Q_{1C}^* Q_{1C}\right) + a^*M\left(\tilde{Q}_2^* \tilde{Q}_2 + \tilde{Q}_{2C}^* \tilde{Q}_{2C}\right)\right]\ , \end{aligned} \tag{2.159}$$

where, in the second step we have used the fact that for a single column vector/matrix Q:

$$Q^\dagger Q = Q^* Q\ ,\quad Q^\dagger = (Q^*)^T\ . \tag{2.160}$$

Now,

$$\begin{aligned}
\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_1\rho_S\right) &= \frac{\delta}{\delta Q_1^*}\left[\left(\frac{\delta H_1}{\delta \tilde{Q}_2^*}\right)\rho_S + H_1\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right] \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(\frac{\delta H_1}{\delta \tilde{Q}_2^*}\right)\right\}\rho_S + \left(\frac{\delta H_1}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta \rho_S}{\delta Q_1^*}\right) + \left(\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta H_1}{\delta Q_1^*}\right) + H_1.\frac{\delta^2 \rho_S}{\delta Q_1^* \delta \tilde{Q}_2^*} \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(a^* M \tilde{Q}_2\right)\right\}\rho_S + \left(a^* M \tilde{Q}_2\right)\frac{\delta \rho_S}{\delta Q_1^*} + \left(aMQ_1\right)\frac{\delta \rho_S}{\delta \tilde{Q}_2^*} + 0\ , \qquad (2.161) \\
&= 0 + a^* M\left(-\frac{\delta}{\delta \tilde{Q}_2^*}.\frac{\delta \rho_S}{\delta Q_1^*}\right) - aM\left(-\frac{\delta}{\delta Q_1^*}.\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right) \qquad (2.162) \\
&= M\left(a^* - a\right)\rho_Q^{ij}\,. \qquad (2.163)
\end{aligned}$$

The last term of Eq. (2.161) vanishes as $Q_{1(C)}^* = \tilde{Q}_{2(C)}^* = 0$, in Eq. (2.162) we used $Q_1 \to \frac{\delta}{\delta Q_1^*}$, $\tilde{Q}_2 \to -\frac{\delta}{\delta \tilde{Q}_2^*}$ as per requirement of Eq. (2.153) and in Eq. (2.163) we used Eq. (2.157).

Now, we will deal with H_2:

$$\begin{aligned}
H_2 = \int d^3x\ \Bigg[&\left\{Q_1^\dagger\left(-\frac{\nabla^2}{2M}\right)Q_1 + Q_{1C}^\dagger\left(-\frac{\nabla^2}{2M}\right)Q_{1C}\right\} \\
&+ \left\{\tilde{Q}_2^\dagger\left(-\frac{\nabla^2}{2M}\right)\tilde{Q}_2 + \tilde{Q}_{2C}^\dagger\left(-\frac{\nabla^2}{2M}\right)\tilde{Q}_{2C}\right\}\Bigg]
\end{aligned}$$

$$\begin{aligned}
= \int d^3x\ \Bigg[&\left\{Q_1^*\left(-\frac{\nabla_x^2}{2M}\right)Q_1 + Q_{1C}^*\left(-\frac{\nabla_x^2}{2M}\right)Q_{1C}\right\} \\
&+ \left\{\tilde{Q}_2^*\left(-\frac{\nabla_y^2}{2M}\right)\tilde{Q}_2 + \tilde{Q}_{2C}^*\left(-\frac{\nabla_y^2}{2M}\right)\tilde{Q}_{2C}\right\}\Bigg] \qquad (2.164)
\end{aligned}$$

Therefore,

$$\begin{aligned}
\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_2\rho_S\right) &= \frac{\delta}{\delta Q_1^*}\left[\left(\frac{\delta H_2}{\delta \tilde{Q}_2^*}\right)\rho_S + H_2\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right] \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(\frac{\delta H_2}{\delta \tilde{Q}_2^*}\right)\right\}\rho_S + \left(\frac{\delta H_2}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta \rho_S}{\delta Q_1^*}\right) + \left(\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta H_2}{\delta Q_1^*}\right) + H_2.\frac{\delta^2\rho_S}{\delta Q_1^*\delta \tilde{Q}_2^*} \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(\frac{\delta H_2}{\delta \tilde{Q}_2^*}\right)\right\}\rho_S + \left(\frac{\delta H_2}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta \rho_S}{\delta Q_1^*}\right) + \left(\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta H_2}{\delta Q_1^*}\right) + 0 \qquad (2.165) \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(-\frac{\nabla_y^2}{2M}\right)\tilde{Q}_2\right\}\rho_S + \left(-\frac{\nabla_y^2}{2M}\tilde{Q}_2\right)\left(\frac{\delta \rho_S}{\delta Q_1^*}\right) + \left(-\frac{\nabla_x^2}{2M}Q_1\right)\left(\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right) + 0 \\
&= 0 + \left(-\frac{\nabla_y^2}{2M}\right)\left(-\frac{\delta}{\delta \tilde{Q}_2^*}.\frac{\delta \rho_S}{\delta Q_1^*}\right) + \frac{\nabla_x^2}{2M}\left(-\frac{\delta}{\delta Q_1^*}.\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right) + 0 \qquad (2.166) \\
&= \left(\frac{\nabla_x^2 - \nabla_y^2}{2M}\right)\rho_Q^{ij} \qquad (2.167)
\end{aligned}$$

The last term of Eq. (2.165) vanishes as $Q^*_{1(C)} = \tilde{Q}^*_{2(C)} = 0$, in Eq. (2.166) we used $Q_1 \to \frac{\delta}{\delta Q_1^*}$, $\tilde{Q}_2 \to -\frac{\delta}{\delta \tilde{Q}_2^*}$ as per requirement of Eq. (2.153) and in Eq. (2.167) we used Eq. (2.157).

Now, we will deal with H_3:

$$H_3 = \frac{1}{2}\int d^3x d^3y \left[-2id(\mathbf{x}-\mathbf{y})N\Big\{j_1^{a0}(\mathbf{x})j_2^{a0}(\mathbf{y})\Big\}\right] \qquad (2.168)$$

From Eq. (2.132) and Eq. (2.133), the time-ordered currents can be written as:

$$\begin{aligned}
Tj_1^{a0} &= \psi_1^\dagger t^a \psi_1 \\
&= Q_1^\dagger t^a Q_1 + Q_{1C} t^a Q_{1C}^\dagger \text{ , using Eq. (2.21)} \\
&= Q_1^* t_{ij}^a Q_1 + Q_{1C} t_{ij}^a Q_{1C}^* \text{ ,} \qquad (2.169)
\end{aligned}$$

and,

$$\begin{aligned} T j_2^{a0} &= -\tilde{\psi}_2^\dagger t_{kl}^a \tilde{\psi}_2 \\ &= -\left(\begin{array}{cc} \tilde{Q}_2^\dagger & \tilde{Q}_{2C} \end{array} \right)_{1\times 2} t_{kl}^a \left(\begin{array}{c} \tilde{Q}_2 \\ \tilde{Q}_{2C}^\dagger \end{array} \right)_{2\times 1} \quad , \text{ using Eq. (2.136)} \\ &= -\tilde{Q}_2^\dagger t_{kl}^a \tilde{Q}_2 - \tilde{Q}_{2C} t_{kl}^a \tilde{Q}_{2C}^\dagger \\ &= -\tilde{Q}_2^* t_{kl}^a \tilde{Q}_2 - \tilde{Q}_{2C} t_{kl}^a \tilde{Q}_{2C}^* \ . \end{aligned} \tag{2.170}$$

So,

$$N\Big\{ j_1^{a0}(\mathbf{x}) j_2^{a0}(\mathbf{y}) \Big\} = \Big(Q_1^* t_{ij}^a Q_1 - Q_{1C}^* t_{ij}^a Q_{1C} \Big) \Big(-\tilde{Q}_2^* t_{kl}^a \tilde{Q}_2 + \tilde{Q}_{2C}^* t_{kl}^a \tilde{Q}_{2C} \Big) \tag{2.171}$$

$$\begin{aligned} &= -\Big(Q_1^* t_{ij}^a Q_1 \Big) \Big(\tilde{Q}_2^* t_{kl}^a \tilde{Q}_2 \Big) + \Big(Q_1^* t_{ij}^a Q_1 \Big) \Big(\tilde{Q}_{2C}^* t_{kl}^a \tilde{Q}_{2C} \Big) \\ &+ \Big(Q_{1C}^* t_{ij}^a Q_{1C} \Big) \Big(\tilde{Q}_2^* t_{kl}^a \tilde{Q}_2 \Big) - \Big(Q_{1C}^* t_{ij}^a Q_{1C} \Big) \Big(\tilde{Q}_{2C}^* t_{kl}^a \tilde{Q}_{2C} \Big) \end{aligned} \tag{2.172}$$

Inserting Eq. (2.172) into Eq. (2.168), we obtain

$$\begin{aligned} H_3 = \int d^3x d^3y \ \Bigg[- id(\mathbf{x}-\mathbf{y}) \Bigg\{ &- \Big(Q_1^* t_{ij}^a Q_1 \Big) \Big(\tilde{Q}_2^* t_{kl}^a \tilde{Q}_2 \Big) + \Big(Q_1^* t_{ij}^a Q_1 \Big) \Big(\tilde{Q}_{2C}^* t_{kl}^a \tilde{Q}_{2C} \Big) \\ &+ \Big(Q_{1C}^* t_{ij}^a Q_{1C} \Big) \Big(\tilde{Q}_2^* t_{kl}^a \tilde{Q}_2 \Big) - \Big(Q_{1C}^* t_{ij}^a Q_{1C} \Big) \Big(\tilde{Q}_{2C}^* t_{kl}^a \tilde{Q}_{2C} \Big) \Bigg\} \Bigg] . \end{aligned} \tag{2.173}$$

Now,

$$\begin{aligned}
\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_3\rho_S\right) &= \frac{\delta}{\delta Q_1^*}\left[\left(\frac{\delta H_3}{\delta \tilde{Q}_2^*}\right)\rho_S + H_3\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right] \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(\frac{\delta H_3}{\delta \tilde{Q}_2^*}\right)\right\}\rho_S + \left(\frac{\delta H_3}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta \rho_S}{\delta Q_1^*}\right) + \left(\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta H_3}{\delta Q_1^*}\right) + H_3.\frac{\delta^2\rho_S}{\delta Q_1^*\delta \tilde{Q}_2^*} \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(\frac{\delta H_3}{\delta \tilde{Q}_2^*}\right)\right\}\rho_S + 0 + 0 + 0 \ , \text{ as } Q_{1(C)}^* = \tilde{Q}_{2(C)}^* = 0 \ , \\
&= -id(\mathbf{x}-\mathbf{y})\left\{-t_{ij}^a Q_1 t_{kl}^a \tilde{Q}_2\right\}\rho_S \\
&= id(\mathbf{x}-\mathbf{y})\left\{t_{ij}^a t_{kl}^a Q_1 \tilde{Q}_2\right\}\rho_S \\
&= id(\mathbf{x}-\mathbf{y})\ t_{ij}^a t_{kl}^a\left\{Q_1\tilde{Q}_2\ \rho_S\right\} \\
&= id(\mathbf{x}-\mathbf{y})\ t_{ij}^a t_{kl}^a\left\{-\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\ \rho_S\right\} \qquad (2.174) \\
&= id(\mathbf{x}-\mathbf{y})\ t_{ij}^a t_{kl}^a\ \rho_Q^{kl}(t,\mathbf{x},\mathbf{y}) \qquad (2.175) \\
&= \frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N_C}\delta_{ij}\delta_{kl}\right) id(\mathbf{x}-\mathbf{y})\ \rho_Q^{kl}(t,\mathbf{x},\mathbf{y}) \qquad (2.176)
\end{aligned}$$

in Eq. (2.174) we used $Q_1 \to \frac{\delta}{\delta Q_1^*}$, $\tilde{Q}_2 \to -\frac{\delta}{\delta \tilde{Q}_2^*}$ as per requirement of Eq. (2.153), in Eq. (2.175) we used Eq. (2.157), and in Eq. (2.176) we used Eq. (A.6) i.e.,

$$t_{ij}^a t_{kl}^a = \frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N_C}\delta_{ij}\delta_{kl}\right) \ , \quad i,j,k,l = 1,2,...,N_C \ . \qquad (2.177)$$

Now, we will deal with H_4:[2]

$$H_4 = \frac{1}{2}\int d^3x d^3y \frac{1}{2T}\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y}).N\left\{\mathbf{j}_{1,\mathrm{NR}}^a j_2^{a0} + j_1^{a0}\mathbf{j}_{2,\mathrm{NR}}^a\right\} \ . \qquad (2.178)$$

From Eq. (2.6), we can write

$$\psi_1 = \begin{pmatrix} Q_1 \\ Q_{1C}^\dagger \end{pmatrix}_{2\times 1} \ , \quad \psi_1^\dagger = \begin{pmatrix} Q_1^\dagger & Q_{1C} \end{pmatrix}_{1\times 2} \qquad (2.179)$$

[2]One can notice that we have not included co-efficients of V and V^* of Eq. (2.144) in H_4 as these terms will vanish and not appear in the master equation of the reduced density matrix.

Using Eq. (2.179), Eq. (2.132) can be written as

$$\begin{aligned}
\mathrm{T}\mathbf{j}^a_{1,\mathrm{NR}} &= \psi_1^\dagger \left[t^a_{ij} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM} \right) \psi_1 \\
&= \begin{pmatrix} Q_1^\dagger & Q_{1C} \end{pmatrix}_{1\times 2} t^a_{ij} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}_{2\times 2} \left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM} \right) \begin{pmatrix} Q_1 \\ Q_{1C}^\dagger \end{pmatrix}_{2\times 1} \\
&= \begin{pmatrix} Q_1^\dagger t^a_{ij} & -Q_{1C} t^a_{ij} \end{pmatrix}_{1\times 2} \left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM} \right) \begin{pmatrix} Q_1 \\ Q_{1C}^\dagger \end{pmatrix}_{2\times 1} \\
&= Q_1^\dagger t^a_{ij} \left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM} \right) Q_1 - Q_{1C} t^a_{ij} \left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM} \right) Q_{1C}^\dagger \,. \qquad (2.180)
\end{aligned}$$

Using Eq. (2.179), we can write

$$\begin{aligned}
\mathrm{T}j_1^{a0} &= \psi_1^\dagger t^a \psi_1 \\
&= Q_1^\dagger t^a_{ij} Q_1 + Q_{1C} t^a_{ij} Q_{1C}^\dagger \,. \qquad (2.181)
\end{aligned}$$

Using Eq. (2.136), Eq. (2.133) can be written as

$$\begin{aligned}
\mathrm{T}\mathbf{j}^a_{2,\mathrm{NR}} &= \tilde{\psi}_2^\dagger \left[t^a_{kl} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM} \right) \tilde{\psi}_1 \\
&= \begin{pmatrix} \tilde{Q}_2^\dagger & \tilde{Q}_{2C} \end{pmatrix}_{1\times 2} t^a_{kl} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}_{2\times 2} \left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM} \right) \begin{pmatrix} \tilde{Q}_2 \\ \tilde{Q}_{2C}^\dagger \end{pmatrix}_{2\times 1} \\
&= \tilde{Q}_2^\dagger t^a_{kl} \left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM} \right) \tilde{Q}_2 - \tilde{Q}_{1C} t^a_{kl} \left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM} \right) \tilde{Q}_{2C}^\dagger \,. \qquad (2.182)
\end{aligned}$$

Using Eq. (2.136), we can also write

$$\begin{aligned}
\mathrm{T}j_2^{a0} &= -\tilde{\psi}_2^\dagger t^a_{kl} \tilde{\psi}_2 \\
&= -\tilde{Q}_2^\dagger t^a_{kl} \tilde{Q}_2 - \tilde{Q}_{2C} t^a_{kl} \tilde{Q}_{2C}^\dagger \,. \qquad (2.183)
\end{aligned}$$

Using Eq. (2.180), Eq. (2.181), Eq. (2.182), and Eq. (2.183), we can write

$$
\begin{aligned}
&N\Big\{\mathbf{j}^a_{1,\mathrm{NR}} j_2^{a0} + j_1^{a0}\mathbf{j}^a_{2,\mathrm{NR}}\Big\} \\
&= \left[Q_1^* t^a_{ij}\left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM}\right) Q_1 + Q^*_{1C} t^a_{ij}\left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM}\right) Q_{1C}\right]\left(-\tilde{Q}_2^* t^a_{kl}\tilde{Q}_2 + \tilde{Q}^*_{2C} t^a_{kl}\tilde{Q}_{2C}\right) \\
&\quad + \left(Q_1^* t^a_{ij} Q_1 - Q^*_{1C} t^a_{ij} Q_{1C}\right)\left[\tilde{Q}_2^* t^a_{kl}\left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM}\right)\tilde{Q}_2 + \tilde{Q}^*_{1C} t^a_{kl}\left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM}\right)\tilde{Q}_{2C}\right] \\
&\quad = -Q_1^* t^a_{ij}\left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM}\right) Q_1 \tilde{Q}_2^* t^a_{kl}\tilde{Q}_2 + Q_1^* t^a_{ij} Q_1 \tilde{Q}_2^* t^a_{kl}\left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM}\right)\tilde{Q}_2 + \ldots.\,,
\end{aligned}
\tag{2.184}
$$

viz., as $Q^*_{1(C)} = \tilde{Q}^*_{2(C)}$, the remaining terms of Eq. (2.184) will not appear in the final result.

Inserting Eq. (2.184), Eq. (2.178) becomes

$$
\begin{aligned}
H_4 &= \int d^3x d^3y \frac{\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y})}{4T}\Bigg[- Q_1^* t^a_{ij}\left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM}\right) Q_1 \tilde{Q}_2^* t^a_{kl}\tilde{Q}_2 \\
&+ Q_1^* t^a_{ij} Q_1 \tilde{Q}_2^* t^a_{kl}\left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM}\right)\tilde{Q}_2 + \ldots.\Bigg]
\end{aligned}
\tag{2.185}
$$

Now,

$$
\begin{aligned}
\frac{\delta}{\delta Q_1^*}.\frac{\delta}{\delta \tilde{Q}_2^*}\left(H_4\rho_S\right) &= \frac{\delta}{\delta Q_1^*}\left[\left(\frac{\delta H_2}{\delta \tilde{Q}_2^*}\right)\rho_S + H_4\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right] \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(\frac{\delta H_4}{\delta \tilde{Q}_2^*}\right)\right\}\rho_S + \left(\frac{\delta H_4}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta \rho_S}{\delta Q_1^*}\right) + \left(\frac{\delta \rho_S}{\delta \tilde{Q}_2^*}\right)\left(\frac{\delta H_4}{\delta Q_1^*}\right) + H_4.\frac{\delta^2\rho_S}{\delta Q_1^*\delta \tilde{Q}_2^*} \\
&= \left\{\frac{\delta}{\delta Q_1^*}\left(\frac{\delta H_4}{\delta \tilde{Q}_2^*}\right)\right\}\rho_S + 0 + 0 + 0 \qquad (2.186) \\
&= \frac{\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y})}{4T}\left[-t_{ij}^a\left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM}\right)Q_1 t_{kl}^a\tilde{Q}_2 + t_{ij}^a Q_1 t_{kl}^a\left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM}\right)\tilde{Q}_2\right] \\
&= \frac{\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y})}{4T}\left[-t_{ij}^a t_{kl}^a\left(\frac{\overrightarrow{\nabla}_x - \overleftarrow{\nabla}_x}{2iM}\right)\rho_Q^{kl} + t_{ij}^a t_{kl}^a\left(\frac{\overrightarrow{\nabla}_y - \overleftarrow{\nabla}_y}{2iM}\right)\rho_Q^{kl}\right] \qquad (2.187) \\
&= \frac{\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y})}{4T}t_{ij}^a t_{kl}^a\left[-\frac{\overrightarrow{\nabla}_x - \overrightarrow{\nabla}_x}{2iM} - \frac{-\overleftarrow{\nabla}_x + -\overleftarrow{\nabla}_y}{2iM}\right]\rho_Q^{kl} \qquad (2.188) \\
&= \frac{\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y})}{4T}t_{ij}^a t_{kl}^a\left(-\frac{\overrightarrow{\nabla}_x - \overrightarrow{\nabla}_y}{iM}\right)\rho_Q^{kl} \qquad (2.189) \\
&= -\frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N_C}\delta_{ij}\delta_{kl}\right)\left\{\frac{\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y})}{4T}.\frac{\overrightarrow{\nabla}_x - \overrightarrow{\nabla}_y}{iM}\right\}\rho_Q^{kl}\ , \qquad (2.190)
\end{aligned}
$$

as $Q_{1(C)}^* = \tilde{Q}_{2(C)}^*$, the remaining terms of Eq. (2.186) vanish. In Eq. (2.187) we used $Q_1 \to \frac{\delta}{\delta Q_1^*}$, $\tilde{Q}_2 \to -\frac{\delta}{\delta \tilde{Q}_2^*}$ as per requirement of Eq. (2.153), and Eq. (2.157); and in Eq. (2.190) we used Eq. (2.177).

Finally, inserting Eq. (2.163), Eq. (2.167), Eq. (2.176), and Eq. (2.190) in Eq. (2.158), we obtain the master equation for a single heavy quark system:

$$
\begin{aligned}
i\frac{\partial}{\partial t}\rho_Q^{ij}(t,\mathbf{x},\mathbf{y}) = &\left\{(a-a^*)M + \left(-\frac{\nabla_x^2 - \nabla_y^2}{2M}\right)\right\}\rho_Q^{ij}(t,\mathbf{x},\mathbf{y}) \\
&+ \frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N_C}\delta_{ij}\delta_{kl}\right)\left\{-id(\mathbf{x}-\mathbf{y}) + \frac{\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y})}{4T}.\frac{\overrightarrow{\nabla}_x - \overrightarrow{\nabla}_y}{iM}\right\}\rho_Q^{kl}(t,\mathbf{x},\mathbf{y})\ .
\end{aligned}
\qquad (2.191)
$$

And for the color-traced reduced density matrix $\rho_Q(t,\mathbf{x},\mathbf{y}) \equiv \rho_Q^{ii}(t,\mathbf{x},\mathbf{y})$, the above master equation becomes

$$i\frac{\partial}{\partial t}\rho_Q(t,\mathbf{x},\mathbf{y}) = \left\{(a-a^*)M + \left(-\frac{\nabla_x^2-\nabla_y^2}{2M}\right)\right\}\rho_Q(t,\mathbf{x},\mathbf{y}) + C_F\left\{-id(\mathbf{x}-\mathbf{y}) + \frac{\overrightarrow{\nabla}_x d(\mathbf{x}-\mathbf{y})}{4T}.\frac{\overrightarrow{\nabla}_x - \overrightarrow{\nabla}_y}{iM}\right\}\rho_Q(t,\mathbf{x},\mathbf{y}) \ , \tag{2.192}$$

where C_F is defined in Eq. (A.5).

Unitarity of the master equation

Now, we will show that the master Eq. (2.192) for a single heavy quark system is trace preserving and, therefore maintains unitarity.

We have

$$\begin{aligned} Tr_x\rho_Q(t) &\equiv \int d^3x\langle\mathbf{x}|\rho_Q(t)|\mathbf{x}\rangle \\ &= \int d^3x\rho_Q(t,\mathbf{x},\mathbf{x}) \ , \end{aligned} \tag{2.193}$$

So,

$$\begin{aligned} i\frac{d}{dt}Tr_x\rho_Q(t) &= \int d^3x d^3y\ \delta(\mathbf{x}-\mathbf{y})\left(i\frac{\partial}{\partial t}\rho_Q(t,\mathbf{x},\mathbf{y})\right) \\ &= \int d^3x\left(i\frac{\partial}{\partial t}\rho_Q(t,\mathbf{x},\mathbf{x})\right) \\ &= \int d^3x\left\{(a-a^*)M - iC_F d(\mathbf{0})\right\}\rho_Q(t,\mathbf{x},\mathbf{x}) \ , \quad \text{using Eq. (2.192)} \\ &= \left\{(a-a^*)M - iC_F d(\mathbf{0})\right\}\int d^3x\rho_Q(t,\mathbf{x},\mathbf{x}) \\ &= \left\{(a-a^*)M - iC_F d(\mathbf{0})\right\}Tr_x\rho_Q(t) \ , \quad \text{using Eq. (2.193)} \end{aligned} \tag{2.194}$$

We also have Eq. (2.145) :

$$a = 1 + \frac{C_F}{2M}\frac{g(T)^2}{4\pi}\left(-\omega_D - iT\right) \ , \tag{2.195}$$

and,

$$a^* = 1 + \frac{C_F}{2M}\frac{g(T)^2}{4\pi}\left(-\omega_D + iT\right) \ . \tag{2.196}$$

Subtracting Eq. (2.196) from Eq. (2.195) yields

$$(a - a^*)M = -iC_F\frac{T\ g(T)^2}{4\pi} \tag{2.197}$$

We also have Eq. (2.147):

$$d(\mathbf{r}) = -\frac{g(T)^2\ \omega_D^2\ T}{2\pi}\int_0^\infty dk\ \frac{\sin(kr)}{r(k^2+\omega_D^2)^2} \tag{2.198}$$

Now, in the limit of $r \to 0$:

$$\int_0^\infty dk\ \frac{\sin(kr)}{r(k^2+\omega_D^2)^2} = \frac{1}{2\omega_D^2} \tag{2.199}$$

Using Eq. (2.199) in Eq. (2.198), we get

$$d(\mathbf{0}) = -\frac{T\ g(T)^2}{4\pi} \ . \tag{2.200}$$

Using Eq. (2.197) and Eq. (2.200), we get

$$(a - a^*)M - iC_F d(\mathbf{0}) = 0 \tag{2.201}$$

Inserting Eq. (2.201) in Eq. (2.194), we get

$$i\frac{d}{dt}Tr_x\rho_Q(t) = 0 \tag{2.202}$$

Above equation ensures trace preserving and unitary properties of the master Eq. (2.192).

2.2.6 Time evolution of the forward correlator

Single heavy quark state

In this section, we will obtain the time evolution of the forward correlator of single heavy quark state. We have the forward correlator of a single heavy quark [11]:

$$G^{>}_{Q,i}(t,\mathbf{x}) = \frac{\delta}{\delta Q_1^{i*}(\mathbf{x})}\rho_S\left[t, Q_1^*, Q_{1C}^*, \tilde{Q}_2^*, \tilde{Q}_{2C}^*\right]\Big|_{Q^*_{1(C)}=\tilde{Q}^*_{2(C)}=0} . \tag{2.203}$$

From Eq. (2.144), we can write

$$\mathbf{H}_{1+2} = \int d^3x \left[aM\left(Q_1^*Q_1 + Q_{1C}^*Q_{1C}\right) + \left\{Q_1^*\left(-\frac{\nabla^2}{2M}\right)Q_1 + Q_{1C}^*\left(-\frac{\nabla^2}{2M}\right)Q_{1C}\right\}\right] + ... \tag{2.204}$$

Now,

$$\begin{aligned}
i\frac{\partial}{\partial t}\rho_S &= \mathbf{H}_{1+2}\ \rho_S \\
\Rightarrow \frac{\delta}{\delta Q_1^*}\left(i\frac{\partial}{\partial t}\rho_S\right) &= \frac{\delta}{\delta Q_1^*}\left(\mathbf{H}_{1+2}\ \rho_S\right) \\
\Rightarrow i\frac{\partial}{\partial t}\left(\frac{\delta}{\delta Q_1^*}\rho_S\right) &= \frac{\delta}{\delta Q_1^*}\left(\mathbf{H}_{1+2}\ \rho_S\right) \qquad (2.205)\\
\Rightarrow i\frac{\partial}{\partial t}G^{>}_{Q,i}(t,\mathbf{x}) &= \frac{\delta}{\delta Q_1^*}\left(\mathbf{H}_{1+2}\ \rho_S\right) \\
&= \left(\frac{\delta \mathbf{H}_{1+2}}{\delta Q_1^*}\right)\rho_S + \mathbf{H}_{1+2}\frac{\delta \rho_S}{\delta Q_1^*} \qquad (2.206)\\
&= \left(aMQ_1 - \frac{\nabla^2}{2M}Q_1\right)\rho_S + 0 \\
&= \left(-\frac{\nabla^2}{2M} + aM\right)Q_1\ \rho_S \qquad (2.207)\\
&= \left(-\frac{\nabla^2}{2M} + aM\right)\frac{\delta}{\delta Q_1^*}\rho_S \qquad (2.208)\\
&= \left(-\frac{\nabla^2}{2M} + aM\right)G^{>}_{Q,i}(t,\mathbf{x})\ , \qquad (2.209)
\end{aligned}$$

where, last term of RHS of Eq. (2.206) vanishes as $Q^*_{1(C)} = \tilde{Q}^*_{2(C)} = 0$; in Eq. (2.207) we make a transformation like $Q_1 \to \frac{\delta}{\delta Q_1^*}$; in Eq. (2.205), and Eq. (2.209) we used Eq. (2.203)

i.e.,

$$\frac{\delta}{\delta Q_1^*}\rho_S = G^{>}_{Q,i}(t,\mathbf{x}) \ . \tag{2.210}$$

so, the time evolution of the forward correlator of single heavy quark state is given by

$$i\frac{\partial}{\partial t}G^{>}_{Q,i}(t,\mathbf{x}) = \left(-\frac{\nabla^2}{2M} + aM\right) G^{>}_{Q,i}(t,\mathbf{x}) \ . \tag{2.211}$$

Heavy quarkonium state

In this section, we will obtain the time evolution of the forward correlator of heavy quarkonium state.

We have the forward correlator for the quarkonium state [11]:

$$G^{>}_{QQ_C,ij}(t,\mathbf{x},\mathbf{y};\mathbf{r}) = \frac{\delta}{\delta Q_1^{i*}(\mathbf{x})}.\frac{\delta}{\delta Q_{1C}^{j*}(\mathbf{y})}\rho_S\left[t, Q_1^*, Q_{1C}^*, \tilde{Q}_2^*, \tilde{Q}_{2C}^*\right]\Big|_{Q^*_{1(C)}=\tilde{Q}^*_{2(C)}=0} \tag{2.212}$$

Using Eq. (2.169), time ordered currents can be written as

$$\begin{aligned} Tj_1^{a0}(\mathbf{x}) &= Q_1^*(\mathbf{x})t_{ij}^a Q_(\mathbf{x}) + Q_{1C}(\mathbf{x})t_{ij}^a Q_{1C}^*(\mathbf{x}) & (2.213)\\ Tj_1^{a0}(\mathbf{y}) &= Q_1^*(\mathbf{y})t_{kl}^a Q_(\mathbf{y}) + Q_{1C}(\mathbf{y})t_{kl}^a Q_{1C}^*(\mathbf{y}) \ . & (2.214) \end{aligned}$$

So,

$$\begin{aligned} &N\Big\{j_1^{a0}(\mathbf{x}), j_1^{a0}(\mathbf{y})\Big\} \\ &= \Big[Q_1^*(\mathbf{x})t_{ij}^a Q_(\mathbf{x}) - Q_{1C}^*(\mathbf{x})t_{ij}^a Q_{1C}(\mathbf{x})\Big] \\ &\qquad \times \left[Q_1^*(\mathbf{y})t_{kl}^a Q_(\mathbf{y}) - Q_{1C}^*(\mathbf{y})t_{kl}^a Q_{1C}(\mathbf{y})\right] \\ &= Q_1^*(\mathbf{x})t_{ij}^a Q_(\mathbf{x})Q_1^*(\mathbf{y})t_{kl}^a Q_(\mathbf{y}) - Q_1^*(\mathbf{x})t_{ij}^a Q_(\mathbf{x})Q_{1C}^*(\mathbf{y})t_{kl}^a Q_{1C}(\mathbf{y}) \\ &\qquad - Q_{1C}^*(\mathbf{x})t_{ij}^a Q_{1C}(\mathbf{x})Q_1^*(\mathbf{y})t_{kl}^a Q_(\mathbf{y}) + Q_{1C}^*(\mathbf{x})t_{ij}^a Q_{1C}(\mathbf{x})Q_{1C}^*(\mathbf{y})t_{kl}^a Q_{1C}(\mathbf{y}) \ . \end{aligned} \tag{2.215}$$

From Eq. (2.144), we write the participating terms of $\mathbf{H}_{1+2}$:

$\mathbf{H}_{1+2}$

$$
\begin{aligned}
&= \int d^3x \left[\left\{ Q_1^*(\mathbf{x}) Q_1(\mathbf{x}) + Q_{1C}^*(\mathbf{x}) Q_{1C}(\mathbf{x}) \right\} + \left\{ Q_1^*(\mathbf{x}) \left(-\frac{\nabla_x^2}{2M} \right) Q_1(\mathbf{x}) + Q_{1C}^*(\mathbf{x}) \left(-\frac{\nabla_x^2}{2M} \right) Q_{1C}(\mathbf{x}) \right\} \right] \\
&+ \int d^3y \left[\left\{ Q_1^*(\mathbf{y}) Q_1(\mathbf{y}) + Q_{1C}^*(\mathbf{y}) Q_{1C}(\mathbf{y}) \right\} + \left\{ Q_1^*(\mathbf{y}) \left(-\frac{\nabla_y^2}{2M} \right) Q_1(\mathbf{y}) + Q_{1C}^*(\mathbf{y}) \left(-\frac{\nabla_y^2}{2M} \right) Q_{1C}(\mathbf{y}) \right\} \right] \\
&\qquad + ... \\
&\quad + \frac{1}{2} \int d^3x d^3y v(\mathbf{x}-\mathbf{y}) \Big[Q_1^*(\mathbf{x}) t_{ij}^a Q_(\mathbf{x}) Q_1^*(\mathbf{y}) t_{kl}^a Q_(\mathbf{y}) - Q_1^*(\mathbf{x}) t_{ij}^a Q_(\mathbf{x}) Q_{1C}^*(\mathbf{y}) t_{kl}^a Q_{1C}(\mathbf{y}) \\
&\qquad - Q_{1C}^*(\mathbf{x}) t_{ij}^a Q_{1C}(\mathbf{x}) Q_1^*(\mathbf{y}) t_{kl}^a Q_(\mathbf{y}) + Q_{1C}^*(\mathbf{x}) t_{ij}^a Q_{1C}(\mathbf{x}) Q_{1C}^*(\mathbf{y}) t_{kl}^a Q_{1C}(\mathbf{y}) \Big] \\
&\qquad + ...
\end{aligned}
\tag{2.216}
$$

Now,

$$
\begin{aligned}
i\frac{\partial}{\partial t}\rho_S &= \mathbf{H}_{1+2}\ \rho_S \\
\Rightarrow \frac{\delta}{\delta Q_1^*(\mathbf{x})} \cdot \frac{\delta}{\delta Q_{1C}^*(\mathbf{y})} \left(i\frac{\partial}{\partial t}\rho_S \right) &= \frac{\delta}{\delta Q_1^*(\mathbf{x})} \cdot \frac{\delta}{\delta Q_{1C}^*(\mathbf{y})} \left(\mathbf{H}_{1+2}\ \rho_S \right) \\
\Rightarrow i\frac{\partial}{\partial t} \left(\frac{\delta}{\delta Q_1^*(\mathbf{x})} \cdot \frac{\delta}{\delta Q_{1C}^*(\mathbf{y})} \rho_S \right) &= \frac{\delta}{\delta Q_1^*(\mathbf{x})} \cdot \frac{\delta}{\delta Q_{1C}^*(\mathbf{y})} \left(\mathbf{H}_{1+2}\ \rho_S \right) \\
\Rightarrow i\frac{\partial}{\partial t} G^{>}_{QQ_C,ij}(t,\mathbf{x},\mathbf{y};\mathbf{r}) &= \frac{\delta}{\delta Q_1^*(\mathbf{x})} \cdot \frac{\delta}{\delta Q_{1C}^*(\mathbf{y})} \left(\mathbf{H}_{1+2}\ \rho_S \right) \qquad (2.217)
\end{aligned}
$$

(2.218)

Hence,

$i\frac{\partial}{\partial t} G^{>}_{QQ_C,ij}(t,\mathbf{x},\mathbf{y};\mathbf{r})$

$$= \left\{ \frac{\delta}{\delta Q_1^*(\mathbf{x})} \left(\frac{\delta \mathbf{H}_{1+2}}{\delta Q_{1C}^*(\mathbf{y})} \right) \right\} \rho_S + \left(\frac{\delta \mathbf{H}_{1+2}}{\delta Q_{1C}^*(\mathbf{y})} \right) \left(\frac{\delta \rho_S}{\delta Q_1^*(\mathbf{x})} \right) + \left(\frac{\delta \mathbf{H}_{1+2}}{\delta Q_1^*(\mathbf{x})} \right) \left(\frac{\delta \rho_S}{\delta Q_{1C}^*(\mathbf{y})} \right) + \mathbf{H}_{1+2} \frac{\delta^2 \rho_S}{\delta Q_1^*(\mathbf{x}) \delta Q_{1C}^*(\mathbf{y})} \quad (2.219)$$

$$= \frac{1}{2} v(\mathbf{x}-\mathbf{y}) \left\{ - t_{ij}^a Q_1(\mathbf{x}) t_{kl}^a Q_{1C}(\mathbf{y}) \right\} \rho_S + \left(aM - \frac{\nabla_y^2}{2M} \right) Q_{1C}(\mathbf{y}) \frac{\delta \rho_S}{\delta Q_1^*(\mathbf{x})} + \left(aM - \frac{\nabla_x^2}{2M} \right) Q_1(\mathbf{x}) \frac{\delta \rho_S}{\delta Q_{1C}^*(\mathbf{y})} + 0 \quad (2.220)$$

$$= \frac{1}{2} t_{ij}^a t_{kl}^a v(\mathbf{x}-\mathbf{y}) \frac{\delta}{\delta Q_1^*(\mathbf{x})} . \frac{\delta}{\delta Q_{1C}^*(\mathbf{y})} \rho_S + \left(aM - \frac{\nabla_y^2}{2M} \right) \frac{\delta}{\delta Q_{1C}^*(\mathbf{y})} \frac{\delta \rho_S}{\delta Q_1^*(\mathbf{x})} + \left(aM - \frac{\nabla_x^2}{2M} \right) \frac{\delta}{\delta Q_1^*(\mathbf{x})} \frac{\delta \rho_S}{\delta Q_{1C}^*(\mathbf{y})} \quad (2.221)$$

$$= \frac{1}{2} t_{ij}^a t_{kl}^a v(\mathbf{x}-\mathbf{y}) G_{QQ_C,kl}^{>}(t,\mathbf{x},\mathbf{y};\mathbf{r}) + \left(aM - \frac{\nabla_y^2}{2M} \right) G_{QQ_C,ij}^{>}(t,\mathbf{x},\mathbf{y};\mathbf{r}) + \left(aM - \frac{\nabla_x^2}{2M} \right) G_{QQ_C,ij}^{>}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \ , \quad (2.222)$$

where, in Eq. (2.217) and Eq. (2.222) we used Eq. (2.212); last term of RHS of Eq. (2.219) vanishes as $Q_{1(C)}^* = \tilde{Q}_{2(C)}^* = 0$; and going to Eq. (2.221) from Eq. (2.220), we make a transformation like $Q_1(\mathbf{x}) \to \frac{\delta}{\delta Q_1^*(\mathbf{x})}$, $Q_{1C}(\mathbf{y}) \to \frac{\delta}{\delta Q_{1C}^*(\mathbf{y})}$.

Therefore, re-arranging Eq. (2.222), and using Eq. (2.177) we get the time evolution of the forward correlator of the quarkonium state:

$$i \frac{\partial}{\partial t} G_{QQ_C,ij}^{>}(t,\mathbf{x},\mathbf{y};\mathbf{r}) = \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM \right) G_{QQ_C,ij}^{>}(t,\mathbf{x},\mathbf{y};\mathbf{r}) - \frac{1}{2} \left(\delta_{il}\delta_{jk} - \frac{1}{N_C} \delta_{ij}\delta_{kl} \right) v(\mathbf{x}-\mathbf{y}) G_{QQ_C,kl}^{>}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \ , \quad (2.223)$$

Projecting the heavy quark state on the color singlet i.e.,

$$G^{>}_{QQ_C,\mathbf{1}}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \equiv G^{>}_{QQ_C,ii}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \ , \tag{2.224}$$

we get

$$i\frac{\partial}{\partial t}G^{>}_{QQ_C,\mathbf{1}}(t,\mathbf{x},\mathbf{y};\mathbf{r}) = \left\{ -\frac{\nabla_x^2+\nabla_y^2}{2M} + 2aM - C_F v(\mathbf{x}-\mathbf{y})\right\}G^{>}_{QQ_C,\mathbf{1}}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \tag{2.225}$$

From Eq. (2.225), the complex color-singlet potential can be read off as

$$v_{\mathbf{1}}(\mathbf{r}) = 2(a-1)M - C_F\ v(\mathbf{r}) \ . \tag{2.226}$$

From Eq. (2.145), we can write

$$\begin{aligned} a &= 1+\frac{C_F}{2M}\frac{g(T)^2}{4\pi}\left(-\omega_D - iT\right) \\ \Rightarrow (a-1) &= \frac{C_F}{2M}\frac{g(T)^2}{4\pi}\left(-\omega_D - iT\right) \\ \Rightarrow 2(a-1)M &= \frac{C_F\ g(T)^2}{4\pi}\left(-\omega_D - iT\right) \ . \end{aligned} \tag{2.227}$$

From Eq. (2.146), we can also write

$$\begin{aligned} v(\mathbf{r}) &= \frac{g(T)^2}{4\pi}.\left\{\frac{e^{-\omega_D r}}{r} - 2iT\omega_D^2\int_0^\infty dk \frac{\sin(kr)}{r(k^2+\omega_D^2)^2}\right\} \\ \Rightarrow -C_F v(\mathbf{r}) &= -\frac{C_F\ g(T)^2}{4\pi}.\left\{\frac{e^{-\omega_D r}}{r} - 2iT\omega_D^2\int_0^\infty dk \frac{\sin(kr)}{r(k^2+\omega_D^2)^2}\right\} \end{aligned} \tag{2.228}$$

Now, inserting Eq. (2.227) and Eq. (2.228), inside Eq. (2.226), we get the color-singlet potential:

$$\begin{aligned} v_{\mathbf{1}}(\mathbf{r}) &= 2(a-1)M - C_F\ v(\mathbf{r}) \\ &= -\frac{C_F\ g(T)^2}{4\pi}\left[\omega_D + \frac{e^{-\omega_D r}}{r} + iT\left\{1 - 2\omega_D^2\int_0^\infty dk\frac{\sin(kr)}{r(k^2+\omega_D^2)^2}\right\}\right] \end{aligned} \tag{2.229}$$

Now, we will project the heavy quark state on the color-octet state such as

$$G^{>}_{QQ_C,\mathbf{8}}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \equiv G^{>}_{QQ_C,ij}(t,\mathbf{x},\mathbf{y};\mathbf{r}) - \frac{\delta_{ij}}{N_C}G^{>}_{QQ_C,kk}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \ . \tag{2.230}$$

From Eq. (2.224) and Eq. (2.225), we can write

$$i\frac{\partial}{\partial t}G^{>}_{QQ_C,kk}(t,\mathbf{x},\mathbf{y};\mathbf{r}) = \left\{ -\frac{\nabla_x^2+\nabla_y^2}{2M} + 2aM - C_F v(\mathbf{x}-\mathbf{y}) \right\} G^{>}_{QQ_C,kk}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \quad (2.231)$$

For next couple of steps, we will use these short notations:

$$\begin{aligned} G^{>}_{QQ_C,ij}(t,\mathbf{x},\mathbf{y};\mathbf{r}) &\equiv G^{>}_{ij} \ , \\ G^{>}_{QQ_C,kl}(t,\mathbf{x},\mathbf{y};\mathbf{r}) &\equiv G^{>}_{kl} \ , \\ G^{>}_{QQ_C,kk}(t,\mathbf{x},\mathbf{y};\mathbf{r}) &\equiv G^{>}_{kk} \ , \\ G^{>}_{QQ_C,\mathbf{8}}(t,\mathbf{x},\mathbf{y};\mathbf{r}) &\equiv G^{>}_{\mathbf{8}} \ . \end{aligned}$$

Writing Eq. (2.230) as

$$G^{>}_{\mathbf{8}} = G^{>}_{ij} - \frac{\delta_{ij}}{N_C}G^{>}_{kk} \quad (2.232)$$

$$\Rightarrow i\frac{\partial}{\partial t}G^{>}_{\mathbf{8}} = i\frac{\partial}{\partial t}G^{>}_{ij} - \frac{\delta_{ij}}{N_C}\, i\frac{\partial}{\partial t}G^{>}_{kk} \quad (2.233)$$

$$= \left(-\frac{\nabla_x^2+\nabla_y^2}{2M} + 2aM\right) G^{>}_{ij} - \frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N_C}\delta_{ij}\delta_{kl}\right) v\ G^{>}_{kl} - \frac{\delta_{ij}}{N_C}\left\{ -\frac{\nabla_x^2+\nabla_y^2}{2M} + 2aM - C_F\ v \right\} G^{>}_{kk} \quad (2.234)$$

$$= \left(-\frac{\nabla_x^2+\nabla_y^2}{2M} + 2aM\right) G^{>}_{ij} - \frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N_C}\delta_{ij}\delta_{kl}\right) v\ G^{>}_{kl} + \left(-\frac{\nabla_x^2+\nabla_y^2}{2M} + 2aM\right)\left(-\frac{\delta_{ij}}{N_C}G^{>}_{kk}\right) + \frac{C_F\ v\ \delta_{ij}}{N_C}G^{>}_{kk} \quad (2.235)$$

$$= \left(-\frac{\nabla_x^2+\nabla_y^2}{2M} + 2aM\right)\left(G^{>}_{ij} - \frac{\delta_{ij}}{N_C}G^{>}_{kk}\right) - \frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N_C}\delta_{ij}\delta_{kl}\right) v\ G^{>}_{kl} + \frac{C_F\ v\ \delta_{ij}}{N_C}G^{>}_{kk} \quad (2.236)$$

$$= \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM \right) G_{\mathbf{8}}^{>} - \frac{1}{2} \left(\delta_{il}\delta_{jk} - \frac{1}{N_C}\delta_{ij}\delta_{kl} \right) v \; G_{kl}^{>} + \frac{C_F \; v \; \delta_{ij}}{N_C} G_{kk}^{>} \tag{2.237}$$

$$= \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM \right) G_{\mathbf{8}}^{>} - \frac{1}{2} v \; \delta_{ij}\delta_{kl} G_{kl}^{>} + \frac{v \; \delta_{ik}\delta_{jl} G_{kl}^{>}}{2N_C} + \frac{C_F \; v \; \delta_{ij}}{N_C} G_{kk}^{>} \tag{2.238}$$

$$= \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM \right) G_{\mathbf{8}}^{>} - \frac{1}{2} v \; \delta_{ij} G_{kk}^{>} + \frac{v \; G_{ij}^{>}}{2N_C} + \frac{C_F \; v \; \delta_{ij}}{N_C} G_{kk}^{>} \tag{2.239}$$

$$= \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM \right) G_{\mathbf{8}}^{>} + \frac{v}{2N_C} \left\{ G_{ij}^{>} - N_C \delta_{ij} G_{kk}^{>} + 2C_F \delta_{ij} G_{kk}^{>} \right\} \tag{2.240}$$

$$= \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM \right) G_{\mathbf{8}}^{>} + \frac{v}{2N_C} \left\{ G_{ij}^{>} - \left(N_C - 2C_F \right) \delta_{ij} G_{kk}^{>} \right\} \tag{2.241}$$

$$= \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM \right) G_{\mathbf{8}}^{>} + \frac{v}{2N_C} \left\{ G_{ij}^{>} - \frac{\delta_{ij}}{N_C} G_{kk}^{>} \right\} \tag{2.242}$$

$$= \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM \right) G_{\mathbf{8}}^{>} + \frac{v}{2N_C} G_{\mathbf{8}}^{>} \tag{2.243}$$

$$= \left(-\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM + \frac{v}{2N_C} \right) G^>_{\mathbf{8}} \ , \tag{2.244}$$

where, in Eq. (2.234) we used Eq. (2.223) and Eq. (2.231); in Eq. (2.237) and Eq. (2.243) we used Eq. (2.232); in Eq. (2.240) we used

$$\delta_{ij}\delta_{kl} G^>_{kl} = G^>_{kk} \ , \ \delta_{ik}\delta_{jl} G^>_{kl} = G^>_{ij} \ ; \tag{2.245}$$

and going to Eq. (2.242) from Eq. (2.241) we used

$$C_F = \frac{N_C^2 - 1}{2N_C} \ . \tag{2.246}$$

Hence we arrive at Eq. (2.244)

$$i\frac{\partial}{\partial t} G^>_{Q\bar{Q}_C,\mathbf{8}}(t,\mathbf{x},\mathbf{y};\mathbf{r}) = \left\{ -\frac{\nabla_x^2 + \nabla_y^2}{2M} + 2aM + \frac{v(\mathbf{x}-\mathbf{y})}{2N_C} \right\} G^>_{Q\bar{Q}_C,\mathbf{8}}(t,\mathbf{x},\mathbf{y};\mathbf{r}) \ . \tag{2.247}$$

From Eq. (2.247), the complex color-octet potential can be read off as

$$v_{\mathbf{8}}(\mathbf{r}) = 2(a-1)M + \frac{v(\mathbf{r})}{2N_C} \tag{2.248}$$

$$= \frac{g(T)^2}{4\pi} \left[-C_F\omega_D + \frac{1}{2N_C}\frac{e^{-\omega_D r}}{r} - iT\left\{ C_F + \frac{\omega_D^2}{N_C}\int_0^\infty dk \ \frac{\sin(kr)}{r(k^2+\omega_D^2)^2} \right\} \right] \tag{2.249}$$

where, we used Eq. (2.146) and Eq. (2.227) in Eq. (2.248).

The value of the integral of Eq. (2.249) is:

$$\int_0^\infty dk \ \frac{\sin(kr)}{r(k^2+\omega_D^2)^2} = \frac{1}{2\omega_D^2} \ , \text{ in the limit } r \to 0 \ , \tag{2.250}$$

$$= 0 \ , \text{ in the limit } r \to \infty \ , \tag{2.251}$$

Using Eq. (2.250) in Eq. (2.229), and Eq. (2.251) in Eq. (2.249) we note that the imaginary parts of complex color-singlet and color-octet potentials are different at the origin:

$$\mathcal{I}m\left[v_{\mathbf{1}}(\mathbf{0})\right] = 0 \ , \tag{2.252}$$

$$\mathcal{I}m\left[v_{\mathbf{8}}(\mathbf{0})\right] = -T\frac{N_C}{2}\frac{g(T)^2}{4\pi} \ , \tag{2.253}$$

while they are the same at infinitely long distance:

$$\mathcal{I}m\Big[v_{\mathbf{1}}(\infty)\Big] = \mathcal{I}m\Big[v_{\mathbf{8}}(\infty)\Big] = -TC_F\frac{g(T)^2}{4\pi} \ . \tag{2.254}$$

Since the imaginary part reflects Landau damping due to the collisions between the heavy quark and the medium gluons and light quarks, it is natural to obtain a vanishing imaginary part in the complex color-singlet potential at the origin and a finite imaginary part in the complex color-octet potential at the origin.

2.3 Conclusion

In this chapter, we review the existing literature for real-time field theory and its application to open quantum system. We derive Lagrangian density for heavy quark in non-relativistic limit. We discuss the basic understanding of real time formalism and develop the perturbation theory for non-equilibrium quantum-statistical mechanics which is known as closed-time path Green function formalism (CTPGF) [58]. Using real time formalism, we obtain the Feynman-Vernon influence functional and an effective action for heavy quark (open quantum system) at leading order of the perturbation expansion in the non-relativistic limit. We then construct a renormalized effective Hamiltonian from the effective action. We derive the master equation for the reduced density matrix for a single heavy quark system. We also obtain the time evolution of the forward correlators for a single heavy quark state and heavy quarkonium state.

Chapter 3

Heavy Quarkonium Dynamics Beyond The Adiabatic Limit

Many prior studies of in-medium quarkonium suppression have implicitly made use of an adiabatic approximation in which it was assumed that the heavy quark potential is a slowly varying function of time. In the adiabatic limit, one can separately determine the in-medium breakup rate and the medium time evolution, folding these together only at the end of the calculation. In this chapter, we relax this assumption by solving the 3d Schrödinger equation in real-time in order to compute quarkonium suppression dynamically. We compare results obtained using the adiabatic approximation with real-time calculations for realistic complex heavy quark potentials. Using the latter, we find that, for the $\Upsilon(1S)$, the difference between the adiabatic approximation and full real-time evolution is at the few percent level, however, for the $\Upsilon(2S)$, we find that the correction can be as large as 18% in low temperature regions. For the J/Ψ, we find a larger difference between the dynamical evolution and the adiabatic approximation, with the error reaching approximately 36% [29].

3.1 Complex KMS potential

We will also present results using a potential model based on high-temperature quantum chromodynamics (QCD) which provides information about the short- and medium-range parts of the heavy-quark potential [75]. For describing finite-temperature states which can have large radii compared to the confinement scale $\Lambda_{\rm QCD}^{-1} \sim 300$ MeV ~ 0.66 fm/c, however, one must supplement the perturbative potential obtained in Ref. [75] by a long range contribution. For this purpose, we will make use of a Karsch-Mehr-Satz type potential based

on the internal energy [66, 50, 110]

$$\begin{aligned}\Re[V] &= -\frac{\alpha}{r}\,(1+m_D\,r)\exp(-m_D\,r)+\frac{2\sigma}{m_D}\left[1-\exp(-m_D\,r)\right] \\ &\quad -\sigma\,r\,\exp(-m_D\,r)-\frac{0.8\,\sigma}{m_Q^2\,r}\,,\end{aligned} \tag{3.1}$$

where the last term is a finite quark mass correction [22], $\sigma = 0.210$ GeV^2 is the string tension, and m_D is the QGP Debye mass $m_D^2 = (1.4)^2 \cdot N_c(1+N_f/6)\,4\pi\alpha_s\,T^2/3$ with $N_c = 3$ and $N_f = 2$. The factor of $(1.4)^2$ takes into account higher-order corrections and has been determined using lattice QCD simulations [64]. We fix the Coulomb interaction strength to $\alpha = 4\alpha_s/3 = 0.385$ to match the low temperature binding energy and mass spectrum of heavy quark states [50].

For the imaginary part of the model potential we use the result obtained from leading-order finite temperature perturbation theory [75]

$$\Im[V] = -\alpha_s T\phi(\hat{r})\,, \tag{3.2}$$

where $\hat{r} \equiv m_D r$, $\alpha_s = 0.289$, and

$$\phi(\hat{r}) \equiv 2\int_0^\infty dz \frac{z}{(z^2+1)^2}\left[1-\frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right]. \tag{3.3}$$

The total complex Karsch-Mehr-Satz (CKMS) potential is a sum of the real (3.1) and imaginary (3.2) parts [1]

$$V(r,t) = \Re[V] + i\,\Im[V]\,, \tag{3.4}$$

where the time-dependence of the potential results from the fact that the temperature and hence the Debye mass m_D depend on the proper-time. All other parameters are held constant.

[1] This potential has also been referred to as the Strickland-Bazow potential in the literature since it was used in Ref. [110].

3.2 Numerical algorithm

Our goal is to solve the time-dependent Schrödinger equation with the CKMS potential (3.4). The complex KMS potential is a purely radial potential which allows us to write the general solution in spherical coordinates as

$$\psi(r,\theta,\phi,t) = \sum_{\ell m} R_{\ell m}(r,t) Y_{\ell m}(\theta,\phi)\,, \tag{3.5}$$

where $Y_{\ell m}$ are spherical harmonics. It is convenient to change variables to $u_{\ell m}(r,t) \equiv r R_{\ell m}(r,t)$ to obtain

$$u(r,\theta,\phi,t) = \sum_{\ell m} u_{\ell m}(r,t) Y_{\ell m}(\theta,\phi)\,, \tag{3.6}$$

where $u(r,\theta,\phi,t) = r\psi(r,\theta,\phi,t)$.

This change of variables allows us to cast the Hamiltonian in one-dimensional form

$$\hat{H}_\ell = \frac{\hat{p}^2}{2m} + V_{\text{eff},\ell}(r,t)\,, \tag{3.7}$$

where $V_{\text{eff},\ell}(r,t) = V(r,t) + \frac{\ell(\ell+1)}{2mr^2}$ and $\hat{p} = -i\frac{d}{dr}$. At a fixed time t the eigenstates satisfy

$$\hat{H}_\ell u_\ell(r,t) = E_\ell(t) u_\ell(r,t)\,, \tag{3.8}$$

where we have used the fact that the eigenvalue equation (3.8) does not depend on the azimuthal quantum number m to simplify $u_{\ell,m} \to u_\ell$. However, for the time evolution, we will keep the azimuthal quantum number for generality:

$$u(r,\theta,\phi,t) = \mathcal{N} \sum_{\ell=\ell_{\min}}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} \frac{1}{\sqrt{2\ell+1}} u_{\ell,m}(r,t) Y_{\ell m}(\theta,\phi)\,, \tag{3.9}$$

where the factor of $(2\ell+1)^{-1/2}$ has been introduced by hand for simplifying the forthcoming normalization of u and $\mathcal{N}$ is a normalization constant to be determined later.

Our goal is to obtain an evolution equation for each of the coefficient functions $u_{\ell,m}(r,t)$. To proceed we use the orthogonality of the spherical harmonics, $\int d\Omega\, Y^*_{\ell' m'}(\theta,\phi) Y_{\ell m}(\theta,\phi) =$

$\delta_{\ell\ell'}\delta_{mm'}$ where $d\Omega = d\phi\, d(\cos\theta)$ is the three-dimensional solid angle differential. Using this, one obtains

$$u_{\ell,m}(r,t) = \sqrt{2\ell+1}\int d\Omega\, Y^*_{\ell m}(\theta,\phi)\, u(r,\theta,\phi,t)\,. \tag{3.10}$$

We will use a pseudo-spectral method. Applying the time evolution operator to u gives

$$\begin{aligned} u(r,\theta,\phi,t+\Delta t) &= \exp(-i\hat{H}\Delta t)u(r,\theta,\phi,t) \\ &= \mathcal{N}\sum_{\ell,m}\frac{1}{\sqrt{2\ell+1}}\exp(-i\hat{H}\Delta t)u_\ell(r,t)Y_{\ell m}(\theta,\phi) \\ &= \mathcal{N}\sum_{\ell,m}\frac{1}{\sqrt{2\ell+1}}Y_{\ell m}(\theta,\phi)\exp(-i\hat{H}_\ell\Delta t)u_\ell(r,t)\,, \end{aligned} \tag{3.11}$$

where the summation limits are implicit. Using Eq. (3.10), one obtains

$$u_{\ell,m}(r,t+\Delta t) = \exp(-i\hat{H}_\ell\Delta t)u_{\ell,m}(r,t)\,, \tag{3.12}$$

which tells us that we can update each of the different ℓ, m states independently using the corresponding Hamiltonian. We note that the time evolution operator Eq. (3.12) is independent of m.

Finally, we need to come up with a prescription for the normalization of the various states. Herein we take

$$\int_0^\infty dr\, u^*_{\ell,m}(r,t)u_{\ell,m}(r,t) = 1\,. \tag{3.13}$$

Using this and requiring that the total wave function is normalized

$$\int dr d\Omega\, u^*(r,\theta,\phi,t)u(r,\theta,\phi,t) = 1\,, \tag{3.14}$$

gives $\mathcal{N} = 1/\sqrt{\ell_{\max}-\ell_{\min}+1}$ and our final expression for the general wave function

$$u(r,\theta,\phi,t) = \frac{1}{\sqrt{\ell_{\max}-\ell_{\min}+1}}\sum_{\ell=\ell_{\min}}^{\ell_{\max}}\sum_{m=-\ell}^{\ell}\frac{1}{\sqrt{2\ell+1}}u_{\ell,m}(r,t)Y_{\ell m}(\theta,\phi)\,. \tag{3.15}$$

3.2.1 Time evolution algorithm

We will use the time evolution operator (3.12) to evolve the wave function given a particular initial condition. One complication is that the function $u_\ell(r,t)$ must vanish at the

origin. In order to enforce this we use real-valued Fourier sine series to describe both the real and imaginary parts of the wave function. In this manner we guarantee that the correct boundary conditions at $r = 0$ are satisfied automatically. The resulting update steps are, $\mathbb{F} \to \mathbb{F}_s$:

1. Update in configuration space using a half-step: $\psi_1 = \exp(-iV\Delta t/2)\psi_0$.

2. Perform Fourier sine transformations on real and imaginary parts separately: $\tilde{\psi}_1 = \mathbb{F}_s[\Re\psi_1] + i\mathbb{F}_s[\Im\psi_1]$.

3. Update in momentum space using: $\tilde{\psi}_2 = \exp\left(-i\frac{p^2}{2m}\Delta t\right)\tilde{\psi}_1$.

4. Perform inverse Fourier sine transformations on real and imaginary parts separately: $\psi_2 = \mathbb{F}_s^{-1}[\Re\tilde{\psi}_2] + i\mathbb{F}_s^{-1}[\Im\tilde{\psi}_2]$.

5. Update in configuration space using a half-step: $\psi_3 = \exp(-iV\Delta t/2)\psi_2$.

3.2.2 Time-dependent basis functions

In the case of the CKMS potential it is not possible to determine the eigenfunctions analytically. Instead, we compute them using a modified "point and shoot" method. For the basis functions, at any time t, one must solve

$$-\frac{1}{2m}\frac{d^2u_\ell(r)}{dr^2} + V_{\text{eff},\ell}(r)u_\ell(r) = Eu_\ell(r)\,, \tag{3.16}$$

where both E and u_ℓ are unknowns and there may be multiple solutions. This equation must be solved subject to the boundary condition

$$\lim_{r\to 0} u_\ell(r) = Ar^{\ell+1}\,, \tag{3.17}$$

where A is an arbitrary constant which can be taken to be one. For a given ℓ we start from a very small value of $r = r_{\min}$ and fix $u_\ell(r_{\min}) = r_{\min}^{\ell+1}$ and $u'_\ell(r_{\min}) = (\ell+1)r_{\min}^{\ell}$. We then choose a trial value for E and integrate Eq. (3.16) from $r_{\min}$ to a large radius $r = r_{\max}$. In

order to be a normalizable solution, u must vanish as $r \to \infty$. For a real-valued potential, one can find all bound state solutions by making steps in E from a negative value to $E = V_{\max}$ and searching for the points where $u(r_{\max}) = 0$. The set of values obtained in this manner, E_n, are the energy eigenvalues and the associated normalized solutions are the eigenfunctions $u_{n\ell}$.

In the case that the potential is complex, the energy eigenvalues become complex and the solutions using the point-and-shoot method become more difficult to obtain. To surmount this difficulty we split the potential into real and imaginary parts and introduce an adjustable parameter δ, i.e.

$$V = \Re[V] + i\,\delta\,\Im[V]\,. \tag{3.18}$$

We then first find the solutions for $\delta = 0$ using the standard point-and-shoot method to identify the eigenvalues. We then increment δ in steps of 0.1 until it takes the value $\delta = 1$, using the previous step's solutions for the eigenvalues as the guess for the current step. In this way we can find the complex energy eigenvalues and eigenfunctions efficiently. These can then be saved to disk for later use.

3.3 Numerical results

We now turn to our numerical results. In what follows we will make comparisons to the "adiabatic approximation". In the adiabatic approximation, one ignores the mixing of various excited states which emerge in a time-dependent potential. The result is that in this approximation the time-evolution of each state is governed solely by the time-dependent imaginary part of the states energy, i.e.

$$N_n(t) = N_n(0) \exp\left(2 \int_{t_0}^{t_f} dt\, \Im[E_n(t)] \right), \tag{3.19}$$

where $N_n(t) = |c_n(t)|^2$ is the modulus-squared overlap states of type n at time t. In this case, one can solve for the time-dependent energy eigenvalues numerically or analytically.

This is expected to be a good approximation if the potential is changing slowly relative to the natural timescale of the quantum evolution.

We made several case studies using the CKMS potential (3.4). In all cases, we assume that the system is undergoing boost-invariant and transversally homogenous one-dimensional Bjorken flow [26] such that

$$T(\tau) = T_0 \left(\frac{\tau_0}{\tau}\right)^{1/3}, \tag{3.20}$$

where T_0 is the initial temperature at the proper time τ_0. We will keep the initial time fixed to be $\tau_0 = 1$ GeV^{-1} = 0.197 fm/c and vary the initial temperature to mimic what is seen in a typical Glauber temperature profile in a LHC 2.76 TeV Pb-Pb collision. In all cases shown below, we terminated the evolution when the temperature fell below the freeze-out temperature $T_{\rm fo}$ =150 MeV. For the CKMS potential, we used a box size of $L = 40$ GeV^{-1} and $N = 1024$ grid points. The spatial step size was $\Delta x = L/(N+1)$ with $r = 0$ excluded. The temporal step size was $\Delta\tau = 1 \times 10^{-3}$ GeV^{-1}.

3.3.1 Bottomonia

In Figs. 3.1-3.3 we collect our results for bottomonium states. In Fig. 3.1 we plot the survival probability of the $\Upsilon(1S)$ state as a function of proper time in panel (a) and the extracted probabilities $|c_n(t)|^2$ for the $1s$, $2s$, and $3s$ states in panel (b). For Fig. 3.1 we assumed an initial temperature of $T_0 = 0.6$ GeV, which is the temperature generated in the center of the QGP in a central Pb-Pb collision at LHC energies [16]. For the initial wave function, we took a pure $\Upsilon(1S)$ wave function which was determined self-consistently at the initial temperature of the QGP using the CKMS potential. Finally, for the bottom quark mass we used $m_b = 4.7$ GeV which is the pole mass reported in the Particle Data Group (PDG) listings [113].

In panel (a) of Fig. 3.1 the black solid line is the result of extracting the $\Upsilon(1S)$ overlap

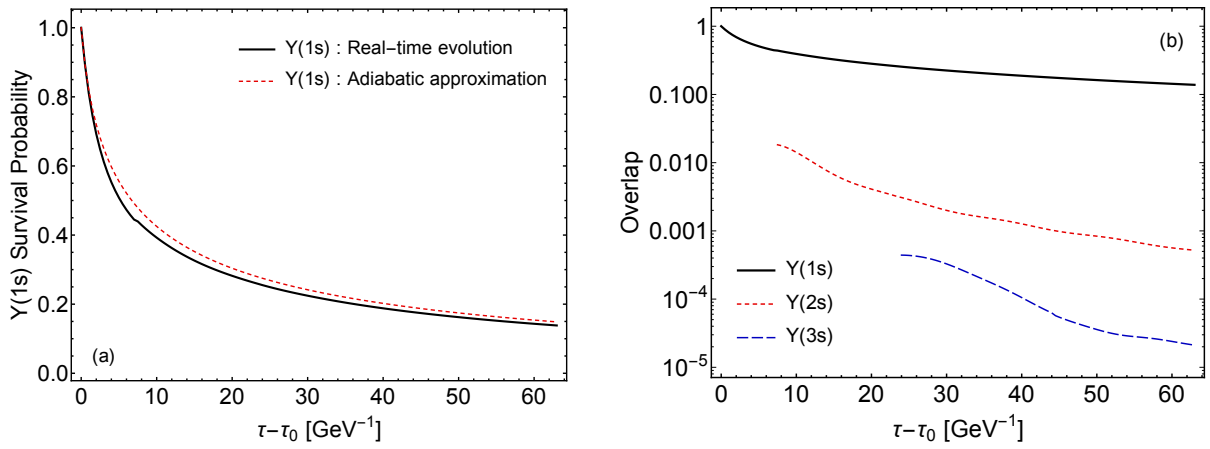


Figure 3.1: Panel (a) shows the survival probability of the $\Upsilon(1S)$ as a function of proper time. In this panel, the solid black line is the result obtained using real-time evolution and the dashed red line is the result obtained using the adiabatic approximation. Panel (b) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black, shorted-dashed red, and long-dashed blue lines are the overlaps computed for the 1s, 2s, and 3s states. In both panels the initial temperature was taken to be $T_0 = 0.6$ GeV at $\tau_0 = 1$ GeV^{-1} and the initial wave function consisted of a 1s state with the wave function determined self-consistently using the CKMS potential.

probability from the real-time solution and the dashed line is the result obtained using the adiabatic approximation (3.19). This panel demonstrates that there are only small corrections to the $\Upsilon(1S)$ survival probability coming from quantum state mixing in the center of the QGP. At the final time shown, the two results differ by approximately 6%. We do see, however, that the real-time evolution predicts enhanced 1s suppression compared to the adiabatic approximation.

Turning to panel (b) of Fig. 3.1 we see quantum state mixing due to the time evolution of the potential which results in the formation of $\Upsilon(2S)$ and $\Upsilon(3S)$ states as the QGP cools. In this panel, the initial times for the 2s and 3s curves are determined by the point in proper time at which these states become bound in the QGP, as determined by the point-and-shoot

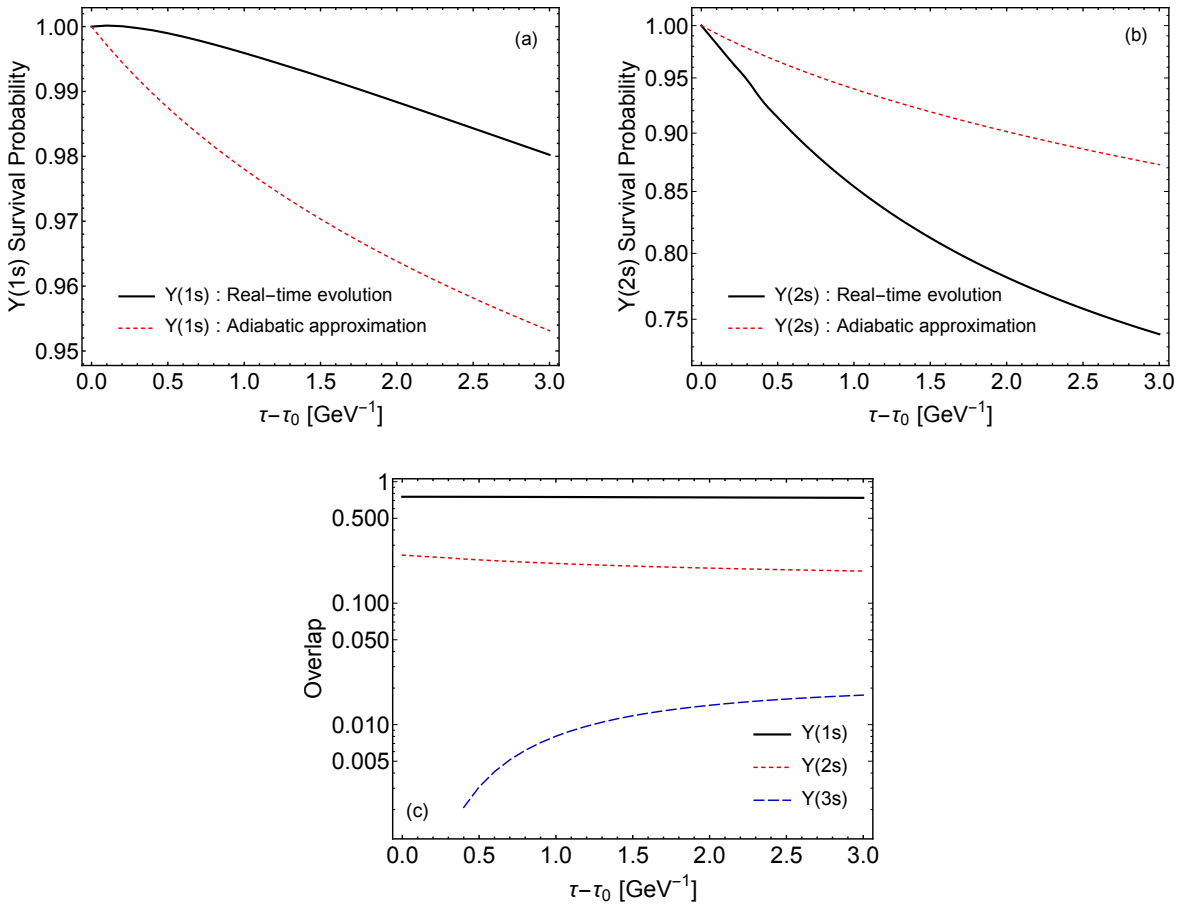


Figure 3.2: Panels (a) and (b) show the survival probability of the $\Upsilon(1S)$ and $\Upsilon(2S)$ as a function of proper time. In these panels, the solid black line is the result obtained using real-time evolution and the dashed red line is the result obtained using the adiabatic approximation. Panel (c) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black, shorted-dashed red, and long-dashed blue lines are the overlaps computed for the 1s, 2s, and 3s states. In both panels the initial temperature was taken to be $T_0 = 0.225$ GeV at $\tau_0 = 1$ GeV^{-1}. The initial wave function consisted of a linear combination of 1s and 2s components with the wave functions determined self-consistently using the CKMS potential.

method described in Sec. 3.2.2. Despite the presence of quantum state mixing, in this case, however, we see only a maximum overlap of approximately 0.02 with the 2s state and 0.0004

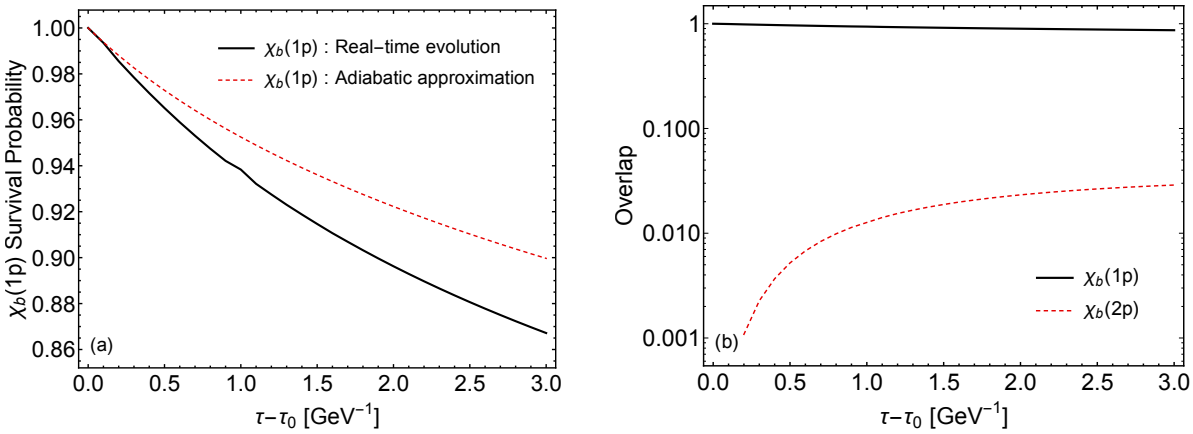


Figure 3.3: Panel (a) shows the survival probability of the $\chi_b(1P)$ as a function of proper time. Panel (b) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black and shorted-dashed red lines are the overlaps computed for the 1p and 2p states, respectively. In both panels the initial temperature was taken to be $T_0 = 0.6$ GeV at $\tau_0 = 1$ GeV^{-1} and the initial wave function consisted of a 1s state with the wave function determined self-consistently using the CKMS potential.

with the 3s state.

In Fig. 3.2 we plot the same quantities as in Fig. 3.1, however, in this case we assumed an initial temperature of $T_0 = 0.225$ GeV. At this initial temperature both the 1s and 2s states are bound, so we take the initial wave function to be a linear combination of the 1s and 2s states with a relative probability of 33% for the 2s. This relative probability is set according to the ratio of the 2s to 1s total production cross sections in pp collisions at 2.76 TeV collision energy, which are 48 nb and 145 nb for $|y| < 2.4$, respectively [70]. As can be seen from panel (a) of Fig. 3.2, for this initial condition there is clearly a difference in the evolution and final results for the $\Upsilon(1S)$ suppression, however, numerically this represents only a 3% difference between the two results shown at the final time. One other thing we notice from panel (a) is that the ordering of the curves is reversed compared to panel (a) of Fig. 3.1, with the real-time solution resulting in less suppression at this lower temperature.

In panel (b) of Fig. 3.2, we present a comparison of the real-time evolution result for the $\Upsilon(2S)$ survival probability with the adiabatic approximation. As can be seen from this panel, there are larger corrections to the adiabatic approximation when considering the $\Upsilon(2S)$, with the final difference between the two approaches being approximately 18% with the real-time evolution method predicting larger suppression. In panel (c) of Fig. 3.2, we present the combined evolution of the overlaps with the 1s, 2s, and 3s states as a function of proper time. Once again we see formation of excited states that were not in the initial wave function. In this case, in contrast to Fig. 3.1(b) we see that the 3s overlap increases monotonically with time. Even with this, one still finds a rather small 3s overlap which is approximately 0.02 at the final time.

To close out this subsection, we turn to Fig. 3.3 which shows results for the $\chi_b(1\mathrm{P})$ and $\chi_b(2\mathrm{P})$ states. The panels are the same as the previous figures and, in this case, for the initial condition we used a pure $\chi_b(1P)$ state determined self-consistently at the assumed initial temperature of $T_0 = 0.225$ GeV. From panel (a) we see that there is an approximately 4% difference between the final $\chi_b(1\mathrm{P})$ suppression when comparing the real-time evolution and the adiabatic approximation. In panel (b) we once again see an excited state $\chi_b(1P)$ overlap which increases monotonically in time, but it remains small, with the maximum overlap being approximately 0.03 at the final time.

3.3.2 Charmonia

In this subsection, we consider charmonia. The only change made to the model parameters was to change the heavy quark mass to the charm pole mass of $m_c = 1.7$ GeV taken from the PDG [113]. In Fig. 3.4 we show the comparison between the adiabatic approximation and the real-time evolution in panel (a) and a plot of the extracted overlaps in panel (b). The initial temperature was taken to be $T_0 = 0.4$ GeV and the initial wave function was self-consistently obtained using the point-and-shoot method with the CKMS potential. This

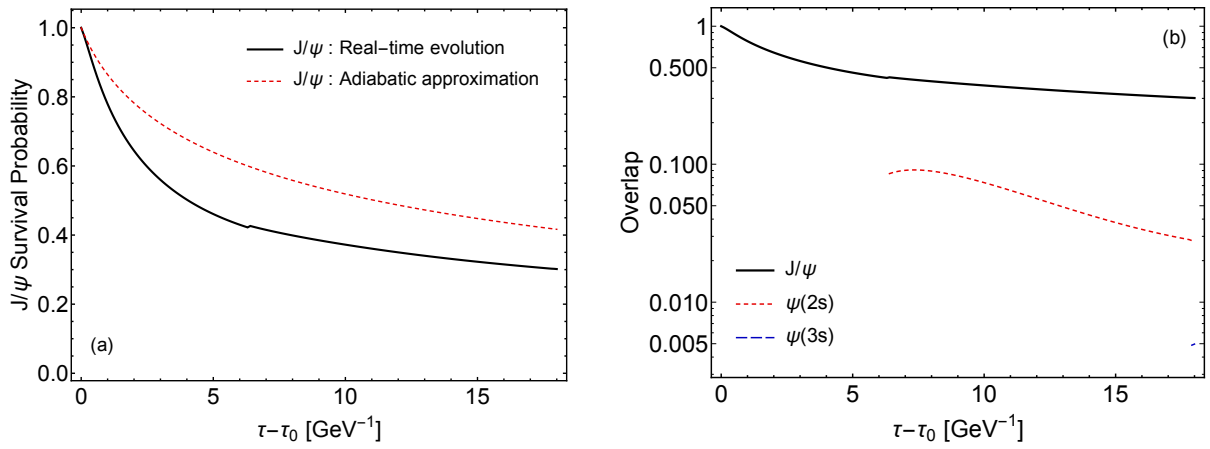


Figure 3.4: Panel (a) shows the survival probability of the J/ψ as a function of proper time. In this panel, the solid black line is the result obtained using real-time evolution and the dashed red line is the result obtained using the adiabatic approximation. Panel (b) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black, shorted-dashed red, and long-dashed blue lines are the overlaps computed for the 1s, 2s, and 3s states. In both panels the initial temperature was taken to be $T_0 = 0.4$ GeV at $\tau_0 = 1$ GeV^{-1} and the initial wave function consisted of a 1s state with the wave function determined self-consistently using the CKMS potential.

temperature corresponds to the central temperature achieved in $b = 12$ fm Pb-Pb collisions at 2.76 TeV. At this initial temperature, the only surviving s-wave state is the $1s$ J/ψ state.

As panel (a) of Fig. 3.4 demonstrates, we observe a larger difference between the real-time evolution result and the adiabatic approximation for the J/Ψ using this initial temperature, with the difference between the two results being approximately 36% at the final time. In panel (b) we show the numerically extracted overlaps with the J/Ψ, $\Psi(2S)$, and $\Psi(3S)$ as a function of proper time. As can be seen from this figure, the 2s and 3s overlaps are approximately 0.03 and 0.005, respectively.

Next, we consider Fig. 3.5 in which we have used a lower initial temperature of $T_0 = 0.2$ GeV. At this lower initial temperature the only s-wave bound state is still the J/Ψ so

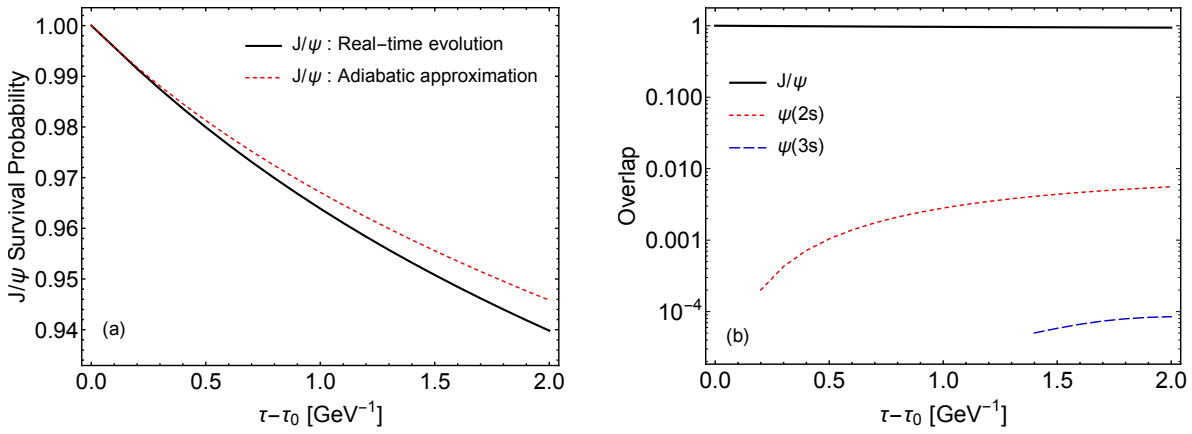


Figure 3.5: Same as Fig. 3.4, but with $T_0 = 0.2$ GeV.

we initialized the wave function as a pure J/Ψ state using the in-medium point-and-shoot method. For this initial temperature we see from panel (a) of Fig. 3.5 that the difference between the real-time and adiabatic results is approximately 0.6%. The overlaps with the 1s, 2s, and 3s states are shown in panel (b) of Fig. 3.5. From this figure we observe that both the 2s and 3s overlaps are monotonically increasing with the overlap being approximately 0.006 and 0.0001, respectively, at the final time.

Finally, in Fig. 3.6 we present the survival probability and overlaps for p-wave charmonium states in panels (a) and (b), respectively. The initial temperature was the same as in Fig. 3.5, namely $T_0 = 0.2$ GeV. As panel (a) demonstrates, in this case one sees a larger deviation between the real-time and adiabatic approximation, with the two results differing by approximately 23%. This difference is similar to that seen in Fig. 3.2(b) for the $\Upsilon(2S)$. Turning to panel (b) of Fig. 3.6, we see that the overlap with the 2p state is monotonically increasing with time and is approximately 0.16 at the final time.

To close out this section in Table 3.1 we summarize the results of the effect on the survival probability at the final time. The left column shows the initial temperature used in GeV, the middle column shows the state considered, and the third column shows the final effect on the primordial survival probability, R_{AA} (ignoring feed down effects). Negative numbers in

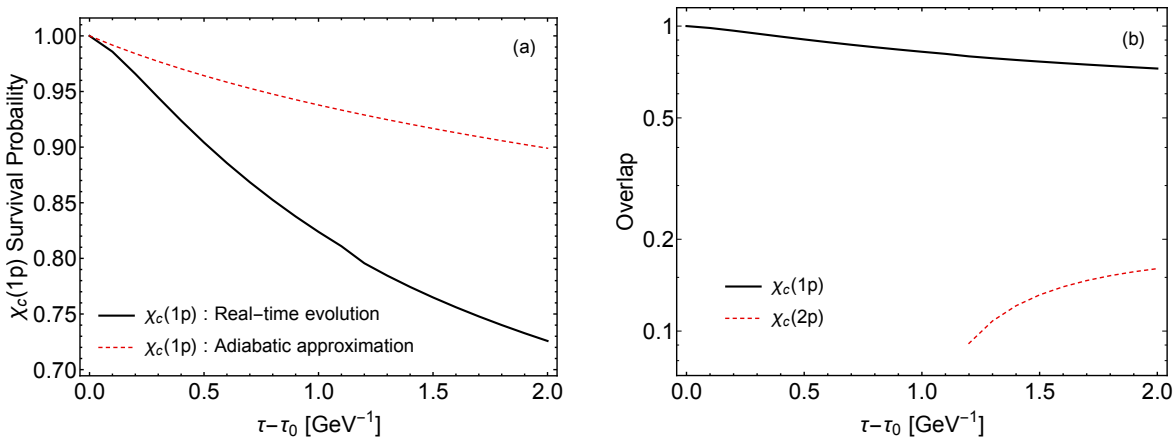


Figure 3.6: Panel (a) shows the survival probability of the $\chi_c(1P)$ as a function of proper time. In this panel, the solid black line is the result obtained using real-time evolution and the dashed red line is the result obtained using the adiabatic approximation. Panel (b) shows the overlap coefficients squared extracted using the real-time evolution. In this panel, the solid black and shorted-dashed red lines are the overlaps computed for the 1p and 2p states. In both panels the initial temperature was taken to be $T_0 = 0.2$ GeV at $\tau_0 = 1$ GeV^{-1} and the initial wave function consisted of a 1p state with the wave function determined self-consistently using the CKMS potential.

the third column indicate that R_{AA} is decreased (more suppresion) while positive numbers indicate the opposite.

3.4 Discussion and conclusion

In this chapter, we assessed the impact of going beyond the adiabatic approximation using a realistic heavy quarkonium potential 3.4. For this purpose, we assumed that the temperature evolved in time using simple boost-invariant and transversally homogeneous Bjorken evolution (3.20). For the bottomonium system we considered initial temperatures of $T_0 = 0.6$ GeV and $T_0 = 0.225$ GeV, with former being the typical temperature achieved in a central Pb-Pb collision at 2.76 TeV/nucleon collision energy and the latter being either

$\mathbf{T_0}$ [GeV]	State	% correction to primordial $\mathbf{R_{AA}}$
0.6	$\Upsilon(1S)$	-6%
0.225	$\Upsilon(1S)$	+3%
	$\Upsilon(2S)$	-18%
	$\chi_b(1P)$	-4%
0.4	J/Ψ	-36%
0.2	J/Ψ	-0.6%
	$\chi_c(1P)$	-23%

Table 3.1: Summary of the CKMS potential test cases considered herein. The quoted percentage change is computed at the final time of evolution and does not take into account feed down corrections. Positive corrections correspond to reduced suppression and negative corrections to enhanced suppression.

the temperature achieved in the periphery of a central Pb-Pb collision or in the center of a peripheral Pb-Pb collision with an impact parameter of $b = 14.7$ fm. For the higher temperature, we found that the corrections to the adiabatic approximation for the $\Upsilon(1S)$ survival probability were on the few percent level, however, for the lower temperature, we found that, although the $\Upsilon(1S)$ still did not receive large corrections, the $\Upsilon(2S)$ exhibited a larger correction when the full real-time evolution was employed.

For charmonium, we analyzed two initial temperatures, $T_0 = 0.4$ GeV and $T_0 = 0.2$ GeV, which were chosen such that, at the higher initial temperature, only the J/Ψ state was bound and, at the lower temperature, only the J/Ψ and $\chi_c(1P)$ were bound. In the charmonium sector, we found larger corrections to the adiabatic approximation, with the real-time evolution predicting stronger suppression in all cases. This can be contrasted with bottomonium states for which we saw less $\Upsilon(1S)$ suppression using the real-time evolution at the lower initial temperature. We also found that for both charmonia and bottomonia, the largest corrections to the adiabatic approximation were seen at low temperatures. In order to draw firmer conclusions about the impact on bottomonium and charmonium observables such as R_{AA}, we need to include a realistic temperature profile and 3+1d viscous hydrodynamic evolution.

This study points to the need for developing efficient routines for solving the real-time Schrödinger evolution of heavy quarkonium states in the QGP that are coupled to realistic hydrodynamical backgrounds and a realistic complex-valued potential. The need is more pressing for charmonium than it is for bottomonium, but in the end a unified framework which can describe both would be preferred. Using the isotropic CKMS potential, it is technically feasible to fold together the real-time evolution with realistic 3+1d viscous hydrodynamics codes similar to what has been done in the case of the adiabatic approximation [107, 110, 108, 73, 74, 72], however, the incorporation of the anisotropic non-equilibrium

corrections [51, 42, 50, 52, 87] to the heavy quark potential will be technically challenging since the problem can no longer be reduced to a one-dimensional effective potential.

Another avenue for future improvement is to include a stochastic noise source similar to what was done in Ref. [94] to treat open quantum systems. In general, one should use the trace- and positivity-preserving Lindblad form [14, 11, 12, 67, 30, 65, 31, 28, 27, 118] for the evolution of the heavy quark system's reduced density matrix. However, if the environment's relaxation time is short compared to the relaxation time of the system, then one can reduce the problem to solving a stochastic Schrödinger equation.

Chapter 4

Stochastic Potential

In our previuos work [29], werelaxed the assumption of adiabatic approximation by solving the 3d Schrödinger equation in real-time in order to compute quarkonium suppression dynamically. We compared results obtained using the adiabatic approximation with real-time evolution for both harmonic oscillator and realistic complex heavy quark potentials. As an extension of ref [29], in this chapter, we will obtain suppression factor for bottomonium and charmonium using stochastic potential with two independent white noise sources. Here we will discuss theoretical advancement of including two independent white noises correspond to quark and anti-quark of the heavy quarkonium state to the stochastic potential.

4.1 Heavy quarkonium as an open quantum system

The heavy quarkonium has been treated as an open quantum system in many works [14, 11, 12, 31]. Inside QGP, the heavy quarkonium behaves as an open quantum system and the environment is composed of light quarks and gluons. The total system (heavy quarkonium and the environment) is described by a hermitian Hamiltonian $H^\dagger = H$, [14, 93, 94, 65]:

$$H = H_{sys} \otimes I_{med} + I_{sys} \otimes H_{med} + H_{int} \ , \quad \frac{d}{dt}\rho(t) = \frac{1}{i\hbar}[H, \rho(t)] \ , \tag{4.1}$$

where H_{int} represents the interaction between the heavy quarkonium and the medium (light quarks and the gluons).

When we focus on the real-time dynamics of the heavy quarkonium in the QGP, we assume some initial condition: $\rho(0) = \rho_{sys} \otimes \rho_{med}$, where $\rho_{sys}(t) \equiv Tr_{med}\{\rho(t)\}$ is the reduced density matrix. Starting with the initial condition, we integrate out the degrres of

freedom corresponds to the light quarks and gluons and then obtain the master equation in Lindblad form [11, 12] for the reduced density matrix ρ_{sys}:

$$\frac{d}{dt}\rho_{sys}(t) = -i[H_{sys}\,,\rho_{sys}] + \sum_n \left(2L_n\rho_{sys}L_n^\dagger - L_n^\dagger L_n\rho_{sys} - \rho_{sys}L_n^\dagger L_n\right)\ , \tag{4.2}$$

where L_n is the Lindblad operator.

The real-time evolution of the reduced density matrix is Markovian and it has three basic properties:

1. the reduced density matrix is hermitian i.e., $\rho_{sys} = \rho_{sys}^\dagger$,
2. it is properly normalized i.e., $Tr(\rho_{sys}) = 1$,
3. it is positive i.e., $\langle\alpha|\rho_{sys}|\alpha\rangle \geq 0$ for any state $|\alpha\rangle$,

during its time evolution.

4.2 Stochastic evolution of heavy quarkonium

In the EFT(effective field theory), we a complex heavy quarkonium potential which consists of a real and imaginary parts i.e., $V_{EFT} = Re[V] + iIm[V]$. Whereas in stochastic approach heavy quarkonium stochastic potential consists of a purely real potential and a term contributed from noise sources i.e., $V_{stochastic} = V(r) + \Theta_{noise}$. Instead of assigning the width to an imaginary part of the complex potential of EFT approach, the open open quantum systems approach deals with a purely real potential is perturbed by the thermal environment. In this open quantum systems approach, the width is then characterized by the strength of the thermal noise $\Gamma(r, r)$, (see figure 4.1).

For instance, we discuss the evolution of bottomonium (see figure 4.2) inside the QGP. For phenomenological purpose, we need to know the abundances of the different bottomonium

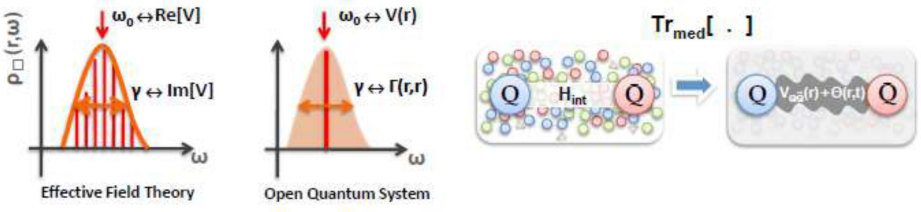


Figure 4.1: Schematic view of the difference between a description on the level of the full system with a hermitian Hamiltonian $H = H_{sys} + H_{med} + H_{int}$ and a stochastic potential description of the heavy quarkonium system after tracing out the medium degrees of freedom (adapted from [93]).

states inside QGP after a certain time t. For this we find the probability $c_{nn}(t)$ of an eigenstate $\phi_n(\mathbf{r})$ of the vacuum Hamiltonian

$$H^{VAC}_{Q\bar{Q}}\phi_n(\mathbf{r}) = E_n\phi_n(\mathbf{r}) \ , \tag{4.3}$$

at time t . The probability $c_{nn}(t)$ is obtained by projecting the current wavefunction $\psi_{Q\bar{Q}}(\mathbf{r},t)$ onto the corresponding vacuum wavefunction $\phi_n(\mathbf{r})$

$$c_{nn}(t) = \int d^3r\, d^3r' \phi^*_n(\mathbf{r}) \langle \psi_{Q\bar{Q}}(\mathbf{r},t)\psi^*_{Q\bar{Q}}(\mathbf{r}',t)\rangle \phi_n(\mathbf{r}') \ . \tag{4.4}$$

This probability depends on the evolution of the density matrix $\rho_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t) = \langle \psi_{Q\bar{Q}}(\mathbf{r},t)\psi^*_{Q\bar{Q}}(\mathbf{r}',t)\rangle$ whose time dependence is sensitive to the off-diagonal elements of $\Gamma(\mathbf{r},\mathbf{r}')$.

4.3 Stochastic potential model

In this section, we review the stochastic potential model described in ref [14]. We consider a wave function based description of the stochastic time evolution of a non-relativistic heavy

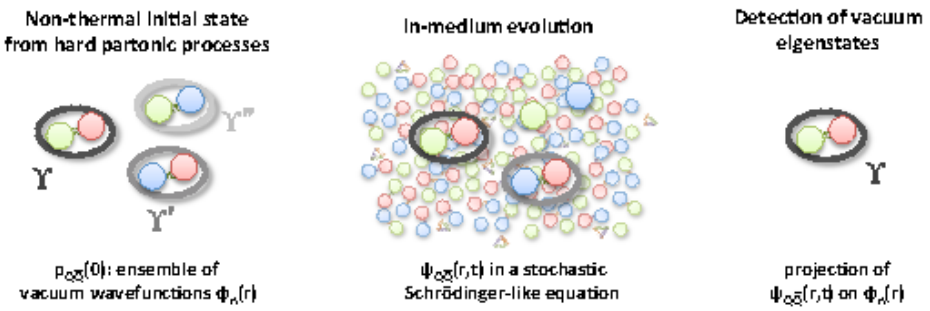


Figure 4.2: Schematic view of the setup used to describe Bottomonium evolution inside QGP (adapted from [94]).

quarkonium state. $\mathbf{X} \equiv (\mathbf{x_1}, \mathbf{x_2})$ and $\mathbf{X'} \equiv (\mathbf{x'_1}, \mathbf{x'_2})$ denote the positions of the quark and anti-quark respectively.The distance between these constituent quarks is

$$\mathbf{\Delta X} \equiv |\mathbf{X} - \mathbf{X'}| = \sqrt{(\mathbf{x_1} - \mathbf{x'_1})^2 + (\mathbf{x_2} - \mathbf{x'_2})^2} \ .$$

We assume that the dynamics described by the stochastic unitary time evolution operator $U_\Theta^{(X)}(t|0)$ is both Markovian and linear in the heavy quarkonium wave function $\Psi_{Q\bar{Q}}(\mathbf{X}, t)$.

The time evolution is given by

$$\Psi_{Q\bar{Q}}(\mathbf{X}, t) = U_\Theta^{(X)}(t|0)\Psi_{Q\bar{Q}}(\mathbf{X}, 0) \ , \tag{4.5}$$

$$U_\Theta^{(X)}(t|0) = \mathcal{T} \exp\left[-\frac{i}{\hbar}\int_0^t dt' \left\{H(\mathbf{X}) + \Theta(\mathbf{X}, t')\right\}\right] \ , \tag{4.6}$$

$$H(\mathbf{X}) \equiv 2M - \frac{\hbar^2 \nabla_X^2}{2M} + V(\mathbf{X}) \ , \tag{4.7}$$

where $\mathcal{T}$ is the time-ordered product and the stochastic term $\Theta(\mathbf{X}, t)$ is the Gaussian white noise.

This noise term has the following properties:

$$\langle \Theta(\mathbf{X},t)\rangle = 0 \ , \quad \langle \Theta(\mathbf{X},t)\Theta(\mathbf{X}',t')\rangle = \frac{\hbar}{\Delta t}\Gamma(\mathbf{X},\mathbf{X}')\,\delta_{tt'} \ , \tag{4.8}$$

where Δt is a time scale during which the light quarks and gluons experience many collisions inside QGP but the heavy quarkonium state does not change considerably. For a uniform and isotropic system, $\Gamma(\mathbf{X},\mathbf{X})$ depends only on $|\mathbf{x_1}-\mathbf{x_2}|$. As the relativistic medium particles have size of $l_{th} \equiv \frac{2\pi\hbar c}{\kappa_B T}$, the transferred energy or the fluctuation is expected to be correlated at $\mathbf{X}$ and $\mathbf{X}'$ if $\Delta X < l_{th}$ and non-correlated i.e., $\Gamma(\mathbf{X},\mathbf{X}') \sim 0$ if ΔX is larger than $\sim l_{th}$.

Using Eq. (4.6), we expand [1] $U_\Theta^{(X)}(t+\frac{\Delta t}{t})$ in Δt up to $\mathcal{O}(t^{3/2})$:

[1] $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + ...$

$$
\begin{aligned}
U_{\Theta}^{(X)}(t+\frac{\Delta t}{t}) &\equiv 1-\frac{i\Delta t}{\hbar}\Big\{H(\mathbf{X})+\Theta(\mathbf{X},t)\Big\}-\frac{(\Delta t)^2}{2\hbar^2}\Theta(\mathbf{X},t)^2 \\
&= 1-\frac{i\Delta t}{\hbar}H(\mathbf{X})-\frac{i\Delta t}{\hbar}\Theta(\mathbf{X},t)-\frac{(\Delta t)^2}{2\hbar^2}\Theta(\mathbf{X},t)^2 \qquad (4.9)\\
&= 1-\frac{i\Delta t}{\hbar}\Big\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})\Big\}+\frac{\Delta t}{2\hbar}\Gamma(\mathbf{X},\mathbf{X}) \\
&- \frac{i\Delta t}{\hbar}\Theta(\mathbf{X},t)-\frac{(\Delta t)^2}{2\hbar^2}\Theta(\mathbf{X},t)^2 \qquad (4.10)\\
&= 1-\frac{i\Delta t}{\hbar}\Big\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})\Big\}+\frac{\Delta t}{2\hbar}.\frac{\Delta t}{\hbar}\langle\Theta(\mathbf{X},t)^2\rangle \\
&- \frac{i\Delta t}{\hbar}\Theta(\mathbf{X},t)-\frac{(\Delta t)^2}{2\hbar^2}\Theta(\mathbf{X},t)^2 \qquad (4.11)\\
&= 1-\frac{i\Delta t}{\hbar}\Big\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})\Big\}-\frac{i\Delta t}{\hbar}\Big[\Theta(\mathbf{X},t)-\frac{i\Delta t}{2\hbar}\Theta(\mathbf{X},t)^2+\frac{i\Delta t}{2\hbar}\langle\Theta(\mathbf{X},t)^2\rangle\Big] \\
&= 1-\frac{i\Delta t}{\hbar}\Big\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})\Big\}-\frac{i\Delta t}{\hbar}\Big[\Theta(\mathbf{X},t)-\frac{i\Delta t}{2\hbar}\Big\{\Theta(\mathbf{X},t)^2-\langle\Theta(\mathbf{X},t)^2\rangle\Big\}\Big] \\
&= 1-\frac{i\Delta t}{\hbar}\Big\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})\Big\}-\frac{i\Delta t}{\hbar}\,\Xi(\mathbf{X},t) \qquad (4.12)\\
&= 1-\frac{i\Delta t}{\hbar}\Big\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})+\Xi(\mathbf{X},t)\Big\} \qquad (4.13)\\
&\equiv \exp\Big[-\frac{i\Delta t}{\hbar}\Big\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})+\Xi(\mathbf{X},t)\Big\}\Big] \quad , \quad \text{upto first order} \qquad (4.14)
\end{aligned}
$$

In Eq. (4.10), we have added and subtracted $\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})$ term and in Eq. (4.11) we have used

$$\langle\Theta(\mathbf{X},t)^2\rangle=\frac{\hbar}{\Delta t}\Gamma(\mathbf{X},\mathbf{X}) \quad , \qquad (4.15)$$

which can be easily derived from Eq. (4.8). In Eq. (4.12) we take the complex noise as

$$\Xi(\mathbf{X},t)=\Theta(\mathbf{X},t)-\frac{i\Delta t}{2\hbar}\Big\{\Theta(\mathbf{X},t)^2-\langle\Theta(\mathbf{X},t)^2\rangle\Big\} \quad , \qquad (4.16)$$

so that $\langle\Xi(\mathbf{X},t)\rangle=0$. From Eq. (4.14), we can read the effective Hamiltonian as

$$H_{\text{eff}}(\mathbf{X},t)=H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})+\Xi(\mathbf{X},t) \quad . \qquad (4.17)$$

Thus the stochastic Schrödinger equation[2] can be written as

$$i\hbar\frac{\partial}{\partial t}\Psi_{Q\bar{Q}}(\mathbf{X},t)=\left\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})+\Xi(\mathbf{X},t)\right\}\Psi_{Q\bar{Q}}(\mathbf{X},t) \quad . \tag{4.18}$$

Now we will derive a master equation for the reduced density matrix:

$$\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t)\equiv\langle\Psi_{Q\bar{Q}}(\mathbf{X},t)\Psi^{*}_{Q\bar{Q}}(\mathbf{X}',t)\rangle_{\Xi} \quad . \tag{4.19}$$

The infinitesimal time evolution of $\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t)$ is:

$$\begin{aligned}
\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t+dt) &= \left\langle U^{(X)}_{\Theta}\left(t+\frac{dt}{t}\right)U^{(X')*}_{\Theta}\left(t+\frac{dt}{t}\right)\right\rangle_{\Xi} \\
&= \left[1-\frac{idt}{\hbar}\left\{H(\mathbf{X})+\Theta(\mathbf{X},t)\right\}-\frac{(dt)^2}{2\hbar^2}\Theta(\mathbf{X},t)^2\right] \\
&\quad\times\left[1-\frac{idt}{\hbar}\left\{H(\mathbf{X}')+\Theta(\mathbf{X}',t)\right\}-\frac{(dt)^2}{2\hbar^2}\Theta(\mathbf{X}',t)^2\right]\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) \\
&= \left[1-\frac{idt}{\hbar}\left\{H(\mathbf{X})+\Theta(\mathbf{X},t)\right\}-\frac{(dt)^2}{2\hbar^2}\Theta(\mathbf{X},t)^2\right. \\
&\quad+\frac{idt}{\hbar}\left\{H(\mathbf{X}')+\Theta(\mathbf{X}',t)\right\}-\frac{(dt)^2}{2\hbar^2}\Theta(\mathbf{X}',t)^2 \\
&\quad\left.+\left(\frac{dt}{\hbar}\right)^2\Theta(\mathbf{X},t)\Theta(\mathbf{X}',t)+...\right]\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) \\
&= \left[1-\frac{idt}{\hbar}\left\{H(\mathbf{X})-\frac{i}{2}\Gamma(\mathbf{X},\mathbf{X})\right\}-\frac{idt}{\hbar}\Xi(\mathbf{X},t)\right. \\
&\quad+\frac{idt}{\hbar}\left\{H(\mathbf{X}')+\frac{i}{2}\Gamma(\mathbf{X}',\mathbf{X}')\right\}+\frac{idt}{\hbar}\Xi(\mathbf{X}',t) \\
&\quad\left.+\frac{dt}{\hbar}.\frac{dt}{\hbar}\,\Theta(\mathbf{X},t)\Theta(\mathbf{X}',t)+...\right]\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) \quad .
\end{aligned} \tag{4.20}$$

Using Eq. (4.20) we obtain master equation for the reduced density matrix as follows:

[2]The expansion of the unitary time evolution operator induces non-unitarity in the stochastic Schrödinger equation.

$$
\begin{aligned}
\frac{\partial \rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t)}{\partial t} &= \Bigg[-\frac{i}{\hbar}\Bigg\{ H(\mathbf{X}) - \frac{i}{2}\Gamma(\mathbf{X},\mathbf{X}) \Bigg\} - \frac{i}{\hbar}\big\langle \Xi(\mathbf{X},t) \big\rangle \\
&\quad + \frac{i}{\hbar}\Bigg\{ H(\mathbf{X}') + \frac{i}{2}\Gamma(\mathbf{X}',\mathbf{X}') \Bigg\} + \frac{i}{\hbar}\big\langle \Xi(\mathbf{X}',t) \big\rangle \\
&\quad + \frac{1}{\hbar}.\frac{dt}{\hbar}\left\langle \Theta(\mathbf{X},t)\Theta(\mathbf{X}',t) \right\rangle + ... \Bigg] \rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) \\
&= \Bigg[-\frac{i}{\hbar}\Bigg\{ H(\mathbf{X}) - \frac{i}{2}\Gamma(\mathbf{X},\mathbf{X}) \Bigg\} + 0 \\
&\quad + \frac{i}{\hbar}\Bigg\{ H(\mathbf{X}') + \frac{i}{2}\Gamma(\mathbf{X}',\mathbf{X}') \Bigg\} + 0 \\
&\quad + \frac{1}{\hbar}\Gamma(\mathbf{X},\mathbf{X}') \Bigg] \rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) \qquad (4.21) \\
&= \left[\frac{1}{i\hbar}\Bigg\{ H(\mathbf{X}) - \frac{i}{2}\Gamma(\mathbf{X},\mathbf{X}) \Bigg\} - \frac{1}{i\hbar}\Bigg\{ H(\mathbf{X}') + \frac{i}{2}\Gamma(\mathbf{X}',\mathbf{X}') \Bigg\} + \frac{1}{\hbar}\Gamma(\mathbf{X},\mathbf{X}') \right] \rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) \\
&= \frac{H(\mathbf{X}) - H(\mathbf{X}')}{i\hbar}\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) + \frac{1}{\hbar}\Bigg\{ \Gamma(\mathbf{X},\mathbf{X}') - \frac{\Gamma(\mathbf{X},\mathbf{X}) + \Gamma(\mathbf{X}',\mathbf{X}')}{2} \Bigg\} \rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) \\
\frac{\partial \rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t)}{\partial t} &= \frac{H(\mathbf{X}) - H(\mathbf{X}')}{i\hbar}\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) + \frac{F(\mathbf{X},\mathbf{X}')}{\hbar}\rho_{Q\bar{Q}}(\mathbf{X},\mathbf{X}',t) \quad , \qquad (4.22)
\end{aligned}
$$

where in Eq. (4.21), we have used

$$
\big\langle \Xi(\mathbf{X},t) \big\rangle = 0 \quad , \quad \big\langle \Xi(\mathbf{X}',t) \big\rangle = 0 \quad , \quad \big\langle \Theta(\mathbf{X},t)\Theta(\mathbf{X}',t) \big\rangle = \frac{\hbar}{dt}\Gamma(\mathbf{X},\mathbf{X}') \ . \qquad (4.23)
$$

In Eq. (4.22), we take

$$
F(\mathbf{X},\mathbf{X}') = \Gamma(\mathbf{X},\mathbf{X}') - \frac{\Gamma(\mathbf{X},\mathbf{X}) + \Gamma(\mathbf{X}',\mathbf{X}')}{2} \ . \qquad (4.24)
$$

Now we write master equation i.e., Eq. (4.22) in the relative coordinates. We define the reduced density matrix in the relative coordinates as:

$$
\begin{aligned}
\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t) &\equiv \mathrm{Tr_R}\big\{ \rho_{Q\bar{Q}}\left(\mathbf{X}_{R,r},\mathbf{X}_{R',r'},t\right) \big\} \\
&= \int d^3R\, \rho_{Q\bar{Q}}\left(\mathbf{X}_{R,r},\mathbf{X}_{R',r'},t\right) \quad , \qquad (4.25)
\end{aligned}
$$

where $\mathbf{X}_{R,r} \equiv \left(\mathbf{R} + \frac{\mathbf{r}}{2}, \mathbf{R} - \frac{\mathbf{r}}{2}\right)$.

From Eq. (4.22), we can write the master equation for $\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t)$ in the relative coordinates as:

$$\frac{\partial\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t)}{\partial t}=\frac{h(\mathbf{r})-h(\mathbf{r}')}{i\hbar}\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t)+\frac{f(\mathbf{r},\mathbf{r}')}{\hbar}\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t)\ , \tag{4.26}$$

where we take

$$h(\mathbf{r})\equiv 2M-\frac{\hbar^2\,\nabla_r^2}{M}+v(\mathbf{r})\ , \tag{4.27}$$

$$f(\mathbf{r},\mathbf{r}')\equiv\gamma(\mathbf{r},\mathbf{r}')-\frac{\gamma(\mathbf{r},\mathbf{r})+\gamma(\mathbf{r}',\mathbf{r}')}{2}\ , \tag{4.28}$$

and define

$$v(r)\equiv V\left(\mathbf{X}_{0,r}\right)\ ,$$

and

$$\gamma(\mathbf{r},\mathbf{r}')\equiv\Gamma\left(\mathbf{X}_{0,r}\ ,\mathbf{X}_{0,r'}\right)\ .$$

The stochastic Schrödinger equation in the relative coordinates is then given by

$$i\hbar\frac{\partial}{\partial t}\psi_{Q\bar{Q}}(\mathbf{r},t)=\left\{h(\mathbf{r})-\frac{i}{2}\gamma(\mathbf{r},\mathbf{r})+\xi(\mathbf{r},t)\right\}\psi_{Q\bar{Q}}(\mathbf{r},t)\ , \tag{4.29}$$

where,

$$\xi(\mathbf{r},t)\equiv\theta(\mathbf{r},t)-\frac{idt}{2\hbar}\left\{\theta(\mathbf{r},t)^2-\langle\theta(\mathbf{r},t)^2\rangle\right\}\ . \tag{4.30}$$

Here, $\theta(\mathbf{r},t)$ satisfies the following properties:

$$\langle\theta(\mathbf{r},t)\rangle=0\quad,\quad\langle\theta(\mathbf{r},t)\theta(\mathbf{r}',t')\rangle=\frac{\hbar}{dt}\gamma(\mathbf{r},\mathbf{r}')\,\delta_{tt'}\ . \tag{4.31}$$

Now we will obtain expressions for the survival probabilities of heavy quarkonia bound states in the vacuum or medium.

Hamiltonian	Eigen function	Eigen value
$h^{vac}(\mathbf{r})$	$\phi_n(\mathbf{r})$	e_n
$h(\mathbf{r})$	$\psi_n(\mathbf{r})$	ϵ_n

Table 4.1: Eigen functions and eigen values of the vacuum Hamiltonian $h^{vac}(\mathbf{r})$ and the medium Hamiltonian $h(\mathbf{r})$.

The expanded reduced density matrix and its expansion coefficient and the bound state survival probability in the vacuum are given by:

$$\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t) = \sum_{nm} c^v_{nm}(t)\phi_n(\mathbf{r})\phi^*_m(\mathbf{r}') \ , \tag{4.32}$$

$$c^v_{nm} = \int d^3r\, d^3r'\, \phi^*_n(\mathbf{r})\langle\psi_{Q\bar{Q}}(\mathbf{r},t)\psi^*_{Q\bar{Q}}(\mathbf{r}',t)\rangle_\xi\, \phi_m(\mathbf{r}') \ , \tag{4.33}$$

$$P^v(t) \equiv \sum_{1\le n\le N^v_b} c^v_{nn}(t) \ , \tag{4.34}$$

where N^v_b is the number of bound states of $h^{vac}(r)$.

Similarly,the expanded reduced density matrix and its expansion coefficient and the bound state survival probability in the medium are given by:

$$\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t) = \sum_{nm} c^m_{nm}(t)\psi_n(\mathbf{r})\psi^*_m(\mathbf{r}') \ , \tag{4.35}$$

$$c^m_{nm} = \int d^3r\, d^3r'\, \psi^*_n(\mathbf{r})\langle\psi_{Q\bar{Q}}(\mathbf{r},t)\psi^*_{Q\bar{Q}}(\mathbf{r}',t)\rangle_\xi\, \psi_m(\mathbf{r}') \ , \tag{4.36}$$

$$P^m(t) \equiv \sum_{1\le n\le N_b^m} c_{nn}^m(t) \ , \tag{4.37}$$

where N_b^m is the number of bound states of $h(r)$.

Now, we obtain the time evolution of the in-medium state admixtures c_{nm}^m.

Putting Eq. (4.28) in Eq. (4.26) we get

$$\frac{\partial \hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t)}{\partial t} = \frac{h(\mathbf{r})-h(\mathbf{r}')}{i\hbar}\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t)+\frac{1}{\hbar}\gamma(\mathbf{r},\mathbf{r}')\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t)-\frac{1}{2\hbar}\Big\{\gamma(\mathbf{r},\mathbf{r})+\gamma(\mathbf{r}',\mathbf{r}')\Big\}\hat{\rho}_{Q\bar{Q}}(\mathbf{r},\mathbf{r}',t) \tag{4.38}$$

Inserting Eq. (4.35) in Eq. (4.38), we get

$$\begin{aligned}\sum_{nm} \dot{c}_{nm}^m(t)\psi_n(\mathbf{r})\psi_m^*(\mathbf{r}') &= \frac{h(\mathbf{r})-h(\mathbf{r}')}{i\hbar}\sum_{nm} c_{nm}^m(t)\psi_n(\mathbf{r})\psi_m^*(\mathbf{r}')+\frac{1}{\hbar}\gamma(\mathbf{r},\mathbf{r}')\sum c_{kl}^m(t)\psi_k(\mathbf{r})\psi_l^*(\mathbf{r}') \\ &\quad -\frac{1}{2\hbar}\Big\{\gamma(\mathbf{r},\mathbf{r})\sum c_{km}^m(t)\psi_k(\mathbf{r})\psi_m^*(\mathbf{r})+\gamma(\mathbf{r}',\mathbf{r}')\sum c_{nk}^m(t)\psi_n(\mathbf{r}')\psi_k^*(\mathbf{r}')\Big\}\end{aligned} \tag{4.39}$$

From Eq. (4.39), we can write the time evolution of c_{nm}^m.

$$\dot{c}_{nm}^m(t) = \frac{\epsilon_n-\epsilon_m}{i\hbar}\, c_{nm}^m(t) + \frac{1}{\hbar}\sum_{kl}\, \gamma_{nk,lm}\, c_{kl}^m(t) - \frac{1}{2\hbar}\sum_k\, \Big\{\gamma_{nk}\, c_{km}^m(t) + c_{nk}^m(t)\gamma_{km}\Big\} \ , \tag{4.40}$$

where

$$\gamma_{nk,lm} \equiv \int\, d^3r\, d^3r'\, \psi_n^*(\mathbf{r})\, \psi_l^*(\mathbf{r}')\, \gamma(\mathbf{r},\mathbf{r}')\, \psi_k(\mathbf{r})\, \psi_m(\mathbf{r}')$$

and

$$\gamma_{nm} \equiv \int\, d^3r\, \psi_n^*(\mathbf{r})\, \gamma(\mathbf{r},\mathbf{r})\, \psi_m(\mathbf{r}) \ .$$

Here $c_{nm}^m(t)$ and γ_{nm} are Hermitian matrices and $\gamma_{nk,lm} = \gamma_{lm,nk} = \gamma_{kn,ml}^*$.

Now we will discuss qualitative feature of $P^m(t)$ by using Eq. (4.40).

- **Case-I:** If we take $\gamma(\mathbf{r},\mathbf{r}') = \gamma$, then we obtain [14]

$$\dot{c}^m_{nm}(t) = \frac{\epsilon_n - \epsilon_m}{i\hbar}\, c^m_{nm}(t) \quad , \quad \dot{P}^m(t) = 0 \ . \tag{4.41}$$

  Therefore in case of spatially uniform fluctuations, the survival of heavy quarkonium is completely determined by the medium Hamiltonian $h(\mathbf{r})$, which is the case of the purely real potential models.

- **Case-II:** If we take $\gamma(\mathbf{r},\mathbf{r}') = \gamma \exp\left\{ -\frac{(\Delta r)^2}{2l^2_{corr}} \right\}$, with $\Delta r \equiv |\mathbf{r}-\mathbf{r}'|$ and $l_{corr} \ll l_{cut}$, we can show [14]

$$\dot{c}^m_{nm}(t) = \frac{\epsilon_n - \epsilon_m}{i\hbar}\, c^m_{nm}(t) - \frac{\gamma}{\hbar} c^m_{nm}(t) \quad , \quad \dot{P}^m(t) = -\frac{\gamma}{\hbar}\dot{P}^m(t) \ . \tag{4.42}$$

  Therefore in case of very localized fluctuation, the damping depends solely on $\gamma = \gamma(\mathbf{r},\mathbf{r})$. Further if $l_{corr} \ll l_{cut}$ then all the diagonal elements of $c^m_{nm}(t)$ decay as $\sim e^{-\frac{\gamma t}{\hbar}}$ and $\mathrm{Tr_r}\big\{\hat{\rho}(\mathbf{r},\mathbf{r}',t)\big\}$ is not conserved. Otherhand if $l_{corr} = l_{th}$ then the states with the scale larger than l_{th} will decay while those smaller than l_{th} will remain undisturbed.

4.4 Stochastic potential with two noise sources

In section 4.3, we took the expression of the noise Θ based on intuition and it was not systematically derived [14, 94]. Ref. [12] suggests the noise Θ should consist of two noise sources for a heavy quark and anti-quark. In this section we systematically derive the noise and review the work of Ref. [65].

Let us consider a heavy quark is situated at $\mathbf{x} = \mathbf{R}+\frac{\mathbf{r}}{2}$ and a heavy anti-quark at $\mathbf{x}' = \mathbf{R}-\frac{\mathbf{r}}{2}$, and the strength of the potential between them is not only weakened by screening but also

fluctuates. Therefore, the Hamiltonian consists of a screened potential $V(\mathbf{r})$ and stochastic noise term $\Theta(\mathbf{r},t)$, i.e.,

$$H(\mathbf{r},t) = -\frac{\boldsymbol{\nabla}^{\mathbf{2}}_{\mathbf{r}}}{M} + V(\mathbf{r}) + \Theta(\mathbf{r},t) \ , \tag{4.43}$$

where,

$$\Theta(\mathbf{r},t) = \theta\left(\mathbf{R}+\frac{\mathbf{r}}{2},t\right) - \theta\left(\mathbf{R}-\frac{\mathbf{r}}{2},t\right) \ . \tag{4.44}$$

Here, $\Theta(\mathbf{r},t)$ is the sum of noise terms for the heavy quark i.e., $\theta(\mathbf{x},t)$ and anti-quark i.e., $\theta(\mathbf{x}',t)$ which satisfy the following correlations:

$$\langle\theta(\mathbf{x},t)\rangle = 0 \ , \tag{4.45}$$

and

$$\langle\theta(\mathbf{x},t)\theta(\mathbf{x}',t')\rangle = D(\mathbf{x}-\mathbf{x}')\delta(t-t') \ . \tag{4.46}$$

From Eq. (4.44), we write

$$\begin{aligned}
\Theta(\mathbf{r},t)\Theta(\mathbf{r}',t') &= \left[\theta\left(\mathbf{R}+\frac{\mathbf{r}}{2},t\right) - \theta\left(\mathbf{R}-\frac{\mathbf{r}}{2},t\right)\right]\left[\theta\left(\mathbf{R}+\frac{\mathbf{r}'}{2},t'\right) - \theta\left(\mathbf{R}-\frac{\mathbf{r}'}{2},t'\right)\right] \\
&= \theta\left(\mathbf{R}+\frac{\mathbf{r}}{2},t\right)\theta\left(\mathbf{R}+\frac{\mathbf{r}'}{2},t'\right) - \theta\left(\mathbf{R}+\frac{\mathbf{r}}{2},t\right)\theta\left(\mathbf{R}-\frac{\mathbf{r}'}{2},t'\right) \\
&\quad -\theta\left(\mathbf{R}-\frac{\mathbf{r}}{2},t\right)\theta\left(\mathbf{R}+\frac{\mathbf{r}'}{2},t'\right) + \theta\left(\mathbf{R}-\frac{\mathbf{r}}{2},t\right)\theta\left(\mathbf{R}-\frac{\mathbf{r}'}{2},t'\right) \ .
\end{aligned} \tag{4.47}$$

Using Eq. (4.46) in Eq. (4.47) and assuming D is an even function i.e., $D(-\mathcal{Z}) = D(\mathcal{Z})$, we obtain the correlation of $\Theta(\mathbf{r},t)$:

$$\begin{aligned}\left\langle \Theta(\mathbf{r},t)\Theta(\mathbf{r}',t')\right\rangle &= D\left(\frac{\mathbf{r}-\mathbf{r}'}{2}\right)\delta(t-t') - D\left(\frac{\mathbf{r}+\mathbf{r}'}{2}\right)\delta(t-t') \\ &\quad -D\left(-\frac{\mathbf{r}+\mathbf{r}'}{2}\right)\delta(t-t') + D\left(-\frac{\mathbf{r}-\mathbf{r}'}{2}\right)\delta(t-t') \\ &= 2\Big[D\left(\frac{\mathbf{r}-\mathbf{r}'}{2}\right) - D\left(\frac{\mathbf{r}+\mathbf{r}'}{2}\right)\Big]\delta(t-t')\ . \end{aligned} \tag{4.48}$$

Now we expand[3] the unitary time evolution operator $e^{-i\Delta t\, H(\mathbf{r},t)}$ in terms of the infinitesimal time Δt:

$$\begin{aligned} e^{-i\Delta t\, H(\mathbf{r},t)} &\simeq 1 - i\Delta t\, H(\mathbf{r},t) - \frac{1}{2}\big\{\Delta t\, H(\mathbf{r},t)\big\}^2 + ... \\ &\approx 1 - i\Delta t\, H(\mathbf{r},t) - \frac{1}{2}\big\{\Delta t\, \Theta(\mathbf{r},t)\big\}^2 + \mathcal{O}\left((\Delta t)^{3/2}\right) \qquad (4.49) \\ &\equiv 1 - i\Delta t\Big[H(\mathbf{r},t) - \frac{i}{2}\Delta t\Theta(\mathbf{r},t)^2\Big] \\ &= 1 - i\Delta t\Big[H(\mathbf{r},t) - \frac{i}{2}\Delta t\Big\{\theta(\mathbf{x},t)^2 + \theta(\mathbf{x}',t)^2 - 2\,\theta(\mathbf{x},t)\theta(\mathbf{x}',t)\Big\}\Big]\ . \qquad (4.50) \end{aligned}$$

Here in Eq. (4.50) we use

$$\Theta(\mathbf{r},t) = \theta\left(\mathbf{x},t\right) - \theta\left(\mathbf{x}',t\right)\ . \tag{4.51}$$

From Eq. (4.50) we extract the effective Hamiltonian as

$$\begin{aligned} \mathrm{H}_{\mathrm{eff}}(\mathbf{r},t) &\simeq H(\mathbf{r},t) - \frac{i}{2}\Delta t\Big\{\langle\theta(\mathbf{x},t)^2\rangle + \langle\theta(\mathbf{x}',t)^2\rangle - 2\,\langle\theta(\mathbf{x},t)\theta(\mathbf{x}',t)\rangle\Big\} \\ &= H(\mathbf{r},t) - \frac{i}{2}\Delta t\Big\{D(\mathbf{0}) + D(\mathbf{0}) - 2\,D(\mathbf{x}-\mathbf{x}')\Big\}\frac{1}{\Delta t} \qquad (4.52) \\ &= H(\mathbf{r},t) - i\Big\{D(\mathbf{0}) - D(\mathbf{x}-\mathbf{x}')\Big\} \\ &= H(\mathbf{r},t) - i\Big\{D(\mathbf{0}) - D(\mathbf{r})\Big\} \qquad (4.53) \\ &\simeq -\frac{\nabla_{\mathbf{r}}^2}{M} + V(\mathbf{r}) - i\Big\{D(\mathbf{0}) - D(\mathbf{r})\Big\} + \Theta(\mathbf{r},t)\ , \qquad (4.54) \end{aligned}$$

[3] $e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + ...$

where in Eq. (4.52), we use

$$\langle \theta(\mathbf{x},t)\theta(\mathbf{x}',t')\rangle = D(\mathbf{x}-\mathbf{x}')\frac{\delta_{tt'}}{\Delta t} \ . \tag{4.55}$$

and in Eq. (4.53) we use the fact that

$$\mathbf{x} = \mathbf{R} + \frac{\mathbf{r}}{2} \quad , \quad \mathbf{x}' = \mathbf{R} - \frac{\mathbf{r}}{2} \Rightarrow \mathbf{x}-\mathbf{x}' = \mathbf{r} \ . \tag{4.56}$$

In Eq. (4.54), we have inserted Eq. (4.43).

As $\langle \Theta(\mathbf{r},t)\rangle = 0$, the averaged effective Hamiltonian can be written as

$$\langle \mathrm{H}_{\mathrm{eff}}(\mathbf{r},t)\rangle = -\frac{\nabla^2_{\mathbf{r}}}{M} + V(\mathbf{r}) - i\Big\{ D(\mathbf{0}) - D(\mathbf{r})\Big\} \ . \tag{4.57}$$

Therefore, the stochastic Schrödinger equation for the noise averaged quarkonium wave function is,

$$\begin{aligned} i\frac{\partial}{\partial}\langle \Psi_{Q\bar{Q}}(\mathbf{r},t)\rangle &= \langle \mathrm{H}_{\mathrm{eff}}(\mathbf{r},t)\rangle\langle \Psi_{Q\bar{Q}}(\mathbf{r},t)\rangle \\ &= \Big[-\frac{\nabla^2_{\mathbf{r}}}{M} + V(\mathbf{r}) - i\Big\{ D(\mathbf{0}) - D(\mathbf{r})\Big\}\Big]\langle \Psi_{Q\bar{Q}}(\mathbf{r},t)\rangle \ . \end{aligned} \tag{4.58}$$

In the $M \to \infty$ limit, Eq. (4.57) becomes

$$\lim_{M\to\infty}\langle \mathrm{H}_{\mathrm{eff}}(\mathbf{r},t)\rangle = V(\mathbf{r}) - i\Big\{ D(\mathbf{0}) - D(\mathbf{r})\Big\} \ . \tag{4.59}$$

In general a complex potential can be expressed as

$$V_{\mathrm{general}} = V_{\mathrm{Re}}(\mathbf{r}) + i\, V_{\mathrm{Im}}(\mathbf{r}) \ . \tag{4.60}$$

Comparing Eq. (4.59) and Eq. (4.60) we get,

$$V_{\rm Re}(\mathbf{r}) = V(\mathbf{r}) \ , \tag{4.61}$$

and

$$V_{\rm Im}(\mathbf{r}) = D(\mathbf{r}) - D(\mathbf{0}) \ , \tag{4.62}$$

From Eq. (4.62) we conclude

$$V_{\rm Im}(\mathbf{r} = \mathbf{0}) = 0 \ . \tag{4.63}$$

We assume the noise correlation vanishes when separated by a long distance i.e.,

$$\lim_{\mathbf{r}\to\infty} D(\mathbf{r}) = 0 \ . \tag{4.64}$$

In the $\mathbf{r} \to \infty$ limit, Eq. (4.62) becomes,

$$\begin{aligned} \lim_{\mathbf{r}\to\infty} V_{\rm Im}(\mathbf{r}) &= \lim_{\mathbf{r}\to\infty} D(\mathbf{r}) - D(\mathbf{0}) \\ &= 0 - D(\mathbf{0}) \qquad (4.65) \\ \Rightarrow D(\mathbf{0}) &= -\lim_{\mathbf{r}\to\infty} V_{\rm Im}(\mathbf{r}) \ , \qquad (4.66) \end{aligned}$$

where in Eq. (4.65) we used Eq. (4.64).

Using Eq. (4.66) in Eq. (4.62), we get the expression for the noise correlation D-function:

$$D(\mathbf{r}) = V_{\rm Im}(\mathbf{r}) - \lim_{\mathbf{r}\to\infty} V_{\rm Im}(\mathbf{r}) \ , \tag{4.67}$$

Eq. (4.67) is a very important expression for the noise correlation D-function. Using Eq. (4.67) we can obtain the noise distribution function for a particular complex potential.

4.5 Noise Correlation D-function for KMS potential

For this purpose, we will make use of a Karsch-Mehr-Satz type potential based on the internal energy [66, 50, 110]

$$\begin{aligned} V_{\rm Re} &= -\frac{\alpha}{r}\,(1+m_D\,r)\exp(-m_D\,r)+\frac{2\sigma}{m_D}\left[1-\exp(-m_D\,r)\right] \\ &\quad -\sigma\,r\,\exp(-m_D\,r)-\frac{0.8\,\sigma}{m_Q^2\,r}\,, \end{aligned} \tag{4.68}$$

where the last term is a finite quark mass correction [22], $\sigma = 0.210$ GeV^2 is the string tension, and m_D is the QGP Debye mass $m_D^2 = (1.4)^2 \cdot N_c(1+N_f/6)\,4\pi\alpha_s\,T^2/3$ with $N_c = 3$ and $N_f = 2$. The factor of $(1.4)^2$ takes into account higher-order corrections and has been determined using lattice QCD simulations [64]. We fix the Coulomb interaction strength to $\alpha = 4\alpha_s/3 = 0.385$ to match the low temperature binding energy and mass spectrum of heavy quark states [50].

For the imaginary part of the model potential we use the result obtained from leading-order finite temperature perturbation theory [75]

$$V_{\rm Im} = -\alpha_s T\phi(\hat{r})\,, \tag{4.69}$$

where $\hat{r} \equiv m_D r$, $\alpha_s = 0.289$, and

$$\phi(\hat{r}) \equiv 2\int_0^\infty dz \frac{z}{(z^2+1)^2}\left[1-\frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right]. \tag{4.70}$$

The total complex Karsch-Mehr-Satz (CKMS) potential is a sum of the real (4.68) and imaginary (4.69) parts [4]

$$V(r,t) = V_{\rm Re} + i\,V_{\rm Im}\,, \tag{4.71}$$

where the time-dependence of the potential results from the fact that the temperature and hence the Debye mass m_D depend on the proper-time.

[4] This potential has also been referred to as the Strickland-Bazow potential in the literature since it was used in Ref. [110].

Now we obtain the expression for the noise correlation D-function for the complex Karsch-Mehr-Satz (CKMS) potential. From Eq. (4.70), we can write

$$\phi\left(m_D\, r\right) = 2\int_0^\infty \frac{z\, dz}{\left(z^2+1\right)^2}\left[1-\frac{\sin\left(z\, m_D\, r\right)}{z\, m_D\, r}\right]\,. \tag{4.72}$$

Also, imaginary part of the potential can be written as

$$\begin{aligned} V_{\rm Im} &= -\alpha_s T\phi(\hat{r}) \\ &= -\alpha_s T\phi(m_D r) \\ &= -\alpha_s T\ 2\int_0^\infty \frac{z\, dz}{\left(z^2+1\right)^2}\left[1-\frac{\sin\left(z\, m_D\, r\right)}{z\, m_D\, r}\right] \end{aligned} \tag{4.73}$$

Now,

$$\begin{aligned} \lim_{\mathbf{r}\to\infty}\phi\left(m_D\, r\right) &= 2\int_0^\infty \frac{z\, dz}{\left(z^2+1\right)^2} \\ &= 2\times\frac{1}{2} \\ &= 1\,. \end{aligned} \tag{4.74}$$

So,

$$\lim_{\mathbf{r}\to\infty} V_{\rm Im}(\mathbf{r}) = -\alpha_s\, T\,. \tag{4.75}$$

Inserting Eq. (4.73),and Eq. (4.75) in Eq. (4.67), we obtain the noise correlation D-Function:

$$\begin{aligned} D(\mathbf{r}) &= V_{\rm Im}(\mathbf{r}) - \lim_{\mathbf{r}\to\infty} V_{\rm Im}(\mathbf{r}) \\ &= -\alpha_s\, T\left[\phi\left(m_D\, r\right)-1\right] \\ &= -\alpha_s\, T\left\{2\int_0^\infty \frac{z\, dz}{\left(z^2+1\right)^2}\left[1-\frac{\sin\left(z\, m_D\, r\right)}{z\, m_D\, r}\right]-1\right\}\,. \end{aligned} \tag{4.76}$$

In our future work we will use Eq. (4.76) to generate noise spectrum for the heavy quarkonium stochastic potential.

Chapter 5

Isotropic Heavy Quarkonium Quantum Dynamics

In this chapter, we introduce a framework called Heavy Quarkonium Quantum Dynamics (HQQD) which can be used to compute the dynamical suppression of heavy quarkonia propagating in the quark-gluon plasma using real-time in-medium quantum evolution [62, 63]. Using HQQD we compute large sets of real-time solutions to the Schrödinger equation using a realistic in-medium complex-valued potential. We sample 2 million quarkonia wave packet trajectories and evolve them through the QGP using HQQD to obtain their survival probabilities. The computation is performed using three different HQQD model parameter sets in order to estimate our systematic uncertainty. After taking into account final state feed down we compare our results to existing experimental data for the suppression and elliptic flow of bottomonium states and find that HQQD predictions are good agreement with available data for R_{AA} as a function of $N_{\rm part}$ and p_T collected at $\sqrt{s_{\rm NN}} = 5.02$ TeV. In the case of v_2 for the various states, we find that the path-length dependence of $\Upsilon(1S)$ suppression results in quite small v_2 for $\Upsilon(1S)$. Our prediction for the integrated elliptic flow for $\Upsilon(1S)$ in the $10-90\%$ centrality class, which now includes an estimate of the systematic error, is $v_2[\Upsilon(1S)] = 0.003 \pm 0.0007 \pm^{0.0006}_{0.0013}$. We also find that, due to their increased suppression, excited bottomonium states have a larger elliptic flow. Based on this observation we make predictions for $v_2[\Upsilon(2S)]$ and $v_2[\Upsilon(3S)]$ as a function of centrality and transverse momentum.

5.1 Phenomenological potential

In vacuum we take the heavy-quark potential to be given by a Cornell potential with a finite string breaking distance

$$V_{\rm vac}(r) = \begin{cases} -\frac{a}{r} + \sigma r & \text{if } r \leq r_{\rm SB} \\ -\frac{a}{r_{\rm SB}} + \sigma r_{\rm SB} & \text{if } r > r_{\rm SB} \end{cases} , \tag{5.1}$$

where $a = 0.409$ is the effective coupling, $\sigma = 0.21$ GeV^2 is the string tension, and $r_{\rm SB} = 1.25$ fm is the string breaking distance. Using this set of vacuum parameters and assuming $M_b = 4.7$ GeV one obtains vacuum masses of $\{9.46, 10.0, 9.88, 10.36, 10.25, 10.13\}$ GeV for $\Upsilon(1S)$, $\Upsilon(2S)$, $\chi_b(1P)$, $\Upsilon(3S)$, and $\chi_b(2P)$, respectively.

For the in-medium potential we use an internal energy based potential. The free energy of a static heavy quark-antiquark pair in an isotropic plasma [80, 66, 99, 98, 110]

$$\begin{aligned} F(r,t) &= -\frac{g^2 C_F}{4\pi r} e^{-m_D r} + \frac{\sigma}{m_D}\left[1 - e^{-m_D r}\right] \\ &= -\frac{a}{r} e^{-m_D r} + \frac{\sigma}{m_D}\left[1 - e^{-m_D r}\right] \quad , \end{aligned} \tag{5.2}$$

where $a = g^2 C_F/4\pi$ and σ is the string tension. We take the in-medium gluonic Debye mass to be $m_D = \lambda\sqrt{4\pi N_c(1 + N_f/6)\alpha_s T^2/3}$. Here we will consider the cases $\lambda \in \{0.8, 1, 1.2\}$, with the central value being the known high-temperature limit of the gluonic Debye mass.

The internal energy, U, of the states which has an entropy contribution is given by

$$\begin{aligned} U &= F + TS \\ &= -\frac{a}{r}(1 + m_D r) e^{-m_D r} + \frac{2\sigma}{m_D}\left[1 - e^{-m_D r}\right] - \sigma r e^{-m_D r} , \end{aligned} \tag{5.3}$$

where we have used $S = -\frac{\partial F}{\partial T}$. Eq.(5.3) gives us the internal energy based real part of the Karsch-Mehr-Satz (KMS) potential

$$V_{\rm KMS}(r) = U = -\frac{a}{r}(1 + m_D r) e^{-m_D r} + \frac{2\sigma}{m_D}\left[1 - e^{-m_D r}\right] - \sigma r e^{-m_D r} . \tag{5.4}$$

We note that, in equilibrium, there are arguments to support the use of the internal energy [99, 98, 97], however, since bottomonium states are not expected to be in thermal equilibrium with their surroundings, it is unclear what the correct choice should be. In order to make sure that we have the correct equilibrium limit we choose to use the internal energy prescription. To match smoothly onto the zero temperature limit we use

$$V_R(r) = \Re[V(r)] = \begin{cases} V_{\rm KMS}(r) & \text{if } V_{\rm KMS}(r) \leq V_{\rm vac}(r_{\rm SB}) \\ V_{\rm vac}(r_{\rm SB}) & \text{if } V_{\rm KMS}(r) > V_{\rm vac}(r_{\rm SB}) \end{cases} . \tag{5.5}$$

In the limit $T \to 0$, Eq. (5.5) reduces to Eq. (5.1).

For the imaginary part of the potential we take the result of the leading-order resummed perturbative QCD calculation of Laine et al

$$V_I(r) = \Im[V(r)] = -C_F \alpha_s T \phi(m_D r) \,, \tag{5.6}$$

with $\phi(\hat{r}) \equiv 1 - 2\int_0^\infty \sin(z)/(z^2+\hat{r}^2)^2$ [75]. We evaluate the strong coupling α_s at the scale $\mu = 2\pi T$ and use three-loop running [89] with $\Lambda_{\overline{MS}} = 344$ MeV. This value of $\Lambda_{\overline{MS}}$ is chosen in order to reproduce the lattice result for the running coupling $\alpha_s(5\text{ GeV}) = 0.2034$ [81]. The resulting final complex-valued potential is of the form

$$V(r) = V_R(r) + iV_I(r) \,. \tag{5.7}$$

5.2 Computation of R_{AA} including feed down

Using the complex potential Eq. (5.7) along with Eq. (5.5) and Eq. (5.6), we then numerically solve the time-dependent Schrödinger equation on a discrete lattice. The DST method used is manifestly unitary for real-valued potentials and based on a split-step pseudospectral method described above [21, 112, 29]. For the discretization we use $N = 4096$ points in r with $L = r_{\rm max} = 19.7$ fm with a corresponding lattice spacing of $a \simeq 0.0048$ fm. We compute the in-medium suppression for $\ell = 0$ and $\ell = 1$ states, separately, following Eq. (3.12).

We choose a Gaussian initial wave-function [62]

$$u_\ell(r, \tau = 0) \propto r^{\ell+1} \exp(-r^2/\Delta^2)\,, \tag{5.8}$$

with $\Delta = 0.04$ fm. This initial conditions mimics the local (delta function) production of bottomonium states in space resulting from their initial hard production. This type of initial state is a quantum superposition of many eigenstates of the Schrödinger equation for a fixed ℓ. Once the wave-function evolves forward in time, we obtain the survival probability of a given vacuum state by computing the overlap of the in-medium quantum wave-function with the vacuum basis states. These overlaps decay in time because of the non-Hermitian nature of the system's Hamiltonian which is fundamentally related to the in-medium breakup of bottomonium states.

Here we solve the 3+1D Schrödinger equation for a large set of trajectories using a realistic 3+1D hydrodynamics background tuned to data [18, 17]. We use smooth optical Glauber initial conditions which provide the initial energy density profile as a function of the impact parameter. For the background evolution we used an initial central temperature of $T_0 = 630$ MeV at $\tau_0 = 0.25$ fm/c and a constant specific shear viscosity of $4\pi\eta/s = 2$, which correspond to the results obtained in a recent aHydroQP fit to soft observables at 5 TeV [18]. We evolve the quantum wave-packets using the vacuum potential starting at $\tau = 0$ fm/c and turn on the in-medium complex potential at $\tau = \tau_{\rm med}$. Finally, when the temperature on a given trajectory drops below the QGP transition temperature, $T_{\rm QGP} = 155$ MeV, we again use the vacuum potential for the wave-function evolution. In the results section we will present our findings when varying $\tau_{\rm med} \in \{0.25, 0.4, 0.6\}$ fm/c

Due to the fact that each sampled wave-packet experiences a different temperature along its trajectory while propagating through the QGP, we numerically solve the time-dependent Schrödinger equation for a large set of bottomonium trajectories (2 million). We assume that the initial transverse spatial distribution for bottomonium production is proportional

State	$\Upsilon(1s)$	$\Upsilon(2s)$	$\chi_{b0}(1p)$	$\chi_{b1}(1p)$	$\chi_{b2}(1p)$	$\Upsilon(3s)$	$\chi_{b0}(2p)$	$\chi_{b1}(2p)$	$\chi_{b2}(2p)$
$\sigma_{exp}(nb)$	57.6	19	3.72	13.69	16.1	6.8	3.27	12.0	14.15
$\sigma_{primordial}(nb)$	37.97	18.27	3.79	20.88	19.53	8.21	3.33	17.28	17.07

Table 5.1: The experimentally observed (σ_{exp}) and primordial ($\sigma_{primordial}$) $pp \to$ bottomonium production cross sections of the various bottomonium states. Data were collected from Refs. [89, 71, 43, 102, 6, 59, 69, 78]. The primordial (direct) cross-sections were obtained from the experimentally measured post feed down cross sections using, $\sigma_{\text{primordial}} = F^{-1}\sigma_{\text{exp}}$.

to the binary overlap profile of the two colliding nuclei, $N_{AA}^{\text{bin}}(x, y)$ and then use Monte-Carlo sampling to generate the initial production points. Here, we consider all bottomonia to have zero rapidity, $y = 0$, in order to simplify the calculation in a manner that takes advantage of the approximate boost-invariance of the QGP. We assume that the transverse momentum (p_T)-distribution is proportional to $p_T/(p_T^2 + \langle M\rangle^2)^2$ for all the states, where $\langle M\rangle$ is the average mass of all states being considered and, once again, we Monte-Carlo sample the p_T for each particle generated. Finally we sample the initial azimuthal angle ϕ from a uniform distribution between 0 and 2π. Once we have the initial position, transverse momentum, and azimuthal angle, we then record the aHydroQP result for the QGP temperature along the (assumed) straight line trajectory followed by the quantum wave-packets.

Once each quantum state propagates along its trajectory, the survival probabilities are converted into particle number by multiplying by

1. The expected number of binary collisions in the centrality bin sampled and

2. The primordial (direct) production cross section for each bottomonium state.

At this point one needs to take into account the late time feed down of excited bottomonium states. This can be done by introducing a feed down matrix F which collects the known information about excited bottomonium state decays available from the Particle

Data Group [89]. In the case of proton-proton collisions one can take the primordial cross sections for production and convert this into the post feed down cross sections (abundances) by applying a feed down matrix to the primordial cross sections $\sigma_{\rm exp} = F\sigma_{\rm primordial}$

$$F = \begin{pmatrix} 1 & 0.2645 & 0.0194 & 0.352 & 0.18 & 0.0657 & 0.0038 & 0.1153 & 0.077 \\ 0 & 0.7355 & 0 & 0 & 0 & 0.106 & 0.0138 & 0.181 & 0.089 \\ 0 & 0 & 0.9806 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.648 & 0 & 0 & 0 & 0.0091 & 0 \\ 0 & 0 & 0 & 0 & 0.82 & 0 & 0 & 0 & 0.0051 \\ 0 & 0 & 0 & 0 & 0 & 0.8283 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0.9824 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.6946 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0.8289 \end{pmatrix}, \tag{5.9}$$

with the vector σ collecting the observed and primordial (direct) cross sections for the $\{\Upsilon(1S), \Upsilon(2S), \chi_{b0}(1P), \chi_{b1}(1P), \chi_{b2}(1P), \Upsilon(3S), \chi_{b0}(2P), \chi_{b1}(2P), \chi_{b2}(2P)\}$. Knowing the experimental values for the production cross-sections $\sigma_{\rm exp}$ (listed in Tab. 5.1), one can compute the primordial cross sections via $\sigma_{\rm primordial} = F^{-1}\sigma_{\rm exp}$. The resulting values for $\sigma_{\rm primordial}$ are listed in Tab. 5.1.

We extract the cross sections of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ from the 5.02 TeV data obtained by the CMS collaboration in the rapidity interval $|y| \leq 2.4$ [102]. The left panel of figure 3 of Ref. [102] presents $B \times d\sigma/dy$, where B is the dimuon branching fraction. Once we take the average over rapidity in the interval presented in that CMS figure, we obtain $B \times d\sigma/dy \approx 1.44$ nb, 0.37 nb, and 0.15 nb, respectively. Dividing by the branching fractions for $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S) \to \mu^+\mu^-$, which are $\approx 2.5\%$, 1.9%, and 2.2% [89], respectively [89], and obtains

$$\langle d\sigma[\Upsilon(1S), \Upsilon(2S), \Upsilon(3S)]/dy\rangle_y = \{57.6, 19, 6.8\} \text{ nb}\,. \tag{5.10}$$

Decay of $\Upsilon(2s)$, $\chi_b(1p)$, and $\Upsilon(3s)$		
Mode	**Fraction (Γ_i/Γ)**	**Feed-down matrix element**
$\Upsilon(2s) \to \Upsilon(1s)\pi^+\pi^-$	17.85%	$f_{12} = (17.85 + 8.6)\% = 0.2654$
$\Upsilon(2s) \to \Upsilon(1s)\pi^0\pi^0$	8.6%	
$\chi_{b0}(1p) \to \gamma\Upsilon(1s)$	1.94%	$f_{13} = 0.0194$
$\chi_{b1}(1p) \to \gamma\Upsilon(1s)$	35.2%	$f_{14} = 0.352$
$\chi_{b2}(1p) \to \gamma\Upsilon(1s)$	18.0%	$f_{15} = 0.18$
$\Upsilon(3s) \to \Upsilon(1s)\pi^+\pi^-$	4.37%	$f_{16} = (4.37 + 2.20)\% = 0.0657$
$\Upsilon(3s) \to \Upsilon(1s)\pi^0\pi^0$	2.20%	
$\Upsilon(3s) \to \Upsilon(2s)$ anything	10.6%	$f_{26} = 10.6\% = 0.106$

Table 5.2: Decay modes of $\Upsilon(2S)$, $\chi_b(1P)$, and $\Upsilon(3S)$ and the calculation of different elements of the feed-down matrix (5.9) using the experimentally measured branching ratios of the various bottomonium states [89].

The χ_b-cross sections can be obtained by through knowledge of $\sigma[\Upsilon(1S)]$, the ratios $\sigma[\chi_{bj}(np)]/\sigma[\chi_{bj'}(np)]$, and the ratios of the 1p and 2p cross-sections to 1s cross sections provided in Ref. [6]. We assume $\sigma[\chi_{b2}(np)]/\sigma[\chi_{b1}(np)] = 1.176$ [59] for both the $1p$ and $2p$ states. This value is consistent with available experimental data [69]. To obtain the $\chi_{bj'}(np)$ cross sections, the values of $\mathcal{R}^{\chi_b(nP)}_{\Upsilon(1S)}$ are taken from the lowest p_T bins of tables 5 and 6 of Ref. [6] (measured at $\sqrt{s} = 7$ TeV and $\sqrt{s} = 8$ TeV, respectively) and extrapolated to $\sqrt{s} = 5$ TeV. Using this extrapolation, $\sigma[\chi_{bj}(np)]$ for j, $n = 1$, 2 can be extracted by using Eq. (1) of Ref. [6]. The $\chi_{b0}(np)$ cross sections are taken to be one quarter of the average of the $\sigma[\chi_{b1}(np)]$ and $\sigma[\chi_{b2}(np)]$ cross-sections [59]. This gives

$$\langle d\sigma[\chi_{b0}(1P), \chi_{b1}(1P), \chi_{b2}(1P)]/dy\rangle_y = \{3.72, 13.69, 16.1\} \text{ nb}\,, \tag{5.11}$$

$$\langle d\sigma[\chi_{b0}(2P), \chi_{b1}(2P), \chi_{b2}(2P)]/dy\rangle_y = \{3.27, 12.0, 14.15\} \text{ nb}\,, \tag{5.12}$$

Note that herein we ignore possible p_T-dependence in the branching ratios which could potentially be important for quantitative understanding of the p_T-dependence of post feed down R_{AA} [78].

To compute the effect of final-state feed-down in heavy-ion collisions, we first construct

a vector $\mathbf{N}_{\rm QGP}$ containing the numbers of each state produced at the end of each simulation (survival probability $\times\langle N_{\rm bin}(b)\rangle \times \sigma_{\rm primordial}$). We then multiply the result by the same feed-down matrix used for pp feed down, i.e. $\mathbf{N}_{\rm final} = F\mathbf{N}_{\rm QGP}$. The use of the same feed down matrix in both cases is related to the fact that feed down occurs on a time scale of $10^{-20} - 10^{-18}$ seconds which is 100-10000 times longer than the QGP lifetime.

The calculation of different elements of the feed-down matrix (5.9) is demonstrated in Table 5.2 and Table 5.3. Since each column of the feed-down matrix (5.9), F, must sum to unity in order to preserve bottom number, one can obtain the diagonal elements by using $f_{ii} = 1 - \sum_j f_{ji}$, where $\{i, j = 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.[1] Using (5.9), the final number of different states produced can therefore be computed as follows

$$\begin{aligned}
N_{\rm final}^{\Upsilon(1S)} &= f_{11}N_{\rm QGP}^{\Upsilon(1S)} + f_{12}N_{\rm QGP}^{\Upsilon(2S)} + f_{13}N_{\rm QGP}^{\chi_{b0}(1P)} + f_{14}N_{\rm QGP}^{\chi_{b1}(1P)} + f_{15}N_{\rm QGP}^{\chi_{b2}(1P)} \quad (5.13)\\
&+ f_{16}N_{\rm QGP}^{\Upsilon(3S)} + f_{17}N_{\rm QGP}^{\chi_{b0}(2P)} + f_{18}N_{\rm QGP}^{\chi_{b1}(2P)} + f_{19}N_{\rm QGP}^{\chi_{b2}(2P)}\,,\\
N_{\rm final}^{\Upsilon(2S)} &= f_{22}N_{\rm QGP}^{\Upsilon(2S)} + f_{26}N_{\rm QGP}^{\Upsilon(3S)} + f_{27}N_{\rm QGP}^{\chi_{b0}(2P)} + f_{28}N_{\rm QGP}^{\chi_{b1}(2P)} + f_{29}N_{\rm QGP}^{\chi_{b2}(2P)}\,, \quad (5.14)\\
N_{\rm final}^{\chi_{b0}(1P)} &= f_{33}N_{\rm QGP}^{\chi_{b0}(1P)}\,, \quad (5.15)\\
N_{\rm final}^{\chi_{b1}(1P)} &= f_{44}N_{\rm QGP}^{\chi_{b1}(1P)} + f_{48}N_{\rm QGP}^{\chi_{b1}(2P)}\,, \quad (5.16)\\
N_{\rm final}^{\chi_{b2}(1P)} &= f_{55}N_{\rm QGP}^{\chi_{b2}(1P)} + f_{59}N_{\rm QGP}^{\chi_{b2}(2P)}\,, \quad (5.17)\\
N_{\rm final}^{\Upsilon(3S)} &= f_{66}N_{\rm QGP}^{\Upsilon(3S)}\,, \quad (5.18)\\
N_{\rm final}^{\chi_{b0}(2P)} &= f_{77}N_{\rm QGP}^{\chi_{b0}(2P)}\,, \quad (5.19)\\
N_{\rm final}^{\chi_{b1}(2P)} &= f_{88}N_{\rm QGP}^{\chi_{b1}(2P)}\,, \quad (5.20)\\
N_{\rm final}^{\chi_{b2}(2P)} &= f_{99}N_{\rm QGP}^{\chi_{b2}(2P)}\,, \quad (5.21)
\end{aligned}$$

where we have discarded terms for which the coefficients were zero in (5.9).

Finally, we compute R_{AA} for each state by dividing the final number produced after feed down by the average number of binary collisions in the sampled centrality class times the

[1] Here we have neglected decay processes with small branching ratios, such as $\Upsilon(2S) \to \Upsilon(1S) + \pi^0$, which has a branching ratio of BR $< 4 \times 10^{-5}$ [89].

Decay of $\chi_b(2p)$			
Mode	**Fraction (Γ_i/Γ)**		**Feed-down matrix element**
$\chi_{b0}(2p) \to \gamma\Upsilon(1s)$	3.8×10^{-3}		$f_{17} = 0.0038$
$\chi_{b1}(2p) \to \omega\Upsilon(1s)$	1.63%	(1.63 + 9.9 = 11.53)%	$f_{18} = 0.1153$
$\chi_{b1}(2p) \to \gamma\Upsilon(1s)$	9.9%		
$\chi_{b2}(2p) \to \omega\Upsilon(1s)$	1.10%	(1.10 + 6.6 = 7.7)%	$f_{19} = 0.077$
$\chi_{b2}(2p) \to \gamma\Upsilon(1s)$	6.6%		
$\chi_{b0}(2p) \to \gamma\Upsilon(2s)$	1.38%		$f_{27} = 0.0138$
$\chi_{b1}(2p) \to \gamma\Upsilon(2s)$	18.1%		$f_{28} = 0.181$
$\chi_{b2}(2p) \to \gamma\Upsilon(2s)$	8.9%		$f_{29} = 0.089$
$\chi_{b0}(2p) \to \pi\pi\chi_{b0}(1p)$	0		$f_{37} = 0$
$\chi_{b1}(2p) \to \pi\pi\chi_{b1}(1p)$	9.1×10^{-3}		$f_{48} = 0.0091$
$\chi_{b2}(2p) \to \pi\pi\chi_{b2}(1p)$	5.1×10^{-3}		$f_{59} = 0.0051$

Table 5.3: Decay modes of $\chi_b(2P)$ and the calculation of different elements of the feed-down matrix (5.9) using the experimentally measured branching ratios of the various bottomonium states [89].

post feed-down pp production cross-section for each state ($\sigma_{\rm exp}$). The momentum anisotropic flow of bottomonium states can be obtained in a similar manner. In this case, we can exploit the fact that the aHydroQP background always has the reaction plane aligned with the x and y axes, i.e. $\Psi_{\rm RP} = 0$, to simply average $\cos(n\phi)$ over all particles produced after feed down is taken into account. In practice we compute the following average over all sampled trajectories in a given centrality and p_T bin, $v_n \equiv \langle\cos(n\phi)\rangle$. For both R_{AA} and v_2, we can directly compute the statistical uncertainty associated with the sum over the sampled quantum wave-packet trajectories. To obtain an estimate of our systematic uncertainty we will vary the HQQD model parameters used in Ref. [62].

5.3 Results

In order to estimate the systematic uncertainty coming from the choice of model parameters we will vary the medium initialization time ($\tau_{\rm med}$) and the pre-factor of the peturbative Debye mass (λ) in the set

$$(\tau_{\rm med}, \lambda) \in \{(0.25 \text{ fm/c}, 1.2), (0.4 \text{ fm/c}, 1), (0.6 \text{ fm/c}, 0.8)\}, \tag{5.22}$$

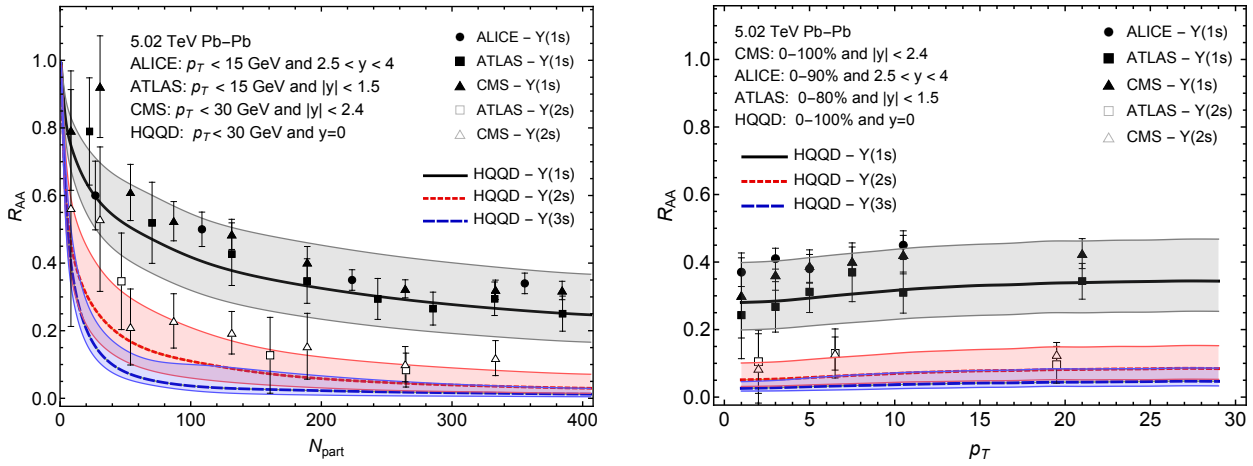


Figure 5.1: Nuclear suppression factor, R_{AA}, of bottomonium s-wave states as a function of $N_{\rm part}$ (left) and p_T (right). The solid, short-dashed, and dashed lines show the predictions of HQQD. The shaded bands indicate the variation of the HQQD result when changing the model parameters in the range specified in Eq. (5.22). The data points are from the ALICE [8], ATLAS [104], and CMS [102] collaborations. Experimental error bars shown were obtained by adding statistical and systematic uncertainties in quadrature.

where the factor λ accounts for higher-order corrections to the perturbative Debye mass

$$m_D = \lambda\sqrt{4\pi N_c(1+N_f/6)\alpha_s T^2/3}\,, \tag{5.23}$$

with $\lambda = 1$ giving the known leading-order result from hard-thermal-loop resummation [19, 55].

We present the HQQD predictions for the suppression of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ states as a function of $N_{\rm part}$ in Fig. 5.1 (left). For the left panel of Fig. 5.1 we applied a transverse momentum cut of $p_T < 30$ GeV in HQQD. The HQQD results are compared with experimental data obtained by the ALICE [8], ATLAS [104], and CMS [102] collaborations, shown as circles, squares, and triangles, respectively. From Fig. 5.1 (left), we see that, for the central choice of model parameters corresponding to $\tau_0 = 0.4$ fm/c and $\lambda = 1$, HQQD does a quite reasonable job in describing the $N_{\rm part}$ dependence of $R_{AA}[\Upsilon(1S)]$. The shaded bands

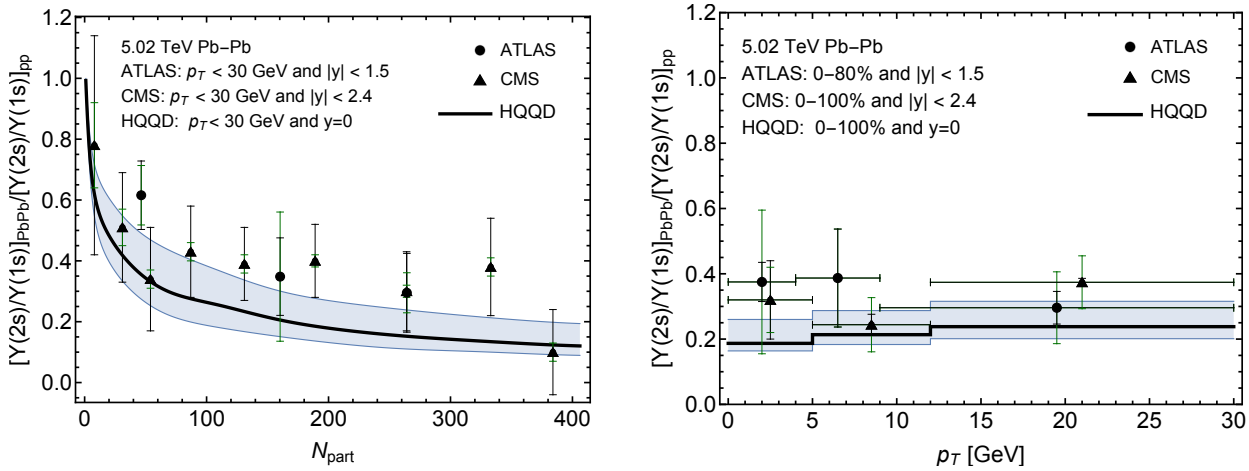


Figure 5.2: Double ratio $[\Upsilon(2S)/\Upsilon(1S)]_{\rm PbPb}/[\Upsilon(2S)/\Upsilon(1S)]_{\rm pp}$ as a function of $N_{\rm part}$ (left panel) and p_T (right panel). The solid line shows the prediction of HQQD. The shaded bands indicate the variation of the HQQD result when changing the model parameters in the range specified in Eq. (5.22). The data points are from the ATLAS [104] and CMS [101] collaborations. Systematic and statistical experimental uncertainties are indicated by the green and black error bars, respectively.

in Fig. 5.1 (left) show the variation of the results obtained when taking the limits indicated in Eq. (5.22), with the first and third cases mapping to stronger and weaker suppression compared to the central line, respectively. In the case of the $\Upsilon(2S)$, one sees that using the central parameters, HQQD seems to under-predict the observed $R_{AA}[\Upsilon(2S)]$.

Turning next to Fig. 5.1 (right), we present the HQQD predictions for the transverse-momentum dependence of $R_{AA}[\Upsilon]$. To obtain the HQQD predictions for this panel, we averaged over centrality with a weight function $w(c) = \exp(-c/20)$, with $c \in [0, 100]$. This weight function allows us to mimic the experimentally observed distribution of the number of Υ states versus centrality [45]. As can be seen from Fig. 5.1 (right), HQQD predicts a very weak dependence of $R_{AA}[\Upsilon]$ on p_T, with only a small decrease in $R_{AA}[\Upsilon]$ at momentum less than the mass scale of the bottomonium states. One can understand the increased

suppression at low-p_T in terms of the average effective lifetime of the quantum wave-packets in the QGP. For low-momentum wavepackets, their effective lifetime in the QGP is longer due to their lower velocities and, hence, they experience stronger suppression than high-momentum wave-packets which can escape from the QGP more quickly.

In Figure 5.2 we plot the HQQD prediction for the $\Upsilon(2S)$ to $\Upsilon(1S)$ double ratio as a function of $N_{\rm part}$(left panel) and p_T (right panel). We compare these predictions with data reported by the ATLAS and CMS collaboration in Refs. [104, 101]. The band shown in the Figure provides an estimate of our systematic uncertainty. As can be seen from both of the panels of this Figure, we see further evidence that HQQD is predicting too much $\Upsilon(2S)$ suppression compared to the experimental data. It is unclear at this moment time what is the source of this discrepancy. We will return to this question in the conclusions where we discuss the possible impact of including stochastic in-medium color and angular momentum transitions in a more complete manner.

In order to gain further insight into the spatial-distribution of bottomonium suppression in the QGP one can construct "tomographic" plots showing the starting positions of the sampled bottomonium states, color-coded by the final, time-evolved R_{AA} for that state. From such plots one can learn where in the QGP bottomonium states are most strongly suppressed and where those with only weak-suppression emerge from. A priori, one would expect trajectories that experience the strongest suppression to be those initially produced in the central of the fireball and trajectories with the weakest suppression to be those initially produced on the periphery of the collision. This expectation is indeed observed in our simulations. In Fig. 5.3 we show such a tomographic plot for the $\Upsilon(1S)$ (left) and the $\Upsilon(2S)$ (right). For both panels in this figure, we considered a Pb-Pb collision with impact parameter of $b \simeq 9.2$ fm. Comparing the left and right panels we firstly see the increased overall suppression of the $\Upsilon(2S)$ relative to the $\Upsilon(1S)$. In both panels, we see that states

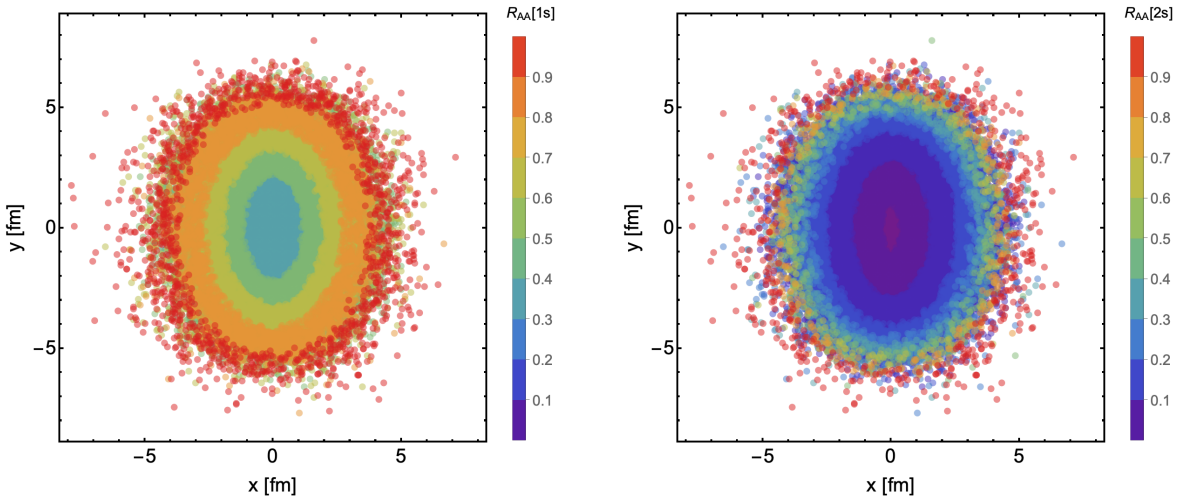


Figure 5.3: Visualization of the starting positions of $\Upsilon(1S)$ (left) and $\Upsilon(2S)$ states (right). In both cases we selected all states with $0 < p_T < 40$ GeV. Color coding shows the R_{AA} for each particle.

with R_{AA} close to one come from initial production near the surface, while those experiencing the strongest suppression come from the center of the fireball as expected.

In order to better visualize the probability density for particles in each R_{AA}-class, in Fig. 5.4 we have plotted each R_{AA}-class in a separate panel with the panels corresponding to 0.1 intervals in R_{AA} in the range $0.4 \leq R_{AA} \leq 1$. In each panel we show the normalized probability distribution function (PDF) for each R_{AA}-class. As in Fig. 5.3 we see that the suppression of the states is correlated with the initial geometry and that states with weak suppression typically come from the surface of the fireball. Statistically, one can see, for example, that there is little overlap between the PDFs for particles with $R_{AA} \geq 0.7$ and $R_{AA} \leq 0.5$. By comparing the production of particles in these two classes, one could investigate potential differences in bottomonium production in the central and peripheral regions of the QGP. For now, however, such plots are mostly useful in making sure that our understanding of suppression dynamics is correct.

In Fig. 5.5, we present the HQQD predictions for centrality dependence of the elliptic

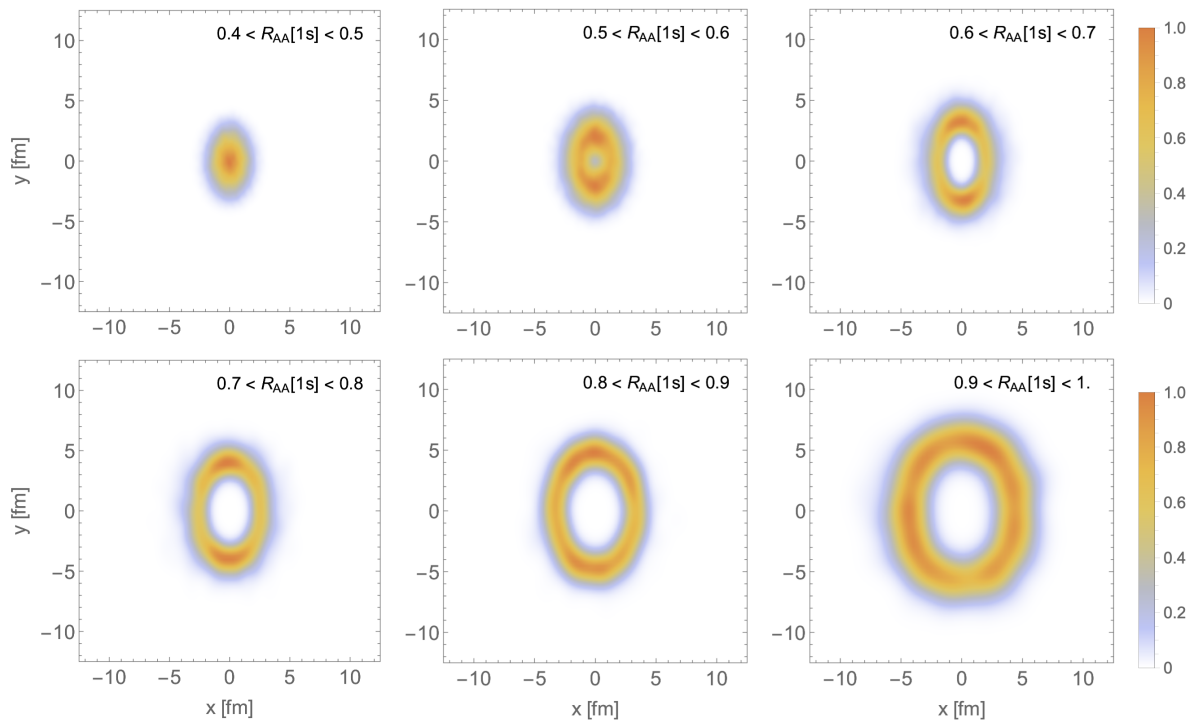


Figure 5.4: Visualization of the initial probability distribution function for the starting positions of $\Upsilon(1S)$. Panels show distribution for states experiencing a given R_{AA} in 10% intervals, e.g. $0.4 < R_{AA} < 0.5$. For this visualization, we selected all states with $0 < p_T < 40$ GeV. Coloring encodes the normalized probability in each R_{AA} range.

flow of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ states. In this figure, we take $p_T < 50$ GeV and compute v_2 in 10 equally spaced centrality bins from 0-100%. The bands show the statistical uncertainty associated with the mean values extracted in each bin. The upper (squares), central (circles), and lower (triangles) lines correspond to the three different cases of varying the model parameters specified in Eq. (5.22), respectively. For each of the three choices of model parameters shown in Fig. 5.5, one sees an ordering of the elliptic flow of the states with the $\Upsilon(3S)$ state having the largest flow and the $\Upsilon(1S)$ the smallest. This result is expected since excited states are more suppressed and hence experience a larger in-medium path-length dependence. For each of the states shown in Fig. 5.5 one notices that the oscillations in v_2 (to

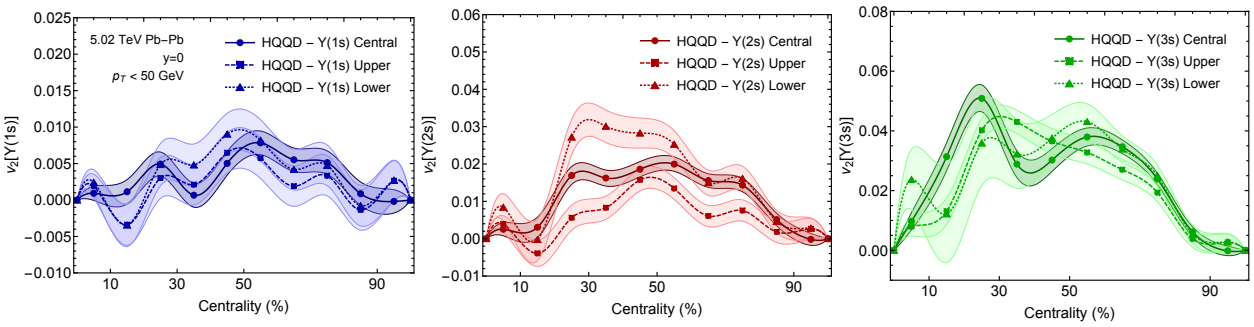


Figure 5.5: Elliptic flow for s-wave bottomonium states as a function of centrality. Solid lines and bands show spline-interpolated results for the mean and statistical uncertainty of the mean obtained from HQQD.The upper (squares), central (circles), and lower (triangles) indicate the variation of the HQQD result when changing the model parameters in the range specified in Eq. (5.22). Points show results obtained in equally spaced bins of 10% centrality from 0-100%.

be discussed later) depend on the choice of the model parameters. This can be attributed to the change in the effective path-length of the quantum wave-packets travel through the medium when changing $\tau_{\rm med}$. Finally, we note that v_2 is more sensitive to these oscillations because it explicitly involves, for example, differences between the survival probability along the short and long sides of the QGP. In R_{AA} one sums over all angles and averages and, as a result, these oscillations are smoothed out in the sum over quantum wave-packet trajectories. On the contrary, since for v_2 different angles contribute with different weights/signs, it is naturally more sensitive to small differences in the survival probability and hence is more sensitive to these oscillations.

From Fig. 5.5, we also observe that the elliptic flow for all states goes to zero for central collisions (centrality = 0%). This is expected using the smooth (optical) Glauber model used herein for the hydrodynamic initial condition. In the future it will be interesting to assess the degree of Υ elliptic flow in the most central classes taking into account also geometrical fluctuations in the initial condition (energy density, flow velocities, etc.). We note, however,

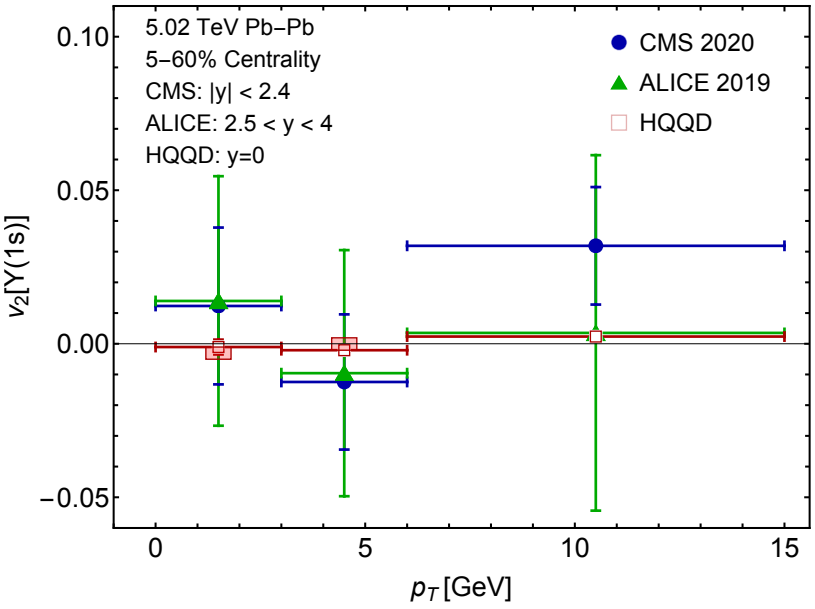


Figure 5.6: The elliptic flow $v_2[\Upsilon(1S)]$ as a function of p_T in three p_T-bins. Open red squares are HQQD predictions and the data are from the ALICE [7] and CMS [103] collaborations. The shaded red boxes associated with each HQQD prediction indicate the systematic variation observed when changing the model parameters in the range specified in Eq. (5.22). Note that, for the highest p_T bin, the systematic error estimate is so small that it is indistinguishable from the horizontal error bar.

that since we use smooth hydrodynamic initial conditions v_2 must go to zero for central collisions. This can be used as a check on our calculation of $v_2[\Upsilon]$ for all three states. We also note that in the opposite limit, that of ultra-peripheral collisions (centrality = 100%), one expects the v_2 for all states to go to zero since the QGP lifetime goes to zero in this limit. This again provides a non-trivial test of our computation of v_2 for bottomonium states.

As alluded to above, from Fig. 5.5 one notices that the dependence of v_2 on centrality can contain oscillations (non-monotonic behavior). These oscillations arise in HQQD due to quantum oscillations in the overlaps of the in-medium bottomonium states with their corresponding vacuum states. Wave-packets which experience different effective path-lengths

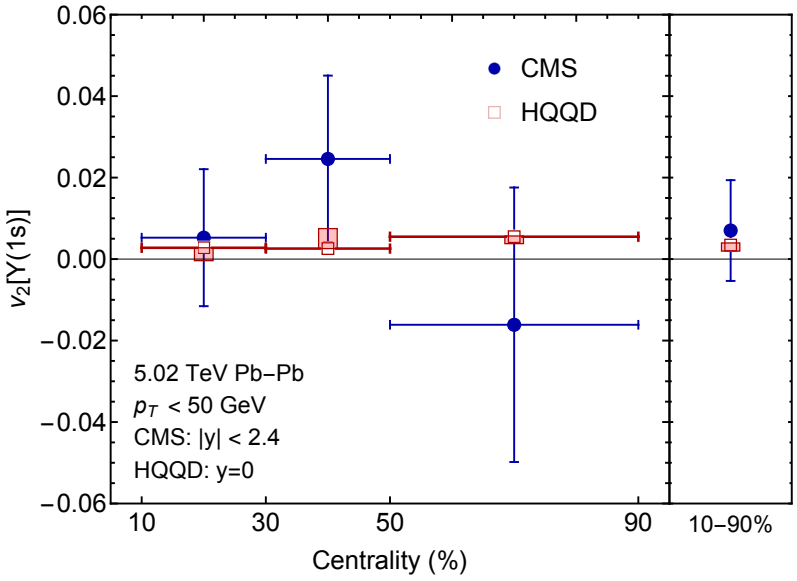


Figure 5.7: Centrality dependence of $v_2[\Upsilon(1S)]$ shown in 10-30%, 30-50%, 50-90%, and 10-90% centrality bins. Open squares are predictions of HQQD. The shaded red boxes associated with each HQQD prediction indicate the systematic variation observed when changing the model parameters in the range specified in Eq. (5.22).

can exit the QGP at different points during this quantum oscillation. Since the characteristic period for these oscillations is on the order of 1 fm/c, this can have an observable effect on the HQQD predictions for v_2 as a function of centrality.

A comparison of HQQD predictions for $v_2[\Upsilon(1S)]$ with experimental data collected by the ALICE [7] and CMS [103] collaborations in three different transverse momentum bins, 0-4, 4-6, and 6-15 GeV, is shown in Fig. 5.6. The HQQD predictions and experimental results are integrated over centrality in the range 5-60%. One sees from this figure that HQQD predicts results consistent with v_2 being zero, slightly negative, and small but positive value in the lowest p_T bin, central p_T bin, and highest p_T bin, respectively. A similar trend has also been predicted earlier by an adiabatic approximation based model in Ref. [25]. According to Ref. [25], the negative v_2 observed in the central p_T bin is related to the transverse expansion

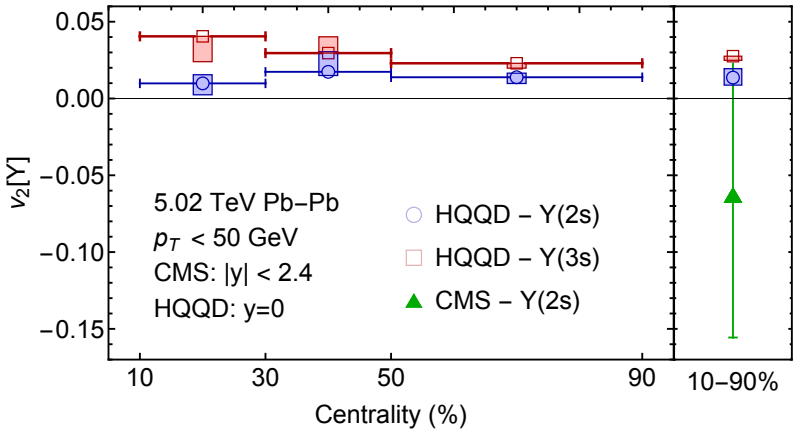


Figure 5.8: Centrality dependence of $v_2[\Upsilon(2S)]$ and $v_2[\Upsilon(3S)]$ in the same centrality bins as Fig. 5.7. Open symbols are predictions of HQQD. In the 10-90% class we include recent data reported by the CMS collaboration for integrated $v_2[\Upsilon(2S)]$ [103]. The shaded red and blue boxes associated with each HQQD prediction indicate the systematic variation observed when changing the model parameters in the range specified in Eq. (5.22).

of the QGP overtaking bottomonia states which have escaped from near the surface of the QGP. To demonstrate that this was indeed the cause for the negative v_2, the authors Ref. [25] considered the case in which the transverse expansion of the QGP was turned off, finding that in this case v_2 was always positive. Returning to the theory to data comparison, Fig. 5.6 demonstrates that HQQD has a reasonable agreement with the available experimental data given current experimental uncertainties. Finally, we note that even given the experimental uncertainties, we see a similar trend in the experimental results as a function of p_T as predicted by HQQD.

In Fig. 5.7, we present the centrality dependence of $v_2[\Upsilon(1S)]$ in 10-30%, 30-50%, 50-90%, and 10-90% centrality bins and compare our HQQD predictions with experimental data from the CMS collaboration [103]. We show a comparison of the HQQD prediction with the experimental result integrated over 10-90% centrality in the right panel. As can be

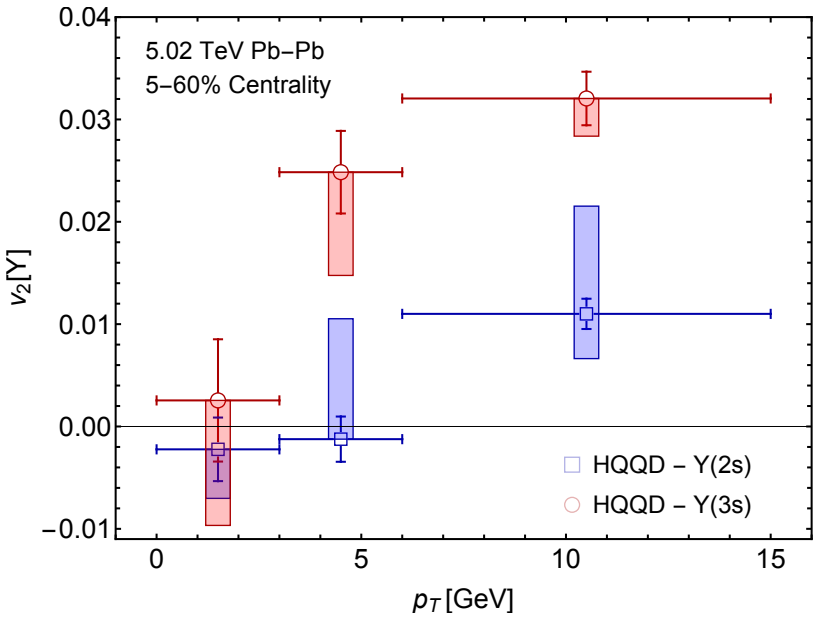


Figure 5.9: HQQD predictions for the elliptic flow of $\Upsilon(2S)$ and $\Upsilon(3S)$ states in the 5-60% centrality bin. The transverse momentum bins are the same as used in Fig. 5.6. The shaded red and blue boxes associated with each HQQD prediction indicate the systematic variation observed when changing the model parameters in the range specified in Eq. (5.22).

seen from this figure, in the case of the integrated $v_2[\Upsilon(1S)]$ in the 10-90% bin (right panel), there is quite reasonable agreement, within experimental uncertainties, between the HQQD prediction and the data reported by the CMS collaboration. In the left panel of the Fig. 5.7, good agreement between HQQD predictions and CMS data can be seen in the 10-30% bin, however, there are larger differences in the other two centrality bins but still within 2σ of the experimental data. Hopefully, higher experimental statistics from future runs will reduce the experimental uncertainties in the near future.

In Fig. 5.8, we show the centrality dependence of $v_2[\Upsilon(2S)]$ and $v_2[\Upsilon(3S)]$ in the same centrality bins as used in Fig. 5.7. One centrality-integrated data point for $v_2[\Upsilon(2S)]$ is currently available from the CMS collaboration and has been shown as a green triangle in the 10-90% panel (right). As can be seen from the right panel of the figure, the integrated result

for $v_2[\Upsilon(2S)]$ is within the experimental uncertainties, however, at the very top end of them. We again hope that more statistics will help in making more constraining comparisons in the near future. Note that for $v_2[\Upsilon(2S)]$ and $v_2[\Upsilon(3S)]$ one sees a larger systematic uncertainty than for $v_2[\Upsilon(1S)]$.

In Fig. 5.9, we present HQQD predictions for the elliptic flow of $\Upsilon(2S)$ and $\Upsilon(3S)$ states as a function of transverse momentum in the 5-60% centrality bin. As can be seen from this figure, a sizable v_2 is predicted by HQQD for the $\Upsilon(3S)$ because of the strong path length dependence of $R_{AA}[\Upsilon(3S)]$ between the short and long sides of the QGP fireball. Similar to the $\Upsilon(1S)$, for the central values of our model parameters, HQQD predicts a negative v_2 for the $\Upsilon(2S)$ in the lowest two p_T-bins. The reason behind this negative v_2 is again that the transverse expansion of the QGP occurs more rapidly along the short side than the long side which has the affect of overtaking bottomonium states which had previously escaped the QGP with $\phi \sim 0$. One can see that HQQD makes a prediction that v_2 is positive in the highest p_T-bin for both states. Turning to the model variation shown as red and blue box in Fig. 5.9, one sees that in the 4-6 GeV p_T bin the variation of the model parameters can result in a positive $v_2[\Upsilon(2S)]$. This large systematic HQQD uncertainty is related to the quantum-mechanical oscillations discussed previously in the context of Fig. 5.5. Given these larger uncertainties, however, one still observes an ordering of the elliptic flow of the excited states with $v_2[\Upsilon(3S)] > v_2[\Upsilon(2S)]$ in the highest two p_T bins.

Finally, in Table 5.4 [2] we compare the predictions of HQQD for various observables with the corresponding experimental results from the ALICE, ATLAS, and CMS collaborations. In each row of the table, the results are integrated over centrality and transverse momentum

[2]In each row, the first column indicates the observable, the second column indicates the source of the experimental result and relevant cuts, and the third column shows the experimental result on the first line and the HQQD prediction on the second line. For both HQQD predictions and the experimental results, the first uncertainty reported is statistical uncertainty and the second is systematic uncertainty. In the case of HQQD, the systematic uncertainities were estimated by varying the HQQD model parameters in the range specified in Eq. (5.22).

in the ranges shown in the middle column, the first line of the third column is the experimental data (if available), and the second line of the third column shows the HQQD prediction. The rapidity cuts are $2.5 < y < 4.0$, $|y| < 1.5$, and $|y| < 2.4$ for the ALICE, ATLAS, and CMS collaborations, respectively. As can be seen from this table, all HQQD predictions are within the combined experimental statistical and systematic uncertainties. However, as can be seen most clearly from our comparisons of the integrated $R_{AA}[\Upsilon(2S)]$ and $\Upsilon(2S)$ to $\Upsilon(1S)$ double ratio, HQQD predicts too much $\Upsilon(2S)$ compared to the current experimental data.

5.4 Conclusions and outlook

In this paper, we have extended our previous work [62] by allowing for variation in the underlying HQQD model parameters in order to assess the impact such variation has on HQQD predictions for $R_{AA}[\Upsilon]$ and $v_2[\Upsilon]$. In addition, herein, we provided more details concerning the numerical method used in HQQD and the method by which final state feed down effects were included. Similar to our previous paper we used real-time quantum evolution averaged over a large set of bottomonium trajectories (2 million). After averaging over all wave-packet trajectories, for each set of model parameters we were able to obtain precise estimates for R_{AA}. For the central values of the model parameters, we found quite reasonable agreement with available experimental data for $R_{AA}[\Upsilon(1S)]$, however, we found that HQQD over-predicts the suppression of $\Upsilon(2S)$ compared to experimental observations. For the case in which we decreased the initialization time and increased the effective Debye mass, we found that the model predicts perhaps too much $\Upsilon(1S)$ suppression as can be seen from the lower limits indicated in Fig. 5.1. For the opposite case in which we increased the initialization time and decreased the effective Debye mass, we found that the model under-predicts the amount of $\Upsilon(1S)$ suppression, while providing better agreement with data available for $R_{AA}[\Upsilon(2S)]$.

Using this same set of model parameters, we then looked at the impact of our model assumptions on $v_2[\Upsilon]$. We found that the results for $v_2[\Upsilon(1S)]$ were relatively insensitive to the choice of model parameters, with the maximum $v_2[\Upsilon(1S)]$ as a function of centrality being on the order of 1% in all three cases considered. We found stronger variations in the HQQD predictions for $v_2[\Upsilon(2S)]$ and $v_2[\Upsilon(3S)]$, however, we still observed a maximal $v_2[\Upsilon(2S)]$ on the order of 3% and $v_2[\Upsilon(2S)]$ on the order of 5%. For each of the sets, we found that the ordering $v_2[\Upsilon(1S)] \lesssim v_2[\Upsilon(2S)] \lesssim v_2[\Upsilon(3S)]$ remains valid. This ordering is expected due to the fact that excited states experience more suppression and, hence, more path-length dependence of their propagation than the ground state. All of our HQQD predictions for integrated suppression and elliptic flow of bottomonium states were finally summarized in Tab. 5.4 which includes estimates of both the systematic and statistical model uncertainties.

With respect to the HQQD over prediction for $\Upsilon(2S)$ suppression, we note that HQQD includes in-medium thermal noise only through the imaginary part of the potential included in the real-time dynamics. This allowed us to extract the survival probability associated with the thermal-noise averaged quantum wave-function, however, does not fully take into account in-medium transitions between different singlet/octet and angular momentum states. There is currently ongoing work to include true stochastic noise at the level of the real-time Schrödinger equation solution, either through explicit noisy potentials [14, 11, 93, 94, 65, 13, 84, 95] or through solution of the Lindblad equation for the reduced density matrix [12, 30, 31, 28, 27]. Preliminary results obtained using the quantum trajectories method to solve the Lindblad equation indicate that more accurate inclusion of noise effects including, e.g., in-medium singlet-octet transitions, will result in only small changes in $R_{AA}[\Upsilon(1S)]$ [39]. Conversely, these studies indicate that the excited states can be less suppressed when including in-medium stochastic singlet-octet transitions. This is expected due to the possibility of "quantum regeneration" in which a state transitions to a finite ℓ octet

state and then back to a singlet $\ell = 0$ state, either through single or multiple in-medium quantum state transitions [39]. It is possible that inclusion of the effect of in-medium quantum jumps will help to reduce the tension between HQQD predictions for $R_{AA}[\Upsilon(2S)]$ and current experimental data.

In closing, we note that herein we made use of a potential whose real part was obtained from the internal energy. There were three reasons for choosing the internal energy over the free energy: (1) it was found in prior studies using the adiabatic approximation that using the free energy resulted in too much suppression for all states (see Ref. [110]), (2) there is some theoretical support for this choice when the system is in equilibrium due to entropy considerations, see e.g. Refs. [99, 98, 97], and (3) this same potential has been used in many prior works that utilized the adiabatic approximation. Another constraint on the potential was that it possesses the correct low temperature limit, i.e. that it should reduce to a real-valued Cornell potential tuned to the known spectrum of bottomonium bound states. This was done because we wanted to (a) have a more realistic phenomenological description of bottomonium excited states and (b) match smoothly onto the known vacuum eigenstates which are used, in practice, to project out the final survival probabilities. Of course, it would also be nice to add some non-perturbative contributions to the imaginary part of the potential, however, on this front things are much less constrained. One possibility would be to use lattice extractions of the imaginary part of the potential similar to what was done in Ref. [72]. Such investigations are planned for the future. An additional possibility is that the choice of the model potential used herein can be improved to include, for example, the effects of non-equilibrium momentum-space anisotropies in the QGP. The inclusion of momentum-space anisotropy has been shown to reduce quarkonium suppression for all states due to a reduction in the effective in-medium Debye mass [79]. We, again, leave this for future work.

Observable	**Source/Cuts**	**Experiment/HQQD**
$R_{AA}[\Upsilon(1S)]$	ALICE 0-90% [8]	0.37 ± 0.02 ± 0.03
	$p_T < 15$ GeV	0.301 ± 0.0002 $\pm^{0.09}_{0.12}$.
$R_{AA}[\Upsilon(1S)]$	ATLAS 0-80% [104]	0.32 ± 0.02 ± 0.05
	$p_T < 30$ GeV	0.3046 ± 0.0001 $\pm^{0.08}_{0.12}$
$R_{AA}[\Upsilon(1S)]$	CMS 0-100% [102]	0.376 ± 0.013 ± 0.035
	$p_T < 30$ GeV	0.312 ± 0.0002 $\pm^{0.085}_{0.12}$
$R_{AA}[\Upsilon(2S)]$	ALICE 0-90% [8]	0.10 ± 0.04 ± 0.02
	$p_T < 15$ GeV	0.05923 ± 0.00005 $\pm^{0.059}_{0.026}$
$R_{AA}[\Upsilon(2S)]$	ATLAS 0-80% [104]	0.11 ± 0.04 ± 0.04
	$p_T < 30$ GeV	0.07920 ± 0.00006 $\pm^{0.058}_{0.039}$
$R_{AA}[\Upsilon(2S)]$	CMS 0-100% [102]	0.117 ± 0.022 ± 0.019
	$p_T < 30$ GeV	0.0679 ± 0.00004 $\pm^{0.059}_{0.026}$
$R_{AA}[\Upsilon(3S)]$	CMS 0-100% [102]	0.022 ± 0.038 ± 0.016
	$p_T < 30$ GeV	0.03622 ± 0.00004 $\pm^{0.032}_{0.012}$
$\frac{[\Upsilon(2S)/\Upsilon(1S)]_{\rm PbPb}}{[\Upsilon(2S)/\Upsilon(1S)]_{\rm pp}}$	ATLAS 0-80% [104]	0.35 ± 0.12 ± 0.12
	$p_T < 30$ GeV	0.193 ± 0.0002 $\pm^{0.08}_{0.04}$
$\frac{[\Upsilon(2S)/\Upsilon(1S)]_{\rm PbPb}}{[\Upsilon(2S)/\Upsilon(1S)]_{\rm pp}}$	CMS 0-100% [101]	0.308 ± 0.055 ± 0.001
	$p_T < 30$ GeV	0.2175 ± 0.0002 $\pm^{0.075}_{0.031}$
$v_2[\Upsilon(1S)]$	ALICE 5-60% [7]	-0.003 ± 0.030 ± 0.006
	$2 < p_T < 15$ GeV	0.0009 ± 0.0009 $\pm^{0.001}_{0.00005}$
$v_2[\Upsilon(1S)]$	CMS 10-90% [103]	0.007 ± 0.011 ± 0.005
	$p_T < 30$ GeV	0.003 ± 0.0007 $\pm^{0.0006}_{0.0013}$
$v_2[\Upsilon(2S)]$	CMS 10-90% [103]	-0.063 ± 0.085 ± 0.037
	$p_T < 30$ GeV	0.0127 ± 0.0008 $\pm^{0.006}_{0.005}$
$v_2[\Upsilon(3S)]$	HQQD 10-90%	N/A
	$p_T < 30$ GeV	0.0264 ± 0.0011 $\pm^{0.0006}_{0.003}$

Table 5.4: Comparison of HQQD predictions for integrated $R_{AA}[\Upsilon]$ and $v_2[\Upsilon]$ with available experimental data.

Chapter 6

Ansotropic Heavy Quarkonium Quantum Dynamics

As an extension of our previuos works [62, 63], in this chapter, we will discuss our development of Ansotropic Heavy Quarkonium Quantum Dynamics (Aniso HQQD) model.

6.1 Real part of the anisotropic KMS potential

In an anisotropic QGP, the distribution function is taken to be of Romatschke-Strickland (RS) form [90, 91]

$$f_{\rm aniso}(\mathbf{p}) \equiv f_{\rm iso}\left(\frac{1}{\lambda}\sqrt{\mathbf{p}^2+\xi(\mathbf{p}\cdot\mathbf{n})^2}\right), \tag{6.1}$$

where $f_{\rm iso}$ is an isotropic distribution which in thermal equilibrium is given by a Bose-Einstein distribution, λ is a temperature-like scale which corresponds to temperature in the equilibrium limit, and the parameter ξ quantifies the degree of momentum-space anisotropy,

$$\xi = \frac{1}{2}\frac{\langle \mathbf{p}_\perp^2\rangle}{\langle p_z^2\rangle} - 1 \ , \tag{6.2}$$

where $p_z \equiv \mathbf{p}.\mathbf{n}$ and $\mathbf{p}_\perp \equiv \mathbf{p} - \mathbf{n}(\mathbf{p}.\mathbf{n})$ denote the particle momenta along and perpendicular to the direction $\mathbf{n}$ of anisotropy[90], respectively.

After considering the "minimal" extension [50] of the internal energy based isotropic KMS potential [66, 110], we write down the real part of the anisotropic KMS potential

$$V_{\rm KMS}^{\rm aniso}(r) = -\frac{a}{r}(1+\mu r)e^{-\mu r} + \frac{2\sigma}{\mu}\left[1-e^{-\mu r}\right] - \sigma r e^{-\mu r} \ , \tag{6.3}$$

where $a = 0.409$ is the effective coupling, $\sigma = 0.21$ GeV^2 is the string tension, and μ is the anisotropic screening scale (anisotropic Debye mass) defined via

$$\mu^2(\xi,T) = f(\xi)m_D^2(T) \ . \tag{6.4}$$

Here, the isotropic Debye mass is defined by

$$m_D(T) = F_{m_D} T \sqrt{N_c \left(1 + \frac{N_F}{6}\right) \pi\alpha} \ , \quad \alpha = \frac{4}{3}\alpha_s(2\pi T) \ , \tag{6.5}$$

where F_{m_D} is an adjustable factor called m_D-factor and $\alpha_s(5\ \text{GeV}) = 0.2034$ [81] is the strong coupling.

The function $f(\xi)$ in Eq. (6.4) is given by [106] (also see in Appendix B)

$$f(\xi) = \frac{\sqrt{2}(1+\xi)\arcsin\left(\sqrt{\frac{\xi}{1+\xi}}\right)}{\sqrt{\xi}\sqrt{1+\xi+\frac{(1+\xi)^2\arcsin\left(\sqrt{\frac{\xi}{1+\xi}}\right)}{\sqrt{\xi}}}} \ , \tag{6.6}$$

which has the following properties

$$\lim_{\xi\to 0} f(\xi) = 1 - \frac{2}{45}\xi^2 + \frac{44}{945}\xi^3 - \frac{8}{189}\xi^4 + \mathcal{O}\left(\xi\right)^{9/2} \ , \tag{6.7}$$

and

$$\lim_{\xi\to\infty} f(\xi) = \sqrt{\pi}\left(\frac{1}{\xi}\right)^{1/4} - \frac{2}{\sqrt{\pi}}\left(\frac{1}{\xi}\right)^{3/4} + \frac{4}{3\sqrt{\pi}}\left(\frac{1}{\xi}\right)^{7/4} + \mathcal{O}\left(\frac{1}{\xi}\right)^{9/4} \ , \tag{6.8}$$

in the small and large-ξ limit respectively.

When ξ is negative, we use

$$f(\xi) = \frac{\sqrt{2}\,\text{arcsinh}\left(\sqrt{\frac{|\xi|}{1-|\xi|}}\right)}{|\xi|^{1/4}\sqrt{\frac{\sqrt{|\xi|}}{1-|\xi|} + \text{arcsinh}\left(\sqrt{\frac{|\xi|}{1-|\xi|}}\right)}} \ , \text{ when } \ \xi < 0 \ . \tag{6.9}$$

6.2 Imaginary part to leading-order in ξ

The isotropic contribution is given by

$$\textbf{Im}\ V_{(0)}(r) = -g^2 C_F \int \frac{d^3\mathbf{p}}{(2\pi)^3}(e^{i\mathbf{p}\cdot\mathbf{r}} - 1)\frac{-\pi\lambda m_D^2}{p\,(p^2+m_D^2)^2} = -\frac{g^2 C_F \lambda}{4\pi}\,\phi(\hat{r}) \ , \tag{6.10}$$

with

$$\phi(\hat{r}) = 2\int_0^\infty dz \frac{z}{(z^2+1)^2}\left[1 - \frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right] \ , \tag{6.11}$$

and $\hat{r} \equiv r\, m_D$. This result has been derived before in Refs. [75, 76]. The term of order ξ can be expressed as

$$\begin{aligned}
\mathbf{Im}\ \xi V_{(1)}(\mathbf{r}) &= -g^2 C_F\, \xi \int \frac{d^3\mathbf{p}}{(2\pi)^3}\, (e^{i\mathbf{p}\cdot\mathbf{r}} - 1) \\
&\times \left[\frac{3\pi\lambda m_D^2}{4p\,(p^2+m_D^2)^2} \sin^2\alpha - \frac{2\pi\lambda m_D^4}{p\,(p^2+m_D^2)^3} (\sin^2\alpha - \frac{1}{3}) \right] \\
&= \frac{g^2 C_F \xi \lambda}{4\pi} \left[\psi_1(\hat{r},\theta) + \psi_2(\hat{r},\theta) \right] \ , \qquad (6.12)
\end{aligned}$$

where θ is the angle between $\mathbf{r}$ and $\mathbf{n}$ and

$$\psi_1(\hat{r},\theta) = \int_0^\infty dz \frac{z}{(z^2+1)^2} \left(1 - \frac{3}{2} \left[\sin^2\theta \frac{\sin(z\,\hat{r})}{z\,\hat{r}} + (1 - 3\cos^2\theta) G(\hat{r},z) \right] \right) \ , \qquad (6.13)$$

$$\psi_2(\hat{r},\theta) = -\int_0^\infty dz \frac{\frac{4}{3}z}{(z^2+1)^3} \left(1 - 3 \left[\left(\frac{2}{3} - \cos^2\theta \right) \frac{\sin(z\,\hat{r})}{z\,\hat{r}} + (1 - 3\cos^2\theta) G(\hat{r},z) \right] \right) \qquad (6.14)$$

with

$$G(\hat{r},z) = \frac{\hat{r}z\cos(\hat{r}z) - \sin(\hat{r}z)}{(\hat{r}z)^3} \ . \qquad (6.15)$$

Note that

$$\begin{aligned}
\phi(\hat{r}) &= -\frac{1}{9}\,\hat{r}^2(-4 + 3\gamma_E + 3\ln\hat{r}) \ , \\
\psi_1(\hat{r},\theta) &= \frac{1}{600}\,\hat{r}^2[123 - 90\gamma_E - 90\ln\hat{r} + \cos(2\theta)(-31 + 30\gamma_E + 30\ln\hat{r})] \ , \\
\psi_2(\hat{r},\theta) &= \frac{1}{90}\,\hat{r}^2(-4 + 3\cos(2\theta)) \ , \qquad (6.16)
\end{aligned}$$

where γ_E is the Euler-Gamma constant.

6.2.1 Angle-averaged imaginary part to leading-order in small-ξ limit

From [52], we can write down the imaginary part of the potential

$$\begin{aligned}
V_I(r) &= Im V_{(0)}(r) + Im\, \xi V_{(1)}(r) \\
&= -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \left[\psi_1(\hat{r},\theta) + \psi_2(\hat{r},\theta) \right] \ , \qquad (6.17)
\end{aligned}$$

where, $\hat{r} \equiv r\, m_D$, θ is the angle between $\mathbf{r}$ and $\mathbf{n}$ (direction of the anisotropy [90]), and

$$\phi(\hat{r}) = 2\int_0^\infty dz \frac{z}{(z^2+1)^2}\left[1 - \frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right] \ , \tag{6.18}$$

$$\psi_1(\hat{r},\theta) = \int_0^\infty dz \frac{z}{(z^2+1)^2}\left[1 - \frac{3}{2}\left\{\sin^2\theta \frac{\sin(z\,\hat{r})}{z\,\hat{r}} + (1-3\cos^2\theta)G(\hat{r},z)\right\}\right] \ , \tag{6.19}$$

$$\psi_2(\hat{r},\theta) = -\int_0^\infty dz \frac{\frac{4}{3}z}{(z^2+1)^3}\left[1 - 3\left\{\left(\frac{2}{3} - \cos^2\theta\right)\frac{\sin(z\,\hat{r})}{z\,\hat{r}} + (1-3\cos^2\theta)G(\hat{r},z)\right\}\right] \ , \tag{6.20}$$

with

$$G(\hat{r},z) = \frac{\hat{r}z\cos(\hat{r}z) - \sin(\hat{r}z)}{(\hat{r}z)^3} \ . \tag{6.21}$$

Now, angle-averaged ψ-functions are obtained as

$$\begin{aligned}
\langle\psi_1\rangle_\theta &= \frac{\int_0^\pi \psi_1(\hat{r},\theta)\sin\theta d\theta}{\int_0^\pi \sin\theta d\theta} \\
&= \int_0^\infty dz \frac{z}{(z^2+1)^2}\left[1 - \frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right] \\
&= \frac{1}{2}\phi(\hat{r})
\end{aligned} \tag{6.22}$$

and

$$\begin{aligned}
\langle\psi_2\rangle_\theta &= \frac{\int_0^\pi \psi_2(\hat{r},\theta)\sin\theta d\theta}{\int_0^\pi \sin\theta d\theta} \\
&= -\int_0^\infty dz \frac{\frac{4}{3}z}{(z^2+1)^3}\left[1 - \frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right] \\
&= -\frac{4}{3}\int_0^\infty dz \frac{z}{(z^2+1)^3}\left[1 - \frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right]
\end{aligned} \tag{6.23}$$

Eq. (6.18) can be simplified as

$$\begin{aligned}
\phi(\hat{r}) &= 2\int_0^\infty dz \frac{z}{(z^2+1)^2}\left[1-\frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right] \\
&= 2\int_0^\infty dz \frac{z}{(z^2+1)^2} - 2\int_0^\infty dz \frac{z}{(z^2+1)^2}.\frac{\sin(z\,\hat{r})}{z\,\hat{r}} \\
&= 2.\frac{1}{2} - 2\int_0^\infty dz \frac{z}{(z^2+1)^2}.\frac{\sin(z\,\hat{r})}{z\,\hat{r}} \qquad (6.24) \\
&= 1 - 2\int_0^\infty dz \frac{z}{(z^2+1)^2}.\frac{\sin(z\,\hat{r})}{z\,\hat{r}} \qquad (6.25) \\
&= 1 - 2\int_0^\infty \frac{dz}{\hat{r}}.\frac{\frac{z}{\hat{r}}}{\left(\frac{z^2}{\hat{r}^2}+1\right)^2}.\frac{\sin(z)}{z} \qquad (6.26) \\
&= 1 - 2\hat{r}^2\int_0^\infty dz\ \frac{1}{\left(z^2+\hat{r}^2\right)^2}\ \sin(z)\ , \qquad (6.27)
\end{aligned}$$

where in Eq. (6.24) we use $\int_0^\infty dz \frac{z}{(z^2+1)^2} = \frac{1}{2}$, and in going from Eq. (6.25) to Eq. (6.26) we use change of variables : $\{z\hat{r}, z, dz\} = \left\{z, \frac{z}{\hat{r}}, \frac{dz}{\hat{r}}\right\}$.

Similarly, Eq. (6.23) can be simplified as

$$\begin{aligned}
\langle\psi_2\rangle_\theta &= -\frac{4}{3}\int_0^\infty dz \frac{z}{(z^2+1)^3}\left[1-\frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right] \\
&= -\frac{4}{3}\left[\int_0^\infty dz \frac{z}{(z^2+1)^3} - \int_0^\infty dz \frac{z}{(z^2+1)^3}.\frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right] \\
&= -\frac{4}{3}\left[\frac{1}{4} - \int_0^\infty dz \frac{z}{(z^2+1)^3}.\frac{\sin(z\,\hat{r})}{z\,\hat{r}}\right] \qquad (6.28) \\
&= -\frac{4}{3}\left[\frac{1}{4} - \int_0^\infty \frac{dz}{\hat{r}}.\frac{\frac{z}{\hat{r}}}{\left(\frac{z^2}{\hat{r}^2}+1\right)^3}.\frac{\sin(z)}{z}\right] \qquad (6.29) \\
&= -\frac{4}{3}\left[\frac{1}{4} - \hat{r}^4\int_0^\infty dz\ \frac{1}{\left(z^2+\hat{r}^2\right)^3}\ \sin(z)\right]\ , \qquad (6.30)
\end{aligned}$$

where in Eq. (6.28) we use $\int_0^\infty dz \frac{z}{(z^2+1)^3} = \frac{1}{4}$, and in going from Eq. (6.28) to Eq. (6.29) we use change of variables : $\{z\hat{r}, z, dz\} = \left\{z, \frac{z}{\hat{r}}, \frac{dz}{\hat{r}}\right\}$.

Hence, in summary, angle-averaged imaginary part of the potential to leading-order in ξ is given by

$$\langle V_I(r)\rangle_\theta = -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \left[\langle\psi_1\rangle_\theta + \langle\psi_2\rangle_\theta\right]\ , \qquad (6.31)$$

where,

$$\phi(\hat{r}) = 1 - 2\hat{r}^2 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^2} \ \sin(z) \ , \tag{6.32}$$

$$\langle \psi_1 \rangle_\theta = \frac{1}{2}\phi(\hat{r}) \ , \tag{6.33}$$

and

$$\langle \psi_2 \rangle_\theta = -\frac{4}{3}\left[\frac{1}{4} - \hat{r}^4 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^3} \ \sin(z)\right] \ . \tag{6.34}$$

Now we will find an expression for $\langle \psi_2 \rangle_\theta$ in terms of $\phi(\hat{r})$ and it's first derivative i.e., $\phi'(\hat{r})$.

$$\begin{aligned}
\phi(\hat{r}) &= 1 - 2\hat{r}^2 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^2} \ \sin(z) \\
\Rightarrow \phi'(\hat{r}) &= 0 - 4\hat{r} \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^2} \ \sin(z) + 2\hat{r}^2 \int_0^\infty dz \ \frac{4\hat{r}}{\left(z^2+\hat{r}^2\right)^3} \ \sin(z) \\
\Rightarrow \phi'(\hat{r}) &= -4\hat{r} \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^2} \ \sin(z) + 8\hat{r}^3 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^3} \ \sin(z) \\
\Rightarrow \frac{1}{6}\hat{r}\phi'(\hat{r}) &= -\frac{2}{3}\hat{r}^2 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^2} \ \sin(z) + \frac{4}{3}\hat{r}^4 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^3} \ \sin(z)
\end{aligned}$$

$$\begin{aligned}
\Rightarrow -\frac{1}{3}\phi(\hat{r}) + \frac{1}{6}\hat{r}\phi'(\hat{r}) = -\frac{1}{3} &+ \frac{2}{3}\hat{r}^2 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^2} \ \sin(z) \\
&- \frac{2}{3}\hat{r}^2 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^2} \ \sin(z) + \frac{4}{3}\hat{r}^4 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^3} \ \sin(z)
\end{aligned}$$

$$\begin{aligned}
\Rightarrow -\frac{1}{3}\phi(\hat{r}) + \frac{1}{6}\hat{r}\phi'(\hat{r}) &= -\frac{1}{3} + \frac{4}{3}\hat{r}^4 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^3} \ \sin(z) \\
&= -\frac{4}{3}\left[\frac{1}{4} - \hat{r}^4 \int_0^\infty dz \ \frac{1}{\left(z^2+\hat{r}^2\right)^3} \ \sin(z)\right] \\
&= \langle \psi_2 \rangle_\theta
\end{aligned} \tag{6.35}$$

Hence, we get

$$\begin{aligned}
\langle \psi_2 \rangle_\theta &= -\frac{1}{3}\phi(\hat{r}) + \frac{1}{6}\hat{r}\phi'(\hat{r}) \\
&= -\frac{1}{3}\left[\phi(\hat{r}) - \frac{1}{2}\hat{r}\phi'(\hat{r})\right] \ .
\end{aligned} \tag{6.36}$$

Inserting Eq. (6.33) and Eq. (6.36) in Eq. (6.31), we obtain angle-averaged imaginary part of the potential

$$\begin{aligned}
\langle V_I(r)\rangle_\theta &= -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \left[\left\langle \psi_1 \right\rangle_\theta + \left\langle \psi_2 \right\rangle_\theta\right] \\
&= -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \left[\frac{1}{2}\phi(\hat{r}) - \frac{1}{3}\phi(\hat{r}) + \frac{1}{6}\hat{r}\phi'(\hat{r})\right] \\
&= -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \left[\frac{1}{6}\phi(\hat{r}) + \frac{1}{6}\hat{r}\phi'(\hat{r})\right] \\
&= -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \frac{1}{6}\phi(\hat{r}) + \alpha_s C_F T \xi \frac{1}{6}\hat{r}\phi'(\hat{r}) \\
&= -\alpha_s C_F \lambda \phi(\hat{r}) \left[1 - \frac{\xi}{6}\right] + \alpha_s C_F \lambda \xi \frac{1}{6}\hat{r}\phi'(\hat{r}) \\
&= -\alpha_s C_F \lambda \left[\left(1 - \frac{\xi}{6}\right)\phi(\hat{r}) - \frac{\xi}{6}\hat{r}\phi'(\hat{r})\right] \qquad (6.37)
\end{aligned}$$

In our model we assume $\langle \psi_2 \rangle_\theta \simeq -\frac{1}{3}\phi(\hat{r})$, and Eq. (6.31) becomes

$$\begin{aligned}
\langle V_I(r)\rangle_\theta &= -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \left[\left\langle \psi_1 \right\rangle_\theta + \left\langle \psi_2 \right\rangle_\theta\right] \\
&\simeq -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \left[\frac{1}{2}\phi(\hat{r}) - \frac{1}{3}\phi(\hat{r})\right] \\
&\simeq -\alpha_s C_F \lambda \phi(\hat{r}) + \alpha_s C_F \lambda \xi \ \frac{1}{6}\phi(\hat{r}) \\
&\simeq -\alpha_s C_F \lambda \phi(\hat{r}) \left[1 - \frac{\xi}{6}\right]. \qquad (6.38)
\end{aligned}$$

We take the final line above as our approximate model of the small ξ behavior

$$\lim_{\xi\to 0} \langle V_I(r)\rangle_\theta^{\text{model}} = -\alpha_s C_F \lambda \phi(\hat{r}) \left[1 - \frac{\xi}{6}\right]. \qquad (6.39)$$

6.2.2 Imaginary part to leading-order in large-ξ limit

We have the static Feynman propagator [87]

$$\lim_{\omega\to 0} \tilde{\mathcal{D}}_F^{00}(p,\xi) = \frac{4i\lambda m_D^2}{\varsigma|\mathbf{p}|[(\mathbf{p}^2+m_\beta^2)(\mathbf{p}^2+m_\alpha^2+m_\gamma^2)-m_\delta^4]^2}\left[\frac{m_\delta^4 - \varsigma(\mathbf{p}^2+m_\alpha^2+m_\gamma^2)^2}{1+\varsigma}E(-\varsigma) - m_\delta^4 K(-\varsigma)\right], \qquad (6.40)$$

where $\varsigma \equiv \xi p_x^2/\mathbf{p}^2$, λ is a temperature-like scale which corresponds to temperature in the equilibrium limit, and K and E are complete elliptic integrals of the first and second kind,

respectively, defined by

$$K(x) \equiv \int_0^{\pi/2} \frac{1}{\sqrt{1-x\sin^2\phi}}\,d\phi\,,$$
$$E(x) \equiv \int_0^{\pi/2} \sqrt{1-x\sin^2\phi}\;d\phi\,. \tag{6.41}$$

And the masses $m^2_{\alpha,\beta,\gamma,\delta}$ are given by in the static limit [90, 51]

$$m_\alpha^2 \equiv -\frac{m_D^2}{2p_x^2\sqrt{\xi}}\left(p_z^2\arctan\sqrt{\xi} - \frac{p_z\mathbf{p}^2}{\sqrt{\mathbf{p}^2+\xi p_x^2}}\arctan\left[\frac{\sqrt{\xi}p_z}{\sqrt{\mathbf{p}^2+\xi p_x^2}}\right]\right)\,, \tag{6.42}$$

$$m_\beta^2 \equiv m_D^2\frac{(\sqrt{\xi}+(1+\xi)\arctan\sqrt{\xi})(\mathbf{p}^2+\xi p_x^2)+\xi p_z\left(p_z\sqrt{\xi}+\frac{\mathbf{p}^2(1+\xi)}{\sqrt{\mathbf{p}^2+\xi p_x^2}}\arctan\left[\frac{\sqrt{\xi}p_z}{\sqrt{\mathbf{p}^2+\xi p_x^2}}\right]\right)}{2\sqrt{\xi}(1+\xi)(\mathbf{p}^2+\xi p_x^2)}\,, \tag{6.43}$$

$$m_\gamma^2 \equiv -\frac{m_D^2}{2}\left(\frac{\mathbf{p}^2}{\mathbf{p}^2+\xi p_x^2} - \frac{\mathbf{p}^2+p_z^2}{\sqrt{\xi}p_x^2}\arctan\sqrt{\xi} + \frac{p_z\mathbf{p}^2(2\mathbf{p}^2+3\xi p_x^2)}{\sqrt{\xi}(\xi p_x^2+\mathbf{p}^2)^{3/2}p_x^2}\arctan\left[\frac{\sqrt{\xi}p_z}{\sqrt{\mathbf{p}^2+\xi p_x^2}}\right]\right), \tag{6.44}$$

$$m_\delta^4 \equiv \frac{\pi^2 m_D^4\xi^2 p_z^2 p_x^2\mathbf{p}^2}{16(\mathbf{p}^2+\xi p_x^2)^3}\,. \tag{6.45}$$

Here, $\mathbf{n}=(0,0,1)$ points along the z-axis and $\mathbf{p}$ lies in the $x-z$ plane; in the general case, one should take $p_z \to \mathbf{p}\cdot\mathbf{n}$ and $p_x \to |\mathbf{p}-(\mathbf{p}\cdot\mathbf{n})\mathbf{n}|$.

With this, we can write down an expression for the real part of the potential which is valid to all orders in ξ [51, 110]

$$\begin{aligned} V(\mathbf{r},\xi) &= -g^2C_F\int\frac{d^3\mathbf{p}}{(2\pi)^3}\left(e^{i\mathbf{p}\cdot\mathbf{r}}-1\right)\tilde{\mathcal{D}}_R^{00}(\omega=0,\mathbf{p},\xi) \\ &= -g^2C_F\int\frac{d^3\mathbf{p}}{(2\pi)^3}\left(e^{i\mathbf{p}\cdot\mathbf{r}}-1\right)\frac{\mathbf{p}^2+m_\alpha^2+m_\gamma^2}{(\mathbf{p}^2+m_\alpha^2+m_\gamma^2)(\mathbf{p}^2+m_\beta^2)-m_\delta^4}\,, \end{aligned} \tag{6.46}$$

Expanding Eq. (6.40) in terms of powers of ξ in the large-ξ limit, the static Feynman propagator takes the form

$$\lim_{\omega\to 0}\tilde{\mathcal{D}}_F^{00}(p,\xi) = -\frac{4im_D^2\lambda\sqrt{\frac{1}{\xi}}}{p^4\,p_x} + \frac{2im_D^4\pi\lambda}{p^6\,p_x\,\xi} + \mathcal{O}\left(\frac{1}{\xi}\right)^{3/2}\,, \tag{6.47}$$

The imaginary part of the heavy-quark potential can be obtained from the Fourier transform of the static limit of the Feynman propagator [87]

$$V_I(\mathbf{r},\xi) \equiv -\frac{g^2C_F}{2}\int\frac{d^3\mathbf{p}}{(2\pi)^3}\left(e^{i\mathbf{p}\cdot\mathbf{r}}-1\right)\tilde{\mathcal{D}}_F^{00}\Big|_{\omega\to 0}\,. \tag{6.48}$$

From this we learn that in the limit $\xi \to \infty$ the magnitude of the imaginary part of the potential will decrease as $1/\sqrt{\xi}$.

6.3 Model for the imaginary part of the potential for general ξ

In order to construct a model that reduces to our model potential in the weak anisotropy limit (6.39) and is proportional to $1/\sqrt{\xi}$ in the large-ξ limit, we construct the following ansatz

$$\langle V_I(r)\rangle_\theta^{\text{model}} = -\frac{\alpha_s C_F \lambda \phi(\hat{r})}{\sqrt{1+\frac{\xi}{3}}}\,, \tag{6.49}$$

with $\hat{r} = m_D(\lambda) r$ and $m_D(\lambda) = F_{m_D}\lambda \sqrt{N_c\left(1+\frac{N_F}{6}\right)\pi\alpha}$.

One can also verify that in the small-ξ limit i.e.,

$$\lim_{\xi\to 0} \frac{1}{\sqrt{1+\frac{\xi}{3}}} = 1 - \frac{\xi}{6} + \mathcal{O}\left(\xi\right)^2\,, \tag{6.50}$$

our model potential (6.49) yields potential in the weak anisotropy limit (6.39) and in the large-ξ limit i.e.,

$$\lim_{\xi\to\infty} \frac{1}{\sqrt{1+\frac{\xi}{3}}} = \sqrt{3}\,\sqrt{\frac{1}{\xi}} + \mathcal{O}\left(\frac{1}{\xi}\right)^{3/2}\,, \tag{6.51}$$

our model potential (6.49) is proportional to $1/\sqrt{\xi}$ which justifies Eq. (6.47).

6.4 The complete anisotropic complex potential

In vacuum we take the heavy-quark potential to be given by a Cornell potential with a finite string breaking distance (5.1). To match smoothly onto the zero temperature limit we use Eq. (5.5). In the limit $T \to 0$, Eq. (5.5) reduces to Eq. (5.1).

For the imaginary part of the potential we use Eq. (6.49)

$$V_I^{\text{aniso}}(r) = \langle V_I(r)\rangle_\theta^{\text{model}}\,. \tag{6.52}$$

The resulting final anisotropic complex-valued potential is of the form

$$V^{\text{aniso}}(r) = V_R^{\text{aniso}}(r) + iV_I^{\text{aniso}}(r)\,. \tag{6.53}$$

6.5 Phenomenological Results

We have performed the numerical analysis similar to previous isotropic HQQD case [62, 63].

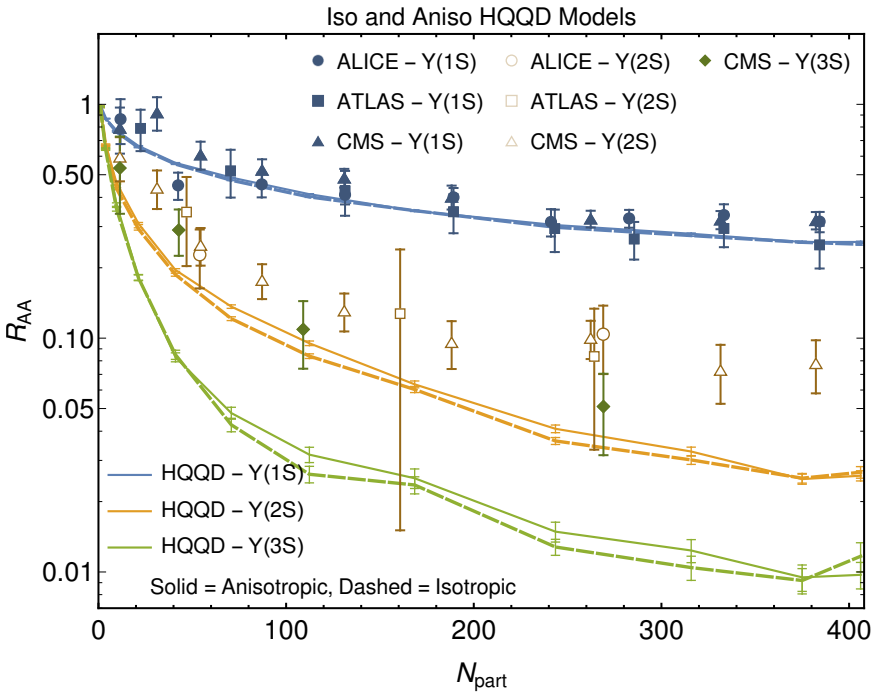


Figure 6.1: The nuclear suppression factor R_{AA} of bottomonia states as a function of $N_{\rm part}$. The solid and dashed lines indicate results obtained with aniso-HQQD and with iso-HQQD potential model, respectively. The experimental measurements shown are from the ALICE [9], ATLAS [104], and CMS [100, 46] Collaborations.

In Fig. 6.1, we present a comparison of the Iso and Aniso HQQD results and experimental data as functions of $N_{\rm part}$. The vertical axis is logarithmic in order to better resolve the suppression of the excited states. In these figure, the solid and dashed lines correspond to the results obtained with Aniso and Iso HQQD models, respectively. The shaded bands indicate the statistical uncertainty associated with the average over quantum and physical trajectories.

Chapter 7

Effective Heavy Quark Potential

In this chapter, we introduce a method for reducing anisotropic heavy-quark potentials to isotropic potentials by using an effective screening mass that depends on the quantum numbers l and m of a given state [49, 48, 61, 60]. We demonstrate that, using the resulting 1D effective potential model, one can solve a 1D Schrödinger equation and reproduce the full 3D results for the energies and binding energies of low-lying heavy-quarkonium bound states to relatively high accuracy. This includes the splitting of different p-wave polarizations. The resulting 1D effective model provides a way to include momentum anisotropy effects in open quantum system simulations of heavy-quarkonium dynamics in the quark-gluon plasma.

7.1 Background

The survival probability of the heavy quarkonium states, such as J/Ψ and Υ has been widely used as a sensitive probe to study the quark-gluon plasma (QGP) formed in relativistic heavy-ion experiments at RHIC and LHC [80, 66]. Due to non-relativistic nature of heavy quarkonium states, one can obtain their in-medium properties, such as masses and decay rates by solving a Schrödinger equation with a complex heavy-quark (HQ) potential. The real part of the HQ potential provides the binding energy, whereas the imaginary part provides information about the decay of a quarkonium state via wave function decoherence [75, 40, 23, 53, 33, 34, 35]. One can obtain the HQ potential at short distances by making use of hard-thermal-loop (HTL) resummed perturbation theory in the weak-coupling limit. Recently, several attempts have been made to develop complex-valued potential models to understand the in-medium properties of quarkonia quantitatively [50, 79, 107, 110, 73, 74, 72, 62, 63, 115].

During the last decade, many prior works have treated the QGP as an anisotropic medium by incorporating momentum-space anisotropies generated by longitudinal expansion into the underlying parton distribution functions. To make a phenomenological study of this effect in heavy-ion collisions, we consider the following spheroidal distribution function ansatz in the local rest frame (LRF) of the QGP [90]

$$f_{\rm aniso}^{\rm LRF}(\mathbf{k}) \equiv f_{\rm iso}\left(\frac{1}{\lambda}\sqrt{\mathbf{k}^2+\xi(\mathbf{k}\cdot\mathbf{n})^2}\right) . \tag{7.1}$$

This form takes into account the rapid longitudinal expansion of the QGP at early times and allows for explicit pressure anisotropies in the LRF [109, 24]. Here, $f_{\rm iso}$ is an arbitrary isotropic distribution function, λ is a temperature-like scale, which becomes the temperature T of the system in the thermal equilibrium limit. The degree of momentum-space anisotropy (ξ) in the range $-1 < \xi < \infty$ is given by

$$\xi = \frac{1}{2}\frac{\langle \mathbf{k}_\perp^2 \rangle}{\langle k_z^2 \rangle} - 1 \,, \tag{7.2}$$

where $k_z \equiv \mathbf{k}\cdot\mathbf{n}$ and $\mathbf{k}_\perp \equiv \mathbf{k} - \mathbf{n}\,(\mathbf{k}\cdot\mathbf{n})$ correspond to the particle momenta along and perpendicular to the direction of anisotropy ($\mathbf{n}$), respectively. Many prior works have studied heavy quarkonium physics by considering the momentum-space anisotropy inside the QGP [110, 114, 72, 62, 63, 50, 90, 109]. Here we focus on how to efficiently take momentum-space anisotropy into account in a one-dimensional effective theory and compare the one- and three-dimensional results for static and dynamical quantities numerically.

In this proceedings contribution, we summarize our previous works where the real part of a 3D anisotropic HQ potential has been reduced to 1D effective potential [49, 48]. This work is organized as follow: In sec. 7.2 we describe the isotropic complex HQ potential model, in sec. 7.3 we obtain the anisotropic complex HQ potential model, in sec. 7.4 we obtain our effective complex HQ potential model, in sec. 7.5 we present our static results, and in sec. 7.6 we present our dynamic results.

7.2 Isotropic Potential Model

The Fourier transform of the real time gluon propagator in the static limit gives the complex HQ potential in an isotropic QGP [57].

$$V(\lambda, r) = -g^2 C_F \int \frac{d^3\mathbf{p}}{(2\pi)^3}(e^{i\mathbf{p}\cdot\mathbf{r}} - 1)D^{00}(p_0 = 0, \mathbf{p}, \lambda)\,. \tag{7.3}$$

7.2.1 Perturbative Contribution

The perturbative contribution to the complex HQ potential can be obtained from HTL resummed perturbation theory. The real and imaginary parts of this perturbative contribution are given by

$$\mathrm{Re}\, V_{\rm pt}(\lambda, r) = -g^2 C_F \int \frac{d^3\mathbf{p}}{(2\pi)^3}(e^{i\mathbf{p}\cdot\mathbf{r}} - 1)\left(\frac{1}{p^2 + m_D^2} - \frac{1}{p^2}\right) \equiv \alpha m_D(\mathcal{I}_1(\hat{r}) - 1)\,, \tag{7.4}$$

$$\mathrm{Im}\, V_{\rm pt}(\lambda, r) = -g^2 C_F \int \frac{d^3\mathbf{p}}{(2\pi)^3}(e^{i\mathbf{p}\cdot\mathbf{r}} - 1)\frac{-\pi\lambda m_D^2}{p(p^2 + m_D^2)^2} \equiv \alpha\lambda(\mathcal{I}_2(\hat{r}) - 1)\,, \tag{7.5}$$

where the integrals $\mathcal{I}_1(\hat{r})$ and $\mathcal{I}_2(\hat{r})$ are

$$\begin{aligned}
\mathcal{I}_1(\hat{r}) &= 4\pi \int \frac{d^3\hat{\mathbf{p}}}{(2\pi)^3} e^{i\hat{\mathbf{p}}\cdot\hat{\mathbf{r}}} \frac{1}{\hat{p}^2(\hat{p}^2 + 1)} = \frac{1 - e^{-\hat{r}}}{\hat{r}}\,, \\
\mathcal{I}_2(\hat{r}) &= 4\pi^2 \int \frac{d^3\hat{\mathbf{p}}}{(2\pi)^3} e^{i\hat{\mathbf{p}}\cdot\hat{\mathbf{r}}} \frac{1}{\hat{p}(\hat{p}^2 + 1)^2} = \phi_2(\hat{r})\,,
\end{aligned} \tag{7.6}$$

with

$$\phi_n(\hat{r}) = 2\int_0^\infty dz \frac{\sin(z\hat{r})}{z\hat{r}} \frac{z}{(z^2 + 1)^n}\,. \tag{7.7}$$

Here, $\hat{\mathbf{p}} \equiv \mathbf{p}/m_D$, $\hat{\mathbf{r}} \equiv \mathbf{r}m_D$, and the strong coupling constant $\alpha = g^2 C_F/(4\pi)$. We also subtracted a term $1/p^2$ in Eq. (7.4) to make the r-independent part finite.

7.2.2 Non-perturbative Contribution

The gluon propagator also contains a non-perturbative string contribution which arises from a dimension two gluon condensate. Its Fourier transform gives us the non-perturbative contributions [57]

$$\mathrm{Re}\, V_{\rm npt}(\lambda, r) = -g^2 C_F m_G^2 \int \frac{d^3\mathbf{p}}{(2\pi)^3}(e^{i\mathbf{p}\cdot\mathbf{r}} - 1)\frac{p^2 + 5m_D^2}{(p^2 + m_D^2)^3} \equiv -\frac{2\sigma}{m_D}(\mathcal{I}_3(\hat{r}) - 1)\,, \tag{7.8}$$

$$\mathrm{Im}\, V_{\rm npt}(\lambda, r) = -g^2 C_F m_G^2 \int \frac{d^3\mathbf{p}}{(2\pi)^3}(e^{i\mathbf{p}\cdot\mathbf{r}} - 1)\frac{4\pi\lambda m_D^2(p^2 - 2m_D^2)}{p(p^2 + m_D^2)^4} \equiv \frac{4\sigma\lambda}{m_D^2}(\mathcal{I}_4(\hat{r}) - 1)\,, \quad (7.9)$$

where $\sigma = \alpha m_G^2/2$ and m_G^2 is a dimensionful constant. The integrals appearing above are

$$\begin{aligned}
\mathcal{I}_3(\hat{r}) &= 4\pi \int \frac{d^3\hat{\mathbf{p}}}{(2\pi)^3} e^{i\hat{\mathbf{p}}\cdot\hat{\mathbf{r}}} \frac{\hat{p}^2 + 5}{(\hat{p}^2 + 1)^3} = (1 + \hat{r}/2)e^{-\hat{r}}\,, \\
\mathcal{I}_4(\hat{r}) &= 8\pi^2 \int \frac{d^3\hat{\mathbf{p}}}{(2\pi)^3} e^{i\hat{\mathbf{p}}\cdot\hat{\mathbf{r}}} \frac{2 - \hat{p}^2}{\hat{p}(\hat{p}^2 + 1)^4} = -2\phi_3(\hat{r}) + 6\phi_4(\hat{r})\,. \qquad (7.10)
\end{aligned}$$

7.2.3 Total Isotropic potential

The sum of the perturbative and non-perturbative contributions give us the total complex isotropic HQ potential

$$\begin{aligned}
\mathrm{Re}\, V_{\rm Iso}(r) &= \mathrm{Re}\, V_{\rm pt}(\lambda, r) + \mathrm{Re}\, V_{\rm npt}(\lambda, r) \\
&= \alpha m_D \left(\frac{1 - e^{-rm_D}}{rm_D}\right) - \alpha m_D - \frac{\sigma}{m_D}(2 + rm_D)\, e^{-rm_D} + \frac{2\sigma}{m_D} - \frac{\alpha}{r}\,, \quad (7.11)
\end{aligned}$$

$$\begin{aligned}
\mathrm{Im}\, V_{\rm Iso}(r) &= \mathrm{Im}\, V_{\rm pt}(\lambda, r) + \mathrm{Im}\, V_{\rm npt}(\lambda, r) \\
&= \alpha\lambda\phi_2\,(rm_D) - \alpha\lambda - \frac{8\sigma\lambda}{m_D^2}\phi_3(rm_D) + \frac{24\sigma\lambda}{m_D^2}\phi_4(rm_D) - \frac{4\sigma\lambda}{m_D^2}\,. \quad (7.12)
\end{aligned}$$

We include a relativistic correction, $-0.8\sigma/(m_{b/c}^2 r)$, in the potential model while solving the Schrödinger equation for charmonia and bottomonia [50], where the masses of the charm and bottom quarks are taken to be $m_c = 1.3\,\mathrm{GeV}$ and $m_b = 4.7\,\mathrm{GeV}$, respectively.

7.3 3D Anisotropic Potential Model

The real and imaginary part of the 3D anisotropic potential model as derived in our previous work [48] are

$$\mathrm{Re}\, V_{\rm Aniso}(r, \theta, \xi) = \alpha m_D^A \left(\frac{1 - e^{-rm_D^R}}{rm_D^R}\right) - \alpha m_D^A - \frac{\sigma}{m_D^A}\left(2 + rm_D^R\right) e^{-rm_D^R} + \frac{2\sigma}{m_D^A} - \frac{\alpha}{r}\,, \quad (7.13)$$

$$\mathrm{Im}\, V_{\rm Aniso}(r, \theta, \xi) = \alpha\lambda^A\phi_2\left(rm_D^I\right) - \alpha\lambda^A - \frac{8\sigma\lambda^A}{\left(m_D^A\right)^2}\phi_3\left(rm_D^I\right) + \frac{24\sigma\lambda^A}{\left(m_D^A\right)^2}\phi_4\left(rm_D^I\right) - \frac{4\sigma\lambda^A}{\left(m_D^A\right)^2}\,, \quad (7.14)$$

where,

$$m_D^A = m_D\left(1-\frac{\xi}{6}\right), \quad \lambda^A = \lambda\left(1-\frac{\xi}{6}\right), \tag{7.15}$$

and

$$m_D^R = m_D\Big[1+\xi\left(0.108\cos 2\theta - 0.131\right)\Big], \quad m_D^I = m_D\Big[1+\xi\left(0.026\cos 2\theta - 0.158\right)\Big]. \tag{7.16}$$

Eq. (7.15) assures a correct asymptotic behavior of the potential. Eq. (7.16) was obtained by matching effective and exact result at $\hat{r}=1$ as described in [48].

7.4 1D Effective Potential Model

Due to the angular dependence in the 3D anisotropic potential model, solving a 3D Schrödinger equation to find various in-medium properties of the quarkonium states is rather time consuming and much more complicated. One possible solution to this problem is to introduce an angle-averaged effective screening mass $\mathcal{M}_{lm}(\lambda,\xi)$ [49]

$$\begin{aligned}\mathcal{M}_{lm}^{R,I}(\lambda,\xi) &= \langle \mathrm{Y}_{lm}(\theta,\phi)|m_D^{R,I}(\lambda,\xi,\theta)|\mathrm{Y}_{lm}(\theta,\phi)\rangle\,,\\ &= \int_{-1}^{1} d\cos\theta \int_0^{2\pi} d\phi \mathrm{Y}_{lm}(\theta,\phi) m_D^{R,I}(\lambda,\xi,\theta)\mathrm{Y}_{lm}^*(\theta,\phi)\,,\end{aligned} \tag{7.17}$$

and where $\mathrm{Y}_{lm}(\theta,\phi)$ refers to the spherical harmonics with azimuthal quantum number l and magnetic quantum number m. The main advantage of using an angle-averaged effective screening mass $\mathcal{M}_{lm}(\lambda,\xi)$ is to utilize the spherical symmetry in the potential model which significantly simplifies the numerics.

The real and imaginary part of the 1D effective potential model as derived in our previuos work [48] are

$$\mathrm{Re}\,V_{\mathrm{Eff}}(r,\xi) = \alpha m_D^A\left(\frac{1-e^{-r\mathcal{M}_{lm}^R}}{r\mathcal{M}_{lm}^R}\right) - \alpha m_D^A - \frac{\sigma}{m_D^A}\left(2+r\mathcal{M}_{lm}^R\right)e^{-r\mathcal{M}_{lm}^R} + \frac{2\sigma}{m_D^A} - \frac{\alpha}{r}\,, \tag{7.18}$$

$$\mathrm{Im}\,V_{\mathrm{Eff}}(r,\xi) = \alpha\lambda^A\phi_2\left(r\mathcal{M}_{lm}^I\right) - \alpha\lambda^A - \frac{8\sigma\lambda^A}{\left(m_D^A\right)^2}\phi_3\left(r\mathcal{M}_{lm}^I\right) + \frac{24\sigma\lambda^A}{\left(m_D^A\right)^2}\phi_4\left(r\mathcal{M}_{lm}^I\right) - \frac{4\sigma\lambda^A}{\left(m_D^A\right)^2}\,, \tag{7.19}$$

	type	l, m values	states
Bottomonium	s-wave	$l = 0, m = 0$	$\Upsilon(1S), \Upsilon(2S), \Upsilon(3S)$
	p-wave	$l = 1, m = -1$	$\chi_{b-1}(1P)$
		$l = 1, m = 0$	$\chi_{b0}(1P)$
		$l = 1, m = +1$	$\chi_{b+1}(1P)$
	type	**l, m values**	**states**
Charmonium	s-wave	$l = 0, m = 0$	$J/\psi\ (1S), \psi(2S), \psi(3S)$
	p-wave	$l = 1, m = -1$	$\chi_{c-1}(1P)$
		$l = 1, m = 0$	$\chi_{c0}(1P)$
		$l = 1, m = +1$	$\chi_{c+1}(1P)$

Table 7.1: l and m values of various quarkonium states.

where,

$$K_{lm} = \frac{2l(l+1) - 2m^2 - 1}{4l(l+1) - 3} \tag{7.20}$$

and

$$\mathcal{M}^R_{lm} = m_D \Big[1 + \xi \left(0.216 K_{lm} - 0.239 \right) \Big] \ , \quad \mathcal{M}^I_{lm} = m_D \Big[1 + \xi \left(0.052 K_{lm} - 0.184 \right) \Big]. \tag{7.21}$$

The l and m values of various quarkonium states are given in Table 7.1.

7.5 Static Results

For the static solutions, we used a previously developed 3D eigensolver called quantumFDTD [111, 47]. Using this code, we compared results obtained with the 1D effective potential and the full 3D anisotropic potential. In Table 7.2, we list the exact results of the eigenenergies ($\mathrm{Re}\, E$), decay widths ($\mathrm{Im}\, E$) and the binding energies (E_{bind}) with the anisotropy parameter $\xi = 1$ for $\Upsilon(1S)$ and J/Ψ.

In the numerical evaluations, we took $\alpha = 0.272$ and $\sigma = 0.215\,\mathrm{GeV}^2$. For the $\Upsilon(1S)$ state, we used a lattice size of $N^3 = 512^3$ with a lattice spacing of $a = 0.020\,\mathrm{GeV}^{-1} \approx 0.004\,\mathrm{fm}$ giving a lattice size of $L = Na \approx 2.05\,\mathrm{fm}$. For the J/ψ, we used a lattice size of $N^3 = 256^3$ with a lattice spacing of $a = 0.085\,\mathrm{GeV}^{-1} \approx 0.017\,\mathrm{fm}$ giving a lattice size of $L = Na \approx 4.35\,\mathrm{fm}$.

$\Upsilon(1S)$	ReE	δReE	$E_{\rm bind}$	ImE	δImE
T_o	182.869	0.611	-662.669	11.838	0.027
$1.1T_o$	174.957	0.593	-570.612	14.830	0.031
$1.2T_o$	166.556	0.573	-493.689	18.190	0.034
$1.4T_o$	148.439	0.531	-372.540	26.004	0.039

J/Ψ	ReE	δReE	$E_{\rm bind}$	ImE	δImE
T_o	439.336	1.230	-406.202	41.980	0.107
$1.1T_o$	422.207	1.163	-323.362	51.467	0.105
$1.2T_o$	404.597	1.095	-255.648	61.698	0.098
$1.3T_o$	386.604	1.028	-199.583	72.564	0.086
$1.4T_o$	368.301	0.963	-152.678	83.958	0.070

Table 7.2: The exact 3D results of the complex eigenenergies (E) and binding energies ($E_{\rm bind}$) for different quarkonium states at various temperatures with $\xi = 1$. δE are the differences in results obtained using 1D effective and 3D anisotropic potentials. Here T_o is 192 MeV and all results are in MeV [48].

7.6 Dynamical Results

In order to solve the 3D Schrödinger equation in real time, we used a split-step pseudospectral method [112] with temporal step size $\Delta t = 0.001$ fm/c. Once again we compare results obtained with the full 3D anisotropic potential to those obtained with the 1D effective potential. We evolve the wave function from $\tau = 0$ fm/c to $\tau = 0.25$ fm/c in the vacuum ($T = 0$). Starting at $\tau = \tau_0 = 0.25$ fm/c, we consider a fixed anisotropy parameter $\xi = 1$ and boost-invariant Bjorken evolution for the hard scale

$$\lambda(\tau) = \lambda_0 \left(\frac{\tau_0}{\tau}\right)^{1/3} . \tag{7.22}$$

Here we take the initial hard scale to be $\lambda_0 = 630$ MeV. Further details of the numerical method can be found in [48].

7.6.1 Bottomonium

For bottomonium states we take the box size to be $L = 2.56$ fm, $m_b = 4.7$ GeV, and use $N = 128$ lattice points in each direction. The top row of fig. 7.1 shows the time evolution of overlaps of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ using a pure $\Upsilon(1S)$ eigenstate as the initial condition. Whereas the bottom row shows the time evolution of the bottomonium p-wave overlaps resulting from initialization with different p-wave polarizations. Results with pure $\Upsilon(2S)$ and $\Upsilon(3S)$ eigenstates and a Gaussian as the initial condition can be found in Ref. [48].

7.6.2 Charmonium

For charmonium states we take $L = 5.12$ fm, $m_c = 1.3$ GeV, and use $N = 128$ lattice points in each direction. The top row of the fig. 7.2 shows the time evolution of overlaps of the J/ψ, $\psi(2S)$, and $\psi(3S)$ by using pure J/ψ eigenstate as the initial condition. Whereas the bottom row shows the time evolution of the charmonium p-wave overlaps resulting from initialization with different p-wave polarizations. The results with pure $\psi(2S)$ and $\psi(3S)$

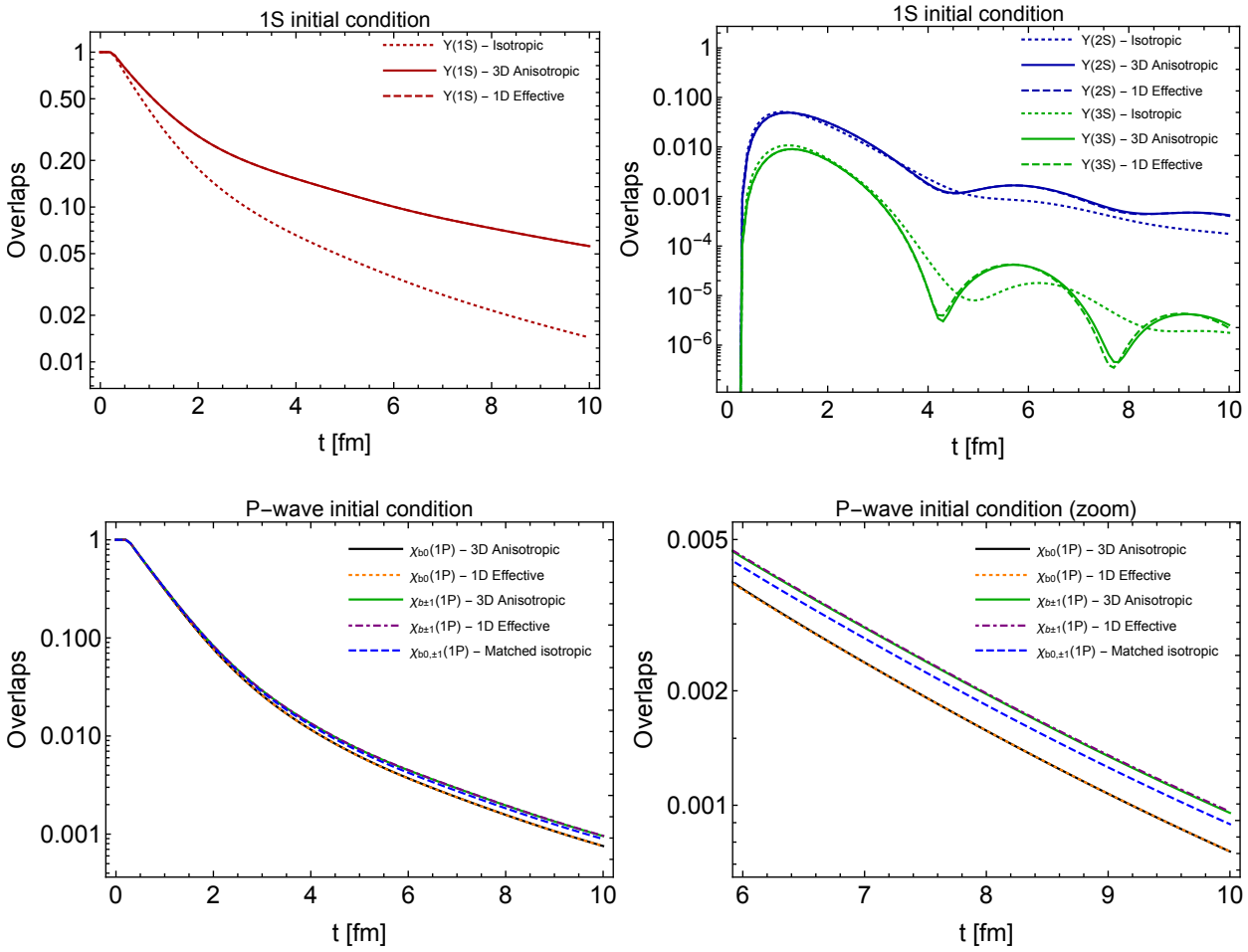


Figure 7.1: The top row shows the overlaps of $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ resulting from real-time solution of the Schrödinger equation. Here we initialized the wave function as pure $\Upsilon(1S)$ eigenstate. The bottom row shows the time evolution of the bottomonium p-wave overlaps resulting from initialization with different p-wave polarizations [48].

eigenstate and Gaussian as the initial condition can be found in Ref. [48].

7.7 Conclusions

We have reduced anisotropic heavy-quark potentials to isotropic ones by introducing an effective screening mass that depends on the quantum numbers l and m of a given state. We demonstrated that, using the resulting 1D effective potential model, one can reproduce the

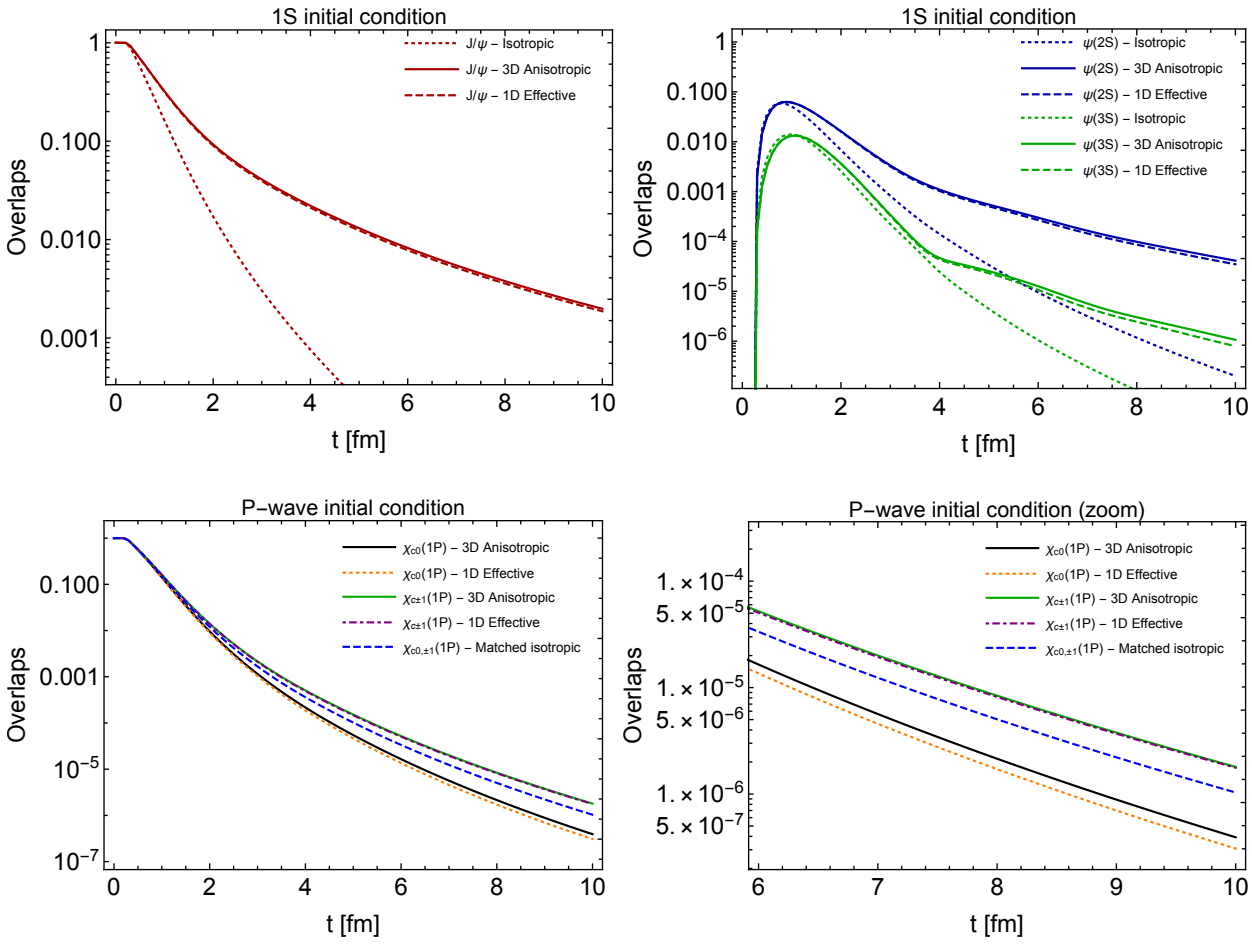


Figure 7.2: The top row shows the overlaps of J/ψ, $\psi(2S)$, and $\psi(3S)$ resulting from real-time solution of the Schrödinger equation. Here we initialized the wave function as pure J/ψ eigenstate. The bottom row shows the time evolution of the charmonium p-wave overlaps resulting from initialization with different p-wave polarizations [48].

full 3D results for the energies and binding energies of low-lying heavy-quarkonium bound states to relatively high accuracy. This finding is important because it can be used to incorporate anisotropy effects into one-dimensional real-time Schrödinger equations which underpin phenomenological calculations of bottomonium suppression in open quantum systems approaches.

Chapter 8

PNRQCD and Open Quantum System Approach

Using the potential non-relativistic quantum chromodynamics (pNRQCD) effective field theory, we derive a Lindblad equation for the evolution of the heavy-quarkonium reduced density matrix that is accurate to next-to-leading order (NLO) in the ratio of the binding energy of the state to the temperature of the medium [36]. The resulting NLO Lindblad equation can be used to more reliably describe heavy-quarkonium evolution in the quark-gluon plasma at low temperatures compared to the leading-order truncation. For phenomenological application, we numerically solve the resulting NLO Lindblad equation using the quantum trajectories algorithm. To achieve this, we map the solution of the three-dimensional Lindblad equation to the solution of an ensemble of one-dimensional Schrödinger evolutions with Monte-Carlo sampled quantum jumps. Averaging over the Monte-Carlo sampled quantum jumps, we obtain the solution to the NLO Lindblad equation without truncation in the angular momentum quantum number of the states considered.

We demonstrate the importance of quantum jumps in the nonequilibrium evolution of bottomonium states in the quark-gluon plasma [37]. Based on nonrelativistic effective field theory and the open quantum system framework, we evolve the density matrix of color singlet and octet pairs. We show that quantum regeneration of singlet states from octet configurations is necessary to understand experimental results for the suppression of both bottomonium ground and excited states. The values of the heavy-quarkonium transport coefficients used are consistent with recent lattice QCD determinations.

8.1 Non-relativistic Effective Field Theories

In this section, we will discuss effective field theories (EFTs) that can be used when there are widely separated energy scales appearing in a problem [56]. In general, an effective field theory of QCD is constructed as an expansion in the ratio of Λ_l/Λ_h, where Λ_l is a low energy scale and Λ_h is a high energy scale. In QCD, there are two sets of quarks

- Heavy quarks (c,b,t) whose masses (m_h) are much larger than the hadronic scale $\Lambda_{\rm QCD}$ i.e., $m_h \gg \Lambda_{\rm QCD}$[1].
- Light quarks (u,d,s) whose masses (m_h) are much smaller than the hadronic scale $\Lambda_{\rm QCD}$ i.e., $m_l \ll \Lambda_{\rm QCD}$.

Here. we discuss EFTs for heavy quarks. These EFTs are called non-relativistic EFTs as the heavy quark mass is the largest scale and also larger than the momentum (p) of the heavy quark in the hadron reference frame i.e., $m_h \gg p$.

8.1.1 Heavy Quark Effective Field Theory

Heavy Quark Effective Field Theory(HQEFT) is the first non-relativistic EFT of QCD that describes the physics of heavy quarks. It is used in studying the properties of hadrons made of a single heavy quark. HQEFT follows from QCD by integrating out modes associated with the heavy quark mass and exploiting the hierarchy $m_h \gg \Lambda_{\rm QCD}$. The HQEFT is obtained as an expansion in $1/m_h$. The HQEFT Lagrangian density for a heavy quark[2] is given by up to order $1/m_h^2$ and including $1/m_h^3$ kinetic operator [56]:

$$\mathcal{L}_{\rm HQEFT} = \mathcal{L}^{\psi} + \mathcal{L}^{l} \tag{8.1}$$

[1]Running coupling $\alpha(E = \Lambda_{\rm QCD}) \approx 1$,
$\Lambda_{\rm QCD} \ll E$: weakly coupled, asymptotic freedom,
$\Lambda_{\rm QCD} \gg E$: strongly coupled, confinement

[2]One can obtain the HQEFT Lagrangian for a heavy antiquark from Eqs. (8.1) and (8.2) by charge conjugation.

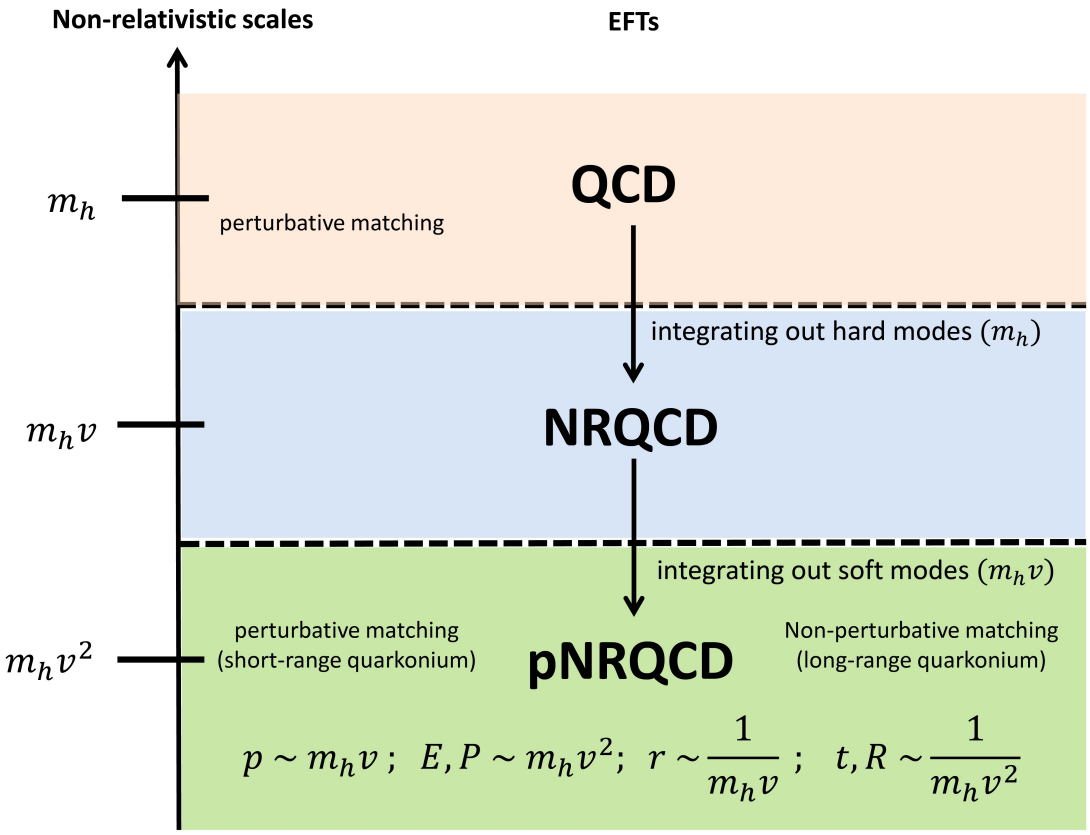


Figure 8.1: Hierarchy of energy scales and EFTs for systems made of a heavy quark and antiquark pair near threshold.

with

$$\mathcal{L}^{\psi} = \psi^{\dagger}\left\{iD_0 + \frac{\mathbf{D}^2}{2m_h} + \frac{\mathbf{D}^4}{8m_h^3} - C_F\frac{\sigma.g\mathbf{B}}{2m_h} - C_D\frac{[\mathbf{D}.,g\mathbf{E}]}{8m_h^2} - iC_S\frac{\sigma.[\mathbf{D}\times,g\mathbf{E}]}{8m_h^2}\right\}\psi \quad (8.2)$$

$$\mathcal{L}^{l} = -\frac{1}{4}F^A_{\mu\nu}F^{A\mu\nu} + \frac{d_2}{m_h^2}F^A_{\mu\nu}D^2F^{A\mu\nu} - \frac{d_3}{m_h^2}gf_{ABC}F^A_{\mu\nu}F^B_{\mu\alpha}F^C_{\nu\alpha} + \sum_{l=1}^{n_l}\bar{q}_l(i\not{D} - m_l)q_l \quad (8.3)$$

where, $[\mathbf{D}.,g\mathbf{E}] = \mathbf{D}.g\mathbf{E} - g\mathbf{E}.\mathbf{D}$ and $[\mathbf{D}\times,g\mathbf{E}] = \mathbf{D}\times\mathbf{E} - g\mathbf{E}\times\mathbf{D}$

$\mathbf{E}^i = F^{i0}$: the chromo-electric field

$\mathbf{B}^i = -\epsilon_{ijk}F^{jk}/2$: chromo-magnetic field

σ : Pauli matrices

q_l : fields of n_l light quarks

m_h : heavy quark pole mass

C_F, C_D, C_S, d_2, d_3 : Wilson co-efficients of the EFT.

The leading order HQEFT Lagrangian is given by [56]:

$$\mathcal{L}^{(0)}_{\rm HQEFT} = \psi^\dagger i D_0 \psi - \frac{1}{4} F^A_{\mu\nu} + \sum_{l=1}^{n_l} \bar{q}_l (i \not{D} - m_l) q_l \tag{8.4}$$

The leading order HQEFT Lagrangian has the heavy-quark symmetry which stands for invariance with respect to the heavy-quark flavor and spin. Moreover, it is exactly renormalizable.

8.1.2 Non Relativistic QCD

Hadrons ($QQ, Q\bar{Q}$) are non-relativistic multiscale systems that follows the following hierarchy of scales:

$$m_h \text{ (hard modes)} \gg m_h v \text{ (soft modes)} \gg m_h v^2 \text{ (ultrasoft modes)} \ . \tag{8.5}$$

Non Relativistic QCD (NRQCD) is the EFT attired to describe systems made of $Q\bar{Q}$ pair near threshold that follows from QCD by integrating out the energy scale (hard modes) associated with the heavy quark mass, m_h. The dynamical scales of NRQCD are at least $m_h v$, $m_h v^2$, and $\Lambda_{\rm QCD}$. NRQCD is a systematic tool for analytical calculations of quarkonium observables.

The NRQCD Lagrangian density for systems made of a $Q\bar{Q}$ pair of equal masses (m_h) up to order $1/(m_h^2)$, and including the $1/m_h^3$ kinetic operator, is given by [56]

$$\mathcal{L}_{\rm NRQCD} = \mathcal{L}^\psi + \mathcal{L}^\chi + \mathcal{L}^{\psi\chi} + \mathcal{L}^l \ , \tag{8.6}$$

where

$\mathcal{L}^\psi$ and $\mathcal{L}^\chi$: the HQEFT Lagrangian densities for the quark and antiquark respectively (see

Eq. (8.2)).

$\mathcal{L}^l$: the Lagrangian density (8.3) for the light degrees of freedom.

$\mathcal{L}^{\psi\chi}$: the four-fermion sector which up to order $1/m_h^2$ reads

$$\begin{aligned}\mathcal{L}^{\psi\chi} &= \frac{f_1(^1S_0)}{m_h^2}\psi^\dagger\chi\chi^\dagger\psi + \frac{f_1(^3S_1)}{m_h^2}\psi^\dagger\sigma\chi.\chi^\dagger\sigma\psi \\ &+ \frac{f_8(^1S_0)}{m_h^2}\psi^\dagger T^A\chi\chi^\dagger T^A\psi + \frac{f_8(^3S_1)}{m_h^2}\psi^\dagger T^A\sigma\chi.\chi^\dagger T^A\sigma\psi\ . \end{aligned} \tag{8.7}$$

Here, m_h is the pole mass and the matrices T^A are the $SU(3)$ generators $\lambda^A/2$. The four-fermion Lagrangian in Eq. (8.7) is made of all possible four-fermion operators of dimension 6. The corresponding Wilson coefficients are $f_1(^1S_0)$, $f_1(^3S_1)$, $f_8(^1S_0)$, and $f_8(^3S_1)$.

- The first four-fermion operator projects on a $Q\bar{Q}$ pair in a color singlet configuration with quantum numbers $^1S_0$.
- The second four-fermion operator projects on a $Q\bar{Q}$ pair in a color singlet configuration with quantum numbers $^3S_1$.
- The third four-fermion operator projects on a $Q\bar{Q}$ pair in a color octet configuration with quantum numbers $^1S_0$.
- The fourth four-fermion operator projects on a $Q\bar{Q}$ pair in a color octet configuration with quantum numbers $^3S_1$.

The leading order NRQCD Lagrangian reads in Coulomb gauge

$$\mathcal{L}^{(0)}_{\rm NRQCD} = \psi^\dagger\left(iD_0 + \frac{\nabla^2}{2m_h}\right)\psi + \chi^\dagger\left(iD_0 - \frac{\nabla^2}{2m_h}\right)\chi - \frac{1}{4}F^A_{\mu\nu}F^{A\ \mu\nu} + \sum_{l=1}^{n_l}\bar{q}_l(i\not{D} - m_l)q_l \tag{8.8}$$

One can notice that the leading order NRQCD Lagrangian contains the heavy quark mass, and therefore violates the heavy-quark flavor symmetry. And that is why the bottomonium binding energy is different, even at leading order, from the charmonium one. Also the NRQCD Lagrangian is, even at leading order, non renormalizable which makes it different from HQEFT.

8.1.3 potential Non Relativistic QCD

Potential Non Relativistic QCD (pNRQCD) is the EFT that follows from QCD by integrating out the soft modes i.e.,$m_h v$ associated with the inverse of the distance r between the heavy quark and antiquark. The pNRQCD is constructed as an expansion in r with Wilson coefficients encoding non-analytic contributions in r. In general, there are two types of pNRQCD depending on the scale $\Lambda_{\rm QCD}$.

Weakly-coupled pNRQCD

If $\Lambda_{\rm QCD} \lesssim m_h v^2$, then the bound states are weakly coupled and the corresponding EFT is called weakly-coupled pNRQCD. In this case, the distance is the order of or smaller than $\frac{1}{m_h v^2}$ and color confinement has not yet set in. Weakly-coupled pNRQCD is well attired to describe tightly bound quarkonia such as bottomonium and charmonium ground states, the B_c ground state, and threshold effects in $t\bar{t}$ production. After integrating out soft gluons and light quarks of energy or momentum $m_h v$ from NRQCD, the remaining degrees of freedom of weakly-coupled pNRQCD are:

- heavy quarks and antiquarks of momentum $m_h v$ and energy $m_h v^2$,
- ultrasoft gluons of momentum and energy $m_h v^2$,
- light quarks of momentum and energy $m_h v^2$.

The pNRQCD Lagrangian is constructed as adouble expansion in $1/m_h$ and r. At order r in the multipole expansion, the weakly-coupled pNRQCD Lagrangian density is given by [56]

$$\mathcal{L}^{\rm weak}_{\rm pNRQCD} = \mathcal{L}^{\rm S,O} + \mathcal{L}^{l} \ , \tag{8.9}$$

with

$$\begin{aligned} \mathcal{L}^{\rm S,O} &= \int d^3r \ {\rm Tr}\left\{ S^\dagger (i\partial_0 - h_s) S + O^\dagger (iD_0 O - h_0) O \right\} \\ &- V_A \ {\rm Tr}\left\{ O^\dagger \mathbf{r}.g\mathbf{E} S + S^\dagger \mathbf{r}.g\mathbf{E} O \right\} - \frac{V_B}{2} \ {\rm Tr}\left\{ O^\dagger \mathbf{r}.g\mathbf{E} O + O^\dagger O \mathbf{r}.g\mathbf{E} \right\} \ , \end{aligned} \tag{8.10}$$

where up to order $1/m_h^2$ and including the $1/m_h^3$ terms in the kinetic energies,

$$h_s = \frac{\mathbf{p}^2}{m_h} + \frac{\mathbf{P}^2}{4m_h} - \frac{\mathbf{p}^4}{4m_h^3} + ... + V_s \ , \tag{8.11}$$

$$h_o = \frac{\mathbf{p}^2}{m_h} + \frac{\mathbf{P}^2}{4m_h} - \frac{\mathbf{p}^4}{4m_h^3} + ... + V_o \ . \tag{8.12}$$

Here,

- $\mathbf{R}$: the center of mass coordinate,
- $\mathbf{r}$: the relative coordinate,
- S: color singlet component of the quark and antiquark field and normalized to $S = \mathbb{1}_{3\times3} S/\sqrt{3}$,
- O: color octet component of the quark and antiquark field and normalized to $O = \sqrt{2}\ O^A T^A$,
- $iD_0 O = i\partial_0 O + g[A_0(\mathbf{R}, t), O]$: the covariant derivative acting on the octet field,
- $\mathbf{P} = -i\mathbf{D}_{\mathbf{R}}$: the center of mass momentum (gauge covariant),
- $\mathbf{p} = -i\nabla_{\mathbf{r}}$: the relative momentum,
- h_s: the Hamiltonian of the color singlet heavy quark-antiquark fields,
- h_o: the Hamiltonian of the color octet heavy quark-antiquark fields,
- the dots in Eqs. (8.11) and (8.12) stand for higher-order terms in the non-relativistic expansion of the kinetic energy,
- the trace in Eq. (8.10) is in spin and color space,
- $\mathcal{L}^l$: the Lagrangian of the light degrees of freedom (8.3),

- V_s, V_o, V_A, V_B: Wilson coefficients of pNRQCD,
- V_s: the color singlet potential and up to leading order $V_s^{(o)} = -\frac{4\alpha_s}{3r}$,
- V_o: the color octet potential and up to leading order $V_o^{(o)} = \frac{\alpha_s}{6r}$.

At leading order in the multipole expansion, the equation of motion for the singlet field is

$$i\partial_0 S = h_s S \ , \tag{8.13}$$

which is the Schrodinger equation that describes the evolution of a nonrelativistic bound state. We have the following power counting of the weakly-coupled pNRQCD:

- $r \sim \frac{1}{m_h v}$
- $t,\ R \sim \frac{1}{m_h v^2}$
- $p \sim m_h v$
- $P \sim m_h v^2$
- ultrasoft gluon and light quark fields $\sim m_h v^2$
- $H_{s,o}^{(o)} = \frac{\mathbf{p}}{m_h} + V_{s,o}^{(o)} \sim m_h v^2$

Strongly-coupled pNRQCD

If $\Lambda_{\rm QCD} \gtrsim m_h v^2$, then the bound states are strongly coupled and the corresponding EFT is called strongly-coupled pNRQCD. In this case, the distance is the order of $\frac{1}{m_h v^2}$ and color confinement has set in. Hence, the only available degrees of freedom are color singlets. Strongly-coupled pNRQCD is well attired to describe higher quarkonium states and quarkonium exotica. If one neglects quarkonia's couplings with light hadrons then the strongly-coupled Lagrangian density has the form [56]

$$\mathcal{L}_{\rm pNRQCD}^{\rm strong} = \int d^3 r \ {\rm Tr}\left\{ S^\dagger (i\partial_0 - h_s) S \right\}, \tag{8.14}$$

The Hamiltonian h_s has the same form as (8.11). The equation of motion is the Scrodinger equation (8.13).

Note that, if $m_h v \gg \Lambda_{\rm QCD} \gg m_h v^2$, the matching to pNRQCD may be done in two steps,

1. first integrating out (perturbatively) $m_h v$

2. then (non-perturbatively) $\Lambda_{\rm QCD}$.

pNRQCD observables will undergo automatic factorization to incorporate contributions originating from both of these scales.

8.2 Derivation of the Lindblad Equation

In this section, we expand the evolution equation obeyed by the heavy quarkonium reduced density matrix to include terms of order E/T and derive a Lindblad equation accurate at this order. In this way, we go beyond the analysis at leading order in E/T presented in refs. [30, 31, 39, 38].

8.2.1 The pNRQCD master equation

The evolution of the reduced density matrix of heavy quarkonium in the regime $Mv \gg T$ was derived in refs. [30, 31]. If one does not make any assumption on the relation between T and E, then the evolution equations that one obtains are

$$\frac{d\rho_s(t)}{dt} = -i[h_s, \rho_s(t)] - \Sigma_s \rho_s(t) - \rho_s(t)\Sigma_s^\dagger + \Xi_{so}(\rho_o(t)), \tag{8.15}$$

$$\frac{d\rho_o(t)}{dt} = -i[h_o, \rho_o(t)] - \Sigma_o \rho_o(t) - \rho_o(t)\Sigma_o^\dagger + \Xi_{os}(\rho_s(t)) + \Xi_{oo}(\rho_o(t)), \tag{8.16}$$

where, assuming the in medium correlators to be isotropic and time-translation invariant and observing the evolution for a time t larger than any other length scale in the problem,

$$\Sigma_s = r_i A_i^{so\dagger}, \tag{8.17}$$

$$\Sigma_o = \frac{1}{N_c^2-1} r_i A_i^{os\dagger} + \frac{N_c^2-4}{2(N_c^2-1)} r_i A_i^{oo\dagger}, \tag{8.18}$$

$$\Xi_{so}(\rho_o(t)) = \frac{1}{N_c^2-1}\left(A_i^{os\dagger}\rho_o(t) r_i + r_i \rho_o(t) A_i^{os}\right), \tag{8.19}$$

$$\Xi_{os}(\rho_s(t)) = A_i^{so\dagger}\rho_s(t) r_i + r_i \rho_s(t) A_i^{so}, \tag{8.20}$$

$$\Xi_{oo}(\rho_o(t)) = \frac{N_c^2-4}{2(N_c^2-1)}\left(A_i^{oo\dagger}\rho_o(t) r_i + r_i \rho_o(t) A_i^{oo}\right), \tag{8.21}$$

with

$$A_i^{uv} = \frac{g^2}{6N_c}\int_0^\infty \mathrm{d}s\, e^{-ih_u s} r_i e^{ih_v s}\langle \tilde{E}_j^a(0,\mathbf{0})\tilde{E}_j^a(s,\mathbf{0})\rangle, \tag{8.22}$$

$$\tilde{E}_i^a(s,\mathbf{0}) = \Omega(s)^\dagger E_i^a(s,\mathbf{0})\Omega(s), \tag{8.23}$$

$$\Omega(s) = \exp\left[-ig\int_{-\infty}^{s}\mathrm{d}s' A_0(s',\mathbf{0})\right]. \tag{8.24}$$

$h_{u,v}$ is the singlet or octet Hamiltonian: $h_{s,o} = \mathbf{p}^2/M + V_{s,o}$ where $V_s = -C_f\alpha_{\rm s}(1/a_0)/r$ and $V_o = \alpha_{\rm s}(1/a_0)/2N_c r$, with $C_f = (N_c^2-1)/2N_c = 4/3$ the Casimir of the fundamental representation, $N_c = 3$ the number of colors, and $\alpha_{\rm s}(1/a_0)$ the strong coupling evaluated at the inverse of the Bohr radius a_0. Details about the derivation of these results can be found in refs. [31, 32].

These equations can be written as the master equation

$$\frac{d\rho(t)}{dt} = -i\left[H,\rho(t)\right] + \sum_{nm} h_{nm}\left(L_i^n\rho(t)L_i^{m\dagger} - \frac{1}{2}\left\{L_i^{m\dagger}L_i^n,\rho(t)\right\}\right), \tag{8.25}$$

where

$$\rho(t) = \begin{pmatrix}\rho_s(t) & 0\\ 0 & \rho_o(t)\end{pmatrix}, \quad H = \begin{pmatrix}h_s + \mathrm{Im}(\Sigma_s) & 0\\ 0 & h_o + \mathrm{Im}(\Sigma_o)\end{pmatrix}, \tag{8.26}$$

$$L_i^0 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} r_i, \quad L_i^1 = \begin{pmatrix} 0 & 0 \\ 0 & \frac{N_c^2-4}{2(N_c^2-1)} A_i^{oo\dagger} \end{pmatrix}, \tag{8.27}$$

$$L_i^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} r_i, \quad L_i^3 = \begin{pmatrix} 0 & \frac{1}{N_c^2-1} A_i^{os\dagger} \\ A_i^{so\dagger} & 0 \end{pmatrix}, \tag{8.28}$$

and

$$h = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{8.29}$$

If the matrix h were a completely positive matrix, its eigenvalues would be positive and we could make a change of variables to bring eq. (8.25) into Lindblad form. However, as the matrix h is not completely positive definite, this manipulation is not possible. Any Markovian, trace preserving evolution that preserves complete positivity of the density matrix can be written as a Lindblad equation. As eq. (8.25) is Markovian and trace preserving but does not preserve complete positivity of the density matrix, it follows that eq. (8.25) allows for negative probabilities, a feature similar to the widely used Caldeira–Leggett master equation [**?**]. We note that complete positivity is necessary for use of the quantum trajectories method.

8.2.2 Lindblad equation to leading-order in E/T

Although, as explained above, the evolution specified in eq. (8.25) does not, in general, take the form of a Lindblad equation, in specific temperature regimes the operators characterizing the medium interaction simplify, and the resulting evolution equation can be rewrtitten in Lindblad form. We consider the case $T \gg E$. In eq. (8.22), the thermal chromoelectric correlator decays rapidly at times $s > 1/T$, while the quarkonium Hamiltonian

$h_{s,o}$ scales as the binding energy E. The arguments of the exponentials in A_i^{uv} thus scale like E/T. In the limit $T \gg E$, the exponentials can be set to 1, and A_i^{uv} can be written in terms of the transport coefficients κ and γ

$$A_i^{uv} = \frac{r_i}{2}\left(\kappa - i\gamma\right) + \cdots, \tag{8.30}$$

where the ellipsis indicates terms of order E/T and higher; κ is the heavy quarkonium momentum diffusion coefficient, and γ is its dispersive counterpart

$$\kappa = \frac{g^2}{6N_c}\int_0^\infty \mathrm{d}s \left\langle \left\{ \tilde{E}_i^a(s,\mathbf{0}), \tilde{E}_i^a(0,\mathbf{0}) \right\} \right\rangle, \tag{8.31}$$

$$\gamma = -i\frac{g^2}{6N_c}\int_0^\infty \mathrm{d}s \left\langle \left[\tilde{E}_i^a(s,\mathbf{0}), \tilde{E}_i^a(0,\mathbf{0}) \right] \right\rangle. \tag{8.32}$$

At leading order in E/T, the L^n are not linearly independent; as a consequence, there exists a rotation of the vector (L^0, L^1, L^2, L^3) that makes it orthogonal to the eigenspaces of the negative eigenvalues of the matrix h_{nm} (see appendix **??** for details). This allows to write the master equation in Lindblad form

$$\frac{d\rho(t)}{dt} = -i\left[H, \rho(t)\right] + \sum_n \left(C_i^n \rho(t) C_i^{n\dagger} - \frac{1}{2}\left\{ C_i^{n\dagger} C_i^n, \rho(t) \right\} \right), \tag{8.33}$$

with Hamiltonian

$$H = \begin{pmatrix} h_s + \mathrm{Im}(\Sigma_s) & 0 \\ 0 & h_o + \mathrm{Im}(\Sigma_o) \end{pmatrix}, \tag{8.34}$$

where

$$\mathrm{Im}\left(\Sigma_s\right) = \frac{r^2}{2}\gamma, \quad \mathrm{Im}\left(\Sigma_o\right) = \frac{N_c^2 - 2}{2(N_c^2 - 1)}\frac{r^2}{2}\gamma, \tag{8.35}$$

and collapse operators

$$C_i^0 = \sqrt{\frac{\kappa}{N_c^2 - 1}} r_i \begin{pmatrix} 0 & 1 \\ \sqrt{N_c^2 - 1} & 0 \end{pmatrix}, \tag{8.36}$$

$$C_i^1 = \sqrt{\frac{\kappa(N_c^2 - 4)}{2(N_c^2 - 1)}} r_i \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{8.37}$$

This equation was first derived and solved in refs. [30, 31] and further studied in refs. [39, 38] using the `QTraj` code of ref. [88].

8.2.3 Lindblad equation to next-to-leading order in E/T

In the previous subsections, we presented the master equation governing the in-medium evolution of Coulombic quarkonium and showed that in the $T \gg E$ limit it takes the form of a Lindblad equation. In this subsection, we consider contributions to A_i^{uv} linear in E/T and investigate the conditions under which a Lindblad equation can still be derived. Making use of the relation

$$i\frac{g^2}{6N_c}\int_0^\infty \mathrm{d}t\, t \left\langle \tilde{E}_i^a(t,\mathbf{0})\tilde{E}_i^a(0,\mathbf{0})\right\rangle = \frac{\kappa}{4T}, \tag{8.38}$$

which follows from the fluctuation-dissipation theorem (see appendix A of [36] for a detailed derivation), we obtain corrections to eqs. (8.35)-(8.37) of order E/T by expanding the exponentials of eq. (8.22) and retaining terms up to order $h_{u,v}$. This results in A_i^{uv} taking the form

$$A_i^{uv} = \frac{r_i}{2}(\kappa - i\gamma) + \kappa\left(-\frac{ip_i}{2MT} + \frac{\Delta V_{uv}}{4T}r_i\right) + \cdots, \tag{8.39}$$

where $\Delta V_{uv} = V_u - V_v$ is the difference of the u and v (singlet or octet) potentials and the ellipsis indicates terms of order $(E/T)^2$ and higher. With these contributions to A_i^{uv}, the operators L_i^1 and L_i^3 in the master equation take the form

$$L_i^1 = \frac{N_c^2-4}{2(N_c^2-1)}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\left[\frac{r_i}{2}(\kappa + i\gamma) + \kappa\frac{ip_i}{2MT}\right], \tag{8.40}$$

$$\begin{aligned} L_i^3 = &\begin{pmatrix} 0 & \frac{1}{N_c^2-1} \\ 0 & 0 \end{pmatrix}\left[\frac{r_i}{2}(\kappa + i\gamma) + \kappa\left(\frac{ip_i}{2MT} + \frac{\Delta V_{os}}{4T}r_i\right)\right] \\ &+ \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\left[\frac{r_i}{2}(\kappa + i\gamma) + \kappa\left(\frac{ip_i}{2MT} + \frac{\Delta V_{so}}{4T}r_i\right)\right], \end{aligned} \tag{8.41}$$

while L_i^0 and L_i^2 keep the form given in eqs. (8.27) and (8.28), respectively.

In contrast to the case at order $(E/T)^0$ considered in subsection 8.2.2, the above master equation cannot be written in Lindblad form as the L_i^n are linearly independent. However, the vector (L^0, L^1, L^2, L^3) can be rotated in such a way that only components of order E/T project on the eigenspaces of the negative eigenvalues of the matrix h_{nm}. Discarding these components amounts to neglecting terms in the evolution equation of order $(E/T)^2$. Hence, we can still write a Lindblad equation that is accurate at order E/T

$$\frac{d\rho(t)}{dt} = -i\left[H, \rho(t)\right] + \sum_{n=0}^{1}\left(C_i^n \rho(t) C_i^{n\dagger} - \frac{1}{2}\left\{C_i^{n\dagger} C_i^n, \rho(t)\right\}\right), \tag{8.42}$$

with Hamiltonian

$$H = \begin{pmatrix} h_s + \mathrm{Im}(\Sigma_s) & 0 \\ 0 & h_o + \mathrm{Im}(\Sigma_o) \end{pmatrix}, \tag{8.43}$$

where

$$\mathrm{Im}\left(\Sigma_s\right) = \frac{r^2}{2}\gamma + \frac{\kappa}{4MT}\{r_i, p_i\}, \quad \mathrm{Im}\left(\Sigma_o\right) = \frac{N_c^2 - 2}{2(N_c^2 - 1)}\left(\frac{r^2}{2}\gamma + \frac{\kappa}{4MT}\{r_i, p_i\}\right), \tag{8.44}$$

and collapse operators

$$\begin{aligned} C_i^0 =& \sqrt{\frac{\kappa}{N_c^2 - 1}} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \left(r_i + \frac{ip_i}{2MT} + \frac{\Delta V_{os}}{4T} r_i\right) \\ &+ \sqrt{\kappa} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \left(r_i + \frac{ip_i}{2MT} + \frac{\Delta V_{so}}{4T} r_i\right), \end{aligned} \tag{8.45}$$

$$C_i^1 = \sqrt{\frac{\kappa(N_c^2 - 4)}{2(N_c^2 - 1)}} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \left(r_i + \frac{ip_i}{2MT}\right). \tag{8.46}$$

At the price of introducing some spurious terms at order $(E/T)^2$, we have derived a Lindblad equation accurate up to order E/T and can thus make use of the quantum trajectories algorithm to solve it. We can understand the anti-commutator term $\{r_i, p_i\}$ multiplying κ

in $\mathrm{Im}\left(\Sigma_{s,o}\right)$ as a drag force. This identification is clear if one uses the procedure described in ref. [28] to obtain a Langevin equation from eq. (8.42).

8.3 The quantum trajectories algorithm at NLO

The Lindblad equation given in eq. (8.42) with the collapse operators of eqs. (8.45) and (8.46) describes the evolution of the heavy quarkonium reduced density matrix at next-to-leading-order accuracy in the E/T expansion. To solve it numerically, we make use of the quantum trajectories algorithm used to solve the LO evolution equations in refs. [38, 39, 88]. In the quantum trajectories approach, one evolves a large set of independently sampled quantum evolutions of the wave function. Observables, e.g., the overlap with vacuum eigenstates, are computed along each sampled quantum trajectory and averaged to obtain the final predictions for the observable in question.

Here we make use of the "waiting time approach" in which one evolves the wave function using the complex effective Hamiltonian until the norm squared of the wave function falls below a random number p_1 uniformly sampled in $[0, 1]$, at which point additional random numbers are generated to determine the outgoing quantum numbers and corresponding collapse operator to apply to the wave function [88]. After application of the selected collapse operator, the wave function is normalized, and the process is repeated until the simulation end time is reached. This method has the benefit that between quantum jumps the evolution proceeds with a fixed color state labeled by c and fixed angular momentum labeled by l. For the description below, we note that as the system is isotropic we can write the Hamiltonian and collapse operators more compactly by using the reduced wave function $u(t, r) \equiv rR(t, r)$, where the three-dimensional wave function for fixed l and m is given by $\psi(t, \mathbf{r}) = R(t, r)Y_{lm}(\theta, \phi)$. In subsections 8.3.1, 8.3.2, and 8.3.3, we give the forms of the collapse operators, widths, and in-medium Hamiltonian acting on the reduced wave function; we refer to quantities acting on the reduced wave function as being in the *reduced spherical*

representation and denote them with an overbar. In subsection 8.3.4, we give the quantum trajectories algorithm as implemented to obtain the phenomenological results in sec. 5.3. In subsection 8.3.5, we describe the procedure used to evolve the wave function between quantum jumps.

8.3.1 Jump operators in the reduced spherical representation

There are six jump operators present in the Lindblad equation. They represent the six possible physical transitions for the quarkonium state inside the QGP. A color singlet state can transition into an octet and vice versa; in addition, the octet state can also transition to another octet state. For each color transition, we have the two additional possibilities of an angular momentum jump up ($l \to l+1$) or down ($l \to l-1$), thus totalling six. When expressed as operators acting on the reduced wave function, the six jump operators are

$$\overline{C}^{\uparrow}_{s\to o} = r - \frac{N_c\alpha_{\rm s}}{8T} + \frac{1}{2MT}\left(\frac{\partial}{\partial r} - \frac{l+1}{r}\right), \tag{8.47}$$

$$\overline{C}^{\downarrow}_{s\to o} = r - \frac{N_c\alpha_{\rm s}}{8T} + \frac{1}{2MT}\left(\frac{\partial}{\partial r} + \frac{l}{r}\right), \tag{8.48}$$

$$\overline{C}^{\uparrow}_{o\to s} = r + \frac{N_c\alpha_{\rm s}}{8T} + \frac{1}{2MT}\left(\frac{\partial}{\partial r} - \frac{l+1}{r}\right), \tag{8.49}$$

$$\overline{C}^{\downarrow}_{o\to s} = r + \frac{N_c\alpha_{\rm s}}{8T} + \frac{1}{2MT}\left(\frac{\partial}{\partial r} + \frac{l}{r}\right), \tag{8.50}$$

$$\overline{C}^{\uparrow}_{o\to o} = r + \frac{1}{2MT}\left(\frac{\partial}{\partial r} - \frac{l+1}{r}\right), \tag{8.51}$$

$$\overline{C}^{\downarrow}_{o\to o} = r + \frac{1}{2MT}\left(\frac{\partial}{\partial r} + \frac{l}{r}\right). \tag{8.52}$$

Since the wave function is normalized after application, the normalization of these operators is arbitrary. In the code, we multiply all of these operators by a factor of T so that they are regular when $T = 0$.

8.3.2 Width operators in the reduced spherical representation

When expressed as operators acting on the reduced wave function, the octet-singlet and octet-octet width operators take the form

$$\overline{\Gamma}^{\uparrow}_{o\to s} = \frac{\hat{\kappa}T^3}{N_c^2-1}\frac{l+1}{2l+1}\left[\left(r+\frac{N_c\alpha_{\rm s}}{8T}\right)^2 - \frac{3}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} - \frac{1}{2MT}\left(\frac{N_c\alpha_{\rm s}}{4T}\right)\frac{1}{r}\right], \tag{8.53}$$

$$\overline{\Gamma}^{\downarrow}_{o\to s} = \frac{l}{l+1}\overline{\Gamma}^{\uparrow}_{o\to s}\,, \tag{8.54}$$

$$\overline{\Gamma}^{\uparrow}_{o\to o} = \hat{\kappa}T^3\frac{N_c^2-4}{2(N_c^2-1)}\frac{l+1}{2l+1}\left[r^2 - \frac{3}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2}\right], \tag{8.55}$$

$$\overline{\Gamma}^{\downarrow}_{o\to o} = \frac{l}{l+1}\overline{\Gamma}^{\uparrow}_{o\to o}\,, \tag{8.56}$$

with

$$\overline{\mathcal{D}}^2 = -\frac{\partial^2}{\partial r^2} + \frac{l(l+1)}{r^2}\,. \tag{8.57}$$

We do not list the singlet-octet width operator because singlet states must always transition to an octet state and the outgoing angular momentum quantum number can be chosen based on the fact that, similar to eq. (8.56), the up and down transitions are related by $\overline{\Gamma}^{\downarrow}_{s\to o}/\overline{\Gamma}^{\uparrow}_{s\to o} = l/(l+1)$.

8.3.3 Effective Hamiltonian in the reduced spherical representation

The effective Hamiltonian for singlet and octet evolution is defined by $H^{\rm eff}_{s,o} = h_{s,o} + {\rm Im}(\Sigma_{s,o}) - i\Gamma_{s,o}/2$ with $\Gamma_s = \sum_{i\in\{\uparrow,\downarrow\}}\Gamma^i_{s\to o}$ and $\Gamma_o = \sum_{i\in\{\uparrow,\downarrow\}}(\Gamma^i_{o\to s} + \Gamma^i_{o\to o})$. When expressed as operators acting on the reduced wave function, the singlet effective Hamiltonian $\overline{H}^{\rm eff}_s$ is given by

$${\rm Re}[\overline{H}^{\rm eff}_s] = \frac{\overline{\mathcal{D}}^2}{M} - \frac{C_f\,\alpha_{\rm s}}{r} + \frac{\hat{\gamma}T^3}{2}r^2 + \frac{\hat{\kappa}T^2}{4M}\{r, p_r\}\,, \tag{8.58}$$

$${\rm Im}[\overline{H}^{\rm eff}_s] = -\frac{\hat{\kappa}T^3}{2}\left[\left(r-\frac{N_c\alpha_{\rm s}}{8T}\right)^2 - \frac{3}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} + \frac{1}{2MT}\left(\frac{N_c\alpha_{\rm s}}{4T}\right)\frac{1}{r}\right], \tag{8.59}$$

where $p_r = -i\partial_r$. Similarly, the octet effective Hamiltonian $\overline{H}_o^{\rm eff}$ is given by

$$\mathrm{Re}[\overline{H}_o^{\rm eff}] = \frac{\overline{\mathcal{D}}^2}{M} + \frac{1}{2N_c}\frac{\alpha_{\rm s}}{r} + \frac{N_c^2-2}{2(N_c^2-1)}\left[\frac{\hat{\gamma}T^3}{2}r^2 + \frac{\hat{\kappa}T^2}{4M}\{r,p_r\}\right], \tag{8.60}$$

$$\begin{aligned}\mathrm{Im}[\overline{H}_o^{\rm eff}] = &-\frac{\hat{\kappa}T^3}{2(N_c^2-1)}\left[\left(r+\frac{N_c\alpha_{\rm s}}{8T}\right)^2 - \frac{3}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2} - \frac{1}{2MT}\left(\frac{N_c\alpha_{\rm s}}{4T}\right)\frac{1}{r}\right] \\ &-\frac{\hat{\kappa}T^3}{4(N_c^2-1)}\left[r^2 - \frac{3}{2MT} + \frac{\overline{\mathcal{D}}^2}{(2MT)^2}\right],\end{aligned} \tag{8.61}$$

where $\hat{\kappa} = \kappa/T^3$ and $\hat{\gamma} = \gamma/T^3$.

8.3.4 Description of the algorithm

The NLO quantum trajectories algorithm proceeds as follows:

1. Initialize the reduced wave function $u(t_0, r) = u_0(r)$ at N points which are uniformly distributed on $(0, L)$ with L being the box size. The wave function is assumed to obey Dirichlet boundary conditions at the boundaries with $u(t, 0) = u(t, L) = 0$ at all times. Additionally, we specify the initial color state $c \in \{0, 1\}$, with 0 being a color singlet and 1 being a color octet state, and the initial orbital angular momentum quantum number $l \geq 0$.

2. Normalize the wave function such that $|u|^2 = \langle u|u\rangle = 1$.

3. Generate a random number p_1 uniformly distributed in $[0, 1]$.

4. Using $\overline{H}_{s,o}^{\rm eff}$ specified in eqs. (8.58)-(8.61), evolve the wave function in time until the jump time t_j. A quantum jump is triggered by the norm squared of the wave function falling below p_1 at time t_j, i.e., $|u(t_j, r)|^2 < p_1$. If the maximum evolution time is reached before a jump is triggered, i.e., $t_j > t_f$, terminate execution at time t_f, otherwise, perform a quantum jump at time t_j by proceeding to step 5.

5. Initiate a *quantum jump* by generating two additional random numbers p_2 and p_3

uniformly distributed in $[0, 1]$. These numbers will be used to select new angular momentum and color states.

6. If the color state is singlet ($c = 0$):

   (a) Set the color state to octet, i.e. $c \to 1$.

   (b) If $p_2 < l/(2l+1)$, $l \to l - 1$, else $l \to l + 1$.

   Else ($c = 1$):

   (a) Evaluate the transition amplitudes $\overline{\Gamma}_1 = \langle \overline{\Gamma}^{\downarrow}_{o\to o} \rangle$, $\overline{\Gamma}_2 = \langle \overline{\Gamma}^{\uparrow}_{o\to o} \rangle$, $\overline{\Gamma}_3 = \langle \overline{\Gamma}^{\downarrow}_{o\to s} \rangle$, and $\overline{\Gamma}_4 = \langle \overline{\Gamma}^{\uparrow}_{o\to s} \rangle$ using the operators listed in eqs. (8.53)-(8.56) above. From these, obtain the normalized amplitudes $\tilde{\Gamma}_i = \overline{\Gamma}_i / \sum_{j=1}^{4} \overline{\Gamma}_j$. The total probability for a downward angular momentum transition is given by $p_\downarrow = \tilde{\Gamma}_1 + \tilde{\Gamma}_3$, and the total probability for an octet-singlet transition is given by $p_{\mathrm{o}\to\mathrm{s}} = \tilde{\Gamma}_3 + \tilde{\Gamma}_4$.

   (b) If $p_2 < p_\downarrow$, $l \to l - 1$, else $l \to l + 1$.

   (c) If $p_3 < p_{\mathrm{o}\to\mathrm{s}}$, $c \to 0$.

7. Based on the choices made above, apply the appropriate collapse/jump operator to the wavefunction $u(r) \to \overline{C} u(r)$ with $\overline{C} \in \{\overline{C}^{\downarrow}_{s\to o}, \overline{C}^{\uparrow}_{s\to o}, \overline{C}^{\downarrow}_{o\to s}, \overline{C}^{\uparrow}_{o\to s}, \overline{C}^{\downarrow}_{o\to o}, \overline{C}^{\uparrow}_{o\to o}\}$ given in eqs. (8.47)-(8.52) above.

8. Repeat starting from step 2.

8.3.5 Evolution between jumps

In step 4 above, we must evolve the wave function forward in time using the complex effective Hamiltonian $\overline{H}^{\mathrm{eff}}_{s,o}$. Because of the appearance of the anticommutator term $\{r, p_r\}$, it is not easy to straightforwardly apply the same split-step pseudospectral method (Suzuki–Trotter) as was used in ref. [88]. Instead, here we use the Crank–Nicolson method to update

the wave function between quantum jumps. With this method, one approximates the infinitesimal time evolution operator as

$$e^{-i\overline{H}^{\text{eff}}_{s,o}\Delta t} \simeq \frac{1 - i\Delta t\overline{H}^{\text{eff}}_{s,o}/2}{1 + i\Delta t\overline{H}^{\text{eff}}_{s,o}/2}\,, \tag{8.62}$$

where the Hamiltonian has implicit time dependence. To proceed, one expresses this as

$$\left(1 + \frac{i}{2}\Delta t\overline{H}^{\text{eff}}_{s,o}\right)\psi(t + \Delta t) = \left(1 - \frac{i}{2}\Delta t\overline{H}^{\text{eff}}_{s,o}\right)\psi(t)\,. \tag{8.63}$$

To evolve the wave function one step forward in time, one applies the terms on the right to obtain the "half updated" wave function $X \equiv \left(1 - i\Delta t\overline{H}^{\text{eff}}_{s,o}/2\right)\psi(t)$. The final step in the Crank–Nicolson algorithm is to solve

$$A_{ij}\psi_j(t + \Delta t) = X_i\,, \tag{8.64}$$

for $\psi_j(t+\Delta t)$ where i, j are spatial coordinates and $A_{ij} = \left(1 + i\Delta t\overline{H}^{\text{eff}}_{s,o}/2\right)_{ij}$. Typically A_{ij} is tridiagonal, in which case there are many optimized solvers. For the results below, we make use of the optimized sparse matrix solver `spsolve` provided by the open-source Armadillo package [20]. The Armadillo package is also used to handle all matrix multiplications required to compute the action of the effective Hamiltonian, width, and jump operators.

8.4 Phenomenological Results

We performed numerical simulations with and without quantum jumps for $\hat{\kappa} \in \{2, 3, 4, 5\}$ and $\hat{\gamma} \in \{-3.5, -2.6, 0, 1\}$. For the background evolution, we made use of previously generated (3+1)-D anisotropic hydrodynamics backgrounds [18] and initialized the QTraj Lindblad equation solver at $\tau_{\text{med}} = 0.6$ fm. The sampling of initial production points and transverse momenta was performed in the same manner as our prior works [39, 38, 36]. We did not include initial state energy loss or other cold nuclear matter effects.

In all cases discussed herein, we generated approximately 100,000 - 200,0000 physical trajectories. When including quantum jumps, we generated approximately 30 quantum trajectories per physical trajectory. We initialized the quantum wave function as a highly-localized Gaussian delta function [39, 38, 36] and terminated the evolution along each physical trajectory when the temperature dropped below $T_F = 190$ MeV. We computed the survival probabilities of the $\Upsilon(1S, 2S, 3S)$ and $\chi_b(1P, 2P)$ states by taking the ratio of the final and initial overlap probabilities obtained using vacuum Coulomb eigenstates. We then took into account final state feed down by applying a feed down matrix constructed from data available from the Particle Data Group [116]. The feed down matrix used was the same as in our prior works [39, 38, 36].

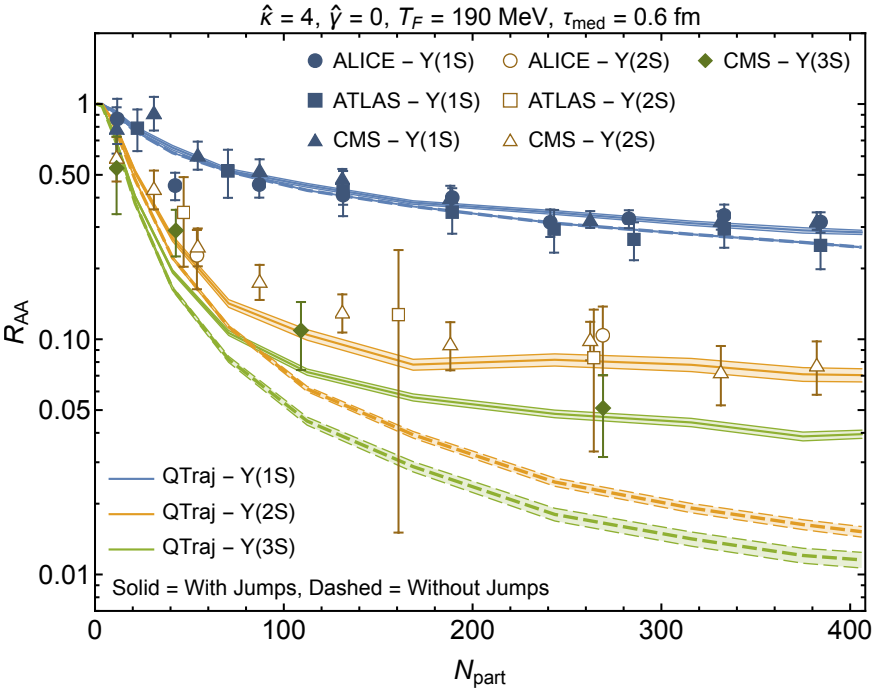


Figure 8.2: The nuclear suppression factor R_{AA} of bottomonia states as a function of N_{part}. The solid and dashed lines indicate results obtained with and without quantum jumps, respectively. The experimental measurements shown are from the ALICE [9], ATLAS [104], and CMS [100, 46] Collaborations.

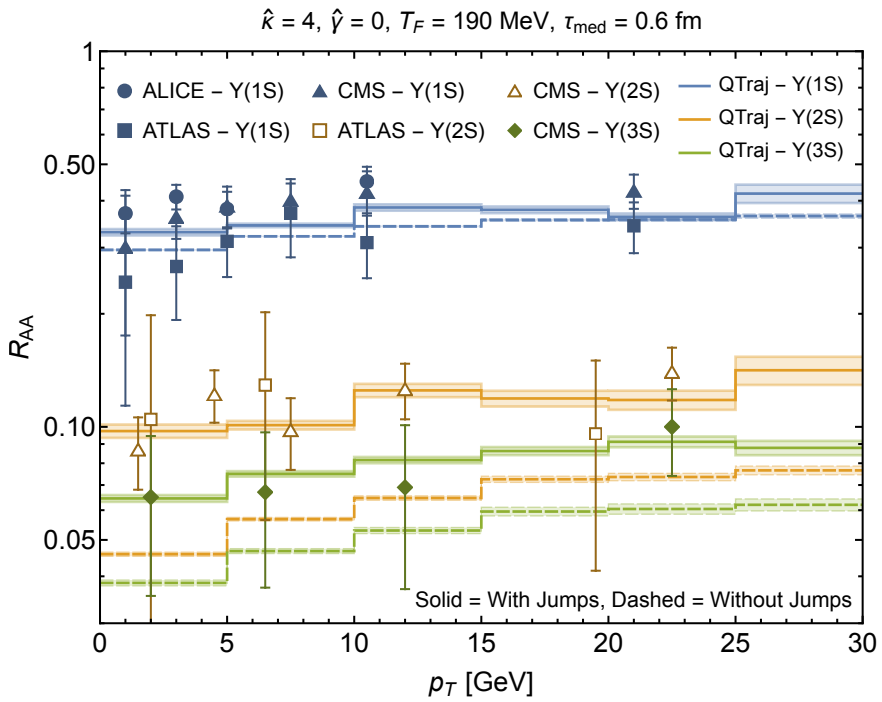


Figure 8.3: R_{AA} of bottomonia states as a function of p_T. Line styles and experimental data sources are the same as Fig. 8.2.

We analyzed the agreement of the resulting NLO QTraj predictions for $R_{AA}(N_{\text{part}})$ with data available from the ALICE, ATLAS, and CMS experiments and found that, without including quantum jumps, it was impossible to simultaneously describe the suppression of the $\Upsilon(1S)$ and $\Upsilon(2S)$ states. Contrarily, when quantum jumps were included, we found that it was possible to quantitatively describe the suppression of both states. In addition, we found that this allowed for a good description of the suppression of the $\Upsilon(3S)$. For details concerning the measure we used to assess agreement with experimental data, we refer the reader to the supplemental material associated with this Letter. Therein, we present plots of the agreement measure as a function of $\hat{\kappa}$ and $\hat{\gamma}$. From our analysis, we concluded that the best description of the data was obtained using the values $\hat{\kappa} \approx 4$ and $\hat{\gamma} \approx 0$ from our original set of simulations. In all figures, we present results obtained with these values of $\hat{\kappa}$ and $\hat{\gamma}$ and assess the effect of including quantum regeneration.

In Figs. 8.2 and 8.3, we present a comparison of the NLO QTraj results and experimental data as functions of $N_{\rm part}$ and transverse momentum, respectively. The vertical axis in both plots is logarithmic in order to better resolve the suppression of the excited states. In these figures and all subsequent figures, the solid and dashed lines correspond to the results obtained with and without quantum jumps, respectively. The shaded bands indicate the statistical uncertainty associated with the average over quantum and physical trajectories. As can be seen from Figs. 8.2 and 8.3, without quantum jumps included, one under predicts R_{AA} for excited states. This is not particular to the case of $\hat{\kappa} = 4$ and $\hat{\gamma} = 0$ and was observed in our prior simulations in which a temperature-dependent $\hat{\kappa}$ was used [39, 38, 36, 15]. When quantum jumps are included, we find excellent agreement with the experimentally measured suppression of both the ground and excited states. For the fully integrated $R_{AA}[1S, 2S, 3S]$, we obtain 0.3598 ± 0.0028, 0.1118 ± 0.0019, and 0.0775 ± 0.0007, respectively.

In Figs. 8.4 - 8.6, we compare our predictions for the 2S to 1S, 3S to 1S, and 3S to 2S double ratios defined via $(n_{AA}[nS]/n_{AA}[n'S])/(n_{pp}[nS]/n_{pp}[n'S])$ as a function of $N_{\rm part}$. In the case of the CMS data, for the 2S to 1S double ratio, we inferred this observable by taking the ratio of the reported $R_{AA}[nS]$ to $R_{AA}[n'S]$ data as no update of the double ratio was reported in Ref. [46]. As a result, for this data set, we do not report the statistical and systematic uncertainties separately. For the ATLAS 2S to 1S double ratio, the black and red error bars correspond to statistical and systematic uncertainties, respectively. In the case of the CMS 3S to 1S double ratio, we inferred this observable from the 3S to 2S double ratio reported by CMS (shown in Fig. 8.6) and the inferred 2S to 1S double ratio presented in Fig. 8.4. For the ATLAS 3S to 1S double ratio, we use their reported combined 2S+3S to 1S double ratio [100]. As Figs. 8.4 - 8.6 demonstrate, this observable is only well explained when quantum regeneration is included. Without quantum regeneration, the 2S to 1S and 3S to 1S double ratios are under predicted and the 3S to 2S double ratio is slightly over

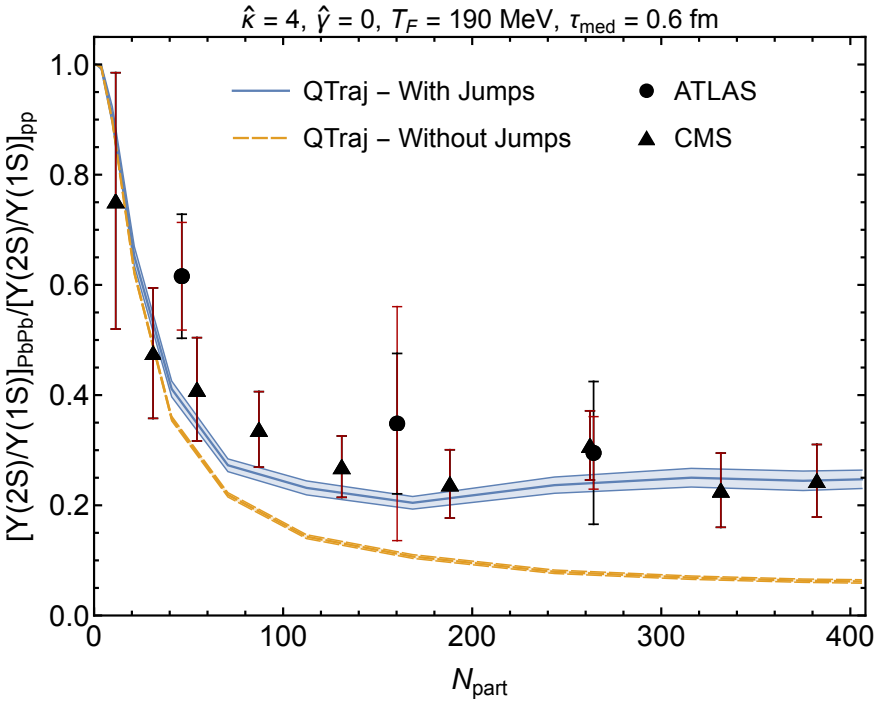


Figure 8.4: The 2S/1S double ratio as a function of $N_{\rm part}$. The solid blue lines and dashed orange lines show the QTraj results with and without quantum jumps, respectively. The experimental measurements shown are from the ATLAS [104] and CMS [100, 46] Collaborations.

predicted.

Finally, in Figs. 8.7 and 8.8, we present predictions for the 2S to 1S and 3S to 2S double ratios as a function of p_T. The data for the 2S to 1S double ratio labeled 'CMS (2023)' were inferred from the reported p_T dependence of $R_{AA}[1S]$ and $R_{AA}[2S]$. For the ATLAS and 'CMS (2019)' data, the Collaborations reported their computed 2S to 1S double ratio directly. As can be seen from these two figures, without jumps, these double ratios are in poorer agreement with the data, again indicating the importance of quantum regeneration for describing the suppression of excited states in heavy-ion collisions. We note, however, that both with and without jumps our NLO QTraj simulations predict a weak dependence of these double ratios on p_T.

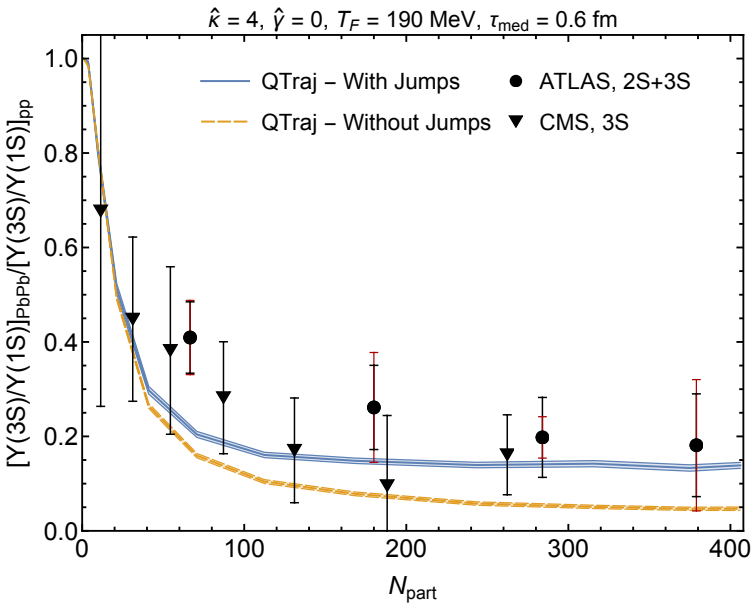


Figure 8.5: The 3S/1S double ratio as a function of $N_{\rm part}$. Line styles and experimental data sources are the same as Fig. 8.4.

8.5 Conclusion

We have studied bottomonium suppression by combining pNRQCD with an open quantum system framework. Working in the regime $Mv \gg T \gg E$, we have gone beyond prior studies [39, 38] by deriving and numerically solving a Lindblad-type evolution equation that is accurate to NLO in E/T. The goal of extending prior calculations to NLO in E/T was (1) to allow the pNRQCD+OQS framework to be applied at lower temperatures than is permitted by a LO truncation and (2) to include terms which are necessary to describe the approach of the system to equilibrium at late times. On the first point, we have shown that when going from LO to NLO there is a sizable correction to the singlet decay width for $1S$ bottomonium and smaller corrections for the $2S$ and $3S$ bottomonium states. The size of the corrections for the $1S$, in particular, stems from the fact that for the $1S$ state E/T is not small at phenomenologically relevant temperatures. Based on our findings in sec. **??**, the inclusion of NLO corrections has allowed us to extend the pNRQCD+OQS treatment down to temperatures in the vicinity of the QCD phase transition; in practice, we have lowered

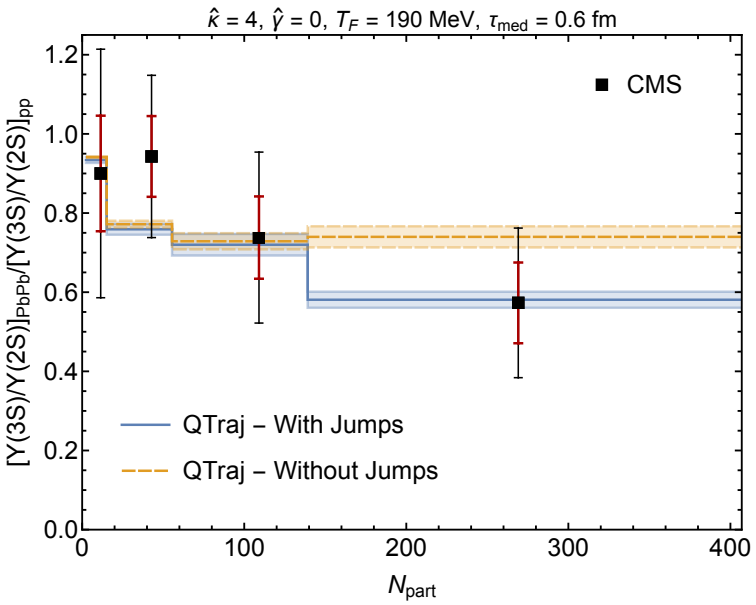


Figure 8.6: The 3S/2S double ratio as a function of $N_{\rm part}$. Line styles are the same as Fig. 8.4. The centrality classes used were 0-30, 30-50, 50-70, and 70-90%. Experimental data are from Ref. [46].

the decoupling temperature from $T_f = 250$ MeV at LO to $T_f = 190$ MeV at NLO. We have shown that the inclusion of quantum regeneration allows for a quite satisfactory description of the world's collected data on suppression of bottomonium ground and excited states in a QGP. Leaving out quantum regeneration considerably drifts predictions away from the data. These results confirm in a fully quantum and rigorous QCD setting semiclassical studies of singlet-octet transitions [27, 118, 117, 119, 83]. We can thus evolve the singlet and octet density matrix considering the effect of medium-induced color and angular momentum transitions between the states. Looking to the future, it will be interesting to examine the effect of temperature dependence in the heavy-quarkonium transport coefficients, the effect of fluctuating hydrodynamical backgrounds [15], and the possibility of further improvements to the underlying pNRQCD treatment.

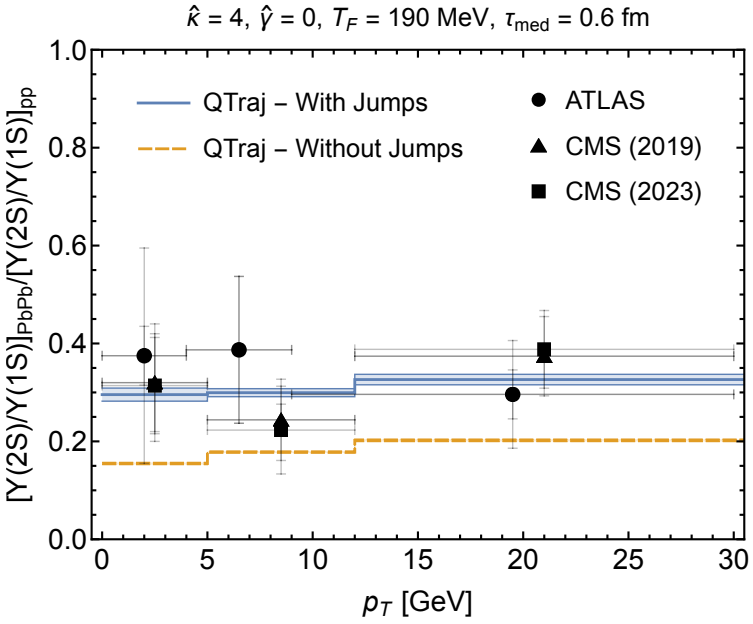


Figure 8.7: The 2S/1S double ratio as a function of p_T. Line styles are the same as Fig. 8.4. The experimental data were inferred from the results of Refs. [100, 46].

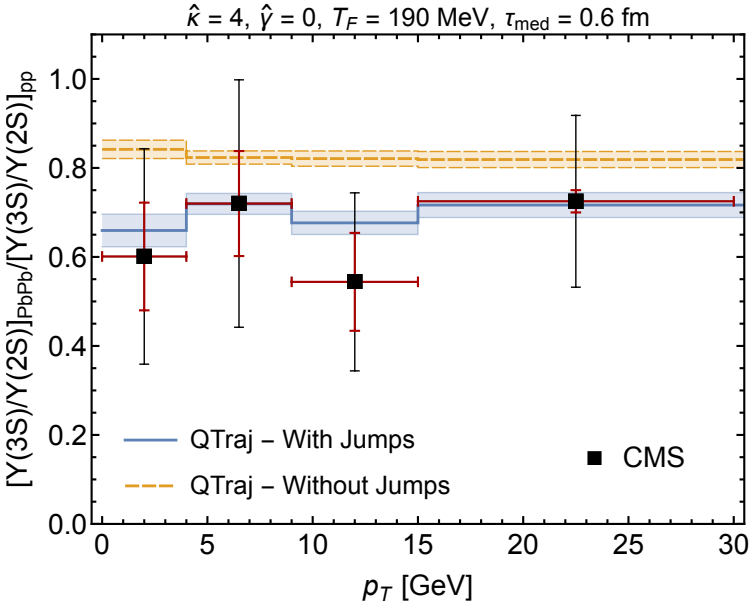


Figure 8.8: The 3S/2S double ratio as a function of p_T. Line styles are the same as Fig. 8.4. The experimental data are from Ref. [46].

Appendix A

SU(N) Generators

(N^2-1) generators of $SU(N)$ group in the defining representation denoted by t^a . The generators satisfy:

$$\left[t^a, t^b\right] = i f_{abc} t^c \ , \quad a, b, c = 1, 2, 3, ..., (N^2-1) \ . \tag{A.1}$$

We also have the following relations:

$$Tr\ (t^a) = 0 \ , \tag{A.2}$$

$$Tr\ (t^a t^b) = \frac{1}{2}\delta_{ab} \ , \tag{A.3}$$

$$t^a t^a = C_F I \ . \tag{A.4}$$

Hence,

$$\begin{aligned} Tr\ (C_F I) &= Tr\ (t^a t^a) \\ \Rightarrow C_F Tr\ (I) &= \frac{1}{2}\delta_{aa} \ , \quad \text{using } Tr\ (t^a t^b) = \frac{1}{2}\delta_{ab} \\ \Rightarrow C_F N &= \frac{1}{2}\left(N^2-1\right) \ , \quad \text{using } Tr\ (I) = N \ , \ \ \delta_{aa} = \left(N^2-1\right) \\ \Rightarrow \boxed{C_F = \frac{N^2-1}{2N}} & \end{aligned} \tag{A.5}$$

An important identity is:

$$\boxed{t^a_{ij} t^a_{kl} = \frac{1}{2}\left(\delta_{il}\delta_{jk} - \frac{1}{N}\delta_{ij}\delta_{kl}\right) \ , \quad i, j, k, l = 1, 2, ..., N} \ . \tag{A.6}$$

Appendix B

Derivation of the $f(\xi)$-function

In anisotropic hydrodynamics (aHydro), the evolution equation for a general moment can be written as [106]

$$\mathcal{M}^{nm}_{\rm aHydro}(\tau) = \frac{\Gamma(n+2m+2)\Lambda^{n+2m+2}(\tau)}{(2\pi)^2}\mathcal{H}^{nm}\big(\alpha(\tau)\big)\,, \tag{B.1}$$

where in case of Bjorken flow, the Minkowski-space components of the four-flow are $u^\mu = (t/\tau, 0, 0, z/\tau)$, with $\tau = \sqrt{t^2 - z^2}$ is the longitudinal proper-time.

Here,

$$\alpha(\tau) = 1/\sqrt{1+\xi(\tau)}\,, \tag{B.2}$$

and

$$\mathcal{H}^{nm}(y) = \tfrac{2y^{2m+1}}{2m+1}\,{}_2F_1(\tfrac{1}{2}+m, \tfrac{1-n}{2}; \tfrac{3}{2}+m; 1-y^2)\,, \tag{B.3}$$

with ${}_2F_1$ being hypergeometric function.

In equilibrium, we have

$$\mathcal{M}^{nm}_{\rm eq}(\tau) = \frac{\Gamma(n+2m+2)T^{n+2m+2}(\tau)}{2\pi^2(2m+1)}\,. \tag{B.4}$$

Now the scaled moments is given by [106]

$$\begin{aligned}\overline{\mathcal{M}}^{nm}_{\rm aHydro}(\tau) &= \frac{\mathcal{M}^{nm}_{\rm aHydro}(\tau)}{\mathcal{M}^{nm}_{\rm eq}(\tau)} \\ &= 2^{(n+2m-2)/4}(2m+1)\frac{\mathcal{H}^{nm}(\alpha)}{[\mathcal{H}^{20}(\alpha)]^{(n+2m+2)/4}}\,.\end{aligned} \tag{B.5}$$

We also have the isotropic Debye mass [110]

$$m_D^2 = -\frac{g^2}{2\pi^2}\int_0^\infty dp\; p^2\frac{df_{\rm iso}}{dp}\;, \tag{B.6}$$

where $p^2 \equiv \mathbf{p}^2 = p_\perp^2 + p_z^2$. Now integrating by parts

$$\int UV dx = U \int V dx - \int \left[\frac{dU}{dx} \int V dx \right] dx \ , \tag{B.7}$$

with $U = p^2$, $V = \frac{df_{\rm iso}}{dp}$; we get

$$\begin{aligned} m_D^2 &= -\frac{g^2}{2\pi^2} \left[p^2 f_{\rm iso} \right]_0^\infty + \frac{g^2}{\pi^2} \int_0^\infty dp \ p f_{\rm iso} \\ &= 0 + \frac{g^2}{\pi^2} \int_0^\infty dp \ p f_{\rm iso} \\ \Rightarrow m_D^2 &= \frac{g^2}{\pi^2} \int_0^\infty dp \ p f_{\rm iso} \ . \end{aligned} \tag{B.8}$$

Now we can write an expression for the anisotropic screen scale or anisotropic Debye mass as

$$\mu^2 = m_{D,\ \rm aniso}^2 = \frac{g^2}{\pi^2} \int_0^\infty dp \ p f(p) \tag{B.9}$$

From Ref. [106], we can also write down the evolution equation for a general moment in anisotropic hydrodynamics (aHydro) as

$$\begin{aligned} \mathcal{M}_{\rm aHydro}^{nm} &= \int dP \ (p.u)^n (p.z)^{2m} f(p) \\ \Rightarrow \mathcal{M}_{\rm aHydro}^{00} &= \int dP \ f(p) && \text{(B.10)} \\ &= \frac{1}{(2\pi)^3} \int d\Omega \int_0^\infty \frac{dp \ p^2}{p} f(p) && \text{(B.11)} \\ &= \frac{1}{(2\pi)^3} \ 4\pi \int_0^\infty \frac{dp \ p^2}{p} f(p) && \text{(B.12)} \\ &= \frac{1}{2\pi^2} \int_0^\infty dp \ p f(p) \ . && \text{(B.13)} \end{aligned}$$

where in going from Eq. B.10 to Eq. B.11, we use $dP = \frac{d^3p}{(2\pi)^3} \frac{1}{p}$ and $d^3p = d\Omega \ dp \ p^2$, in Eq. B.12 we use $\int d\Omega = 4\pi$.

Inserting Eq. B.13 in Eq. B.9, we get

$$\mu^2 = 2g^2 \mathcal{M}_{\rm aHydro}^{00} \ . \tag{B.14}$$

Similarly, we can write

$$m_D^2 = m_{D,\ \rm eq}^2 = 2g^2 \mathcal{M}_{\rm eq}^{00} \ . \tag{B.15}$$

Hence,

$$\frac{\mu^2}{m_D^2} = \frac{\mathcal{M}^{00}_{\rm aHydro}}{\mathcal{M}^{00}_{\rm eq}} = \overline{\mathcal{M}}^{00}_{\rm aHydro} \ . \tag{B.16}$$

With the help of Eq. B.16 and Eq. B.5, we extract the $f(\xi)$-function as follows

$$\begin{aligned} f(\xi) &= \frac{\mu^2}{m_D^2} \\ &= \overline{\mathcal{M}}^{00}_{\rm aHydro}(\tau) \\ &= \overline{\mathcal{M}}^{00}_{\rm aHydro}(\xi) \\ &= \frac{\sqrt{2}(1+\xi)\arcsin\left(\sqrt{\frac{\xi}{1+\xi}}\right)}{\sqrt{\xi}\sqrt{1+\xi+\frac{(1+\xi)^2\arcsin\left(\sqrt{\frac{\xi}{1+\xi}}\right)}{\sqrt{\xi}}}} , \end{aligned} \tag{B.17}$$

which is Eq. 6.6.

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